in appearance to ordinary ellipticals, but their brightness declines more slowly with distance from the centre, making them look as if they have diffuse halos.

d-: Symbol of dextrorotatory.

d-BLOCK ELEMENTS: The block of elements in Group 3 (IIIB) of the Periodic Table consisting of scandium, yttrium and lanthanum together with the three periods of transition elements: titanium to zinc, zirconium to cadmium and hafnium to mercury. These elements all have two outer s-electrons and have d-electrons in their penultimate shell. Although Lutetium and Lawrencium are in the d-block, they are not considered transition metals but a lanthanide and an actinide, respectively, according to IUPAC. Group 12 (IIB) elements are also in the d-block but are considered post-transition metals if their d-subshell is completely filled. The d-block contains those elements which have partially filled d-orbitals when in their ground states. The elements are notable, in that horizontal trends across the periods do become important. In the s-block and p-block of the periodic table, trends in properties across the periods are generally not observed: the most important trends tend to be vertical (down groups).

D-LINES: Two close lines in the yellow region of the visible spectrum of sodium, having wavelengths of 589.0 and 589.6 nm. They are prominent and easily recognised hence they are used as a standard in spectroscopy. The sodium D-lines are intrinsically strong in relatively cool stars, including the Sun and are also present in the spectra of very distant stars due to absorption by sodium atoms in interstellar space.

d-ORBITALS: A group of five degenerate orbitals, which begin in the third energy level and always have four lobes per energy level. The five d-orbitals include $3d_{xy}$ orbital, $3d_{xz}$ orbital, $3d_{yz}$ orbital, $3d_{x^2-y^2}$ orbital and $3d_{z^2}$ orbital. The d-orbitals are higher in energy than s and p orbitals of the same energy level. The name "d" stands for "diffusion", which describes the characteristics of the orbital spectroscopic lines used to describe the electron configurations. Some others are: p-orbital (Principal), s-orbital (Sharp) and f-orbital (Fundamental).

D-TRANSITION ELEMENTS (TRANSITION METALS): Any of the metallic elements in the Periodic Table with atomic numbers 21-30, 39-40 and 71-80 that have an incomplete inner electron shell and that serve as transitional links between the most and the least electropositive in a series of elements. They are characterised by multiple valences, by high densities, high melting points and low vapour pressures, the ability to form stable complex ions and coloured compounds. They include iron, nickel, cobalt, zinc, copper, gold, mercury, cadmium, tungsten, chromium, manganese and titanium.

D'ALEMBERT'S PRINCIPLE: A principle introduced by French mathematician, Jean le Rond d'Alembert in 1742, which permits the reduction of a problem in dynamics to one in statics. This is accomplished by introducing a fictitious force equal in magnitude to the product of the mass of the body and its acceleration and directed opposite to the acceleration. The result is a condition of kinetic equilibrium. The principle shows that Newton's third law of motion applies to bodies free to move as well as to stationary bodies.

D'ALEMBERT'S RATIO TEST (GENERALIZED RATIO TEST): The test for the convergence of an infinite series by testing whether there is a $k < 1$ such that the ratio of the absolute values of a term to its predecessor is less than k for all terms after some term. If this ratio is always greater than unity, then the series diverges, allowing the limit to be placed in the preceding limit by an appropriate limit inferior or limit superior. The ratio test is given by $\frac{U_{k+1}}{U_k} = p$, if p<1 the series converges, the series diverges when p>1 and inconclusive if p = 1.

D2D (DISK-TO-DISK): A backup solution that copies data from one hard drive such as from the primary hard drive in the computer to another hard drive such as another internal or external hard drive connected to the same computer. The primary disk is one that is being copied from. The secondary disk or backup disk is the one being copied. This system of backup enables a very fast restore capability compared with tape backup.

DABBLE: Bird behaviour in which they forage for food with their beaks or bills from shallow water.

DACITE: An igneous, volcanic rock with high iron content. Dacite lava is most often light grey, but can be dark grey to black, composed mainly of between 63-68% silica (SiO_2). It consists mostly of plagioclase feldspar, pyroxene and amphibole. Dacite generally erupts at temperatures of between 800 - 1000 °C.

DAEMON: In multitasking computer operating systems it refers to a program that is constantly running and triggers actions when it receives certain periodic services such as incoming e-mails by sending the mails to the appropriate mailboxes or in the case of a web server running the HTTP daemon, it sends data to users when they access Web pages.

DAIRY: A place such as a farm, where produce such as butter, cheese, yoghurt and milk are made, kept or sold. **Dairy cattle** are cows kept purposely for producing milk, but not meat.

DAISYWHEEL PRINTER: An impact printing technology used in some electric typewriters, word processors, computer systems and printers. It uses interchangeable pre-formed type elements, each with typically 96 glyphs, to generate high-quality output. The printer rotates the disk to each character and then strikes it onto an ink ribbon to create the character on paper and then moves to the next location. Daisy wheel printers print typewriter-like quality from 10 to 75 characters per second (cps). It was invented in 1969 by David S. Lee at Diablo Data Systems; has now been superseded by dot matrix and laser printers.

DALBERGIA: A large genus of tropical trees in the family, Fabaceae that is native to tropical regions of Central and South America, Africa, Madagascar and southern Asia. The species have pinnate leaves and paniculate flowers and are cultivated commercially for their grained, decorative and fragrant timbers. Some of the species have a high oil content that is used as an essential oil for fragrance cosmetics and for use in medicines. An example is *Dalbergia nigra*, commonly known as Brazilian Rosewood, Bahia Rosewood and Rio Rosewood, occurring in the Brazilian Atlantic Forest, growing to a height of about 40 metres. Due to nitrogen fixing bacteria and fungi in its roots, it can survive in nutrient deficient soils. The wood has a high oil content used as an essential oil for fragrance cosmetics and for use in medicines. Its heavy and strong timber is highly resistant to insect attack and decay and therefore, used in flooring, structural beams and wall panelling, etc. The timber is also highly resonant and used to make musical instruments.

DALTON: 1. Also known as **atomic mass unit**, it is a non-SI unit of mass, used to express the masses of atoms and molecules. It is equivalent to the mass of a hydrogen atom (1.66 x 10^{-24}). It was named after John Dalton (1766-1844), British chemist and physicist. The symbols are u for unified atomic mass unit and Da for Dalton. 2. An ab initio quantum chemistry software program, which is capable of calculating various molecular properties using the Hartree-Fock, MP2, MCSCF and coupled cluster theories. Version 2.0 of Dalton added support to the density functional theory calculations.

DALTON, JOHN (1766-1844): British chemist and physicist, noted for developing the atomic theory upon which modern physical science is founded. He received his early education from his father and at a Quaker school. He was taught mathematics, meteorology and botany by the scientist John Gough. His theory, first advanced in 1803 that matter is composed of atoms of differing weights and combine in simple ratios by weight became very important to modern physical science. In 1808, he listed the atomic weights of a number of known elements relative to the weight of hydrogen, although not very accurate, they form the basis for the modern periodic table of the elements. He discovered the law of partial pressures of mixed gases, that the total pressure exerted by a mixture of gases is equal to the sum of the separate pressures that each of the gases would exert if it alone occupied the whole volume, known as **Dalton's law**. He was the first to prove the validity of the concept that rain is precipitated by a decrease in temperature, not by a change in atmospheric pressure. He is also remembered for his research on colour blindness, a condition called **Daltonism**, of which he and his brother suffered from. Dalton was made a fellow of the Royal Society in 1822 and was awarded the society's gold medal in 1826; and was made an associate of the French Academy of Sciences in 1830, one of the eight foreigners to be given that honour.

DALTON'S ATOMIC THEORY: A theory of chemical combination first stated by British chemist John Dalton (1766 - 1844). According to this theory, matter is composed of indivisible particles called atoms and hence chemical reaction must take place between atoms or group of atoms. Atoms of the same element are similar but differ from atoms of different elements.

DALTON'S LAW (LAW OF PARTIAL PRESSURES): It states that the total pressure of a mixture of ideal gases or vapour is equal to the sum of the partial pressures of its components. It may be expressed as $P = P(a) + P(b) + P(c) + \ldots$; where P is the total pressure of the mixture, $P(a)$ the partial pressure for gas A, $P(b)$ the partial pressure for gas B, etc. This formulation is strictly valid only for mixtures of ideal gases. For real gases, the total pressure is not the sum of the partial pressures, except in the limit of zero pressure because of interactions between the molecules.

DAM: 1. A structure built to hold back water for purposes such as to prevent flooding and to provide water for storage and irrigation. 2. A structure built across a river to control its flow for the purpose of turning turbines to generate hydroelectric power. Such a dam is usually built in a gorge or canyon. An example is the Akosombo Hydroelectric plant in Ghana. 3. The female parent of an animal, especially of four-legged domestic livestock.

DAMPED HARMONICS: A simple harmonic motion in which amplitude diminishes with time as a consequence of a small damping impeding the motion.

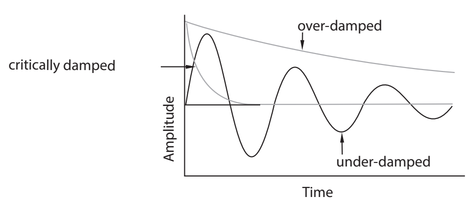
DAMPED NATURAL FREQUENCY: The frequency at which a damped system will oscillate in a free vibration situation.

DAMPED OSCILLATION (DAMPED VIBRATION; DAMPED HARMONIC MOTION): A decrease in the amplitude of an oscillation with time as a result of energy being drained from the oscillating system to overcome frictional or other resistive force. For instance, a pendulum soon comes to rest unless it is supplied with energy from an outside source. In a pendulum clock, energy is supplied through an escapement from a wound spring or a falling mass to compensate for the energy lost through friction.

DAMPED WAVE: A wave with amplitude of oscillation decreasing with time and finally reaching zero.

DAMPER: Any device such as a magnetic needle that eliminates or progressively diminishes vibrations or oscillations. Applications include controlling vibrations in the strings of musical instruments, in swinging pendulums and electric motors to reduce speed pulsations.

DAMPING: The force that causes the amplitude of an oscillation device such as a swinging pendulum or a plucked violin string to decrease as a result of energy being drained from the oscillating system to overcome frictional or other resistive forces. The main damping forces are frictional, electrical or magnetic. A mechanism may be deliberately damped to facilitate reading an oscillating needle or to slow down an oscillating balance. Types of damping include the following: (i) **critical damping**, in which oscillation is stopped in the shortest possible time; (ii) **overdamped system**, in which the system returns or exponentially decays to equilibrium without oscillating; (iii) **underdamped system**, in which the system oscillates at reduced frequency with the amplitude gradually decreasing to zero.



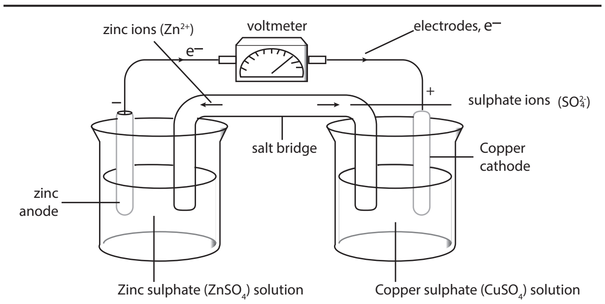
Various degrees of damping

DAMPING OFF: A common, fatal fungal disease that attacks seedlings, grown under very damp conditions, weakening the stems right at the soil level. Infected seedlings usually die. It is caused by various fungi, which attack and rot the collar of the seedling, causing it to fall. There is no cure, but prevention includes starting seeds in a sterile soil mix, using clean containers and tools and providing good air circulation.

DANCE OF THE BEES: A form of communication performed by honeybee workers on returning to the hive after a successful foraging expedition. It is a celebrated example of communication in animals, first investigated by Karl von Frisch (1886–1982). For example the waggle dance (a term used in beekeeping and ethology for a particular figure-eight dance of the honey bee), which is characterised by tail-wagging movements, indicates the direction of a food source at a distance of more than 100 metres. Other workers, sensing vibrations from the dance, follow the instructions to find the food source. By performing this dance, successful foragers can share with others information about the direction and distance to patches of flowers yielding nectar and pollen, to water sources or to new housing locations. It is, therefore, a mechanism whereby successful foragers can recruit other bees in their colony to good locations for collecting various resources.

DANDELION (*Taraxacum officinale*): A herbaceous perennial plant in the family, Compositae (Asteraceae), considered as weed and grows on lawns, roadsides, water banks and other areas which have moist soils. Dandelion develops from unbranched taproots and produces many stems which are about 0.4 metres tall, with sparsely short hairs on them. The leaves, about 0.45 metres long, are petiolated, oblanceolate, oblong or obovate in shape and are either unwinged or narrowly winged. The margins of the leaves are shallowy lobed to deeply lobed and usually lacerate or serrated. The plant has milky latex. The flowers are bisexual and ligulate with yellow heads and are borne on hollow stalks. The flower heads lack receptacle bracts. The florets produce fruits called cypselae which are olive-green or olive-brown to straw-coloured to greyish. The fruits are oblanceoloid in shape. The silky puppus produced by the fruits forms a parachute making it easier to be dispersed by wing since it is light and dry. Dandelions are eaten fresh as salad or used as medicine.

DANIELL CELL (GRAVITY CELL; CROWFOOT CELL): A type of disposable primary voltaic cell invented in 1836 by John Frederic Daniell, a British chemist and meteorologist. It contains a positive copper electrode immersed in a solution of copper sulfate and a negative zinc electrode immersed in dilute sulfuric acid or zinc sulfate solution. The Daniell cell was a great improvement over the voltaic pile used in the early days of battery development. Its theoretical voltage is 1.1 volts and the chemical reaction is $Zn_{(s)} + Cu^{2+}_{(aq)} \rightarrow Zn^{2+} + Cu_{(s)}$.



The Daniell cell

DANIELL INTEGRATION: A type of integration approach to Lebesgue integral using extensions of positive linear functionals.

DAPHNIA: A genus of crustaceans belonging to the class Branchiopoda and order Cladocera (water fleas), which reproduce by parthenogenesis (without mating and the unfertilised egg develops into an adult). Their species have a transparent carapace and a protruding head with a pair of highly branched antennae for swimming and a single median compound eye. The five pairs of thoracic appendages form an efficient filter-feeding mechanism.

DARBOUX INTEGRAL: The limit of a Darboux sums as the lengths of the subintervals tends to zero. It was named after the Czech analyst Bernhard Bolzano (1781-1848).

DARBOUX PROPERTY: In Bolzano's theorem, also known as intermediate value theorem, the intermediate value $f(x)$ and $f(y)$ for at least one argument between x and y value property, which derivatives also possess by virtue of the mean-value theorem.

DARBOUX SUM: An upper sum or lower sum for an integral.

DARCY: The centimetre–gram–second (cgs) unit of permeability, named after Henry Darcy (1803-1858), a French engineer. Its symbol is D. It is used mainly in geology to describe the permeability of rock to substances such as oil, gas or water. The International System of Units (SI) is the currently used system of units, and the square metre (m^2) has replaced darcy, though it is still used in some fields such as petroleum engineering and geology.

DARCY'S LAW: A law in geology that was formulated in 1856 by Henry Darcy (1803-1858), a French engineer that describes the flow of a fluid through a porous medium. It states that the velocity of flow of a liquid through a porous medium due to difference in pressure is proportional to the pressure gradient in the direction of flow. Its equation is given by: $Q = \left(\frac{-kA(P_b - P_b)}{\mu L}\right)$, where Q is the volume of water passing in unit time (m^3/s), k is soil coefficient (m^2) depending on the nature of the soil, A is the cross sectional area of the flow in m^2, L is the length in metres, P_b is the initial pressure in P_a, P_a pressure in P_a and μ is the viscosity in P_a.

DARK ADAPTATION: The ability of the eye to adjust to various levels of darkness and light when an animal moves from a brightly lit environment to a relatively dark one to enable objects to be seen clearly. The eye takes approximately 20 to 30 minutes to fully adapt from bright sunlight to complete darkness and become ten thousand to one million times more sensitive than at full daylight. In this process, the eye's perception of colour changes as well. It takes approximately five minutes for the eye to adapt to bright sunlight from darkness. This is due to cones obtaining more sensitivity when first entering the dark for the first five minutes but the rods take over after five or more minutes.

DARK ENERGY: Energy in the universe associated with the fact that the expansion of the universe is accelerating and the cosmological constant could have a nonzero value. Analysis of data from WMAP indicates that about 70% of the energy of the universe is in the form of dark energy. It is detected by its effect on the rate at which the universe expands. Its nature is not well understood, but it differs from matter in being gravitationally repulsive. One of the explanations for dark energy is that it is an energy density inherent to empty space.

DARK GALAXY: A hypothetical galaxy-sized object containing very few or no stars held together by dark matter. It is composed largely of dark matter, from which the name "dark" is derived from; and may also contain gas and dust. The existence of a dark galaxy has not been confirmed to date but there is some evidence for the existence of such galaxies. It is thought that they should be very common, particularly since theories of large scale structure work much better if the existence of plentiful dark galaxies is assumed.

DARK MATTER: A non-luminous gravitational component of the Universe invoked to explain the internal motions of galaxies and the motions of galaxies within clusters of galaxies. It could take either of two forms: weakly interacting particles (cold dark matter) or high-energy randomly moving particles created soon after the Big Bang (hot dark matter). it does not appear to interact with observable electromagnetic radiation, such as light, and is thus invisible (or 'dark') to the entire electromagnetic spectrum, making it extremely difficult to detect using usual astronomical equipment.

DARK PERIOD: The period considered to be critical in the responses of plants to changes in day length. It is believed that such responses, which include the onset of flowering, are determined by the length of the period of darkness that occurs between two periods of light.

DARK REACTIONS (LIGHT INDEPENDENT REACTIONS): Chemical reactions that can occur in the dark, which convert carbon dioxide and other compounds into glucose. The process is usually associated with light, such as the dark reactions of photosynthesis, where carbon dioxide is reduced to carbohydrate in a metabolic pathway. There are three phases to the light-independent reactions, collectively called the Calvin cycle: carbon fixation, reduction reactions and ribulose 1,5-bisphosphate (RuBP) regeneration. These reactions occur in the stroma, the fluid-filled area of a chloroplast outside of the thylakoid membranes. These reactions take the products of the light-dependent reactions and perform further chemical processes on them. Despite its name, this process occurs only when light is available.

DARK-GROUND ILLUMINATION: A microscopical technique used to examine living cells or microorganisms. In dark-ground illumination, direct light is prevented from forming an image by illuminating the specimen from the side so only diffracted light from the specimen passes through into the objective. The image of the specimen then appears luminous against a completely dark background.

DARMSTADTIUM: A radioactive transactinide with the symbol Ds and atomic number 110. It has several isotopes, with the most stable being 281Ds, with a half-life of about 1.6 minutes. It can be produced by bombarding a plutonium target with sulfur nuclei or by bombarding a lead target with nickel nuclei. It is the eighteenth of the synthetic transuranium elements and the eighth element in the 6d shell. The various isotopes of element 110 have been reliably documented by the observation of at least two atoms with half-lives in the range of milliseconds, a time too short for the application of chemical methods, verifying its position in the periodic system. Its chemical properties probably resemble those of platinum. Element 110 should be a heavy homologue of the elements platinum, palladium and nickel (group 10).

DART (ARROWHEAD): A special kind of quadrilateral that has one reflex angle.

DARWIN, CHARLES (1809-82): British naturalist, who studied medicine at the University of Edinburgh and biology at Cambridge. He was a naturalist on "HMS Beagle", which went on a long scientific survey expedition to South America and the South Seas (1831-36). His zoological and geological discoveries on the voyage resulted in numerous important publications and formed the basis of his theories of evolution. He laid the foundation of modern evolutionary theory with his concept of the development of all forms of life through the slow-working process of natural selection. He published a book, "The Origin of Species (1859)" on the subject.

DARWIN'S FINCHES (GALAPAGOS FINCHES; GEOSPIZINAE): The 14 species of Passerine birds (finches) now placed in the tanager family rather than the true finch family, which are unique to the Galapagos Islands, off the shores of Ecuador in South America that Charles Darwin studied during his journey on the "HMS Beagle". The birds are all about the same size (10–20 cm). The most important differences between species are in the size and shape of their beaks and the beaks are highly adapted to exploitation of different food sources. The birds are all dull-coloured. These and other scientific observations formed the basis of the formulation of the theory of origin of species. The term, "Darwin's Finches" was first applied by Percy Lowe in 1936 and popularised in 1947 by David Lack in his book "Darwin's Finches".

DARWINIAN FITNESS: 1. A relative measure of the contribution of the reproductive success of an organism or individual to the gene pool of the next generation. It describes how successful an organism has been at passing on its genes. The more likely the organism or individual is able to survive and live longer to reproduce, the higher the fitness of that individual. It is depended on and applied in kin selection because it considers the role relatives play when evaluating genetic fitness of a given organism. A parent shares half of its genes with each progeny, so a gene that promotes parental altruism is favoured by natural selection. 2. A biological condition in which a competing variant is increasing in frequency relative to other competing variants in a population. 3. The capability of the body to distribute inhaled oxygen to muscle tissue during increased physical effort. It ensures efficient respiration and eliminates oxygen debt cutting down on production of lactic acid that results in muscular cramps.

DARWINISM: The theory of evolution proposed by Charles Darwin in 1850, which postulated that present day species have evolved from simpler ancestral types through the process of natural selection, acting on the variability found within populations. It suggests that mutation or a change in genetic material caused changes in the characteristics inherited by organisms. Darwin developed the concept that evolution is brought about by the interplay of three principles: variation (present in all forms of life), heredity (the force that transmits similar organic form from one generation to another) and the struggle for existence (which determines the variations that will be advantageous in a given environment, thus altering the species through selective reproduction). Present knowledge of the genetic basis of inheritance has contributed to scientists' understanding of the mechanisms behind Darwin's ideas, in a theory known as **neo-Darwinism**.

DASHPOT: A mechanical damping device which depends on the fact that when a body moves through a fluid medium, viscous forces are set up which dampen the motion of the body.

DATA: 1. Facts, figures and symbols, especially as stored in computers or collection of numbers, characters, images and other outputs from devices that collect information that are unprocessed or raw unprocessed facts as distinct from information to which a measuring or interpretation has been applied. It can be the basis of graphs, images or observations of a set of variables. Data is often viewed as the lowest level of abstraction from which information and knowledge are derived. 2. In biology, data is thought of as documented information or evidence of any kind that lends itself to biological interpretation. Data may be quantitative or qualitative.

DATA ACCESS: A user's ability to access or retrieve data stored within a database or other repository. Users, who have data access can store, retrieve, move or manipulate stored data, which can be stored on a wide range of hard drives and external devices.

DATA ACCESS LAYER (DAL): A data access layer in computer software is a layer of a computer program, which provides simplified access to data stored in persistent storage of some kind, such as an entity-relational database. It can be used to hide access of complexity.

DATA CAPTURE: Collecting information for computer processing and analysis.

DATA CENTRE (DATA FARM; SERVER FARM): A facility that houses various equipment such as networked computers, servers, switches routers, data storage devices for the remote storage, processing or distribution of large amounts of data. The term may also refer to a facility used for the storage, management and dissemination of data and information organised around a particular body of knowledge or pertaining to a particular business.

DATA CHANNEL: A path between parts of devices and a digital computer for the transfer of data between them.

DATA COMMUNICATIONS EQUIPMENT (DCE): In computing, it is a modem that is connected to the computer and used to establish, maintain and terminate communication network sessions between a data source and its destination. DCE is connected to the data terminal equipment (DTE) and data transmission circuit (DTC) to convert transmission signals.

DATA COMPRESSION: In computing, any of the formats used to reduce the number of bits needed to represent data so that it takes up less computer memory and can be transmitted more rapidly and economically via telecommunications systems. The two most commonly used file formats for compressing pictures on the World Wide Web are JPEG (Joint Photographic Experts Group) and Graphics Interchange Format (GIF), because pictures compressed in these formats take up relatively small amount of space and help greatly reduce the file size. **Compression techniques** can be divided into two main types: lossy and non-lossy. With non-lossy compression, there is no loss in the quality of the data. With lossy compression, the compression is greater but the quality of the data is reduced; when the data is uncompressed it will be slightly different from before it was compressed. Lossy compressed is used chiefly for images, video and music.

DATA DICTIONARY: A dictionary that describes the structure of data in database.

DATA ENTRY: 1. In computing, entering information into a computer or data-recording system using an electronic or mechanical device. It includes keyboard entry, scanning and voice recognition. 2. Data entry is a job where an employee inputs data into a computer from data captured on a paper or other non-electronic forms of data. Nowadays, online data entry jobs available require that the employee enters the data into an online database.

DATA ENCRYPTION: The process of making data unreadable by unauthorised people for transmission over a public network. Encrypted data is generated using an encryption program such as Pretty Good Privacy (PGP), encryption machine or a simple encryption key to turn plaintext into a coded equivalent called ciphertext. At the receiving end, the ciphertext is decrypted by the correct decryption key into plaintext.

DATA FIELD (FIELD): A specific item of data, which is usually part of a record, which in turn is part of a file. It is a place where data can be stored.

DATA FLOW DIAGRAM (DFD): A diagram which depicts the flow of information or data within an information system.

DATA INTEGRITY: Relating to the accuracy and consistency of data.

DATA LOGGING: In computing, it is the process of collecting and recording a sequence of data or values of a system, network or IT environment for later processing and analysis. For example, the level in a water-storage tank might be automatically logged every hour over a seven-day period, so that a computer can produce an analysis of water use. Usually the logging is done electronically by a datalogger that is battery-powered equipped with an internal microprocessor, data storage and sensors connected to the computer via an interface.

DATA MIGRATION: The process of transferring data between computers, storage systems or formats. It is a key consideration for any system implementation, upgrade or consolidation. Data migration is categorised as storage migration, database migration, application migration and business process migration. The reasons of data migration includes server or storage equipment replacements, maintenance or upgrades, application migration, website consolidation, disaster recovery, and data centre relocation.

DATA MILE: A unit of distance used in radar technology, which equals exactly 1828.8 metres.

DATA MINING: Searching data warehouses for patterns and related information; or searching through very large amounts of data for useful information.

DATA MODELING: A structured set of techniques for defining and recording business information requirements. It is a depiction of the user's view of the data needs of the organisation in a consistent and rigorous fashion. The data model eventually serves as the basis for translation to computer system databases.

DATA PACKET: A small amount of computer data for transmission over a network. Most data communications sent over a network is sent in small packets. Anytime a file such as e-mail message or HTML file is sent over an internet, the Transmission Control Protocol (TCP) layer of TCP/IP divides the file into blocks of efficient size for routing. Each of these packets is separately numbered and includes the Internet address of the destination. For example, an Ethernet packet can be from 64 to 1,518 bytes in length.

DATA PROCESSING: The use of computers for manipulating data to produce information such as analysis of scientific data, stock control, payroll and dealing with orders.

DATA PROTECTION: Safe-guarding of information stored on a computer or other storage devices to ensure privacy. In computer science, it is the process of minimising the amount of data that requires storage.

DATA RECOVERY: The process of retrieving or restoring accidentally deleted or erased files, corrupted or damaged data to a desktop, laptop, CDs, digital photo memory cards, server or external storage system from a backup or inaccessible storage data. Data may be damaged through accidents, natural disasters, power surges and defective electronics. Some software platforms allow end users to restore lost files themselves, while restoration of a corrupted database from a tape backup is a more complicated process that requires specialist's assistance. Data recovery can also be provided as service. The best data recovery method is to back up data on another storage device either on the same computer, a network server or the Internet.

DATA REDUCTION: The process of minimising the amount of data that requires storage.

DATA SECURITY: In computing, security measures taken to prevent the loss, corruption and misuse of data, whether accidental or deliberate to computers, databases and websites. Some of the measures include ensuring that only authorised personnel gain entry to a computer system or file and carrying out regular procedures such as storing and backing up data in an external storage medium to enable files to be retrieved or recreated in the event of loss, theft or damage. Examples of data security technologies include encryption using mathematical algorithms to scramble data into unreadable text; backups, data masking, authentication and data erasure.

DATA STREAM: 1. All information (data and control commands) sent over a data link usually in a single read or write operation. 2. A continuous stream of data elements being transmitted or intended for transmission, in character or binary-digit form, using a defined format.

DATA TAMPERING (TAMPERING): A term used to describe any type of unauthorised modification of data.

DATA TERMINATOR (FLAG VALUE; ROGUE VALUE; SENTINEL VALUE; SIGNAL VALUE; TRIP VALUE): In computing, it is a special value in the context of an algorithm that is used to mark the end of a list of input data items. The computer must be able to detect that the data terminator is different from the input data in some way, for example, a negative number such as -1 might be used as the sentinel value as the computation will never encounter that value as a legitimate processing output; otherwise the presence of such values would prematurely signal the end of the data. A lot of old programs used data terminators and many programmers used 00 as a year data terminator, leading to problems in the year 2000.

DATA TRANSFER RATE (BIT RATE): The number of data transmitted per second between one device to another in a given time. It is often measured in bits per second (bps) or bytes per second (Bps). Because the volume of transfer is usually large, metric prefixes such as kilo (K), mega (M), giga (G) and tera (T) are often added. For example, USB 2 has a maximum data transfer rate of 480 Mbps (Megabits per second) or 60MB per second (megabytes per second), whilst USB 3.0 also known as SuperSpeed USB has a maximum data transfer rate of 5 Gbps (gigabits per second) or 640 MBps (megabytes per second)

DATA TRANSMISSION: The physical transfer of data points in or between different computer systems.

DATA TYPE: In computer science, it is a classification identifying one of various types of data, such as floating-point, integer or Boolean, stating the possible values for that type, the operations that can be done on that type and the way the values of that type are stored.

DATABASE: Information consisting of a file or set of files that can be broken down into records, each of which consists of one or more fields. It is usually stored in a computer in such a way that it can be extracted under a number of different categories of headings or a structured collection of data.

DATABASE MANAGEMENT SYSTEM (DBMS): A software system that uses a standard method of cataloguing, retrieving and running queries on data. There are many different types of DBMSs, which include small systems that run on personal computers to huge systems that run on mainframes. Examples of database applications are computerized library systems, automated teller machines and flight reservation systems.

DATING: The science of determining the age of geological structures, rocks and fossils and placing them in the context of biological time.

DATING TECHNIQUES: The methods of estimating the true age of rocks, palaeontological specimens, archaeological sites, etc. For example, stratigraphy is used to establish the succession of fossils.

DATIVE BOND (COORDINATION BOND; DIPOLAR BOND; SEMIPOLAR BOND): A covalent bond formed between two atoms, in which one of the atoms provides both electrons that form the bond. An example is the N → B bond in $H_3N \rightarrow BH_3$. Dative bonds are distinguished by their significant polarity, lesser strength and greater length. The distinctive feature of dative bonds is that their minimum-energy rupture in the gas phase or in inert solvent follows the heterolytic bond cleavage path.

DAUGHTER: The offspring of an organism or animal. In the case of mammals, it is usually the female offspring.

DAUGHTER CELL: Either of the two cells that result from the reproductive division of one cell during mitosis or meiosis.

DAUGHTER ELEMENT (DAUGHTER PRODUCT; DECAY PRODUCT; DAUGHTER ISOTOPE; DAUGHTER NUCLIDE): Any of the elements produced when an atom divides by nuclear fission. Radioactive decay often involves a sequence of steps called a decay chain. For example, U-238 decays to Th-234, which decays to Pa-234, which continues to decay until Pb-206, which is stable is achieved. In this example, Th-234 is the daughter of the parent U-238 and Pa-234 is the granddaughter of U-238.

DAUGHTER ION: An electrically charged product of reaction of a particular parent ion. The reaction need not necessarily involve fragmentation. It could, for example, involve a change in the number of charges carried. Thus, all fragment ions are daughter ions, but not all daughter ions are necessarily fragment ions.

DAUGHTERBOARD (DAUGHTER CARD; PIGGYBACK BOARD; MEZZANINE BOARD; RISER CARD): In computing, it is a small printed circuit board that connects directly into a motherboard to give it new and added capabilities, such as increased memory.

DAVISSON-GERMER EXPERIMENT: An experiment conducted by American physicists, Clinton Davisson and Lester Germer in 1927, which confirmed the de Broglie hypothesis that particles of matter such as electrons have wave properties. The experiments showed that, like electromagnetic waves, electrons can produce interference patterns and be diffracted by crystals; that electrons can show wavelike behaviour and that all matter can show wavelike behaviour since electrons are one of the subatomic particles that make up all matter. This demonstration of wave-particle duality was important historically in the establishment of quantum mechanics and of the Schrödinger equation.

DAVY LAMP: An oil-burning miner's safety lamp invented by Sir Humphry Davy in 1816, when investigating methane explosions in coal mines. It has a metal gauze surrounding the flame, which cools the hot gases by conduction and prevents ignition of gas outside the gauze. If methane is present it burns within the gauze cage. Lamps of this type are still used for testing for gas.

DAVY, SIR HUMPHRY: British chemist (1778–1829), who studied gases and later discovered the anaesthetic properties of nitrous oxide (dinitroge oxide) at the Pneumatic Institute in Bristol. He isolated potassium and sodium by electrolysis at the Royal Institution, London. He also prepared barium, boron, calcium and strontium as well as proving that chlorine and iodine are elements. He invented the Davy lamp in 1816.

DAY: The time it takes Earth to spin once around on its axis relative to some external reference such as the Sun, known as the **solar day**; and a star, known as the **sidereal day**. **Solar day** is equal to 24 hours; and **sidereal day** is equal to 23 hours, 56 minutes, 4.1 seconds.

DAY-NEUTRAL PLANT: A plant in which flowering or other such activity can occur irrespective of the day-length. This is because they have limited access to light and have adapted to this environment. Examples are maize, rice and cucumber. Other environmental factors that modify an organism's responses include temperature and nutrition.

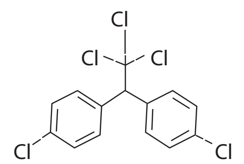
DBASE: A family of microcomputer programs used for manipulating large quantities of data.

DCE: 1. In computing, it is an acronym for **Data Communication Equipment**, a modem that is connected to the computer and used to establish, maintain and terminate communication network sessions between a data source and its destination. DCE is connected to the data terminal equipment (DTE) and data transmission circuit (DTC) to convert transmission signals. 2. An acronym for **Distributed Computing Environment**, a suite of technology services software developed by The Open Group for setting up and managing computing and data exchange in a system of distributed computers.

DCOM: In computing, it is an acronym for **Distributed Component Object Model**, the network version of Microsoft's Component Object Model, which provides a set of interfaces allowing clients and servers to communicate within the same computer.

DDE: In computing, it is the acronym for **Dynamic Data Exchange**, a form of interprocess communication used in Microsoft Windows that uses shared memory as a common exchange area to provide exchange of commands and data between two applications.

DDT (DICHLORODIPHENYLTRICHLOROETHANE; 1,1,1-TRICHLORO-2,2-DI(4-CHLOROPHENYL)ETHANE): A colourless organic crystalline compound, with the molecular formula $C_{14}H_9Cl_5$, density 0.99 g/cm³ (solid), melting point 108.5 °C (381.65 K) and decomposes at boiling point of 260 °C (533 K). It is made by the reaction of trichloromethanol with chlorobenzene. It is the best known of a number of chlorine-containing pesticides used extensively in agriculture. It has also been used to successfully control mosquitoes. It is, however, harmful to animal life as some animals can bioaccumulate it in their bodies. It is chemically stable and persists in the environment and in living tissues for several years. Its use has been banned worldwide.



Chemical structure of Dichlorodiphenyltrichloroethane (DDT)

DE BROGLIE EQUATION (DE BROGLIE RELATION): The mathematical expression of the relationship in which the de Broglie wave associated with the motion of a free particle of matter and the electromagnetic wave in a vacuum associated with a photon, has a wavelength equal to Planck's constant divided by the particle's momentum and a frequency equal to the particle's energy divided by Planck's constant. In other words, the de Broglie relations show that the wavelength is inversely proportional to the momentum of a particle and the frequency is directly proportional to the particle's kinetic energy. Mathematically, de Broglie wavelength or wavelength of matter is given by $\lambda = \frac{h}{mv}$, where h is the Planck constant, m is the mass of the particle and v its velocity. It was suggested by the French physicist, Louis De Broglie, (1892-1987) that all particles (not just photons) have both particle and wave properties (theory of wave-particle). A matter wave or de Broglie wave is the wave (wave-particle duality) of matter.

DE BROGLIE WAVELENGTH: The wavelength associated with a moving particle and is calculated from the relation $\lambda = \frac{h}{mv}$, where h is the Planck constant, m is the mass of the particle and v its velocity.

DE BROGLIE, LOUIS-VICTOR PIERRE RAYMOND (1892-1987): French physicist, who was educated at the University of Paris and taught for 34 years at the Sorbonne in Paris and also professor of theoretical physics at the University of Paris in 1928. He is best known for his 1924 theory of wave-particle duality, which reconciled the corpuscular and wave theories of light and proved important in quantum theory; for which he was awarded the 1929 Nobel Prize. He was elected to the Academy of Sciences in 1933 and to the French Academy in 1943. He was made adviser to the French Atomic Energy Commission in 1945.

DE DICTO: In logic, relating to the expression of a belief, possibility, etc., rather than to the entities referred to. For example, "the number of planets is the number of satellites of the Sun", is necessary de dicto, as its truth is not dependent upon which number in fact that is.

DE MOIVRE, ABRAHAM (1667-1754): A French mathematician, who settled in England, he used trigonometry in complex numbers. He is also noted for his works in probability. His Doctrine of Change was discovered in 1718 and he examined numerous problems and developed a number of principles, such as the notation of independent events and the product law.

DE MOIVRE'S THEOREM: The *n*th power of the quantity $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$ for any integer n, where i is the square root of -1. Mathematical induction can be used to prove De Moivre's Theorem is valid for any positive integer n. The formula is valid when n = 1 and n = 2.

DE MORGAN, AUGUSTUS (1806-1871): English mathematician, who was responsible for the complete geometrical interpretation of the square root of minus one. He also discovered symbolic approach to algebra and well-noted for his algebraic laws concerning sets.

DE MORGAN'S LAWS: Two laws in Boolean algebra and set theory; which state that AND and OR or union and intersection, are dual. They are used to simplify the design of electronic circuits. The laws can be expressed in Boolean logic as: NOT (a AND b) = NOT a OR NOT b; NOT (a OR b) = NOT a AND NOT b. In terms of set notations, it is given by: $(P \cup Q)' = P' \cap Q'$ and (P Q)' = P' Q'.

DE NOVO PATHWAY: Any metabolic pathway in which a biomolecule is synthesised from single precursor molecules. Acetic acid pathway is an example of a biochemical pathway that starts from elementary substrates (acetic acid and acetyl Co A) and ends in the synthesis of terpenoid compounds such as cholesterol, citronella and camphor.

DE NOVO SYNTHESIS: The synthesis of complex molecules from simple molecules such as sugars or amino acids, as opposed to their being recycled after partial degradation. For example, nucleotides are not needed in the diet as they can be constructed from small precursor

D

molecules such as formate and aspartate. De novo synthesis also refers to DNA replication.

DE RE: In logic, it is a Latin phrase, literally meaning "about the thing", it relates to an actual individual mentioned, rather than to the expression of a belief, possibility, etc., concerning it. For example, "the number of planets is a perfect square", is necessary *de re* since its truth is dependent upon which number that in fact is.

DEACETYLATION: The removal of an acetyl group (-$COCH_3$) from a molecule. It is an important reaction in several chemical pathways, including the Krebs cycle and in the reversible condensation of chromatin. On the other hand, acetylation (ethanoylation) describes a reaction that introduces an acetyl functional group into a chemical compound.

DEACON PROCESS: A former process for making chlorine by oxidising hydrogen chloride in air at 450 °C (723 K), using a copper chloride catalyst. Invented by Henry Deacon in 1874, the process was based on the reaction: $4HCl + O_2 \rightarrow 2Cl_2 + 2H_2O$. It was a secondary process used during the manufacture of alkalis (the initial end product was sodium carbonate) by the Leblanc process. Hydrogen chloride gas was converted to chlorine gas which was then used to manufacture a commercially valuable bleaching powder and at the same time the emission of waste hydrochloric acid was curtailed. Most chlorine today is produced by using electrolytic processes. However, recent developments with new catalysts based on Ruthenium(IV) oxide were developed by Sumitomo.

DEAD LOAD: The weight of components of a structure such as a bridge or building that remain relatively constant over time and comprise, for example, the weight of beams, walls and roof. The term can also apply to the weight of a vehicle, excluding the weight of passengers or goods.

DEADLY NIGHTSHADE (BELLADONNA): Common names for ***Atropa belladonna***, a poisonous perennial herbaceous plant in the nightshade family, Solanaceae that grows up to 1.5 metres. It has purplish-red bell-shaped flowers which bear black berries. All parts of the plant contain tropane alkaloids. Atropine, a poisonous crystalline alkaloid, with the chemical formula $C_{17}H_{23}NO_3$ is isolated from its leaves and root. Medicinally, the uses of atropine include to check secretions and spasms, to relieve pain or dizziness and as a cardiac and respiratory stimulant.

DEACTIVATION: 1. A reduction in the reactivity of a molecule as a result of the introduction or removal of some functional group. 2. A reduction in the rate of a reaction or the activity of a biological system as a result of changes in the medium, temperature or pressure or the addition of an external agent such as a strong acid or base, a concentrated inorganic salt, an organic solvent such as alcohol or chloroform or heat. If proteins in a living cell are denatured, this results in disruption of cell activity and may cause cell death.

DEAFNESS: It is the impairment to hearing. In other words, it is a partial or complete hearing loss. It is caused by many factors such as noise, drugs and toxins. Deafness can also result from inherited disorders.

DEAMINATION: The removal of the amino group (-NH_2) from a compound or the removal of an amine group from a molecule. In biological systems, this happens when too much protein has been taken in. The amino group is converted to ammonia, uric acid or urea, depending on the type of animal and excreted in the urine. Enzymatic deamination occurs in the liver and is important in amino acid metabolism.

DEATH: The termination of the biological functions that define a living organism or the cessation of all life functions or the end of life so that the molecules and structures associated with life become irreversibly disorganised and disintegrated. The word refers both to a particular process and to the condition that results.

DEATH OF A STAR: The final collapse of a star resulting in the formation of a black hole.

DEATH RATE (OF A POPULATION): The ratio of total deaths to total population, especially in the human population in a specified community or area over a specified period of time, measured as the number of deaths in one year per 1000 of the population.

DEBEAKING: The partial removal of the beak of poultry, especially layer hens and turkeys; it may also be performed on quails and ducks.

DEBIAN: An open-source and free operating system (OS) that uses the Linux operating system and other numerous software packages. Currently, it has more than 42550 software packages. It was packaged by a group of individuals participating in the Debian Project.

DEBRIS AVALANCHE: A moving mass of rock, soil and snow that occurs when the flank of a mountain or volcano collapses and slides down slope. As the moving debris rushes down a volcano and into river valleys, it incorporates water, snow, trees, bridges, buildings and anything else in the way. Debris avalanches may travel several kilometres before coming to rest or they may transform into more water-rich lahars, which travel many tens of kilometres downstream.

DEBT-FOR-NATURE SWAP: An agreement under which a proportion of a country's debts are written off in exchange for a commitment by the debtor country to undertake projects for environmental protection. It is a financial transaction in which a percentage of a developing nation's foreign debt is written-off in exchange for local investments in conservation measures. The concept was first conceived by Thomas Lovejoy of the World Wildlife Fund in 1984 as an opportunity to deal with the problems of developing countries' indebtedness and its consequent deleterious effect on the environment.

DEBUGGING: Finding and removing errors from a computer program or system. A debugger or debugging tool is a computer program that is used to test and debug other programs called the **target program**. Some debuggers offer two modes of operation: full or partial simulation, to limit this impact.

DEBYE: A centimetre–gram–second system (cgs) of unit for electric dipole moment in the electrostatic system with the symbol D, used to express dipole moment of molecules. It is equivalent to 1×10^{-18} statcoulomb-centimetre.The International System of Units (SI) is the currently used system of units, and the coulomb-metre (C. m) has replaced the debye. The debye was named in honour of the Dutch-American theoretical physicist, Peter J. W. Debye, (1884-1966), who received the 1936 Nobel Prize in Chemistry for his studies in molecular structure, dipole movements and the diffraction of x-rays and electrons in gases.

DEBYE LENGTH: Characteristic length in plasma corresponding to the distance within which an electron will be influenced by the electric field of a given positive ion.

DEBYE MODEL: A method for estimating the phonon contribution to the specific in a solid, developed by Peter Debye in 1912. It treats the vibrations of the atomic lattice (heat) as phonons in a box, in contrast to the Einstein model, which treats the solid as many individual, non-interacting quantum harmonic oscillators. It correctly predicts that the specific heat capacity of solids is proportional to T^3, where T is the thermodynamic temperature. Like the Einstein model, it also recovers the Dulong-Petit law at high temperatures, but due to simplifying assumptions, its accuracy is suspect at intermediate temperatures.

DEBYE SHEATH: The region of strong electric field in front of a material surface in contact with plasma. Its characteristic thickness is the Debye length and it is caused by Debye shielding of the negative surface charge resulting from electrons flowing to the surface much faster (initially) than the ions. The lost electrons leave behind a region of net positive charge, which gradually diminishes the strength of the electric field over the Debye length.

DEBYE TEMPERATURE: A parameter θ that appears in the Debye theory of specific heat, defined by $k\theta = h\nu$, where k is the Boltzmann constant, h is Planck's constant and ν the Debye frequency.

DEBYE THEORY OF SPECIFIC HEAT (CHARACTERISTIC TEMPERATURE): A theory of the specific heat capacity of solids in which it was assumed that the specific heat is a consequence of the lattice of the solid. Debye postulated that there is a continuous range of frequencies that cuts off at a maximum frequency, which is characteristic of a particular solid. It concludes that the specific heat capacity of solids is proportional to T^3, where T is the thermodynamic temperature. The temperature θ arising in the computation of the Debye specific heat, is defined by $k\theta = h\nu$, where k is the Boltzmann constant, h is Planck's constant and ν is the Debye frequency.

DEBYE WALLER FACTOR (B FACTOR; TEMPERATURE FACTOR): A quantity that characterises the effect of lattice vibrations on the scattering intensity in x-ray diffraction by crystals. Named after Peter Debye and Ivar Waller, it is used in condensed matter physics to describe the attenuation of x-ray scattering or coherent neutron scattering caused by thermal motion. Often, "Debye-Waller factor" is used as a generic term that comprises the Lamb-Mössbauer factor of incoherent neutron scattering and Mössbauer spectroscopy.

DEBYE-HÜCKEL ONSAGER THEORY: A theory providing quantitative results for the conductivity of ions in dilute solutions of strong electrolytes, which enables the Kohlrausch equation to be derived.

DEBYE-HÜCKEL THEORY: A theory that explains variations from the ideal behaviour of electrolytes in terms of inter-ionic interaction and assumes that electrolytes in solution are completely dissociated into charged ions with each ion surrounded by an ionic atmosphere of oppositely charged ions, whose behaviour retards the movement of the ion when current is passed through the medium. Developed and named after Peter Debye and Erich Hückel, it provides one way to obtain activity coefficients (γ). Activities, rather than concentrations, are needed in many chemical calculations because solutions that contain ionic solutes do not behave ideally even at very low concentrations. The activity is proportional to the concentration by a factor known as the **activity coefficient** and takes into account the interaction energy of ions in the solution.

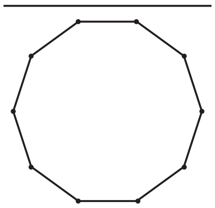
DEBYE–SCHERRER METHOD: A method of x-ray diffraction in which a sample, consisting of a powder stuck to a thin fibre or contained in a thin-walled silica tube, is rotated in a monochromatic beam of x-rays and the diffraction pattern is recorded on a cylindrical film whose axis is parallel to the axis of rotation of the sample. Since the powder consists of very small crystals of the material in all possible orientations, the diffraction pattern is a series of concentric circles. This type of pattern allows the unit cell to be found with great precision.

DEBYE, PETER JOSEPH WILHELM (1884-1966): A Dutch-American theoretical physicist who received the 1936 Nobel Prize in chemistry for his work in molecular structure, dipole movements and the diffraction of X rays and electrons in gases. He was born at Maastricht in the Netherlands. He was professor at the universities of Zürich, Utrecht, Göttingen, Leipzig and Berlin. In 1940, he went to the United States and served as professor of chemistry at Cornell University (1940-52). In 1912 Debye modified the specific-heat theory put forth by Albert Einstein by calculating the probability of any frequency of molecular vibration up to a maximum frequency.

DECA- (DEKA-): A prefix used in the metric system to denote a factor of ten. Its symbol is Da. For example, decametre denotes 10 metres.

DECADE: A group or set of ten or any sequence of ten successive terms.

DECADE NUMERALS: A numeral among 10, 20, 30, 40, 50, 60, 70, 80, or 90 that can be expressed as one, two, three, four, five, six, seven, eight, or nine tens and zero ones, respectively.



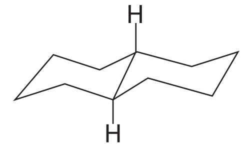
Decagon (regular polygon)

DECAGON: A ten-sided polygon. Each interior angle of a regular decagon is 144°.

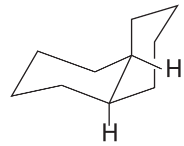
DECAHEDRON: A geometrical three-dimensional figure with ten sides and angles.

DECAHYDRATE: A crystalline hydrate containing ten moles of water per mole of compound. An example is sodium carbonate, Na_2CO_3 or $Na_2CO_3.10H_2O$, also known as washing soda or soda ash.

DECALIN (DECAHYDRONAPHTHALENE; BICYCLO[4.4.0] DECANE): A liquid bicyclic hydrocarbon with the molecular formula $C_{10}H_{18}$. It is a colourless liquid with an aromatic odour, used as a solvent for many resins or fuel additive. It is the saturated analogue of naphthalene and can be prepared from it by hydrogenation in a fused state in the presence of a catalyst. It easily forms explosive organic peroxides upon storage in the presence of air. It occurs in two isomeric forms, cis and trans as shown in the diagrams.



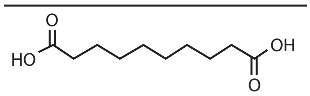
Trans-decalin



Cis-decalin

DECANE: An isomeric liquid alkane, with the molecular formula $C_{10}H_{22}$, density 730 g/cm³, melting point -29 °C (242.7 K) and boiling point 174 °C (447 K). It is an alkane hydrocarbon with a surface tension of 0.0238 $N \cdot m^{-1}$ having 75 isomers, all of which are flammable liquids. It is a component of petrol and like other alkanes, it is nonpolar and therefore insoluble in polar liquids such as water.

DECANEDIOIC ACID (1,8-OCTANEDICARBOXYLIC ACID; SEPATIC ACID): A naturally-occurring white crystalline dicarboxylic acid, with the molecular formula $C_{10}H_{18}O_4$, density 1.209 g/cm³ (at 25 °C, 298 K), melting point 131-134.5 °C (404-407.65 K) and boiling point 294.4 °C (567.55 K) at 100 mmHg. It is obtained from castor oil. In its pure state it is a white flake or powdered crystal. The product is described as non-hazardous, though in its powdered form can be prone to flash ignition which is a typical risk in handling fine organic powders. Industrially, its homologues such as azelaic acid can be used in plasticizers, lubricants, hydraulic fluids, cosmetics, candles, etc., as well as an intermediate for aromatics, antiseptics and painting materials.



Chemical structure of Decanedioic Acid

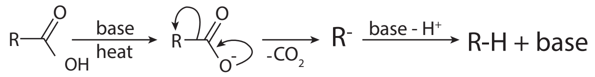
DECANOIC ACID (CAPRIC ACID; *n*-CAPRIC ACID; *n*-DECANOIC ACID; DECYLIC ACID; *n*-DECYLIC ACID): A white crystalline straight-chain saturated carboxylic acid, with the chemical formula $C_{10}H_{20}O_2$, density 0.893 g/cm³, melting point 31.6 °C (304.8 K) and boiling point 269 °C (542 K). Salts and esters of decanoic acid are called **decanoates**. It occurs naturally in coconut oil and palm kernel oil, as well as in the milk of various mammals and to a lesser extent in other animal fats. It can be prepared from oxidation of primary alcohol decanol, by using chromium trioxide (CrO_3) oxidant under acidic conditions in an acetone solvent. It is used industrially in the manufacture of perfumes, lubricants, greases, rubber, dyes, plastics, food additives/flavourings and pharmaceuticals as well as in organic synthesis.

DECANTATION: The process of separating a liquid from a settled solid suspension or from a heavier immiscible liquid by carefully pouring it into a different container. A centrifuge may be useful in successfully decanting a solution. It causes the precipitate to be forced to the bottom of the container and if the force is high enough, the precipitate may form a hard single solid. The liquid can then be poured away more easily, as the precipitate will likely remain in its compressed form. For example, the oil and water extracted from olives may be decanted to obtain the olive oil. A mixture of kerosene and water can also be separated through decantation and a mixture of sand and sugar solution can also be separated through decantation.

DECAPODA: An order of higher crustaceans distributed worldwide in mainly marine habitats containing nearly 15,000 species in around 2,700 genera, with about 3,300 fossil species. They belong to the class Malacostraca and are mostly scavengers. Examples are shrimps, prawns crabs, lobsters and crayfish.

DECARBOXYLASE: Any enzyme that catalyses the removal of carbon dioxide from a substrate. An example is the enzyme pyruvate decarboxylase, which catalyses the formation of acetaldehyde from pyruvate, a reaction in the alcoholic fermentation of glucose. Decarboxylases have a tightly bound coenzyme, often biotin or thiamin pyrophosphate, which acts as an intermediate carboxyl carrier.

DECARBOXYLATION: The removal of carbon dioxide from a molecule or chemical compound, usually with hydrogen replacing it. Decarboxylation is an important reaction in many biochemical processes such as the Krebs cycle.



Decarboxylation

DECATRON (DEKATRON): A neon-filled tube with a central anode surrounded by ten cathodes and associated transfer electrodes. They were used in computers, calculators and other counting-related products during the 1950s and 1960s. It is now a generic trademark or brand name used by the British Ericsson Telephones Limited (ETL). They generally have a maximum input frequency in the high kilohertz range (100 kHz -1 MHz). These frequencies are obtained in hydrogen-filled fast dekatrons. However, dekatrons filled with inert gas are inherently more stable and have a longer life, but their counting frequency is limited to 10 kHz; 1–2 kHz is more common.

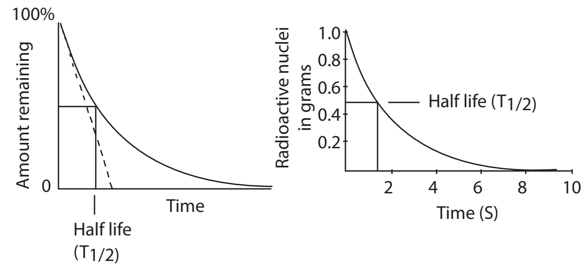
DECAY: 1. Also known as **decompose** or **rot**, it is the process by which tissues of a dead organism break down into simpler forms of matter or the process by which decomposers use the organic matter of dead organisms as a source of energy. Bodies of living organisms begin to decompose shortly after death. It may be divided into two stages by the types of end products. The first stage is characterised by the formation of liquid materials, flesh or plant matter begins to decompose. The second stage is limited to the production of vapours. Besides the two stages mentioned, historically the progression of decomposition of the flesh of dead organisms has been viewed also as four phases: fresh (autolysis); bloat (putrefaction); decay (putrefaction and carnivores) and dry (diagenesis). The process is essential for new growth and development of living organisms because it recycles the finite matter that occupies physical space in the biome. The science which studies such decomposition generally is called **taphonomy**. 2. **Radioactive decay** is the process by which an unstable atomic nucleus loses energy by emitting ionising particles or radiation. In other words, it is the spontaneous disintegration of the nucleus of an unstable nuclei to produce more stable nuclide with emission of radiation and energy. The emission is spontaneous in that the nucleus decays without collision with another particle. The SI unit of activity is the becquerel (Bq) and it is defined as one transformation or decay per second. 3. Reduction in the intensity of a signal or a wave with distance.

DECAY ACCELERATING FACTOR (DAF): Plasma protein that regulates complement cascade by blocking the formation of the C3bBb complex (the C3 convertase of the alternate pathway). It is a protein of most blood as well as endothelial and epithelial cells.

DECAY CHAIN (DECAY SERIES; RADIOACTIVE CHAIN): A series of nuclides in which each member transforms into the next through nuclear decay until a stable nuclide has been formed. For example, the decay chain that begins with uranium-238 (U-238) ends in lead-206 (Pb-206), after forming isotopes, such as uranium-234 (U-234), thorium-230 (Th-230), radium-226 (Ra-226) and radon-222 (Rn-222).

DECAY CONSTANT: The probability per second that an individual nucleus will decay. Its symbol is λ and expressed mathematically as $\lambda = \frac{\ln 2}{t_{1/2}} = \frac{0.693}{t_{1/2}}$ and its unit is per second (s^{-1}). It is the proportionality constant relating the activity of a radioactive substance to the number of decaying particles. It is the probability of a nuclear decay within a time interval divided by that time interval. The decay constant is inversely proportional to the radioactive half-life.

DECAY CURVE: A graph showing the relative amount of radioactive substance remaining after any time interval.



A radioactive decay curve of an element

DECAY DISTANCE: The distance through which waves travel after leaving the generating area.

DECAY ENERGY: The energy released by a radioactive decay. Radioactive decay is the process in which an unstable atomic nucleus loses energy by emitting ionising particles and radiation. This decay or loss of energy, results in an atom of one type, called the **parent nuclide** transforming to an atom of a different type, called the **daughter nuclide**.

DECAY PRODUCT: A radionuclide produced as a result of a parent radionuclide's decay. The decay products may be derived from their immediate predecessor or through several other decays in a decay chain series. An example is radon decay products, where Ra-222 decays to products including Polonium 218 through alpha decay and then Lead-214, also through alpha decay.

DECAY RATE: The reduction over time of the amplitude of a decaying quantity or the time rate of disintegration of radioactive material, generally accompanied by emission of particles or gamma radiation.

DECAY SCHEME: A graphical representation of the energy levels of the members of a decay chain showing the way along which nuclear decay occurs.

DECAY SEQUENCE: A plot of neutron number (N) against proton number (Z) for nuclides, belonging to a particular radioactive series.

DECELERATION: To reduce speed or to retard the motion of an object.

DECI-: A prefix used in the metric system to denote one tenth. It is denoted by d.

DECIBEL: A non-SI unit of measurement used to compare two power levels usually applied to sound or electrical signals and voltages; or a unit for measuring the relative intensities of sound or the relative amounts of acoustic or electric power. Since it requires about a tenfold increase in power for a sound to register twice as loud to the human ear, a logarithmic scale is useful for comparing sound intensity. Thus, the threshold of human hearing (absolute silence) is assigned the value of 0 dB and each increase of 10 dB corresponds to a tenfold increase in intensity and a doubling in loudness. A related unit is bel, which is equal to 10 dB.

DECIDABLE: 1. In logic, of a statement, able to be shown either to be true or to be false. In other words, an algorithm that can and will return a Boolean true or false value, instead of looping indefinitely. Also, of a well-formed formula of a given theory, either provable or having a provable negation, in the given theory; equivalently, either the formula or its negation is a theorem. For example, the continuum has been shown not to be decidable. Logical systems such as propositional logic are decidable if membership in their set of logically valid formulas or theorems can be effectively determined. A theory (set of sentences closed under logical consequence) in a fixed logical system is decidable if there is an effective method for determining whether arbitrary formulas are included in the theory. 2. Of a formal theory, having the property that each theorem is recursive; equivalently, by Church's thesis, that it is possible to determine by a mechanistic procedure whether or not any given well-formed formula is a theorem. For example, sentential calculus is decidable, but predicate calculus is not.

DECIDUOUS: 1. A term used to describe plants which shed all their leaves at the end of each growing season or during a dry season to reduce transpiration. 2. Referring to animal parts such as antlers in deer, which are dropped or shed and replaced each year; or deciduous teeth, also known as baby teeth, in some mammals including human children.

DECIDUOUS TEETH (MILK TEETH; BABY TEETH; TEMPORARY TEETH; PRIMARY TEETH): The first of two sets of teeth of mammals. These teeth are smaller and fewer in numbers than the permanent teeth that replace them. They develop during the embryonic stage of development and become visible in the mouth, during infancy.

DECILE: Any of the nine values that divides the total number of items in a frequency distribution into ten groups, each containing an equal number of items. In descriptive statistics, it is any of the nine values that divides the sorted data into ten equal parts, so that each part represents one tenth (10th) of the sample or population. For example, the 1st decile cuts off the lowest 10% of data, i. e., the 10th percentile; the 5th decile cuts off lowest 50% of data, i. e., the 50th percentile; the 2nd quartile or median and the 9th decile cuts off lowest 90% of data, i.e., the 90th percentile. The formula for calculating Decile is $L_1 + [\frac{(N.d) - \Sigma f_1}{f_d}]c$, where L_1 = lower boundary of the class of the decile, N = total number of observations, $d = \frac{1}{10}$ of the first decile, Σf_1 = sum of frequencies lower than the class of the decile, f_d = frequency of the decile class and c = the decile class size or class width.

DECIMAL (DECIMAL NUMBERS): Relating to base 10 and counted in units of ten. Each digit in a decimal number represents a multiple exponent of ten. The position of a digit determines what exponents of ten is to be multiplied by. For example, $43425 = 4\times10^4 + 3\times10^3 + 4\times10^2 + 2\times10 + 5\times1.$

DECIMAL FRACTION: A fraction in which the denominator is any higher power of 10. Thus $\frac{3}{10}$, $\frac{51}{100}$ and $\frac{23}{1000}$ are decimal fractions and are normally expressed as 0.3, 0.51, 0.023 or any number written in the form of an integer followed by a decimal point then a (possibly infinite) string of digits.

DECIMAL NOTATION (BASE TEN; DENARY STANDARD NOTATION): The normal way in which whole numbers, integers, and decimals are written. It is base-ten place-value numeration. The decimal notation system has ten as its base and it is the numerical base most widely used in the world. In some contexts, especially mathematics education, the term decimal can refer specifically to decimal fractions. In such cases, a single decimal fraction is called a "decimal" and non-fractional numbers, even when written in base 10, are not considered "decimals".

DECIMAL NUMBER SYSTEM (DENARY NUMBER SYSTEM; BASE TEN): Relating to a number system that has 10 as its base. It is the most commonly used number system to the base ten.

DECIMAL PLACE: The number of digits to the right of a decimal point required to specify a real number to a certain accuracy. For example, the number 4.893302 to three decimal places is 4.893. To approximate a number to a given number of n decimal places, n + 1 digit to the right of the decimal point is considered. The $(n + 1)^{th}$ digit is rounded up to 1, if it is greater than or equals to 5, 1 is added to the n^{th} digit. Thus, the numbers 239.705 and 239.706 are both 239.71 to two decimal places, whereas the number 239.704 to two decimal places is 239.70.

DECIMAL POINT: The period in a decimal number separating the integer part from the fractional part.

DECIMAL TREE DIAGRAM: The visual means of representing a series of events for which there are two possible outcomes at each stage.

DECIMETRE: A unit of length in the metric system, which is equal to one tenth of a metre. It is denoted by dm.

DECISION PROBLEM: In logic, the problem, for any given theory, of whether there exists a decision procedure for questions such as whether an expression is a theorem. Also, it determines whether some potential solution to a question is actually a solution or not. It is a question in some formal system which demands a 'yes' or 'no' answer, depending on the values of some input parameters.

DECISION PROCEDURE: In logic, an algorithm for solving a decision problem or to determine whether any given well-formed formula of a formal theory is a theorem. For example, a decision procedure for the decision problem "given two numbers x and y, does x evenly divide y?" would give the steps for determining whether x evenly divides y, given x and y. Not all consistent theories possess such a procedure; for example, truth-tables provide a decision procedure for sentential calculus, but predicate calculus can be proved to have no such procedure. Those for which one does exist are called **decidable**.

DECISION TABLE: In computing, it is a table that shows decisions and the criteria that triggers them and what action should be followed. It is used in programs to control flow and processing. It is often used as an aid in systems design.

DECISION THEORY: The branch of statistics and game theory that deals with decision-making under conditions of uncertainty in such a way as to maximise expected utility.

DECISION TREE: A chart that is used to exhibit the process of decision analysis problem. Various notations are used to represent the different types of nodes or vertices.

DECISION VARIABLE: The variables to be found in linear programming and other constrained optimisation programmes.

DECLARATIVE PROGRAMMING: In computing, it is a program that describes the logical structure of a problem, but not how to solve it. Examples of declarative programming languages are logic programming language PROLOG; functional programming languages, such as Haskell and ML; dataflow programming languages, such as Lucid; and descriptive markup languages, such as HTML and SGML.

DECLINATION: 1. The angle measured in a clockwise direction from the positive direction of the x-axis to a given line. The declination of a line has opposite sign to the conventional anticlockwise measurement of angles. 2. The angle between the magnetic meridian and the geographic meridian at a point on the surface of the Earth. 3. The angular distance of a celestial body north (positive) or south (negative) of the celestial equator. 4. The coordinate on the celestial sphere (imaginary sphere surrounding the Earth) that corresponds to latitude on the Earth's surface. In astronomy, declination (abbrev. dec or **δ**) is one of the two coordinates of the equatorial coordinate system, the other being either right ascension or hour angle. It is comparable to geographic latitude, but projected onto the celestial sphere. Declination is measured in degrees north and south of the celestial equator. Points north of the celestial equator have positive declinations, while those to the south have negative declinations. An object on the celestial equator has a declination of 0°, at the celestial North Pole it has a declination of +90° and at the celestial South Pole it has a declination of -90°. Any unit of angle can be used for declination, but it is often expressed in degrees, minutes and seconds of arc.

DECOCTION: A solution made by boiling material, such as substances in water, followed by filtration. It is a method of extraction by boiling of dissolved chemicals or herbal or plant material, which may include stems, roots, bark and rhizomes. It involves first mashing and then boiling in water to extract oils, volatile organic compounds and other chemical substances. Decoctions differ from most tea infusions or tisanes, in that they are usually boiled.

DECODER: 1. An electronic circuit used to select one of several possible data pathways in computing. 2. In navigation, it is an electronic circuit designed to respond only to certain signals and to reject others. 3. A box containing electronic circuit connected to a television set to translate encoded signals into its original signals to view on the screen.

DECODING: To convert data that had been encoded into its original form.

DECOHERENCE (DEPHASING): A process by which a quantum mechanical state of a system is altered by the interaction between the system and its environment in a thermodynamically irreversible way to exhibit probabilistically additive behaviour. It is used to clarify discussions of the foundations of quantum mechanics.

DECOLOURISATION OF BROMINE WATER: A reaction used to detect the presence of an unsaturated aliphatic carbon-carbon multiple bonds. Bromine water is prepared by adding bromine to water to form a brown solution.

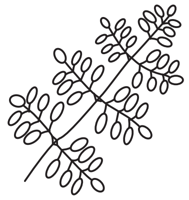
DECOMMISSIONING: A general term applied to situations, where nuclear plant or any other plant containing sources of ionising radiations comes to the end of its useful life. Decommissioning is then required in order to dismantle the plant and recover radioactive materials for either reuse, recycling or disposal. Decommissioning uses an extremely controlled approach and on larger plants such as nuclear power stations, will require full safety cases and regulator attention.

DECOMPOSER (SAPROTROPH; SAPROBE): Any organism such as bacteria and fungi that breaks down dead animals and plants or an organism that causes the decay of materials from dead animals and plants. They are important because they return essential nutrients to the soil. Like herbivores and predators, decomposers are heterotrophic, meaning that they use organic substrates to get their energy, carbon and nutrients for growth and development. **Decompose**, also known as **rot** is to decay into soil-building materials and nutrients or to become bad or rotten. **Decompose** is also to separate, for example, light into its various parts or to breakdown into separate smaller parts. When **decompose** is used with regards to chemical compounds, it is to break down or cause to break down into component elements or simpler constituents; many chemicals decompose rapidly under high temperature.

DECOMPOSITION: 1. In mathematics, it is the breakdown of a quantity, object or expression into a number of simpler components, for example, 48 = 2 x 2 x 2 x 2 x 3 is the decomposition of 48 into its prime factors. Some other examples include the expression of a polynomial function as the product of its factors; a number as a product of primes; a set as a canonical union of appropriate disjoint subsets; and a vector as the resultant of orthogonal components. 2. The breakdown of a single phase into two or more phases or the process, in which a chemical compound is reduced to its component substances. Chemical decomposition, analysis or breakdown is the separation of a chemical compound into elements or simpler compounds. There are broadly three types of decomposition reactions: thermal, electrolytic and catalytic. It is sometimes defined as the exact opposite of a chemical synthesis. Chemical decomposition is exploited in several analytical techniques, notably mass spectrometry, traditional gravimetric analysis and thermogravimetric analysis. Chemical decomposition is often an undesired chemical reaction. The stability that a chemical compound ordinarily has is eventually limited when exposed to extreme environmental conditions like heat, radiation, humidity or the acidity of a solvent. 3. In biology, decomposition or **rotting** is the disintegration of dead organisms either by chemical reduction or by the action of a decomposer or the process by which tissues of a dead organism break down into simpler forms of matter. The process is essential for new growth and development of living organisms, because it recycles the finite matter that occupies physical space in the biome.

DECOMPOSITION REACTION: The breakdown of a compound into two or more components. Chemical decomposition, analysis or breakdown is the separation of a chemical compound into elements or simpler compounds. It is sometimes defined as the exact opposite of a chemical synthesis. A broader definition of the term decomposition also includes the breakdown of one phase into two or more phases. There are broadly three types of decomposition reactions: thermal, electrolytic and catalytic. The generalised reaction formula for chemical decomposition is: AB → A + B, for example, the electrolysis of water to gaseous hydrogen and oxygen: $2H_2O_{(l)} \rightarrow 2H_{2(g)} + O_{2(g)}$.

DECOMPOUND: 1. Compounded or consisting of things or parts that are already compound. 2. In botany, of a leaf, having divisions that are themselves compound or a compound leaf having leaflets consisting of several distinct parts, as shown in diagram. 3. In chemistry, a rarely used synonym of decompose.



Decompound leaves

DECOMPRESSION SICKNESS (DIVERS' DISEASE; BENDS; CAISSON DISEASE): A condition that is caused by a sudden and substantial change in atmospheric pressure. It is mostly experienced by deep-sea divers who surface too quickly, causing nitrogen bubbles to be dissolved into the bloodstream and tissues because of a sudden drop in the surrounding pressure. It is characterised by breathing difficulties, joint and muscle pain, cramps, numbness, nausea and paralysis.

DECONFINEMENT TEMPERATURE: The hypothesis that free quarks can never be seen in isolation. It is a result of quantum chromodynamics, in which the property of asymptotic freedom means that the interactions between quarks get weaker as the distance between them gets smaller and tends to zero as the distance between them tends to zero. Conversely, the attractive interactions between quarks get stronger as the distance between them gets greater and the quark-confinement hypothesis is that the quarks cannot escape from one another. It is possible that at very high temperatures, such as those in the early universe, quarks may become free. The temperature at which this occurs is called the **deconfinement temperature**. The hypothesis of quark confinement has not been proved theoretically in a conclusive way, although there is a lot of evidence for it.

DECONGESTANT: A drug used to relieve the stuffy nose and rhinitis associated with common cold or hay fever. Decongestants reduce swelling and secretion of affected mucous membrane and lessen symptoms, but prolonged use leads to chronic changes in mucous membranes.

DECONTAMINATION: A general expression applied to the removal of undesirable or harmful substances such as noxious chemicals, harmful bacteria or other organisms or radioactive material from exposed individuals, rooms, buildings or the exterior environment.

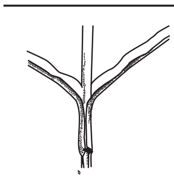
DECONTAMINATION FACTOR: A measure of the effectiveness of decontamination, measured as the ratio of the original contamination to the remaining radiation, after decontamination; or the ratio of initial specific radioactivity to final specific radioactivity resulting from a separation process.

DECREASING FUNCTION: The function $f(x)$ is decreasing if $f(c) < f(d)$ for any interval c and d, when $c < d$. A function can also be described as decreasing on the interval (c,d) if $f(x)<0$ for $c < x < d$; $f(x) = 6 - 2x$ is decreasing because $f'(x) = -2$.

DECREPITATION: Unusual features that accompany the thermal decomposition of some crystals such as lead(II) nitrate. When these are heated they split and crackle and may jump out of the test tube before they decompose. In geophysics, it is the breaking up of mineral substances when exposed to heat, usually accompanied by a crackling noise.

DECUMBENT: A term used to describe a stem that lies along theground. Decumbent stems often have an upturned tip.

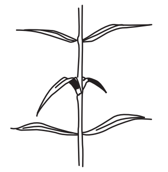
DECURRENT: Of leaf bases, extending down the stem beyond their point of insertion, forming a wing on the stem, as shown in the diagram. An example is found in the leaves of the common mullein (*Verbascum thapsus*). The term is also used of the gills of



Decurrent leaves

basidio-mycete fungi that run down the stem, as seen, for example, in the genus *Clitocybe*.

DECUSSATE: 1. Intersecting in the form of an X. 2. In botany, arranged in pairs each at right angles to the next pair above or below, such as those found in leaves, especially in many Hebe species.



Decussate leaves

DEDEKIND COMPLETE (COMPLETE; ORDER COMPLETE): Of a partially ordered set, such that every subset has a supremum and infimum, for example, the reals are not complete but the interval [0,1] is. A partially ordered set is complete for every countable non-empty set S of F that has an upper bound in F, a supremum exits in F and for countable non-empty set S of F, which has a lower bound in F, the supremum exits in F, an infimum exits in F.

DEDEKIND CUT: A set partition of the rational numbers into two nonempty subsets S_1 and S_2 such that all members of S_1 are less than those of S_2 and such that S_1 has no greatest member. It is used to define the real numbers in terms of pairs of sequences of rationals, as opposed to using a metric completion. For example, √2 is defined as the pair $\langle \{x : x^2 > 2\}, \{x : x^2 < 2\} \rangle$.

DEDEKIND DOMAIN (DEDEKIND RING): In abstract algebra, it is an integral domain in which every non-zero proper ideal factors into a product of prime ideals.

DEDEKIND–PEANO AXIOMS (PEANO'S AXIOMS): A set of axioms that describe the set of natural numbers, , including zero by defining a first member and a unique successor to each member, by excluding loops and by providing for mathematical induction. The axioms include the following: (i) There exists a natural number zero (0). (ii) For each n , there exists a natural number that is the successor of n, denoted by n'. (iii) Zero is not the successor of any natural number: for each , $n' \neq 0$. (iv) For all natural numbers n, m , if $n' = m'$, then $n = m$. (iii) Given any predicate P on the natural numbers, if $P(0)$ is true and $P(k)$ implies $P(k')$ for any k , then $P(n)$ is true for all n . It is possible, therefore, to regard the natural numbers as excluding zero while the whole numbers include zero.

DEDEKIND, JULIUS WILHELM RICHARD (1831-1916): German mathematician, who studied under Gauss and Dirichlet. He discovered the Dedekind cut to define the real numbers. He also provided important support for Georg Cantor's set theory and made important contributions to abstract algebra, particularly ring theory in the sense of Dedekind ring and the definition of ideal and algebraic number theory.

DEDICATED COMPUTER: A microprocessor built into another device such as washing machines, digital watches and cars for the purpose of performing a single, data processing function.

DEDICATED SERVER: A single computer in a network reserved for serving the needs of the network such as hosting applications and/or services, as well as for data storage and backup services. For example, some networks require that one computer be set aside to manage communications between all the other computers. In the Web hosting business, a dedicated server is typically a rented service and exclusive use of a computer that includes a Web server, related software and connection to the Internet, housed in the Web hosting company's premises.

DEDIFFERENTIATION: The loss of the specialised features and reversal to a meristematic state of a differentiated cell. Dedifferentiation often occurs immediately prior to further reorganisational changes as in the production of a secondary meristem. The process may also be brought about by wounding or stimulation by growth substances. It is thought to involve the removal of inhibitors that normally prevent the expression of the complete genetic complement.

DEDUCTION: 1. The act or process of deducting or subtracting. 2. A series of logical steps in which a conclusion is reached directly from a set of initial statements. Alternatively, it is the conclusion drawn from a set of initial statements known as the premises.

DEDUCTION THEOREM: In logic, the property of many formal systems that if an argument is valid, then its premises imply its conclusion; that is, it is true that if the premises of the given argument are true then so is its conclusion. Formally, the conditional statement is derived from an argument by taking the conjunction of the premises as antecedent and the conclusion as consequent.

DEDUCTIVE REASONING: Observance of an event occurring on a repeated basis that leads one to believe that a certain probability is attached to the occurrence of that event. For example, if there are a red ball and a blue ball in a bag and each colour of ball is drawn one-half of the time, we come to believe that each colour of ball has a one-half probability of being drawn at any one time.

DEEP FASCIA: A thin, but dense and strong, bluish membrane that lies under the superficial fascia and is attached to it by fibrous strands. The deep fascia clothes the muscles, investing them so closely that it forms a tight sheath around the limb and preserves the contours of the limb.

DEEP FREEZING: A method of preserving food by rapid freezing and storage at -18°C (255 K) or below.

DEEP INELASTIC SCATTERING: A process used to probe the insides of hadrons, particularly the baryons such as protons and neutrons, using electrons, muons and neutrinos. It provided the first convincing evidence of the reality of quarks, which up until that point had been considered by many to be a purely mathematical phenomenon. It is a relatively new process, first attempted in the 1960s and 1970s. It is conceptually similar to Rutherford scattering, but with important differences.

DEEP OCEAN TRENCH (DEEP-SEA TRENCH; OCEANIC TRENCH): A long narrow steep-sided depression in the ocean floor with depths of about 7,300 to over 11,000 metres, that typically forms in locations where one tectonic plate subducts another. These trenches are the deepest parts of the ocean floor. Examples are the Cayman Trench in the Caribbean Sea with a depth of 7,686 metres and the Puerto Rico Trench in the Atlantic Ocean with a depth of 8,605 metres. The deepest trench on Earth is the Challenger Deep of the Mariana Trench in the Pacific near Guam, with a depth of 11,034 metres below sea level.

DEEP SPACE: Any region of space beyond the Moon's orbit; or any region beyond the orbit of Mars.

DEEP ZONE TILLAGE, DZT (VERTICAL TILLAGE): A type of reduced-tillage that relies on the residue of a cover crop to protect the soil surface and help improve soil health over time. When combined with the use of cover crops, it reverses the deterioration of the soil, improves soil drainage, increases soil water and nutrient holding capacity, and allows beneficial soil organisms to thrive.

DEEP-ROOTED PLANTS: Plants having roots that grow deep into the soil. The deepest roots are generally found in deserts, dry arid regions and temperate coniferous forests. The main function of deep roots is to enhance water uptake. Examples of deep-rooted plants are Shepherd's tree (Boscia albitrunca) found in the Kalahari Desert reaching depths of up to 68 metres; Acacia erioloba in the Kalahari Desert (up to 60 metres) and Eucalyptus sp. found in Australia reaching depths of 61 metres.

DEER: A hoofed ruminant mammal such as elk and reindeer of the family Cerridae, the males of which have distinctive solid branched antlers, which are deciduous or shed each year.

DEFAULT: In computing, it is a term used to describe a preset value for some option in a computer program, when none is specified by the user. A **default programme** is an application that opens a file by double-clicking it.

DEFECATION (BOWEL MOVEMENT): The final act of digestion by which organisms eliminate solid, semisolid or liquid waste material or faeces from the digestive tract via the anus. Waves of muscular contraction known as **peristalsis** in the walls of the colon move faecal matter through the digestive tract towards the rectum. Undigested food may also be expelled in this way. This process is called **ingestion** or **diarrhoea**.

D

DEFECT: 1. Also known as **crystal defect** or **lattice defect**, it is a discontinuity in crystal lattice. Crystalline solids have a very regular atomic structure; the local positions of atoms with respect to each other are repeated at the atomic scale. However, most crystalline materials are not perfect: the regular pattern of atomic arrangement is interrupted by crystallographic defects. The defect could be a pair of oppositely charged "holes" in the crystal or a vacant site and an ion which should have occupied that site. 2. The **spherical defect** of a spherical triangle is the difference between the sum of the internal angles of the given triangle and 3π. Also, if a, b and c are the sides of a spherical triangle, then the spherical defect D, is $D = 2\pi - (a + b + c)$.

DEFECT OF THE EYE: A fault in the eye, which causes the formation of an imperfect image. The imperfect image can be improved by wearing the appropriate spectacle length. In **myopia** (**short sightedness**), rays parallel to the axis of the eye are focused in the front of the retina. A diverging lens is, therefore, used to correct this defect. Since the diverging lens brings the rays to a focus on the retina. In **hypermetropia** (**long sightedness**), rays parallel to the axis of the eye are focused behind the retina. A converging lens is, therefore, recommended for an individual with this defect. The converging lens brings the rays to a focus on the retina.

DEFECT STATE: A quantum mechanical state that exists due to the pressure of crystal defect.

DEFERRED APPROACH TO THE LIMIT (RICHARDSON IMPROVEMENT; RICHARDSON EXTRAPOLATION): An extrapolation method for improving the rate of convergence of a sequence, using two computed values, h_{LARGER} and h, of the form $E(h) = \dfrac{F(h_{LARGER}) - F(h)r^n}{1-r^n}$, where r is the ratio of h_{LARGER} to h and F is an O(h”) approximation to some quantity. If the truncation error for F is of higher order $O(h^n)$, then the error in the extrapolation $E(h)$ will be of this order. The process can then be repeated with m replacing n. It is named after Lewis Fry Richardson (1881-1953), English mathematician and physicist.

DEFIBRILLATION: In medicine, it is the use of controlled electrical stimulation to restore the normal rhythm of the heart. Fibrillation may occur in people having some heart diseases with the heart muscle contracting irregularly resulting in extremely rapid, irregular heartbeat. If defibrillation is not done soon, the heart may stop pumping blood to the body, leading to brain damage and/or cardiac arrest. It involves attaching an electronic device to the chest that sends an electric shock to the heart to restore the normal heart rhythm.

DEFICIENCY DISEASE: A disease caused by the lack of an essential nutrient element needed for optimum growth. Examples are scurvy, caused by the lack of vitamin C; kwashiorkor, caused by the lack of protein; and night blindness, caused by the lack of vitamin A. In botany, for example, nitrogen deficiency results in stunting of the plant and chlorosis of older leaves.

DEFICIENT: In biology, it is the lack of enough quantity of a required element or compound such as nutrient, vitamin or water that a plant may need to grow well. For example, in arid and desert regions, water deficiency is the main problem that affects the growth of many plants. In humans, vitamin C deficiency in diet leads to scurvy disease and severe protein and vitamin deficiencies cause kwashiorkor in children.

DEFICIENT NUMBER (DEFECTIVE NUMBER): A positive integer that is larger than the sum of its proper divisors. It is a number n for which $\sigma(n) < 2n$. Here, $\sigma(n)$ is the sum-of-divisors function: the sum of all positive divisors of n, including n itself. The value $2n - \sigma(n)$ is called the **deficiency of** n. The first few deficient numbers are: 1, 2, 3, 4, 5, 6, 7, 8, 9 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27. An example is the number 21 whose divisors are 1, 3, 7 and 21 and their sum is 32. Because 32 is less than 2 × 21, the number 21 is deficient. Its deficiency is 2 × 21 − 32 = 10.

DEFINED MEDIUM: A growth medium whose quantitative and chemical composition is exactly known; suitable for the in vitro cell culture of human or animal cells.

DEFINIENDUM: Latin for a term that is defined or requires a definition. A definition consists of a definiendum and definiens, which is a fragment of text that defines the definiendum. Also, the term at the head of a dictionary entry that is to be defined. 2. In logic, an expression to be defined in terms of another expression previously defined.

DEFINIENS: Latin for a term of which a definition, especially an explicit definition, is given. Definiens defines definiendum.

DEFINITE DESCRIPTION: 1. In logic, an expression capable of having a unique reference; for example, 'the woman in white' or 'Rosemary's baby'. Also, an analogous plural expression; for example, 'the dogs of war'. 2. The **theory of definite descriptions**, is the analysis, proposed by Bertrand Russell, of singular definite descriptions in which a sentence of the form 'the F' is 'G' is said to be equivalent to 'there is one and only one F and it is G'.

DEFINITE GROWTH: In botany, it is a form of growth, where a maximum size is reached beyond which there is no further increase or growth is limited. It is seen in annual and biennial plants and in many plant organs, such as internodes, which do not continue to lengthen throughout the life of the plant. The term is also used for cymose inflorescences, in which growth is terminated by the production of a terminal flower bud.

DEFINITE INTEGRAL: An integral with upper and lower limits that determines the difference in values; used in finding the area under a curve. The definite integral of $g(x)$ is written as $\int_b^a g(x)\mathrm{d}x = G(a) - G(b)$. For example, $\int_1^3 (2x+3)dx$ is a definite integral. To evaluate $\int_1^3 (2x+3)dx$, the solution is

$$\int_1^3 (2x+3)dx = \left[\frac{2x^2}{2} + 3x\right]_1^3 = [x^2 + 3x]_1^3$$

$= [3^2 + 3(3)] - [1^2 + 3(1)] = [9 + 9] - [1 + 3]$
$= 18 - 4 = 14.$

DEFINITION: Abbreviated **def**, **defn** or **df**, it is a precise statement of the meaning of an expression (the definiendum) in terms equivalent to it. An explicit definition is an identity between the definiendum and another expression (the definiens) all the terms of which are already well understood. For example, the empty set can be defined in terms of negation, identity and set membership by the definition
$=_{\text{df}} \{x : x \neq x\}$. This permits the definiendum to be replaced by the definiens wherever it occurs. An implicit or contextual definition is an identity between complex expressions in only one of which the definiendum occurs.

DEFINITIVE METHOD: A method of exceptional scientific status, which is sufficiently accurate to stand alone in the determination of a given property for the certification of a reference material. Such a method must have a firm theoretical foundation so that systematic error is negligible relative to the intended use. Analyte masses (amounts) or concentrations must be measured directly in terms of the base units of measurements or indirectly related through sound theoretical equations. Definitive methods, together with certified reference materials, are primary means for transferring accuracy or establishing traceability.

DEFINITIVE NUCLEUS: In botany, it is the diploid nucleus found in the centre of the embryo sac after the fusion of the two haploid polar nuclei. A male gamete released from the pollen tube may fuse with this to form the triploid primary endosperm nucleus from which the endosperm develops.

DEFINITIVE PLUMAGE: The plumage of a bird attained after the shedding of all previous feathers that do not change significantly in colour or pattern for the rest of the bird's life.

DEFLAGRATION: A type of explosion in which the shock wave arrives before the reaction is complete, because the reaction front moves more slowly than the speed of sound in the medium. It describes subsonic combustion that usually propagates through thermal conductivity (hot burning material heats the next layer of cold material and ignites it). Most fires found in daily life, from flames to explosions, are technically deflagration. Deflagration is different from detonation, which is supersonic and propagates through shock compression.

DEFLATION: In soils, it is the removal of dust, sand and loose rocks from the soil by wind.

DELAYED PLUMAGE MATURATION: A common phenomenon seen in male birds where the definitive colour and pattern of plumage is delayed until after the first potential breeding period.

DEFLECTION: 1. To cause something to turn from its direction of movement. 2. In physics, it is the event, where an object collides and bounces against a plane surface. In such collisions involving a sphere and a plane, the collision angle formed with the surface (the incidental angle) must equal the bounce angle (the accidental angle β): = β. **Magnetic deflection** refers to Lorentz forces acting upon a charged particle moving in a magnetic field. An object's deflective efficiency can never equal or surpass 100%. For example, a mirror will never reflect exactly the same amount of light cast upon it. Also, a ball in free-fall (no forces are acting on it other than gravity), upon hitting the ground, will never bounce back up to the spot, where it first started to descend. This is due to the **Laws of Thermodynamics**, which state that, in every action, some energy being used in that action escapes into the environment.

DEFLECTION ANGLE: The angle by which a light ray is curved by the gravitational field of a massive body. General relativity gives a value twice as large as that which Newtonian physics would provide, assuming that photons have non-zero mass.

DEFLECTION CIRCUIT: A circuit, which alters the path of the electron beam in a cathode-ray tube (CRT) by the use of the deflection yoke's electromagnets. It synchronises the influence exerted by each of the yoke's two magnets and so controls the direction of the beam towards specific parts of the screen. A **horizontal deflection circuit** generates the horizontal deflection pulses fed to the horizontal deflection coils. This makes sawtooth-shaped current flow through the horizontal deflection coil; it is used to control the picture tube electron beam to make horizontal sweeping (scanning) from the left towards the right side of the screen. A **vertical deflection circuit** adjusts the position of an image on the screen of the CRT vertically in order to compensate for an image-position deflection originating from deflection of a beam spot of the CRT or for the purpose of properly correcting the image position as desired.

DEFLECTION COEFFICIENT: Quotient of the voltage to the magnitude of the deflection produced by this voltage.

DEFLEXED: Referring to a structure that is bent sharply downwards.

DEFLOCCULATION: 1. The separation of individual components of compound particles by chemical (using, for example, sodium carbonate) or physical (using, for example, mixers) or both. A **deflocculant** is a chemical that is added to prevent a colloid from coming out of suspension; in this case it is a desired effect. The undesired effect is, for example, an activated sludge basin if the sludge is subjected to high-speed mixing. In such a case, deflocculation can be prevented or reduced by applying gentle mixing, for example, by using submersible propeller mixers that utilise large/wide propeller blades and operate at low rotational speed.

DEFOLIANT: Any chemical such as a herbicide sprayed or dusted on plants to cause its leaves to fall off. An example is Agent Orange, which was used widely by the U.S. military during the Vietnam War from 1961 to 1970.

DEFOLIATION: The loss of leaves from a plant, especially as a result of using a herbicide or because of disease or other stress.

DEFORESTATION: Destruction of forest for timber, fuel, charcoal burning and clearing for agriculture and extractive industries such as mining, without planting new trees to replace those lost or working on a cycle that allows the natural forest to regenerate. Rates of deforestation are particularly high in the tropics, where the poor quality of the soil has led to the practice of routine clear-cutting to make new soil available for agricultural use. It can lead to erosion, drought, loss of biodiversity through extinction of plant and animal species and increased atmospheric carbon dioxide. Many nations have undertaken afforestation or reforestation projects to reverse the effects of deforestation or to increase available timber.

DEFORMATION: A change in the shape of a material or substance, often described as strain, especially one that suggests disfigurement or damage or a change in shape resulting from the application of stress. This can be as a result of tensile (pulling) forces, compressive (pushing) forces, shear, bending or torsion (twisting). **Continuous deformation is a** transformation of the shape or dimensions of a body by stretching but not tearing, resulting from stress. T(p) from A to B is a deformation if there is a continuous function F(p, t), for t between 0 and 1, such that $F(p, 0) = p$ and $F(p, 1)$ = T(p). T is then said to deform A into B.

DEFORMATION GRADIENT: A measure of the extent of deformation caused by the motion of a body.

DEFRAGMENTATION (DEFRAGMENTATION PROGRAM): In computing, it is the process of locating the noncontiguous fragments of data into which a computer file may be divided as it is stored on a hard disk and rearranging the fragments and restoring them into fewer fragments or into the whole file using disk optimisation software such as Microsoft Windows utility. Defragmentation reduces data access time and allows storage to be used more efficiently. Large files may be broken into thousands of fragments, causing the read/write head to move back and forth many times to read the data.

DEGAUSSING: The process of removing the magnetisation of a permanent magnet or other object by withdrawing it slowly from a strong oscillating magnetic field, so that it is repeatedly magnetised in opposite directions but to progressively smaller degrees. Due to magnetic hysteresis it is generally not possible to reduce a magnetic field completely to zero, so degaussing typically induces a very small "known" field referred to as bias. It was named after Carl Friedrich Gauss.

DEGENERACY: 1. The number of degenerate quantum states of a particular orbital, degree of freedom, energy level, etc. 2. In genetics, it is the presence in the genetic code of multiple codons for the same amino acid.

DEGENERACY PRESSURE (DEGENERATE ELECTRON PRESSURE): The pressure in a degenerate gas of fermions caused by the Pauli Exclusion Principle and the Heisenberg uncertainty principle. White dwarfs and neutron stars are supported against collapse under their own gravitational fields by the degeneracy pressure of electrons and neutrons respectively.

DEGENERATE: 1. Denoting energy levels for different quantum states that have the same energy level but a different wave function from another state of the system. For example, the five d-orbitals in an isolated transition-metal atom have the same energy, although they have different spatial arrangements and are thus degenerate. 2. In mathematics, it is a limiting case in which a class of object changes its nature so as to belong to another, usually simpler, class. For example, the point is a degenerate case of the circle as the radius approaches 0 and the circle is a degenerate form of an ellipse as the eccentricity approaches 0. 3. In linear programming problems, if one or more basic variables turn out to be zero, the basic feasible solution is said to degenerate; if all the basic variables are positive, the basic feasible solution is nondegenerate.

DEGENERATE CONFIGURATION: Magnetic field configuration in which the magnetic lines of force close exactly on themselves after passing around the configuration a finite number of times.

DEGENERATE ELECTRON PRESSURE: The pressure produced by closely packed electrons as a result of the Pauli Exclusion Principle. It is what supports white dwarfs against further gravitational collapse.

DEGENERATE MATTER: Matter in which all the available quantum energy states allowed by the Pauli Exclusion Principle have been occupied. Degenerate matter occurs, for example, in stars that have collapsed under their own gravitational pull following the exhaustion of their useable nuclear fusion energy reserves. Electron degenerate matter is found in white dwarfs and baryon degenerate matter is found in neutron stars.

DEGENERATE ORBITALS: Atomic or molecular orbitals that have the same energy or orbitals whose energy levels are equal in the absence of external fields. These are important in physical chemistry because they affect the ways electrons fill atoms (Hund's rule). For example, all the 3p orbitals have same energy level and so do all the 5d orbitals. Degenerate orbitals play an important role in molecular orbital theory.

DEGENERATE REARRANGEMENT: A rearrangement of a molecule in which the product is chemically indistinguishable from the reactant. It is detected by using isotopic labelling.

DEGENERATE SEMICONDUCTOR: A heavily doped semiconductor in which the Fermi level is located in either the valence band or the conduction band causing the material to behave as a metal. A degenerate semiconductor still has far fewer charge carriers than a true metal so that its behaviour is in many ways intermediary between a semiconductor and a metal.

DEGENERATE STAR: A star, such as a white dwarf or neutron star, which is composed of degenerate matter.

DEGENERATE STATES: Quantum states of a system that have the same energy level but a different wave function from another state of the system.

DEGENERATION: Changes in cell tissues or organs due to, for example, a disease that can result in an impairment or loss of function and possibly death or breakdown of the affected part. The term also refers to the reduction in size or complete loss of organs during evolution. For example, the human appendix has undergone this process and performs no function in man. True degeneration is said to occur when there is actual chemical change of the tissue itself and infiltration is said to occur when the change consists of the deposit of abnormal matter in the tissues.

DEGENERATION GAS: A gas in which, because of density, the particle concentration is so high that the Maxwell-Boltzmann distribution does not apply and the behaviour of the gas is governed by quantum statistics.

DEGENERATIVE LEVEL: An energy level of a quantum mechanical system that corresponds to more than one quantum. The number of different states at a particular energy level is called the level's degeneracy and this phenomenon is generally known as quantum degeneracy. The usage comes from the fact that degenerate eigenstates correspond to identical eigenvalues of the Hamiltonian. Since eigenvalues correspond to roots of the characteristic equation, degeneracy here has the same meaning as the common mathematical usage of the word.

DEGLUTITION (SWALLOWING): A reflex action initiated by the presence of food in the pharynx. It is a succession of muscular contractions from up to downward or from the front to backward, propelling food from the mouth toward the stomach. The action is generally initiated at the lips, then proceeds back through the oral cavity and the food is moved automatically along the dorsum of the tongue. When the food is ready for swallowing, it is passed back through the fauces or between the back of the mouth and the pharynx. Once the food is beyond the fauces and in the pharynx, the soft palate closes off the nasopharynx and the hyoid bone and larynx are elevated upward and forward. This action keeps food out of the larynx and dilates the esophageal opening so that the food may be passed quickly toward the stomach by peristaltic contractions. The separation between the voluntary and involuntary characteristics of this wave of contractions is not sharply defined. At birth the process is already well established as a highly coordinated activity (the swallowing reflex).

DEGRADATION: 1. The erosion of the Earth's land surface by wind, ice or water. 2. The breakdown of a chemical compound into simpler compounds or atoms. 3. The process by which the energy available for doing work is irreversibly decreased. Degradation in other words could be **biodegradation**, which is the processes by which organic substances are broken down by living organisms; **chemical degradation**, which is the degradation of chemical compounds; and **environmental degradation**, which in ecology is the damage to the ecosystem and loss of biodiversity.

DEGRADED ENERGY: Energy that is transformed into some form which is transferred to the surroundings as thermal energy and is less available for doing work.

DEGREE: 1. **Degree of an arc** is a unit of measurement for angle or a unit used in numerical geometry and trigonometry, which is the measure of the central angle, subtended by an arc and is equal to 1/360 degree of the circumference of the circle. One degree is divided into 60 minutes (3600 seconds). A full circle is divided into 360 degrees. For example, a right angle is 90 degrees. A degree has the symbol ° and so ninety degrees is written as 90°. Another unit of angle measure is the radian. 2. A division on a temperature scale. 3. **Degree of a term** is the power to which a variable is raised or the exponent of a variable. For example, the degree of $7x^5$ is 5. 4. The highest power to which the derivation of the highest order is raised in a differential equation. 5. In topology, the **degree** of a continuous mapping between two compact oriented manifolds of the same dimension is a number that represents the number of times that the domain manifold wraps around the range manifold under the mapping. The degree is always an integer, but may be positive or negative depending on the orientations.

DEGREE DAY: A measure of heating or cooling, calculated against a standard reference temperature of 18° Celsius to assess the climate and the heating and cooling needs for different locations, or demand for energy needed to cool or heat a building. It is also used to plan the planting of crops and management of pests and pest control timing. There are two types of degree days: cooling and heating degree days. Cooling degree day (CDD) is how hot the temperature is on a given day or over a period of days. It is the number of degrees that the average temperature over a day is above 18 °C, for example, a day with an average temperature of 20 °C counts as 2 cooling degree days (CDD). Heating degree day (HDD) is how cold the temperature is on a given day or over a period of days, calculated against a standard reference temperature of 18° Celsius. For example, a day with a mean temperature of 16 °C has 2 heating degree days (HDD).

DEGREE OF A POLYNOMIAL: The degree of a polynomial in one variable is the greatest exponent of its variable. For example, the degree of the polynomial $x^3 + 4x^2 - 6x - 7$ is 3, because the highest exponent is 3.

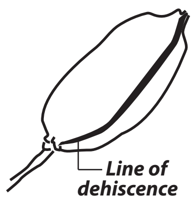
DEGREE OF ACCURACY: The level of approximation or closeness of measurements of a quantity to its actual or true value.

DEGREE OF FREEDOM: 1. The least number of independent variables required to specify the state of a system in the phase rule. In this sense, a gas has two degrees of freedom: temperature and pressure. 2. The number of independent parameters required to specify the configuration of a system. In kinetic theory it represents the number of independent ways in which an atom or molecule can take up energy. 3. An independent variable in a statistical measure or frequency distribution or the number of values in the final calculation of a statistic that are free to vary. 4. In mechanics, it is the independent displacements and/or rotations that specify the orientation of the body or system.

DEGREE OF POLYMERISATION: The number of monomeric units in a macromolecule or oligomer molecule, a block or a chain.

DEGREE OF REACTION: Extent of reaction divided by the maximum extent of reaction.

DEHISCENCE: The spontaneous and often violent split-opening of a fruit, along certain lines to release and disperse the seed or pollens, for example, capsules and pods. Structures that open in this way are said to be **dehiscent**. Structures, such as fruit, that do not open are called **indehiscent**.



DEHORNING: The removal of the horn of an animal by chemical means using caustic or by electrocautery (to remove a body tissue by burning or searing using electric current) when very young or by surgical amputation with a dehorner or saw; most commonly performed early in an animal's life. Animals are often dehorned for economic and safety reasons, for example, to prevent injury to humans or other animals and to prevent them from entangling in objects, preventing feeding. **Disbudding** is the method of destroying the horn buds before they grow into horns.

DEHUSKING: The removal of the outer covering of some fruits and seeds such as coconut, maize, rice, etc.

DEHYDRATING AGENT (DEHYDRATOR): A substance, which is used to remove water from other substances or for drying food. Such substances always have an attraction for water. They may be used in desiccators. Some examples are sodium sulfate and silica gel.

DEHYDRATION: 1. The removal of water or moisture from a substance. 2. An elimination reaction in which molecules lose both a hydroxyl group (OH) and a hydrogen atom (H) bonded to an adjacent atom or a chemical reaction in which a compound loses hydrogen and oxygen in the ratio 2:1. For instance, ethanol passed over hot pumice undergoes dehydration to ethene: $CH_3CH_2OH \xrightarrow{-H_2O} CH_2 = CH_2$. Substances such as concentrated sulfuric acid, which can remove H_2O in this way, are known as **dehydrating agents**. For example, with sulfuric acid, methanoic acid gives carbon monoxide: $HCOOH \xrightarrow{-H_2O} CO$.

DEHYDROARENES: Species, usually transient, derived formally by the abstraction of a hydrogen atom from each of two ring atoms of an arene. The name for specific compounds requires the numerical prefix di-, for example 1,8- didehydronaphthalene or naphthalene-1,8-diyl. 1,2-Didehydroarenes are called arynes and are commonly represented with a formal triple bond.

DEHYDROFREEZING: A process for preservation of fruits and vegetables by evaporation of 50-60% of the water before freezing. The texture and flavour are claimed to be superior to those resulting from either dehydration or freezing alone and rehydration is more rapid than with dehydrated products.

DEHYDROGENASE (DHO): Any enzyme that catalyses the removal of hydrogen atoms (dehydrogenation) in biological reactions or an enzyme that accelerates oxidation of a substrate by transferring one or more hydrides (H^-) to an acceptor, usually NAD^+/$NADP^+$ or a flavin coenzyme such as FAD or FMN. Examples include aldehyde dehydrogenase, acetaldehyde dehydrogenase, alcohol dehydrogenase, glutamate dehydrogenase, lactate dehydrogenase, pyruvate dehydrogenase, glucose-6-phosphate dehydrogenase and glyceraldehyde-3-phosphate dehydrogenase.

DEHYDROGENATION: A type of chemical reaction in which a molecular hydrogen is removed from the same molecule or compound to yield a double or triple bond. It converts organic single carbon-carbon bonds into double bonds. The reaction is strongly endothermic and, therefore heat must be supplied to maintain the reaction temperature. When the detached hydrogen is immediately oxidised, the conversion of reactants to products is increased because the equilibrium concentration is shifted toward the products (law of mass action) and the added exothermic oxidation reaction supplies the needed heat of reaction. Dehydrogenation converts saturated fats to unsaturated fats. It is also used to produce styrene. **Hydrogenation** is a chemical reaction that involves the removal of hydrogen from a molecule. It is the reverse of dehydrogenation.

DEIMOS: One of the two moons of Mars, the other being Phobos. It is the outermost moon and the smaller of the two and also one of the smallest known moons in the Solar System, with a length of only 11 km.

DEIONISATION: An ion-exchange process in which all charged species or ionisable organic and inorganic salts are removed from solution.

DEIONISED WATER: Water from which dissolved ionic salt has been removed by ion-exchange technique. Purified water is water from any source that is physically processed to remove impurities. Distilled water and deionised (DI) water have been the most common forms of purified water, but water can also be purified by other processes including reverse osmosis, carbon filtration, microporous filtration, ultrafiltration, ultraviolet oxidation or electrodialysis. Purified water has many uses, largely in science and engineering laboratories and industries and is produced in a range of purities.

DEL OPERATOR (DIFFERENTIAL OPERATOR; NABLA): The differential vector operator, with the symbol Δ, used in vector analysis and defined as $i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}$, where i, j, k are unit vectors in the direction of the x-, y- and z-axes, respectively and $\partial/\partial x$, $\partial/\partial y$ and $\partial/\partial z$ are the respective partial derivatives of the given function of x, y and z.

DELAY LINE: A component in an electronic circuit that is introduced to provide a specified delay in transmitting a signal. It is specifically defined as a single-input-channel device, in which the output channel state at a given instant, t, is the same as the input channel state at the instant $t-n$, where n is a number of time units (the input sequence undergoes a delay of n time units, such as n femtoseconds, nanoseconds or microseconds).

DELAYED COINCIDENCE: The occurrence of two or more events separated by a short but measurable time interval.

DELAYED COKE: A commonly used term for a primary carbonisation product (green or raw coke) from high-boiling hydrocarbon fractions (heavy residues of petroleum or coal processing) produced by the delayed coking process.

DELAYED COKING PROCESS: A thermal process which increases the molecular aggregation or association in petroleum-based residues or coal tar pitches leading to extended mesophase domains. This is achieved by holding them at an elevated temperature, usually 750 – 765 K, over a period of time (12 to 36 h). It is performed in a coking drum and is designed to ultimately produce delayed coke. The feed is rapidly pre-heated in a tubular furnace to about 760 K.

DELAYED FLUORESCENCE: There are three types of delayed fluorescence: (i) **E-type delayed fluorescence**, in which the first populated by a thermally activated radiationless transition from the first excited triplet state. Since in this case the populations of the singlet and triplet states are in thermal equilibrium, the lifetimes of delayed fluorescence and the concomitant phosphorescence are equal. (ii) **P-type delayed fluorescence** in which the first excited singlet state is populated by interaction of two molecules in the triplet state (triplet-triplet annihilation) thus producing one molecule in the excited singlet state. In this biphotonic process the lifetime of delayed fluorescence is half the value of the concomitant phosphorescence. (iii) **Recombination fluorescence**, in which the first excited singlet state becomes populated by recombination of radical ions with electrons or by recombination of radical ions of opposite charge.

DELAYED LUMINESCENCE: Luminescence decaying more slowly than that expected from the rate of decay of the emitting state. Examples are: (i) **triplet-triplet** or **singlet-singlet annihilation**

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to form one molecular entity in its excited singlet state and another molecular entity in its electronic ground state (sometime referred to as P type); (ii) **thermally activated delayed fluorescence** involving reversible intersystem crossing (sometimes referred to as E type); and (iii) Combination of oppositely charged ions or of an electron and a cation. For emission to be referred to in this case as delayed luminescence at least one of the two reaction partners must be generated in a photochemical process.

DELAYED NEUTRONS: The small proportion of neutrons that are emitted with a reasonable time delay in a nuclear fission process. Analogously, delayed emission of protons and alpha particles is also observed, but the known delayed neutron emitters are more numerous and some of them have practical implications. In particular, they are of importance in the control of nuclear chain reactors.

DELETED NEIGHBOURHOOD (PUNCTURED NEIGHBOURHOOD): In mathematics, it is a neighbourhood of a point, without the point itself; that is the point has been deleted. It is usually written with a prime to indicate deletion, as $N'(\delta, a)$ in a metric space or $U'(a)$ more generally in a topological space.

DELETION: 1. In computing, it is a process of removing or erasing texts from a document and also entire files or folders from a hard drive. 2. In genetics, it is also called **gene deletion**, which is a chromosome mutation that arises from an inequality in crossing-over during meiosis such that one of the chromatids loses more genetic information than it receives. In other words, it is a mutation or a genetic aberration in which a part of a chromosome or a sequence of DNA is missing during replication. Deletions can be caused by errors in chromosomal crossover during meiosis. The nucleotides deleted may be small, involving a single missing DNA base pair or large, involving a piece of a chromosome. Deletion causes a shift in the reading frame of the codons in the mRNA and may eventually lead to the alteration in the amino acid sequence at protein translation. **Point mutation** is when an error occurs in a single nucleotide, causing the entire base pair to be missing or just the nitrogenous base on the master strand. **Chromosome deletion** is when an entire section of a chromosome is deleted. This can involve any number of base pairs as long as it is more than one (that would be a point deletion). Sometimes an entire chromosome can be deleted as well. Deletion of DNA can cause some genetic diseases or disorders such as colour blindness, cystic fibrosis, sickle-cell disease, Haemophilia, Turner Syndrome and Down Syndrome.

DELETION MUTATION: A type of gene mutation in which the deletion or addition of one or more nucleotides causes a shift in the reading frame of the codons in the mRNA. This may eventually lead to the alteration in the amino acid sequence at protein translation.

DELIAN ALTAR PROBLEM (DELIAN PROBLEM; DOUBLING THE CUBE): The traditional geometric problem of constructing a cube having a volume which is double that of a given cube, using a straight-edge and compasses only. It was named after the oracle at Delos. The problem was solved using conics by Apollonius in third century BC, but it was not shown to be impossible using Euclidean constructions until the 18th century, since $\sqrt[3]{2}$ is not a constructible number.

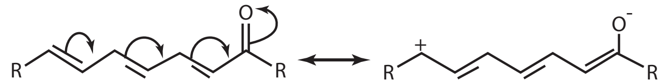
DELIMITER: 1. A character used to indicate the beginning and end of a character string. 2. A flag that separates and organises items of data. 3. A character that groups or separates words or values in a line of input.

DELINEATE: To represent visually using something such as a graph or chart.

DELIQUESCENT: 1. Substances that are able to absorb so much moisture from the atmosphere that they ultimately dissolve in it to form solutions or the absorption of atmospheric water vapour by a crystalline solid until the crystal eventually dissolves into a saturated solution. If the substance merely forms a crystalline hydrate it is termed **hygroscopic**. An example is sodium hydroxide. This behaviour is also well known for certain salts such as hydrated calcium chloride, $CaCl_2 \cdot 6H_2O$ and zinc chloride, $ZnCl_2$, but it is a property of all soluble salts in air of sufficiently high humidity. Thermodynamically, the condition for deliquescence is that the partial pressure of the water vapour in the air exceeds the vapour pressure (aqueous tension) of the water in the saturated solution of the salt. The speed at which the process takes place depends upon the rate of diffusion of water vapour into the crystal lattice, crystal size and other factors. The process will stop when the water vapour in the atmosphere is depleted to the point at which its partial pressure equals that of the saturated solution. Sugar, for example, deliquesces above 85% humidity. 2. A form of branching in angiosperms that has no well-defined axis and not arranged in any regular form. This can be seen in the American elm (*Ulmus Americana*) and many oaks.

DELIST: In zoology, it is removing an animal species from the list of endangered, threatened and vulnerable wildlife list.

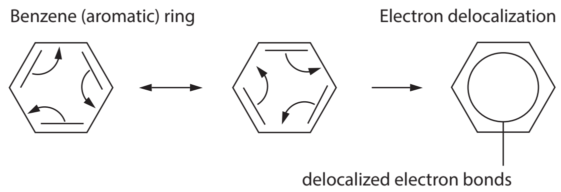
DELOCALISATION (DELOCALIZATION): Redistribution of the valence-shell electron density throughout a molecular entity as compared with some localised models (individual atoms in their valence states, separated bonds or fragments). Different topological modes of the electron delocalisation include: (i) **ribbon delocalisation** of either π- or σ-electrons (*i.e.*, electrons occupying respectively π- and σ-orbitals); (ii) **surface delocalisation** of σ-electrons occurring through an overlap of radially oriented σ-orbitals of a cyclic molecule, as is the case of cyclopropane; and (iii) **volume delocalisation** of σ-electrons through an overlap of σ-orbitals directed inside a molecular polyhedron, as is the case in tetrahedrane. The electrical conductivity of metals is due to the presence of delocalised electron(s) in the atoms. In the benzene delocalised structure shown, the inner circle indicates that the valence electrons are shared equally by all six carbon atoms (that is, the electrons are delocalised or spread out, over all the carbon atoms). It is understood that each corner of the hexagon is occupied by one carbon atom and each carbon atom has one hydrogen atom attached to it. Any other atom or groups of atoms substituted for a hydrogen atom must be shown bonded to a particular corner of the hexagon.



Delocalization

DELOCALISATION/DELOCALIZATION ENERGY (DE): The difference between the actual π-electron energy of a molecular entity and the π-electron energy of a hypothetical species with a localised form of the π-electron system. These energies are normally evaluated within Hückel molecular orbital theory.

DELOCALISED/DELOCALIZED ELECTRONS: Electrons in an atom, ion, molecule or solid metal that are not associated with individual atoms or identifiable chemical bonds but are shared by more than two atoms that are bonded together. In the structure of solid metals, the d-subshell interferes with the s-subshell. Delocalised electrons contribute to the conductivity of the atom, ion or molecule. Materials with many delocalised electrons tend to be highly conductive. Delocalised electrons can be found in conjugated systems and mesoionic compounds and it is increasingly being realised that electrons in sigma bonding levels are also delocalised. For example, in methane, the bonding electrons are shared by all five atoms equally. Pervasive existence of delocalisation is implicit in molecular orbital theory. Another example of delocalised electrons can be found in a carboxylate group, wherein the negative charge is centred equallyon the two oxygen atoms.



Resonance and electron delocalization in benzene (aromatic) ring

DELOCALISED MOLECULAR ORBITALS: Molecular orbitals that are not confined between two adjacent bonding atoms but actually extend over three or more atoms.

DELPHINIDAE: A group of small and toothed marine mammals such as dolphins and relative species that belong to the family Delphinidae and the Order Cetacea, found in all oceans. Some others include killer whales, short-finned pilot whales and several others. The killer whale is the largest member of the family. Delphinidaes are the most abundant and varied of all Cetacea.

DELTA: 1. A tract of land at a river's mouth, often triangular in shape composed of silt or soil deposited as the water slows on entering the sea. 2. A small change in the value of a variable, with the symbol Δ. 3. A code word for the letter "D" used in international radio communications. 4. The 4th brightest star in the constellation.

DELTA BONDING: Chemical bonding involving delta (δ) orbitals. Delta orbital is so called because it resembles a d-orbital when viewed along the axis of a molecule and has two units of orbital angular momentum around the internuclear axis. Delta bonds (δ bonds) are chemical bonds of the covalent type, where four lobes of one involved electron orbital overlap four lobes of the other involved electron orbital. In sufficiently-large atoms, occupied d-orbitals are low enough in energy to participate in bonding. Delta bonds are usually observed in organometallic species. Some ruthenium and molybdenum compounds contain a quadruple bond, which can only be explained by invoking the delta bond.

DELTA CURVE: A curve that can be turned inside an equilateral triangle while continuously making contact with all three sides. There are an infinite number of delta curves, but the simplest are the circle and lens-shaped **delta-biangle**. All the delta curves of height h have the same perimeter, $\frac{2\pi h}{3}$.

DELTA FUNCTION (δ FUNCTION; DIRAC DELTA FUNCTION): The function, $\delta(x)$, introduced by theoretical physicist, Paul Dirac that has the value zero for all non-zero real x, except at the origin, where it is infinite, with integral of this function having the value 1. It satisfies the identity $\int_{-\infty}^{\infty} \delta(x)\,\mathrm{d}x = 1.$

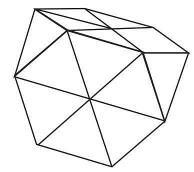
DELTA METAL: An alloy of copper (55%) and zinc (43%) and a small amount of iron and other metals. It is suitable for making valves and bearings.

DELTA RAY (KNOCK-ON ELECTRON): An electron that is ejected from an atom when it is struck by a high-energy particle. It is characterised by very fast electrons produced in quantity by alpha particles or other fast energetic charged particles knocking orbiting electrons out of atoms. Collectively, these electrons are defined as delta radiation when they have sufficient energy to ionise further atoms through subsequent interactions on their own. Delta ray is used in high energy physics to describe single electrons in particle accelerators that are exhibiting characteristic deceleration.

DELTA VALUE: A quantity that measures the shift in nuclear magnetic resonance (NMR).

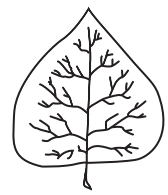
DELTA-BRASS: A strong hard type of brass that contains, in addition to copper and zinc, a small percentage of iron. It is mainly used for making cartridge cases.

DELTAHEDRON: A polyhedron that has triangular faces or a polyhedron whose faces are all equilateral triangles. There are infinitely many deltahedra, but of these only eight are convex, having 4, 6, 8, 10, 12, 14, 16 and 20 faces.



Deltahedron

DELTOID: 1. Also known as **Steiner's hypocycloid** or **tricuspoid**, it is a plane curve traced by a point on a circle while the circle rolls along the inside of another circle whose radius is three times as large. 2. A thick triangular muscle covering the shoulder joint that serves to raise the arm laterally. 3. A leaf having the shape of an equilateral triangle, as shown in the diagram.



A deltoid leaf

DELTOID MUSCLE: The thick triangular muscle of the shoulder region that forms the rounded flesh of the outer part of the upper arm and passes up and over the shoulder joint. The wide end of the deltoid is attached to the scapula (shoulder blade) and the clavicle (collarbone). The muscle fibres converge to form the apex of the triangle, which is attached to the humerus (upper-arm bone) about halfway down its length. The central, strongest part of the muscle raises the arm sideways. The front and back parts of the muscle twist the arm.

DEMAGNETISATION (DEMAGNETIZATION): The removal of the ferromagnetic properties of a body by disordering the domain structure. Some of the processes include the following: applying intense magnetic field in a direction opposite to that of the existing magnetisation; by subjecting the magnet to intense heating to a temperature above its Curie's point, then returning it to room temperature, in the absence of any external magnetic field; and by hitting the magnet repeatedly on the floor. The harder the material is, the harder it is to demagnetise.

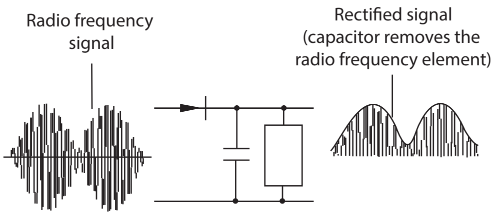
DEME: A group of organisms in the same taxon. The term is used with various prefixes that denote how they differ from other groups, for example, **ecodeme** occurs in a particular ecological habitat, **cytodemes** differ from each other cytologically; and **genodemes** differ genetically. In biology, it refers to a local population of organisms of one species that actively interbreed with one another and share a distinct gene pool. In evolutionary computation, a deme often refers to any isolated subpopulation subjected to selection as a unit rather than as individuals. When demes are isolated for a very long time they can become distinct subspecies or species.

DEMENTIA: A group of symptoms affecting memory, thinking and social abilities resulting in memory loss and impaired judgment or language, personality changes and the inability to perform some daily tasks. It is caused by a number of disorders, the most common being Alzheimer's disease; and damage due to the brain due to stroke or some other disease.

DEMINERALISATION (DEMINERALIZATION): The loss of minerals, such as calcium and phosphorus, from bone.

DEMISTER: 1. Apparatus made of wire mesh or glass fibre which is used to help remove acid mist as in the manufacture of sulfuric acid. Demisters are also components of wet arrestment plants. 2. A device often fitted to vapour liquid separator vessels to enhance the removal of liquid droplets entrained in a vapour stream. Demisters may be a mesh type coalescer, vane pack or other structure. It is intended to aggregate the mist into droplets that are heavy enough to separate from the vapour stream. 3. A heating device used in a motor vehicle to free the windscreen or windows of condensation. It consists of a series of ducts in automobiles arranged so that hot, dry air directed from the heat source is forced against the interior of the windscreen or windshield to prevent condensation.

DEMODULATION: A circuit that is used in amplitude modulation (AM) and frequency modulation (FM) receivers in order to separate the information that was modulated onto the carrier from the carrier itself. A demodulator or detector is the analogue part of the modulator. A modulator puts the information onto a carrier wave at the transmitter end and then a demodulator pulls it so it can be processed and used on the receiver end. The term is normally used in connection with radio receivers, but many other systems use many kinds of demodulators, the common one is in a modem, which is a contraction of the terms **mod**ulator/**dem**odulator. Amplitude modulation (AM) refers to any method of modulating an electromagnetic carrier frequency by varying its amplitude in accordance with the message intelligence that is to be transmitted. There are many ways to demodulate AM, the simplest being a diode detector. It operates by detecting the envelope of the incoming signal, by rectifying the signal. The AM demodulator includes a capacitor at the output. Its purpose is to remove any radio frequency components of the signal at the output. The value is chosen so that it does not affect the audio base-band signal. Frequency demodulation or detection can be obtained directly by using the phase lock loop (PLL).



A simplified demodulator for AM signal

DEMOGRAPHY: A branch of sociology, which studies the distribution, composition and internal structure of human populations. It draws on many disciplines such as genetics, psychology, economics and geography. The prime concerns of demography are birth rate, emigration and immigration.

DEMONSTRATION: Showing something by giving proof or evidence or showing and explaining how something works. For example, in mathematics, showing an explicit statement of the steps of the derivation of a mathematical theorem.

DEMULCENTS: Agents and substances that soothe, soften and relieve irritations of the mucus membranes and other surfaces. Demulcents include herbs such as aloe vera, marshmallow leaf or root, corn silk, and licorice.

DEMYELINATION: A medical condition in which the myelin sheath insulating some nerve fibres is damaged; leading to weakness and loss of function in the parts of the body served by the damaged nerves. The damage may be caused by inflammation, infection with certain viruses, metabolic problems, loss of oxygen and physical compression. Demyelinization occurs in a number of diseases, particularly multiple sclerosis and also in severe diabetes and some neurological diseases.

DENARY: Relating to a number system that has ten as its base, as in the decimal system.

DENATURATION: 1. Irreversible changes occurring in the structure and functioning of proteins including enzymes, when subjected to extreme temperature or pH. It is a biochemical process which modifies a protein's natural configuration and it involves breaking many weak hydrogen and hydrophobic bonds that maintain the protein's highly ordered structure. This usually results in loss of biological activity such as loss of an enzyme's ability to catalyse reactions. It can be brought about by heating, treatment with alkalis, acids, urea or detergents, as well as a vigorous shaking of the protein solution. It can be reversed in some cases such as serum albumin and haemoglobin, if conditions favourable to the protein are restored, but not in others. 2. The process of adding methanol or other poisonous substances to ethanol rendering it unfit to drink. 3. The process of adding another isotope to a fissile material to make it unsuitable for use.

DENGUE FEVER (BREAKBONE FEVER; DANDY FEVER; DENGUE): An infectious viral disease spread by mosquitoes that occurs in tropical regions. Symptoms include fever, headache, vomiting, swollen glands, rash, and severe pain in joints and the back. Subsequent infections of the virus may lead to dengue haemorrhagic fever (DHF) and dengue shock syndrome (DSS), which may be fatal.

DENDRIFORM: A branch in the form of a tree. **Dendritic** refers to such a form or possessing dendrites.

DENDRIMER (DENDRITIC POLYMER; ARBOROL; CASCADE MOLECULE): A type of macromolecule in which a number of chains radiate out from a central atom or cluster of atoms. Dendritic polymers have a number of possible applications. They are repeatedly branched, roughly spherical large molecules or typically symmetric around the core and often adopts a spherical three-dimensional morphology. On the other hand, a dendron usually contains a single chemically addressable group called the focal point.

DENDRITE: The branched part at the end of a nerve cell or neuron. It is a slender filament projecting from the cell body, which receives incoming messages or impulses from many other nerve cells and passes them on to the cell body. It plays a critical role in integrating synaptic inputs and in determining the extent to which action potentials are produced by the neuron. It has also been discovered that dendrites can support action potentials and release neurotransmitters, a property that was originally believed to be specific to axons. The long outgrowths on dendritic cells are also called dendrites but they do not process electrical signals.

DENDRITIC CELL: One of the main types of antigen-presenting cells in the immune system, responsible for presenting antigen to naive T cells and inducing them to become effective components of the adaptive immune response. Their main function is to process antigen material and present it on the surface to other cells of the immune system, thus functioning as antigen-presenting cells. They also act as messengers between the innate and adaptive immunity. Once activated, they migrate to the lymphoid node, where they interact with T cells and B cells to initiate and shape the adaptive immune response. They are present in small quantities in tissues that are in contact with the external environment, mainly the skin and the inner lining of the nose, lungs, stomach and intestines.

DENDROCHRONOLOGY: A scientific dating technique using the growth rings of trees or analysis of the annual rings of trees to date past events. The method can date the time at which tree rings were formed in many types of wood, to the exact calendar year. The technique is applied in paleoecology, where it is used to determine certain aspects of past ecologies (most prominently climate); archaeology, where it is used to date old buildings, etc.; and radiocarbon dating, where it is used to calibrate radiocarbon ages.

DENDROGRAM: A diagram similar to a family tree that indicates some type of similarity between different organisms. It is used to illustrate the arrangement of the clusters produced by hierarchical clustering or the clustering of genes/samples.

DENDRON: Any of the major cytoplasmic processes that arise from the cell body of a motor neurone or part of a neurone carrying a nerve impulse towards a cell body. They usually branch into dendrite.

DENIAL OF SERVICE ATTACK (DoS ATTACK): An attack on a network when the Network is flooded with so many requests that it is overwhelmed and unable to respond and users are therefore, unable to access the website. This type of attack only involves the service and does not destroy anything.

DENISE MATRIX: A term used to describe a matrix with a high percentage of non-zero entries.

DENITRIFICATION: The chemical process occurring naturally in soil, whereby bacteria break down nitrate to give nitrogen gas or molecular nitrogen (N_2), which returns into the atmosphere. It is a microbially facilitated process of dissimilatory nitrate reduction that may ultimately produce molecular nitrogen (N_2) through a series of intermediate gaseous nitrogen oxide products. The preferred nitrogen electron acceptors in order of most to least thermodynamically favourable include nitrate, nitrite, nitric oxide and nitrous oxide. In terms of the general nitrogen cycle, denitrification completes the cycle by returning N_2 to the atmosphere. The chemical equation for the reduction of nitrate to N_2 is as follows:
NO_3^- NO_2^- NO N_2O N_2. The process is mainly performed by heterotrophic bacteria, although autotrophic denitrifiers have also been identified and other enzymatic pathways have been identified in the reduction process. Complete denitrification produces some intermediates such as nitric oxide and nitrous oxide, which are significant greenhouse gases that react with sunlight and ozone to produce nitric acid, a component of acid rain.

DENOMINATOR: The number below the line in a fraction showing how many parts a whole is divided into. For example, 4 is the denominator in ¼.

DENSE: 1. Of liquids or vapour that is not easily seen through, for example, dense smoke. 2. Very heavy in relation to each unit in volume. 3. A group of organisms in the same taxon. 4. Of a set in an ordered space, having the property that between any two comparable elements a third can be interposed. The rational numbers are dense, since for any rationals a and b, 1/2 $(a + b)$ lies between them and is also a rational. 5. In mathematics, of a set in a topology, having a closure that contains a given set. One set is dense in another if the second is contained in the closure of the first. For example, the rationals are dense in the reals, since the latter are contained in the closure of the former.

DENSE BODY: One of the numerous structures found in the fibres of the involuntary muscle to which the thin filaments are anchored within the muscle cell.

DENSE IN ITSELF: Of a set in a topological space, such that every punctured neighbourhood of every element of the set intersects the set. An example of such a set which is not closed is the subset of irrational numbers, because every neighbourhood of an irrational number x contains at least one other irrational number $y \neq x$.

DENSIFICATION: Removal of impurities and the elimination of pores from a xerogel to give a material as near bulk density as possible.

DENSITOMETER: An instrument used to measure the photographic density of an image on a film or photographic print. There are two types: transmission densitometers that measure transparent materials; and reflection densitometers that measure light reflected from a surface. It determines the density of a sample placed between the light source and the photoelectric cell from differences in the readings.

DENSITY: 1. In biology, it is also known as **population density** and it is the number of individuals belonging to the same species that live in the same region at the same time per unit area. Human population density is the number of people per unit area usually per square kilometre. When resources are plentiful and environmental conditions are favourable, populations can increase rapidly. Low population densities may cause extinction and lead to further reduced fertility, called the Allee effect. **Allee effect** is a decline in individual fitness at low population size or density that can result in critical population thresholds below which populations crash to extinction. Examples of the causes of low population densities are density-independent factors such as availability of food, parasitism, predation, disease and migration; and density-independent factors such as weather, forest fire and pollutants. Different species have different expected densities. R-selected species commonly have high population densities, while K-selected species may have lower densities. 2. The degree of optical opacity of a medium or material such as photographic negative. 3. Logarithm to the base 10 of the reciprocal of transmittance. 4. Also known as **density function** or **probability density function** (PDF), in statistics, a function whose integral is calculated to find probabilities associated with a continuous random variable. 5. The ratio between the mass of an object and its volume. In other words, it is a measure of the compactness of a substance, which is equal to its mass per unit volume. In SI units it is measured in kgm^{-3}. The term is applicable to mixtures and pure substances and to matter in the solid, liquid, gaseous or plasma state. Density of all matter depends on temperature, however, the density of a mixture may depend on its composition and the density of a gas on its pressure. Common units of density are grams per cubic centimetre.

Substance	Density (kgm^{-3})	Relative density
Gold	19300	19.3
Mercury	13600	13.6
Lead	11370	11.37
Brass	8900	8.9
Iron	7870	7.87
Aluminium	2650 (2700)	2.65 (2.7)
Glass (Flint)	3700	3.7
Glass (Crown)	2500	2.5
Cork	250	0.25
Rubber	950	0.95
Wood (varies)	750	0.75
Petrol	700	0.7
Fresh water	1000	1.0
Sea water	1030	1.03
Ice	920	0.92

Densities of some common substances

DENSITY ALTITUDE: The altitude at which a given density is found in the standard atmosphere. It is used in aviation and computed from the station pressure at takeoff and the virtual temperature at the particular altitude under consideration.

DENSITY CURRENT: Any current in either a liquid or a gas that is kept in motion by the force of gravity acting on small differences in density.

DENSITY DEPENDENT INHIBITION OF GROWTH: The phenomenon exhibited by most normal animal cells in culture that stop dividing once a critical cell density is reached. The critical density is considerably higher for most cells than the density at which a monolayer is formed.

DENSITY FUNCTION: 1. A density function for a measure *m* is a function which gives rise to *m* when it is integrated with respect to some other specified measure. 2. In statistics, a function whose integral is calculated to find probabilities associated with a continuous random variable. Its graph is a curve above the horizontal axis that defines a total area. The area under the curve is 1. The percentage of this area included between any two values coincides with the probability that the outcome of an observation described by the density function falls between those values. Every random variable is associated with a density function. If there exists a function $y = f(x)$ such that $\lim_{\delta x \to 0} \frac{P(x < \overline{x} < x + \delta x}{\delta x} = f'(x)$, then $f(x)$ is a density function.

DENSITY FUNCTIONAL THEORY (DFT): A theory demodulation or quantum mechanical theory used to describe many-fermion systems in which the energy is functional of the density of fermions. It has been used extensively in the theory of electrons in atoms, molecules and solids and the theory of nucleons in nuclei.

DENSITY INVERSION: The interchange of the denser and less dense phases due to changes in solute concentration. Phase inversion is often used in this context but is ambiguous.

DENSITY OF GASES: The mass of a unit volume of a gas at a stated temperature and pressure. Examples of the density of some gases are: oxygen 1.4280, hydrgen 0.0896, carbon dioxide 1.9800, ammonia 0.769 and air 1.293.

DENSITY OF MATTER: It refers to the quantity of mass per unit volume. For gases, density involves the number of atoms and molecules per unit volume.

DENSITY OF SOLIDS AND LIQUIDS: The mass per unit volume of a material at a specified temperature.

DENSITY OF STATES: A system, which describes the number of states per interval of energy at each energy level that are available to be occupied. Unlike isolated systems, like atoms or molecules in gas phase, the density distributions are not discrete like a spectral density but continuous. A high DOS at a specific energy level means that there are many states available for occupation. A DOS of zero means that no states can be occupied at that energy level. In general, a DOS is an average over the space and time domains occupied by the system. Local variations, most often due to distortions of the original system, are often called local density of states (LDOS). If the DOS of an undisturbed system is zero, the LDOS can locally be non-zero due to the presence of a local potential.

DENSITY PARAMETER: The density parameter (Ω) is the ratio of the actual mean density of mass-energy in the universe (ρ) to the critical density (ρc). A value of Ω greater than 1 implies that the universe will eventually collapse and corresponds to the case of a closed universe (i.e. one with positive curvature like that of a sphere). A value less than 1 suggests eternal expansion and corresponds to the case of an open universe (i.e. with negative curvature like that of the surface of saddle). The actual value remains to be determined but is thought to be close to unity (equivalent to a flat universe).

DENSITY-DEPENDANT FACTOR: Any biotic factor that limits the size of a population and whose effect is dependent on the number of individuals in the population. Usually these factors arise because the populations of the species have to compete for the same resources in a limited geographical niche. As the population increases, food become scarce, infectious diseases can spread easily and many of its members emigrate. Examples of such factors are the availability of food, parasitism, predation, disease and migration.

DENSITY-GRADIENT CENTRIFUGATION: The separation by centrifugation, of mixtures or molecules according to their density, in a gradient varying in solute concentration. More-dense components of the mixture migrate away from the axis of the centrifuge, while less-dense components of the mixture migrate towards the axis. The effective gravitational force on a test tube may be increased so as to more rapidly and completely cause the precipitate or pellet to gather on the bottom of the tube. The remaining solution which is called the supernate or supernatant liquid is then decanted from the tube without disturbing the precipitate or withdrawn with a Pasteur pipette. It is used in industry and in laboratory settings.

DENSITY-INDEPENDENT FACTOR: Any factor such as earthquake, which limits the size of a population whose effect is not dependent on the number of individuals in the population or any factor that affects the birth rate or mortality rate of a population in ways that are independent of the population density.

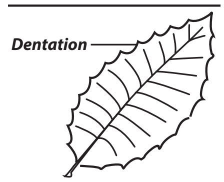
DENTAL CARIES: Tooth decay, which involves the destruction of the enamel layer of the tooth by acid produced by the action of bacteria on sugar. It is caused by bacteria that colonise the tooth surface and, under certain conditions, produce sufficient acids to demineralise the enamel covering of the tooth crown or the cementum covering the root and then the underlying dentin, with the breakdown of the organic components. The bacteria then invade the dead tissue and enter the pulp chamber, infecting the pulpal tissue. This may develop into toothache, with the infection ultimately destroying the pulpal tissue and extending through the apical openings of the roots and into the surrounding periodontal tissues.

DENTAL FORMULA: The method of showing the number and types of teeth in a mammal or the formula describing the dentition of a mammal. A dental formula consists of eight numbers, four above and four below a horizontal line. Letters represent the various types of teeth: I for incisor; C for canine; P for premolar; and M for molar. Each letter is followed by a horizontal line. Numbers above the line represent maxillary teeth; those below, mandibular teeth. The human dental formula for deciduous teeth in one jaw is calculated as $I\frac{2}{2}C\frac{1}{1}PM\frac{0}{0}M\frac{2}{2}=10$. For permanent teeth in one jaw, it is $I\frac{2}{2}C\frac{1}{1}PM\frac{2}{2}M\frac{3}{3}=16$. The total number of teeth in both jaws is, therefore, obtained by adding up all the numbers in the dental formula and multiplying by 2. For example, for an adult human, the total number of teeth is represented by: $I\frac{2}{2}C\frac{1}{1}PM\frac{2}{2}M\frac{3}{3}=16\times 2=32$.

DENTAL NERVE: Either of two nerves that supply the teeth, which are branches of the trigeminal nerve. The inferior dental nerve supplies the lower teeth and for most of its length consists of a single large bundle. The superior dental nerve, which supplies the upper teeth, breaks into separate branches at some distance from the teeth.

DENTAL FLOSS: A tuft of thin filaments used to remove food particles and dental plaque from in between the teeth.

DENTARY: A membrane bone, present in the lower jaw of the vertebrates that supports the teeth. The dentary is the sole bone of the lower jaw in mammals.



Dentate leaf

DENTATE: Of a leaf margin, toothed, with outward-pointing notches. Leaf margins finely toothed in this way are termed **denticulate**. **Dentation** is the state of being dentate, such as a tooth along a margin in a leaf.

DENTICITY: In a coordination entity the number of donor groups from a given ligand attached to the same central atom.

DENTICLE: A small tooth or tooth-like projection.

DENTINE: The bony material that forms the bulk of teeth or the portion of the tooth that lies subjacent to the enamel and cementum. It originates from the mersoderm and consists of an organic matrix on which mineral (calcific) salts are deposited, pierced by tubules containing filamentous protoplasmic processes of the odontoblasts that line the pulpal chamber and canal.

DENTIST: A person who has been professionally trained to treat ailments of the teeth. **Dentistry** is the study of care and treatment of the teeth and gums. **Orthodontics** is concerned with the irregularities of the teeth for aesthetic and clinical reasons, for example, by using braces; and **periodontics** deals with the prevention, diagnosis and treatment of diseases that affect the gums and supporting structures of the teeth.

DENTITION: The number, types and arrangement of teeth in the buccal cavity of a mammal, as described by a dental formula. It reflects an animal's diet. Animals, whose teeth are all of the same type, as occurs in most non-mammalian vertebrates, are said to have **homodont dentition** and those whose teeth differ morphologically are said to have **heterodont dentition**. The dentition of animals such as humans with two successions of teeth called deciduous and permanent is referred to as **diphyodont**, while the dentition of animals with only one set of teeth throughout life is **monophyodont**. The dentition of animals in which the teeth are continuously discarded and replaced throughout life is termed **polyphyodont**.

DETRITIVORE: Organisms that feed on dead, decomposed or organic waste.

DETRITUS: A mixture of dead or decomposing particulate organic material such as plant materials, fragments of dead organisms as well as faecal material.

DENUDATION: The natural loss of soil or raw debris, blown away by wind or washed away by running water that lays bare the rock below or the process by which rock disaggregates by means of erosion, mass wasting and weathering, which eventually leads to a reduction in elevation and relief of landforms and landscapes. Together with deposition, denudation is the major process, which creates the Earth's landscape. **Denudate** means naked, bare or without covering.

DENUDER SYSTEM: An apparatus used to separate gases and aerosols over a given diameter, which is based upon the difference in diffusion velocity between gases and aerosol particles. Usually, it is a tube containing an internal wall coating, which removes the gaseous compounds at the wall.

DENUMERABLE (COUNTABLY INFINITE; ENUMERABLE): A set is denumerable if its elements can be put in one-to-one correspondence with the set of natural numbers.

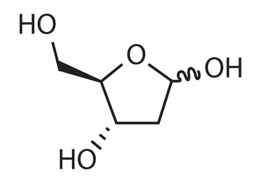
DEOBSTRUENTS: Agents and substances containing properties to remove obstructions by opening the natural passages or pores of the body.

DEODORISER (DEODORIZER): Equipment for the removal of noxious gases and odours, which may consist of combustion, absorption or adsorption units.

DEONTIC LOGIC: The branch of modal logic to represent the relationships of the concepts of obligatoriness and permissibility. It introduces the primitive symbol O for 'it is obligatory that', from which symbols P for 'it is permitted that' and F for 'it is forbidden that' are defined: $PA = {\sim}O{\sim}A$ and $FA = O{\sim}A$.

DEOXYRIBONUCLEIC ACID (DNA): A nucleic acid composed of two polynucleotide strands wound around a central axis to form a double helix, the repository of genetic information. It functions as the physical carrier of inheritance for 99% of all species. The molecule is double-stranded and composed of two strands in an antiparallel and complementary arrangement. The basic unit, the nucleotide, consists of a molecule of deoxyribose sugar, a phosphate group and one of four nitrogenous bases. Alternatively, it can be defined as the genetic material of most living organisms, which is a major constituent of the chromosomes within the cell nucleus and plays a central role in the determination of hereditary characteristics by controlling protein synthesis in cells. It is also found in chloroplasts and mitochondria.

DEOXYRIBOSE (2-D-DEOXYRIBOSE): A five-carbon simple sugar that is a structural component of DNA. With assistance of enzymes, it is formed in living cells by reduction of ribonucleoside di- or triphosphate. The four deoxyribose nucleotides, containing adenine, guanine, cytosine and thymine, are the main constituents of the deoxyribonucleic acids (DNA), which control the hereditary characteristics of all living organism.



Chemical structure of deoxyribose

DEPAUPERATE: Impoverished as a result of bad environmental conditions; or small and usually poorly developed.

DEPENDENT EQUATIONS: An equation is said to be dependent on a set of equations if every set of values of the unknowns that satisfy the set of equations also satisfy this equation. For example, a set of functions $\{g_1,...g_n\}$ are dependent if one can be expressed as a function of the others. For instance, $6a + 12b + 13$ and $a + 2b$ are dependent since $6a + 12b + 13 = 6(a + 2b) + 13$. A set of matrices or vectors $\{y_1,...y_n\}$ are said to be linearly dependent if there exists a linear relation $b_1y_1 + b_2y_1 + ... + b_ny_n = 0$, provided one of the coefficients is not zero.

DEPENDENT EVENT: Two events A and B, are dependent if the outcome of event A affects the outcome of event B. The probability of two dependent events occurring is found as follows:
P(A B) = P(A).P(B|A), where P(B|A) means probability of B given that A has already occurred. This is a conditional probability.

DEPENDENT VARIABLE: 1. A variable whose value depends upon independent variables. For example, in $y = 2x + 3$, y is the dependent variable and x is the independent variable. The value of y depends on the value chosen for x. Usually the dependent variable is isolated on one side of an equation. 2. In statistics, also known as **response variable** or **predicted variable**, it is the variable whose values are measured or observed in an experiment for different values of the independent variable.

DEPLETED: Denoting a material that contains less of a particular isotope than it normally contains, especially a residue from a nuclear reactor or isotope separation plant, containing fewer fissile atoms than natural uranium. For example, **depleted uranium** (DU) is uranium consisting mostly of isotope uranium-238 (U-238), obtained as a product of enriching natural uranium by separating the isotopes by mass. Uranium is enriched in U-235 by separating the isotopes by mass and the by-product of enrichment, called depleted uranium (DU), contains less than one third as much U-235 and U-234 as natural uranium. DU is also found in reprocessed spent nuclear reactor fuel; however, it can be distinguished from DU produced as a by-product of uranium enrichment by the presence of U-236. The external radiation dose from DU is about 60% of that from the same mass of natural uranium. Natural uranium is about 99.27 percent U-238, 0.72 percent U-235 and 0.0055 percent U-234. It has a high density of 19.1 g/cm^3, about 68.4% denser than lead and is used in amour-piercing shells, counterweights in aircraft, radiation shielding in medical radiation therapy and industrial radiography equipment. U-235 is used for fission in nuclear reactors and nuclear weapons.

DEPLETED SOIL: Soil that has lost most of its available nutrients, biological diversity, or structural quality. This may be caused by over harvesting, chemical pollution, pesticides and other harmful farming practices or improper extractive practices.

D

DEPLETION LAYER (DEPLETION REGION; DEPLETION ZONE; JUNCTION REGION; SPACE CHARGE REGION): A region in a semiconductor that has a lower-than usual number of mobile charge carriers. This region occurs at the interface between two dissimilar regions of conductivity such as a p–n junction. In semiconductor physics, it is an insulating region within a conductive, doped semiconductor material, where the mobile charge carriers have been forced or diffused away by an electric field. Ionised donor or acceptor impurities are the only elements left in the depletion region. The depletion layer is important in explaining modern semiconductor electronics: diodes, bipolar junction transistors, field-effect transistors and variable capacitance diodes all rely on depletion layer phenomena.

DEPOLARISATION (DEPOLARIZATION): 1. In physics, it is the prevention of polarisation in a primary cell. Polarisation in a primary cell is the chemical action that occurs in the cell, while the current is flowing, causing hydrogen to accumulate on the cathode. 2. In biology, it is a reduction in the difference in electrical potential that exists across the cell membrane of a nerve or muscle cell or a change in a cell's membrane potential, which makes it more positive or less negative. In neurons and some other cells, a large enough depolarisation may result in an action potential in neurons and some other cells. A neuron membrane is depolarised if a stimulus decreases its voltage from the resting potential of –70 mV in the direction of zero voltage. It is often caused by influx of cations, such as Ca^{2+} through Ca^{2+} channels or Na^+ through Na^+ channels. However, the efflux of K^+ through K^+ channels inhibits depolarisation, as does influx of Cl^- (an anion) through Cl^- channels. If a cell has K^+ or Cl^- currents at rest, the inhibition of those currents will also lead to a depolarisation. Because depolarisation is a change in membrane voltage, in measuring it, current clamp techniques are used. In voltage clamp, the membrane currents giving rise to depolarisation are either an increase in inward current or a decrease in outward current. The opposite of depolarisation is hyperpolarisation, which inhibits the rise of an action potential.

DEPOLARISATION (DEPOLARIZATION) CURRENT: The current through a short circuit established between two electrodes in contact with an insulating medium after electrification by direct voltage for some time. The depolarisation current is usually measured after electrification for so long a time that the polarisation current is negligible.

DEPOLARISATION (DEPOLARIZATION) OF SCATTERED LIGHT: The phenomenon, due primarily to the anisotropy of the polarisability of the scattering medium, resulting from the fact that the electric vectors of the incident and scattered beams are not coplanar and that, therefore, light scattered from a vertically (horizontally) polarised incident beam contains a horizontal (vertical) component.

DEPOLARISER (DEPOLARIZER): 1. Oxidising agent used in an electrochemical or dry cell or negative electrode in water. It takes up electrons during discharging the cell. This prevents the building up of gas which would otherwise impair the efficiency of the battery. 2. In biology, it is the region of the synaptic knob which receives nerve impulses for onward transmission along the axon, as occurs, for example, in nerve impulse transmission across a synapse.

DEPOLYMERISATION (DEPOLYMERIZATION): The process of converting a polymer into a monomer or a mixture of monomers. **Unzipping** is depolymerisation occurring by a sequence of reactions, progressing along a macromolecule and yielding products, usually monomer molecules at each reaction step, from which macromolecules similar to the original can be regenerated.

DEPOPULATION: 1. The decline of the population of a given area, usually caused by emigration, rather than by an increase in the death rate or decrease in birth rate. 2. The planned destruction of all animals in, for example, a herd because of a disease such as foot-and-mouth disease that can be spread to other herd of animals in an area. If not done, it can pose serious economic threats to the livestock industries.

DEPOSIT-FEEDER: A heterotroph, such as an earthworm, that eats its way through detritus, salvaging bits and pieces of decaying organic matter.

DEPOSITION: 1. The building up of a layer of a substance on a surface. For example, the deposition of a metal occurs at the cathode during electroplating; and deposition of sediments on the ocean floor builds up strata that become sedimentary rocks. **Dry deposition** is the process by which aerosols and gases in the air are deposited on the surface of the earth such as soil, water, rock and plants; even when the receptor surface is moist. **Wet deposition** is that process which involves the transport of chemicals to the surface of the earth by water droplets or snow crystals which scavenge pollutants as they form and fall through the atmosphere. Together with denudation, deposition is the major process which creates the Earth's landscape. **Glacial deposition** is the laying-down of sedimentary material carried by glacier and left behind when the ice melts. 2. Also known as **desublimation**, it is a thermodynamic process, a phase transition, in which a gas transforms into a solid directly. When state of a matter changes from solid state to gas directly by absorbing heat then it is known as **sublimation**.

DEPRECATED: In computing, the term refers to functions or elements that are in the process of being replaced by newer ones.

DEPRECIATION: In mathematics, it is the loss in value of an item with time. It can be calculated by the relation, $A = P\left(1 - \frac{d}{100}\right)^n$, where A is the value of the article after a period of time, p is the value of the article when new, d is the rate of depreciation and n is the number of years over which the value is lost. For example, if a new printing machine costs 200 dollars and its value depreciates by 5% each year, its value after 3 years, i.e., $d = 5$, $p = 200$ and $n = 3$, then the solution is $A = 200\left(1 - \frac{5}{100}\right)^n = 200$ dollars.

DEPRESSANT: An agent that depresses any organ, especially the functioning of the central nervous system. Examples of depressants are alcohol, barbiturates and sedatives. Some effects of depressants are reduction of the level of consciousness and impaired control of breathing, when taken in excess.

DEPRESSION: 1. A region of relatively low atmospheric or barometric pressure in the mid and high attitudes that is shown on a weather map, surrounded by several closed isobars. The term is often applied to a certain stage in the development of a tropical cyclone, but it also applies to migratory lows and troughs and also to upper air lows and troughs that are only weakly developed. 2. In psychiatry, depression is a symptom of mood disorder characterised by intense feelings of loss, sadness, hopelessness, failure and rejection. The main types are unipolar disorder called major depression; and bipolar or manic-depressive disorder, which causes radical emotional changes of alternating states of emotion and extreme elation to the sufferer.

DEPRESSION OF FREEZING POINT: The lowering of the freezing point of a pure liquid, which is a solvent, when another substance is dissolved in it, resulting in the solution having a lower freezing point than a pure solvent. The lowering of the freezing point is proportional to the number of dissolved particles (molecules or ions) and does not depend on their nature. The phenomenon of freezing point depression is used in technical applications to avoid freezing. For example, in the case of water, ethylene glycol or other forms of antifreeze is added to cooling water in internal combustion engines to make the mixture stay liquid at temperatures below its normal freezing point. The use of freezing-point depression through "freeze avoidance" happens in animals which live in environmentally cold places through permanently high concentration of physiologically inert substances such as sorbitol or glycerol to increase the molality of fluids in cells and tissues and thus decrease the freezing point. Examples include some species of arctic-living fish, such as rainbow smelt, which need to be able to survive in freezing temperatures for a long time. Also in the spring peeper frog (*Pseudacris crucifer*), the molality is increased temporarily as a reaction to cold temperatures. In the case of the peeper frog, this happens by massive breakdown of glycogen in the frog's liver and subsequent release of massive amounts of glucose. The phenomenon may also be observed in sea water, which due to its salt content, remains liquid at temperatures below 0 °C, the freezing point of pure water.

DEPRESSOR: 1. Also known as **depressor muscle**, it is a muscle whose contraction pulls down a part of the body. 2. Also known as **depressor nerve**, it is a nerve whose stimulation results in a lowering of blood pressure by dilating the arteries or lowering the heartbeat. 3. An instrument used to press down or to push aside an organ or part. An example is a tongue depressor.

DEPSIDES: A class of compounds formed by condensation of phenolic acid with a similar compound, the reaction being between the carboxylic acid group on one molecule and a phenolic OH group on the other. They are similar to esthers except that the OH group is linked directly to an aromatic ring. They are often found in lichens. They have also been isolated from higher plants, such as the Papaveraceae, Ericaceae, Lamiaceae and Myrtaceae. They have antioxidant, antibiotic, anti-HIV and anti-proliferative activities. As inhibitors of prostaglandin synthesis and leukotriene B_4 biosynthesis, they are powerful non-steroidal anti-inflammatories.

DEPSIPEPTIDES: A peptide in which one or more of the amide (-CONHR-) bonds are replaced by ester (COOR) bonds. They are found in nature as natural products. Depsipeptides have often been used in research to probe the importance of hydrogen bond networks in protein folding kinetics and thermodynamics.

DEPTH DIVERSITY GRADIENT: The increase in species' richness with increasing water depth until about 2,000 metres below the surface, where species' richness begins to decline.

DEPTH OF FIELD (DOF): The range of distance in front of or behind an object that is being focused by an optical instrument such as a microscope or camera, within which other objects will be in focus. In film and photography industry particular, it is the portion of a scene that appears acceptably sharp in the image.

DEPTH PERCEPTION: The ability of the eyes to locate the position of objects in three-dimensional objects. The retina is two-dimensional, so information about depth is created in the brain. The brain uses factors such as linear perspective, parallax, relative size and the slightly different view each eye has of the object.

DEPTH PROFILE: Dependence of concentration on depth perpendicular to the surface in a solid sample. It can be obtained by a simultaneous or sequential process of erosion and surface analysis or by measurement of the energy loss of primary backscattered particles produced by nuclear reactions.

DEPTH RESOLUTION: The distance between the 84 and 16 per cent levels of the depth profile of an element in a perfect sandwich sample with an infinitesimally small overlap of the components. These limits correspond to the 2-value of the Gaussian distribution of the signal at the interface.

DERACEMIZATION (ASYMMETRIC TRANSFORMATION): The conversion of a racemate into a pure enantiomer or into a mixture in which one enantiomer is present in excess or of a diastereoisomeric mixture into a single diastereoisomer or into a mixture in which one diastereoisomer predominates.

DERANGEMENT: A permutation such that no element occurs in its original or ordered position. For example, the only derangements of {1,2,3} are {2,3,1} and {3,1,2}.

DERELICT: Referring to land which has been damaged and made ugly by mining or other industrial processes or which has been neglected and is not used for anything.

DEREPRESSIBLE ENZYME: Enzyme produced in the absence of a specific inhibitory compound.

DERIVATION: 1. The act or process of deducing some expression from one result to another or a record of the steps of this process. 2. The process of finding the derivative of a function. In other words, A function which gives the slope of a curve; that is, the slope of the line tangent to a function. The derivative of a function f at a point x is commonly written $f(x)$. 3. An additive mapping on a commutative ring that satisfies the analogue of the product rule.

DERIVATION RULES: The transformation rules or rules of inference of a system of formal logic by which theorems may be recursively derived from axioms.

DERIVATIVE: 1. A compound that is formed from a related compound or that can be imagined to arise from another compound, if one atom is replaced with another atom or group of atoms. In biochemistry, the derivative is used for compounds that can be formed from the precursor compound, at least in theory. 2. In mathematics, also known as **differential coefficient**, it is the limiting value of the ratio of the change in a function to the corresponding change in its independent variable. The derivative of a variable with respect to another variable, for example, $dy/dx = y'$ is the differential coefficient of y with respect to x.

DERIVATIVE NOTATION: The derivative notation is $\frac{dy}{dx}$ or $f'(x)$ and is read "the derivative of y with respect to x".

DERIVATIVE POTENTIOMETRIC TITRATION: A titration that involves measuring, recording or computing the first derivative of the potential of a single indicator electrode with respect to the volume or otherwise added amount of reagent.

DERIVED COHERENT UNIT: A unit such as mol kg^{-1} and kg m^{-3} constructed exclusively from base units.

DERIVED QUANTITY: A kind of quantity characterised by an equation between basic quantities, for example, mass, length and time. Examples of some derived quantities are (i) Force (F) with a formula of mass (kg) × acceleration (ms^{-2}) and a derived unit of kg ms^{-2} named as Newton, symbol N; (ii) Pressure (P) with a formula of Force/Area and a derived unit of kg/m/s^{-2} named Pascal, Pa; (iii) Work (W) with a formula of Force × Displacement and a derived unit of N × m named joule, symbol J; (iv) electric charge (Q) with a formula of Current (A) × Time (S) and a derived unit of Ampere second (A s), named Coulomb with a symbol of C; and (v) Power (P) with a formula of Work (Joule) ÷ Time (S) and a derived unit of J/s, named watt, symbol W.

DERIVED PROPERTY: A characteristic that is not essential to a definition and is consequent to it.

DERIVED SAVANNAH: Forests converted into grass savannah by the action of man. Examples are abandoned forest zone farm lying near the forest-savannah boundary invaded and colonised by grasses and fire-tender forest trees destroyed by the action of man.

DERIVED SERIES: The series defined inductively starting with a given group, G, by taking each $G^{(n+1)}$ to be the derived subgroup of the preceding member $G^{(n)}$.

DERIVED SET: The set of all the cluster points of a given set. The second derived set is the derived set of the derived set. This process can be continued transfinitely. The Cantor-Bendixson theorem asserts that on the real line this process terminates countably, in that every closed set is the disjoint union of a perfect set and a finite or denumerable set.

DERIVED SUBGROUP (COMMUTATOR SUBGROUP): The subgroup generated by the set of commutators of a given group. It is denoted by G′.

DERIVED UNIT: An SI unit of measurement which is acombination of the base units. Derived units are expressed algebraically in terms of base units by means of the mathematical symbols of multiplication and division. Examples of derived units in the SI system are hertz for frequency derived as 1/s (s^{-1}); joule for energy or work derived as N m (kg m^2 s^{-2}); Newton (N) for force derived as kg m ($kg \cdot m \cdot s^{-2}$), watt for power (P) derived as J/s ($kg \cdot m^2 \cdot s^{-3}$); pascal (Pa) for pressure derived as N/m^2 ($kg \cdot m^{-1} \cdot s^{-2}$); tesla (T) for magnetic flux density and magnetic field strength derived as Wb/m^2 ($kg \cdot s^{-2} \cdot A^{-1}$); Ohm (Ω) for electrical resistance, impedance and reactance derived as V/A ($kg \cdot m^2 \cdot s^{-3} \cdot A^{-2}$); sievert (Sv) for equivalent dose (ionising radiation) derived as J/kg (m^2 s^{-2} and becquerel (Bq) for radioactivity (decays per unit time) 1/s (s^{-1}).

DERMAL: Of or relating to skin, especially to the dermis.

DERMAL PAPILLA (PAPILLA OF CORIUM): Any of the conical superficial projections of the the fibres, capillary blood vessels and sometimes nerves that interlock with recesses in the overlying epidermis, contain vascular loops and specialised nerve endings and are arranged in ridgelike lines most prominent in the hand and foot.

DERMAL TISSUE (DERMAL TISSUE SYSTEM): The outer protective covering of plants including the roots, stems, leaves, flowers, fruits and seeds, consisting generally of a single layer of tightly packed epidermal cells covering young plant organs formed by primary growth. It protects the plant from injury and water loss.

DERMAL TOXICITY: Adverse health effects resulting from skin exposure to a substance.

DERMAPTERA: An order of small to medium-sized, slender insects comprising the earwigs, characterised by cylindrical bodies, incomplete metamorphosis, short forewings, cerci, chewing mouthparts and a pair of stout curved forceps at the tip of the abdomen used to catch preys and in courtship.

DERMATITIS: A skin disorder (inflammation of the skin) with redness, swelling, itching and blistering, caused by the deficiency of riboflavin, niacin or biotin. The various types include **contact dermatitis**, which appears at the site of contact with an irritating substance or allergen; **atopic dermatitis**, which has patches of dry skin and occurs in infants, children and young adults with genetic hypersensitivities (atopy); **stasis dermatitis**, which affects the ankles and lower legs because of chronic poor blood flow in the veins; **seborrheic dermatitis**, which appears as scales in skin and is most often found on the scalp (dandruff) and areas rich in sebaceous glands; and **neurodermatitis**, which is the repeated scratching of an itchy skin area.

DERMATOGEN: The outer primary meristem of a plant or plant part that according to the histogen theory gives rise to epidermis. According to the histogen theory, it is one of the three histogens, the others being periblem and plerome.

DERMATOLOGY: The branch of medicine concerned with the structure and functions of the skin for prevention, diagnosis and treatment of disorders affecting it including the hair, nails oral cavity and genitals. A specialist in dermatology is called dermatologist.

DERMATOPHILUS INFECTION (CUTANEOUS STREPTOTHRICHOSIS; LUMPY WOOL; STRAWBERRY FOOTROT): A bacterial disease, *Dermatophilus congolensis*, which affects sheep, cattle and horses causing ulcers. It also affects people, who attend to infected animals if elementary hygiene measures are not observed. It causes severe skin infections.

DERMIS: The thicker and innermost layer of the skin of vertebrates. It is beneath the epidermis, lying between the epidermis and subcutaneous tissues and it is composed of two layers, the papillary and reticular dermis. The structural components of the dermis are collagen, elastic fibres and extrafibrillar matrix. The dermis contains nerve endings, sweat glands and oil (sebaceous) glands, hair follicles and blood vessels. The nerve endings sense pain, touch, pressure and temperature. The sweat glands produce sweat in response to heat and stress. The sebaceous glands secrete sebum into hair follicles. The hair follicles produce the various types of hair found throughout the body.

DERRIS: Any of various usually woody vines of the genus *Derris*, whose roots yield the insecticide rotenone, which is a powdered insecticide extracted from the root and used against fleas, lice and aphids.

DESALINISATION (DESALINIZATION; DESALINATION): Removal of salt, usually from sea water to produce fresh water for irrigation or drinking or any of several processes that remove excess salt and other minerals from water. Generally, however, it may refer to the removal of salts and minerals as in soil desalination, for example. Desalination is used on many seagoing ships and submarines, however, the use of the process is focused on developing cost-effective ways of providing fresh water for human use in regions, where the availability of fresh water is limited.

DESARGUES' THEOREM: A theorem in geometry that if the lines joining corresponding vertices of two triangles in three-dimensional space pass through a common point then the extended corresponding lines of the two triangles meet in three points, all on the same line. It was named after Gerard Desargues (1591-1661), French mathematician and engineer, a founder of modern geometry.

DESCARTES, RENÉ (1596-1650): French scientist, philosopher and mathematician who founded analytic geometry and introduced exponential notation, Cartesian coordinates and methods of solving polynomial equations. He is often credited with being the "Father of Modern Philosophy.

DESCARTES' RULE OF SIGNS: It states that the number of positive roots of a polynomial equation will equal the number of sign changes among the coefficients or that number minus a multiple of 2. To count the sign changes, the polynomial terms must be arranged in a descending order by the power of x and any zero coefficient is ignored. For example, consider the third-degree polynomial with roots a, b and c: $(x-a)(x-b)(x-c)=x^3-(a+b+c)x^2+(ab+bc+ac)x-abc$. Here are just two of the possibilities. If a,b and c are all positive, then the sign pattern of the coefficients of the polynomial will be +,-,+,- (three changes) and there are three positive roots. If a, b and c are all negative, then the sign pattern will be +,+,+,+ (no sign changes), there are no positive roots.

DESCENDANT: An element related to a given one by a chain of instances of a given relation. For example, 5 is a descendant of 3 under the successor relation on the whole numbers. Also, in a tree, a node further from the root than a given node, such that there is a branch including both. Let T be a rooted tree with root r_T and t a node of T. A descendant node s of a t is a node such that t is in the path from s to r_T. In other words, , the descendant nodes of t are all the nodes of T of which t is an ancestor node.

DESCENDING: In botany, it refers to parts going nearer to the ground at a moderate angle or at an angle of divergence of between 136-165°.

DESCENDING AORTA: The part of the aorta beginning at the aortic arch that runs down through the chest and abdomen. It is the largest artery in the body; divided into two portions, the thoracic and abdominal, in correspondence with the two great cavities of the trunk in which it is situated. Within the abdomen, the descending aorta branches into the two common iliac arteries which serve the pelvis and eventually legs.

DESCENDING CHAIN CONDITION: A finiteness condition for ascending or descending chains in a partially ordered set. In other words, the condition on submodules that no descending chain $M_1 \supseteq M_2 \supseteq M_3 \supseteq \dots$ has more than a finite number of distinct members; for every such chain there is an n such that $M_n = M_m$ for all $m \geq n$. Equivalently, every non-empty set of submoduless has a minimal element. Similar conditions are defined for rings, groups, etc.

DESCENDING ELUTION/DEVELOPMENT: A mode of operation in which the mobile phase is supplied to the upper edge of the paper or plate and the downward movement is governed mainly by gravity.

DESCENDING NODE: The point in an orbit, where the orbiting body crosses a reference plane, such as the ecliptic or the celestial equator, going from north to south.

DESCENT METHODS: The class of numerical optimisation methods that proceed iteratively in order to reduce continually the value of some given function, called the descent function $f(x^*) = f(x)$, where f is some function of the variables $(x_1, ..., x_n)$. This is often done in a predetermined direction, called the descent condition. The iterative sequence $\{x_k\}$ of the descent method is computed by the formula $x^{k+1} = x^k + a_k g^k$, where g^k is a vector indicating some direction of decrease of f at x^k and α^k is an iterative parameter, the value of which indicates the step-length in the direction g^k.

DESCENT PATH: The path followed by a spacecraft during its descent to Earth, particularly after re-entry the atmosphere.

DESCRIPTION: 1. The process of drawing a line or figure, the tracing of such a figure by the path of some object or to follow the shape of a line, curve or figure. For example, in a non-resistant medium, a projectile describes a parabola. 2. An expression containing a predicate and capable of replacing a name as the subject of a sentence. 3. The communication or portrayal of the meaning of something using verbal or written explanations, concrete models, gestures, assistive devices, etc.

DESCRIPTIVE GEOMETRY: The branch of geometry concerned with the representation of three-dimensional objects on a plane, the basis of architectural and technical drawing. There are two main systems of projection: the **perspective view**, in which the object is projected from point onto a plane; and **orthographic projection**, where the object is projected from one plane onto another.

DESCRIPTIVE INVESTIGATION: The use of careful observations to collect qualitative and quantitative data which measures or describes an organism, substance, reaction, or natural process. It should provide factual, accurate and systematic descriptions of phenomena without attempting to infer causal relationships. It is one of the three main types of investigations used by scientists; the others are comparative investigation and experimental investigation.

DESCRIPTIVE STATISTICS: Descriptive statistics is the study of ways to summarise data. For example, the mean, median and standard deviation are descriptive statistics that summarise some of the properties of a list of numbers.

DESERT: A dry arid area such as the Sahara Desert, without sufficient rainfall and subsequently vegetation to support human life; or a large sandy piece of land where there is very little rain and not much plant life. They often have an average annual precipitation of less than 250 millimetres per year and more water is lost by evapotranspiration than falls as precipitation in such areas. They may be classified as hot desert (*BWh, Köppen-Geiger-Pohl climate classification of hot deserts*); temperate desert (*BWk denotes a cold desert climate*); or as arid megathermal climates. The Sahara Desert is the largest desert in the world, encompassing almost all of northern Africa, covering an area of about 8.6 million sq km.

DESERT BIOME: An area characterised by dry conditions and plants and animals that have adapted to those conditions. It is found in areas where local or global influences block rainfall.

DESERTIFICATION: The creation of deserts by changes in climate or by human aided processes. The processes leading to it include overgrazing, destruction of forest belts, exhaustion of the soil by intensive cultivation without restoration of fertility, over-drafting of groundwater and diversion of water from rivers for human consumption and industrial use.

DESICCANT: A hygroscopic substance added to other substance to remove water molecules from them to make them dry. It induces or sustains dryness in its local vicinity in a moderately-well sealed container. The various types include anhydrous calcium sulfate, anhydrous calcium chloride, concentrated sulfuric acid, phosphorous(V) oxide, lime, silica and solid sodium hydroxide. Most pre-packaged desiccants are solids, which work through absorption or adsorption of water or a combination of both. Those used for specialised purposes may be in forms other than solids and function through principles such as chemical bonding of water molecules. Desiccant is the opposte form of humectant.

DESICCATION: 1. A method of preserving materials by removal of its water content or a state of extreme dryness or the process of extreme drying. Cells and tissues can be preserved by desiccation after subjecting the samples to freezing temperatures and stored at room temperature. 2. A state of extreme dryness or the process of extreme drying.

DESICCATOR: An airtight vessel made of glass in which materials may be stored either to dry them or to prevent them from absorbing moisture or substances which need a dry atmosphere. It usually contains a dehydrating agent which keeps the air dry. They are often used to protect chemicals, which are hygroscopic or which react with water from humidity. In view of this they are inappropriate for storing chemicals which react quickly or violently with atmospheric moisture such as the alkali metals. A glovebox or Schlenk-type apparatus may be more suitable for such purposes. Desiccators are also used to remove traces of water from an almost-dry sample.

DESIGNATED: In logic, of a truth value in a valuation system, functioning as the analogue of truth in a two-valued system or having one of the analogous values in a many-valued logic. It is often convenient to regard all the designated values as species of truth and all the anti-designated values as species of falsehood, at which level the law of excluded middle may then hold or else there may still remain a truth-value gap between designated and anti-designated values.

DESIGNATED AREA: An area that may be used for work with carcinogens, reproductive toxins or substances that have a high degree of acute toxicity. A designated area may be the entire laboratory, an area of a laboratory or a device such as a laboratory hood.

DESKTOP COMPUTER (DESKTOP PC): A computer that is designed to stay in a single location, usually on a desk or table. This restriction is imposed by its size and power requirement. Desktop computers cannot be powered from an internal battery and therefore must remain connected to a wall outlet.

DESKTOP PUBLISHING (DTP): The use of microcomputers for small scale typesetting and page make up. Desktop publishers use programs like Adobe InDesign and QuarkXpress to create page layouts for documents they want to print. These desktop publishing programs can be used to create books, magazines, newspapers, flyers, pamphlets and many other kinds of printed documents. Publishers may also use programs like Adobe Photoshop and Illustrator to create printable images. Word processing programs like Microsoft Word can be used for basic desktop publishing purposes.

DESMIDS: A group of unicellular green algae belonging to the order *Desmidiales* and the division Chlorophyta, with an elaborate chloroplast as in spirogyra. Their cells are characteristically split into two halves joined by a narrow neck, with each half being a mirror image of the other. There are around 40 genera and 5,000 to 6,000 species, found mostly but not exclusively in fresh water. Most are unicellular and are divided into two compartments separated by a narrow bridge or isthmus. Each compartment has one chloroplast and no flagella. Sexual reproduction occurs through conjugation.

DESMODIUM (BEGARWED; TICK CLOVER): Any herb or small shrub in the flowering plant family, Fabaceae that occurs in a variety of habitats from forests to grasslands and in secondary and disturbed vegetation. The plants grow wild in the Amazon rainforest of Peru and other South American countries and on the West Coast of Africa. These inconspicuous legumes have bright or large flowers and produce loment fruits. Some of the species are used medicinally to treat various ailments, for mulching and as manure to improve soil fertility. ***Desmodium adscendens*** commonly known as amor seco and beggar-lice grows to about 50 centimetres tall and produces numerous light-purple flowers and green fruits in small, bean-like pods. The plant contains alkaloids; flavonoids such as astragalin and cosmosin; soyasaponins such as dehydrosoyasaponin; and bioamine (tyramine). Medicinally, it has been used traditionally to treat various aiments such as hepatitis, asthma, bronchitis, jaundice, inflammation, muscle cramps and backache. ***Desmodium intortum*** with the common names greenleaf desmodium and beggarlice is a large trailing and scrambling perennial with a strong taproot. Its long trailing stems can root at the nodes if in contact with moist soil. It has trifoliate leaflets, with reddish-brown flecking on the upper surface. It bears deep pink flowers on a compact terminal raceme. The plant is used in irrigated pastures, for conservation as hay and silage, for cut-and-carry systems and as ground cover where the abundant leaf fall and slow decomposition result in a deep duff layer under the plants. ***Desmodium paniculatum***, known commonly as panicled-leaf tick trefoil grows up to about 1 metre tall. The trifoliate leaves alternate, with multiple elliptic to oblong, pointed green leaflets growing singly on the stem. The flowers are purple to pink growing on a stem and produce leguminous seedpods. The plant enriches the soil through nitrogen fixation.

DESMOSOME (MACULA ADHEREN): Patch-like junctions found between adjacent cells in epithelial tissues, randomly arranged on the lateral sides of plasma membranes that help strengthen the tissue by binding the cells together, while allowing movement of material within the intercellular space. They are specialised for cell-to-cell adhesion and help to resist shearing forces and are found in simple and stratified squamous epithelium. They are also found in muscle tissue, where they bind muscles' cells to one another.

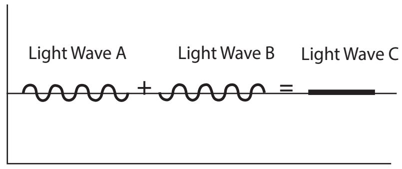
DESOLVATION: The process of removal or dissociation of the solvent component from the particle as a method of drying a sample or material in solution. In other words, it is the removal of solvents from material in solution. An example is the process of dissociating or releasing water molecules electrostatistically bound to a particle in aqueous solution. Desolvation is an initial major step in atomic absorption spectroscopy where a liquid sample is turned into an atomic gas. The other steps include vapourisation and atomisation.

DESORPTION: The removal of adsorbed atoms, molecules or ions from a surface. It occurs in a system being in the state of sorption (adsorption and absorption) equilibrium between bulk phase (fluid, i.e. gas or liquid solution) and an adsorbing surface (solid or boundary separating two fluids). When the concentration or pressure of a substance in the bulk phase is lowered, some of the sorbed substance changes to the bulk state. The more a chemical desorbs, the less likely it will adsorb, thus instead of sticking to the stationary phase, the chemical moves up with the solvent front. In chromatography, however, it is the ability for a chemical to move with the mobile phase.

DESTINATION FILE: The document into which an object created in a source file is inserted.

DESTRUCTIVE DISTILLATION: A process of heating solid substances in the absence of air to decompose them in order to obtain useful products from the vapour and residue or the process of pyrolysis conducted in a distillation apparatus such as a retort, to form volatile products, which are collected. In this way, the building blocks of many natural materials were deduced from the fragments generated from their thermal degradation. Most materials crack to give complex mixtures. Unlike distillation, which is a unit operation, it is a set of chemical reactions which entails the cracking or breaking up of macromolecules into smaller, more volatile, components. It has led to the discovery of many chemical compounds before such compounds could be prepared synthetically. It is also a viable route to many compounds and it has emerged as a possible route for recycling of monomers derived from waste polymers.

DESTRUCTIVE INTERFERENCE: The interference of two waves of equal frequency, which are out of phase resulting in cancellation, where the displacements at any instant are equal and opposite to each other. For light waves to interfere destructively, two conditions must be met. Firstly, the beam must be coherent, so that there is a perfect match of phase along the two light beams. For this to happen, both light beams must come from the same source. The second condition for destructive interference to occur is that the two light beams must be out of phase when they meet. Destructive interference is altogether different from no interference. Two candles give twice the brightness of one; whenever their lights coincide, the two beams of light neither cancel nor enhance because light frrom these two different candles is not coherent and it will neither interfere. Referring to the diagram, if light waves a and b combine out of phase their combined amplitudes are less and may even totally cancel each other, which is destructive interference.



Destructive interferance

DESTRUCTIVE PLATE BOUNDARY (CONVERGENT PLATE BOUNDARY; TENSIONAL PLATE MARGIN): A type of plate boundary where plate material is subducted. This occurs when two plates converge or move towards each other. The denser and heavier plate is forced under the lighter plate. Friction causes melting with the molten crust becoming magma which may be forced to the surface of the earth causing a volcanic eruption. One of the converging plates must be an oceanic plate for this to occur. If two plates of equal density collide, mountain ranges, such as the Himalayas are formed.

DESTRUCTIVE PROCESSES: Forces that destroy or break down physical features on Earth. Some of these processes occur slowly, called **slow destructive processes**, which can take millions of years to have an effect on the Earth. The two main forces that break down land slowly are weathering and erosion. Examples of the effects of **slow destructive processes** are the creation of a cliff by weathered and eroded mountains; creation of a valley from erosion by rivers or glaciers; and the creation of canyons or gorges, such as the Colorado River in the Grand Canyon by the wearing away of land. The other type of destructive processes occur quite quickly, called **quick destructive processes**, brought about by forces like earthquakes, floods, tsunamis, etc. These forces can change the surface of the Earth in seconds. A volcanic eruption can lead to landslides, ash falls, mud-flows, pyroclastic flows, etc., that can change the surface of that portion of the Earth where it occurs in seconds.

DESTRUCTIVE SUBSTANCE: Any explosive substance, flammable material, infernal machine or other chemical, mechanical or radioactive device or matter of a combustible, contaminative, corrosive or explosive nature.

DESULFURIZATION: The removal of compounds of sulfur from petroleum, natural gas or other fossil fuel before use. It may involve one of many techniques including elutriation, froth flotation, laundering, magnetic separation, chemical treatment, etc. It prevents the release of the acidic pollutant sulfur dioxide as well as allows the sulfur which is reduced to be collected.

DESYMMETRIZATION: The modification of an object, which results in the loss of one or more symmetry elements, such as those which preclude chirality (mirror plane, centre of inversion, rotation-reflection axis), as in the conversion of a prochiral molecular entity into a chiral one. **Desymmetrization step** is the removal of the smallest possible number of symmetry elements from a molecule with or without the restriction to pointwise substitution.

DETACH: In logic, to derive (an unconditional statement) by modus ponens or rule of detachment. **Detachment** is another word for **rule of detachment** or **modus ponens**.

DETAILED BALANCE: The cancellation of the effect of one process by another process that operates at the same time with the opposite effect.

DETECTION AND ATTRIBUTION (CLIMATE CHANGE DETECTION AND ATTRIBUTION): Detection of climate change is the process of demonstrating that climate has changed in some defined statistical sense, without providing a reason for that change. Attribution of causes of climate change is the process of establishing the most likely causes for the detected change with some defined level of confidence.

DETECTION EFFICIENCY: The ratio between the number of particles or photons detected and the number of similar particles or photons emitted by the radiation source.

DETECTION LIMIT: The minimum single result which, with a stated probability, can be distinguished from a suitable blankvalue. The limit defines the point at which the analysis becomes possible and this may be different from the lower limit of the determinable analytical range.

DETECTOR: 1. Any device that senses the presence of a signal or stimulus such as heat or pressure or light or motion and responds to it in a distinctive manner. 2. An electronic equipment that detects the presence of radio signals or radioactivity. 3. Also called a **demodulator**, it is a device such as a rectifier that extracts signal carrying information contained in a modulated wave or a a radio carrier wave. 4. In physics, a **particle detector** is a device for detecting, measuring and analysing particles and other forms of radiation entering it. Such devices play an important role not only in basic research, as in the study of elementary particles, but also in numerous applications of physics, from uses of radioactive tracers in medicine and biology to prospecting for natural ores that exhibit radioactivity. Almost all instruments used for detecting are based either on the ionisation of matter caused by radiation or on the luminescence it can cause in certain materials. 5. **Detector coil** is a coil used in magnetic resonance imaging as an antenna to record radiofrequency emissions of stimulated nuclei such as body coil and head coil.

DETECTOR EFFICIENCY: The ratio of the number of detected photons or particles to the number of photons or particles of the same type which are incident on the detector in the same time interval.

DETERGENT: A surfactant or a mixture containing one or more surfactants, having cleaning properties in dilute solutions (soaps are surfactants and detergents) or a substance that modifies the cleaning properties of water. Detergents are used for cleaning since they lower the surface tension of the water and emulsify fats and oils, making them go into solution with water and not easily forming a scum with any of the substances that cause hardness of water. They can be soapy or soapless detergents.

DETERMINANT: A scalar quantity denoted by |A| or det (A) which represents a particular sum of defined products of a square matrix. For a 2 × 2 matrix, the determinant is given by the difference between the products of the diagonal terms, i.e., the product of the elements of the principal diagonal minus the product of the elements of the second diagonal. For example, if $A=\begin{pmatrix} 13 & 4 \\ 6 & 2 \end{pmatrix}$ then det(A) = (13 x 2) - (4 x 6), therefore, det(A) = 26 - 24, which implies that det(A) = 2. For a 3 x 3 matrix the determinant can be found by using the evaluation of the row or column with the format $\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$ depending on the number of rows or columns needed.

Forexample, $A=\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}, |A|=a\begin{vmatrix} e & f \\ h & i \end{vmatrix}-b\begin{vmatrix} d & f \\ g & i \end{vmatrix}+c\begin{vmatrix} d & e \\ g & h \end{vmatrix}.$

DETERMINATE (SYMPODIAL GROWTH): A type of growth which results from a terminal bud growing to form lateral buds, the size of which is limited by cessation of meristematic activity during the year. The terminal flower matures first and the others start maturing from the bottom of the stem. This stops the main axis from growing long.

DETERMINATE CLEAVAGE: A type of embryonic development in protostomes that rigidly casts the developmental fate of each embryonic cell very early.

DETERMINATION: 1. Also known as **cell determination**, it is the committment of an undifferentiated embryonic cell to a particular path of differentiation, even though there may be no morphological features that reveal this determination. It is generally irreversible, but in the case of imaginal discs of drosophila that are maintained by serial passage, transdetermination may occur. 2. **Sex determination** is the establishment of the sex of an organism, usually by the inheritance at the time of fertilisation of certain genes commonly localised on a particular chromosome. In animals, it is determined by the presence or absence of the Y chromosome. The sex of the foetus before birth can be performed by examination of foetal fluids obtained by a process called amniocentesis.

DETERMINE: 1. In mathematics, to fix or define the position, form or configuration of, for example, any two points determine a straight line; a definite integral is determined up to a constant. 2. In logic, to explain or limit by adding differentiating characteristics.

DETONATOR: A small explosive charge used to trigger off a main charge of high explosive. It can be chemically, mechanically or electrically initiated, the latter two being the most common. Often, the primary explosive is a material called ASA compound, which is made from lead azide, lead styphnate and aluminium and is pressed into place above the base charge, usually TNT or tetryl in military detonators and PETN in commercial detonators.

DETOXIFICATION (DETOXICATION; DETOX): The process by which harmful compounds such as drugs and poison are converted to less toxic compounds in the body. It can be achieved by decontamination of poison ingestion and the use of antidotes as well as techniques such as dialysis and chelation therapy.

DETRITIVORE (SAPROPHAGES; DETRITUS FEEDERS): An animal that obtains nutrients by feeding on detritus or decomposing organic matter. Some examples are maggots, worms, blowflies, millipedes, woodlice, dung flies, sea stars, fiddler crabs and some sedentary polychaetes such as amphitrites and other terebellids. They can survive on any soil with an organic component and even live in marine ecosystems. They contribute to decomposition and the nutrient cycle and are important aspects of many ecosystems. In a food chain, detritivores are primary consumers.

DETRITUS: The organic debris produced during the decomposition of animals and plants or fragments of rock that have been worn away. It is the non-living particulate organic material as opposed to dissolved organic material. In terrestrial ecosystems, it is encountered as leaf litter and other organic matter intermixed with soil, which is referred to as **humus**, while in aquatic ecosystems it is encountered as organic material suspended in water (marine snow).

DETRUSOR MUSCLE: The smooth muscle of the bladder wall which is innervated with sympathetic fibre and contracts as a result of a reflex action when the bladder wall tension reaches a certain level. For example, it contracts when urinating to squeeze out urine but remains relaxed to allow the bladder to fill.

DEUTERATED COMPOUND: A compound in which some or all of the hydrogen atoms have been replaced by deuterium atoms.

DEUTERIUM (HEAVY HYDROGEN): An isotope of hydrogen containing one neutron and one proton in its nucleus, with the symbol D. It reacts with water to form heavy water with the formula D_2O. It is a stable isotope of hydrogen with a natural abundance in the oceans. It accounts for approximately 0.0156% of all naturally occurring hydrogen in the oceans on Earth. The nucleus of deuterium, called a **deuteron**, contains one proton and one neutron, whereas the far more common hydrogen nucleus contains no neutron.

DEUTERIUM OXIDE (HEAVY WATER): Water containing a higher-than-normal proportion of the hydrogen isotope deuterium, either as deuterium oxide, D_2O or ^2H_2O or as deuterium protium oxide, HDO or

H²HO. Heavy water is highly enriched in deuterium, up to as much as 100% D_2O and about 11% denser than water, but otherwise, physically and chemically very similar to water. In water, the deuterium-to-hydrogen ratio is about 156 ppm. Deuterium oxide boils at a slightly higher temperature and is harder to separate by electrolysis than protium water. Electrolysis can be used to drive off the protium water in a sample of regular water and the liquid left behind gets heavier and heavier as the concentration of deuterium oxide rises. The isotopic substitution with deuterium alters the bond energy of the water's hydrogen-oxygen bond, altering the physical, chemical and, especially, the biological properties of the pure or highly-enriched substance to a degree greater than is found in most isotope-substituted chemical compounds. Pure heavy water is not radioactive. Scientists use deuterium to follow the movement of materials in chemical and biochemical research, it is also an essential component in the design of some nuclear reactors, either for generating electric power or for producing nuclear-weapons isotopes, such as plutonium-239. The adult human body naturally contains deuterium equivalent to that in five grams of heavy water, which is thought to be harmless.

DEUTEROMYCOTA (FUNGI IMPERFECT; IMPERFECT FUNGI; ANAMORPHIC FUNGI; MITOSPORIC FUNGI): A taxon, usually given the rank of a phylum that is used in some classification to indicate all fungi in which sexual reproduction is apparently absent (they do not fit into the commonly established taxonomic classifications of fungi that are based on biological species concepts or morphological characteristics of sexual structures). Their sexual form of reproduction has never been observed. Only their asexual form of reproduction is known, implying that this group of fungus are known to produce their spores only asexually. There are about 25,000 species that have been classified in the deuteromycota. Fungi producing the antibiotic penicillin and those that cause athlete's foot and yeast infections are imperfect fungi. In addition, there are a number of edible imperfect fungi. The term is now used only informally, to denote species of fungi that are asexually reproducing members of the fungal phyla Ascomycota and Basidiomycota.

DEUTERON: A nucleus of a deuterium atom (the atom of heavy hydrogen, ^2H), consisting of a proton and a neutron bound together, whereas the far more common hydrogen nucleus contains no neutron. It is the simplest multinucleon nucleus. Its binding energy (the amount of energy which must be added to a deuteron for it to dissociate into a proton and a neutron) is 2.227 MeV. Deuterons are much used as projectiles in nuclear bombardment experiments.

DEUTEROSTOMES (DEUTEROSTOMIA; ENTEROCOELOMATES): Animals in which the first opening that appears in the embryo becomes the anus while the mouth appears at the other end of the digestive system. The main groups include chordates and echinoderms. They are distinguished by their embryonic development in the sense that in deuterostomes, the first opening (the blastopore) becomes the anus, while in protostomes it becomes the mouth. Their coelom develops through enterocoely. The extant phyla of deuterostomes are Phylum Chordata (vertebrates and their kin), Phylum Echinodermata (sea stars, sea urchins, sea cucumbers, etc.), Phylum Hemichordata (acorn worms and possibly graptolites) and Phylum Xenoturbellida (2 species of worm-like animals).

DEUTOPLASM: The nutritional material found in the yolk of eggs or in the cytoplasm of an ovum or other cell.

DEVARDA'S ALLOY: An alloy of copper (50%), aluminium (45%) and zinc (5%), used as a reducing agent in chemical tests for the nitrate ion (in alkaline solutions it reduces nitrate to ammonia). Prior to the development of ion chromatography which is currently the predominant analysis method, it was often used in the quantitative or qualitative analysis of nitrates in agriculture and soil science. It was named after the Italian chemist Arturo Devarda (1859-1944), who synthesised it for analysing nitrate in Chile saltpetre.

DEVELOPMENT: 1. In biology, the process whereby a living thing transforms itself from a single cell into a complicated multicellular organism. It involves gradual changes in size, shape and function which translate its genetic potentials (genotype) into functioning mature systems (phenotype). It includes growth but not repetitive chemical changes (metabolism) or changes over more than one lifetime (evolution). DNA directs the development of a fertilised egg so that cells become specialised structures that carry out specific functions. In humans, development progresses through the embryo and foetus stages before birth and continues during childhood. In both animals and plants, growth is greatly influenced by hormones. Also, factors within individual cells probably also play a role. 2. In photography, the process that produces a visible image on exposed photographic film involving the treatment of the exposed film with a chemical developer.

DEVELOPMENTAL PSYCHOLOGY: The scientific study of physical, cognitive and social change of human beings development of cognition and behaviour from birth to adulthood.

DEVELOPMENTAL RESPONSE: The development of morphological and physiological qualities of an organism in response to prolonged or changing environmental conditions.

DEVIATION: 1. The difference between one of an observed set of values and the true value, usually represented by the mean of all the observed values. In the set of values x_1, x_2, ..., x_n, the mean of these values is denoted by μ. The deviation of a given value x_n with respect to mean is given by: deviation = $(x_i - \mu)$. 2. The angle formed between a ray of light falling on a surface or transparent body and the ray leaving it.

DEVIATORIC: Of second-order Cartesian tensor, having zero trace. The deviatoric part of a second-order Cartesian tensor, **T**, is

$\mathbf{T}-\frac{1}{3}(\mathrm{tr}\mathbf{T})\mathrm{I}$, where I is the identity.

DEVICE: An invention or contrivance, especially a mechanical or electrical one, made or adapted for a special purpose. Examples include pulleys, electric motors, etc.

DEVICE DRIVER: In computing, software or program that controls a particular type of device such as printers, video adapters, network cards and CD-ROM readers that is attached to a computer.

DEVIL'S CURVE (DEVIL ON TWO STICKS): A curve with the Cartesian equation $y^4 - a^2y^2 = x^4 - b^2x^2$ and the polar equation $r^2(\sin^2\theta - \cos^2\theta) = a^2\sin^2\theta - b^2\cos^2\theta$. For $a = 25/24$, the curve is called the **electric motor curve**.

DEVITRIFICATION: Loss of the amorphous nature of glass as a result of crystalliation. In the course of firing fused glass, the surface of the glass develops a whitish scum, crazing or wrinkles instead of a smooth glossy shine, as the molecules in the glass change their structure into that of crystalline solids. The causes, commonly referred to as "devit" include holding a high temperature for too long, resulting in the nucleation of crystals. The presence of foreign residue such as dust on the surface of the glass or inside the kiln prior to firing can be key nucleation points where crystals can easily propagate. Also, the chemical composition of some glasses makes them more liable to devitrification than others. For example, a high lime content can be a factor in inducing this condition. In general, opaque glass can devit easily as crystals present in the glass gives it its opaque appearance and thus the higher the chance that it might devit. Clear uncoloured glass is least likely to devit. Methods used in avoiding it include cleaning the glass surfaces of dust or unwanted residue and allowing rapid cooling once the piece reaches the desired temperature, until the temperature approaches the annealing temperature. Devit spray can be applied to the surfaces. Even though this condition is normally undesired in glass art, it is sometimes used as a deliberate artistic technique.

DEVOLATILIZER: A material added to a sample to decrease its volatilisation or that of some component of it.

DEVONIAN PERIOD: A period of geological time 408-360 million years ago, the fourth period of the Palaeozoic era, characterised by the dominance of fishes, flourishing of plants and the advent of amphibians and ammonites. The Devonian Period has three subdivisions: Early, Middle and Late. It is named after Devon, England, where rocks from this period were first studied.

DEW: Precipitation in the form of moisture that collects on the ground as a result of condensation of atmospheric water vapour exposed to surfaces that cool during the night. It forms after the temperature of the ground has fallen below the dew point of the air in contact with it or fallen sufficiently to saturate air in contact with the surface, when the surface temperature drops to below atmospheric dew-point temperature. It is obvious in the early morning after a calm, cool, clear night, usually as beads of liquid water on the outside and upward-facing surfaces of trees, buildings, etc. When the surface cools to below freezing temperature, frost occurs.

DEW POINT: The temperature at which the air becomes saturated with water vapour, when cooled without addition of moisture or change of pressure. Any further cooling causes condensation. For example, fog and dew are formed in this way. On the other hand, **frost point** is the corresponding temperature of saturation with respect to ice.

DEW PONDS: A pond made to provide water supply, especially to serve as drinking ponds for farm animals in arid regions such as chalk or limestone region. The sides and bottom are lined with impermeable clay to retain the water, which generally comes from rainfall rather than from dew.

DEW-POINT HYGROMETER (COOLED SURFACE CONDENSATION): An instrument that measures relative humidity by means of the dew point. A small amount of ether is placed in a highly polished, thin, metallic cup and the evaporation of the ether, accelerated by blowing air through it, lowers the temperature of the cup. When the dew point of the surrounding air is reached, a film of moisture suddenly appears on the surface of the cup. The temperature is read by means of a thermometer and a table accompanying the instrument gives the relative humidity in terms of the atmospheric and dew-point temperatures.

DEWAR BENZENE (BICYCLO[2.2.0]HEXA-2,5-DIENE): An isomer of benzene, C_6H_6, which is a nonplanar bicyclic molecule. The compound is named after James Dewar ,who was believed to have proposed this structure for benzene in 1867.

DEWAR FLASK: A vessel for storing hot and cold liquid so that they maintain their temperature from their environment. It has double walls, the space between, being evacuated and the surfaces facing the vacuum being heat reflective. It was invented in 1892 by James Dewar as a container for liquid oxygen.

DEWAR STRUCTURE: A proposed structure of benzene, having a hexagonal ring of six carbon atoms with two opposite atoms joined by a long single bond across the ring and with two double C–C bonds, one on each side of the hexagon. Dewar structures contribute to the resonance hybrid of benzene.

Dewar Structure

DEWAR, SIR JAMES (1842-1923): British chemist and physicist and best known for his work with low-temperature phenomena. In 1875, he became a professor at Cambridge University and professor of chemistry at the Royal Institution of Great Britain in 1877. He began studying gases at low temperatures and in1872, invented the Dewar flask, the first vacuum or thermos bottle. In 1891, together with Frederick Abel (1827-1902), he developed the smokeless propellant explosive cordite and in 1898 was the first to liquefy hydrogen.

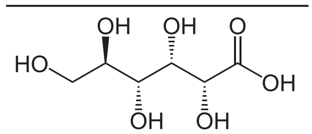
DEWLAP: Fold of loose skin hanging below the throat of some vertebrate animals. Examples are found in many reptiles such as lizards, in some birds, dogs, rabbits, cattle and some humans.

DEXTRAN: A glutinous glucose polymer, produced by fermenting sucrose or a complex, branched glucan polysaccharide, made of many glucose molecules. It is composed of chains of varying lengths from 10 to 150 kilodaltons. It is used as a thickening agent or a stabiliser in ice cream. It is also used as a substitute for plasma in blood transfusions or as an antithrombotic (anti-platelet), to reduce blood viscosity as well as a volume expander in anaemia.

DEXTRIN: An intermediate polysaccharide compound resulting from the hydrolysis of starch in maltose by amylase enzymes or a group of white, yellow or brown, low-molecular-weight carbohydrates, which is partially or fully water-soluble powder, produced by the hydrolysis of starch. They are mixtures of polymers of D-glucose units linked by -(1,4) or -(1,6) glycosidic bonds. During roasting under acid condition the starch hydrolyses and short chained starch parts partially rebranches with -(1,6) bonds to the degraded starch molecule. They yield optically active solutions of low viscosity and most can be detected with iodine solution, giving a red colouration, i.e. one can distinguish erythrodextrin (dextrin that colours red) and achrodextrin (giving no colour). They can be produced from starch using enzymes such as amylases, as happens during digestion in the human body and during malting and mashing. They are produced industrially by applying dry heat under acidic conditions (pyrolysis or roasting). This latter process also occurs on the surface of bread during the baking process, contributing to flavour, colour and crispness. Dextrins produced by heat are also known as **pyrodextrins** and white; and yellow dextrins from starch roasted with little or no acid is called **British gum**.

D

DEXTRONIC ACID (GLUCONIC ACID; D-GLUCONIC ACID): The acid formed by oxidation of the hydroxyl group on carbon-1 of glucose to a carboxylic acid group, with the chemical formula $C_6H_{12}O_7$ and melting point of 131 °C (404 K). It is very soluble in water and occurs naturally in fruit, honey and wine. It is used as an an acidity regulator. It is also used in cleaning products where it dissolves mineral deposits especially in alkaline solution.



Chemical Strucure of Dextronic Acid

DEXTROROTATORY: A term describing an optically active compound that causes the plane of polarised light to rotate in a clockwise direction.

DEXTROROTATORY AND LEVOROTATORY FORMS (DL-FORM): Identification of enantiomers with a *D*- or *L*- prefix because of the direction they rotate polarised light. Enantiomers that rotate polarised light in a clockwise direction are known as **dextrorotatory** (**right-handed**) **molecules** and enantiomers that rotate polarised light in a counter clockwise direction are known as **levorotatory** (**left-handed**) **molecules**. A racemic mixture is one that has equal amounts of left- and right-handed enantiomers of a chiral molecule and therefore does not rotate the plane of incident polarised light.

DEXTRORSE: Growing upward spirally from left to right, a characteristic exhibited by twining stems. On the other hand, **Sinistrorse** means growing spirally from right to left.

DEXTROSE (DEXTROGLUCOSE; D-GLUCOSE; GRAPE SUGAR): A type of glucose or sugar with the chemical formula, $C_6H_{12}O_6{\cdot}H_2O$. It is produced during cellular metabolism in plant and animal tissue. It is found in many fruits, especially grapes; and also a major component of honey and corn syrup. It is derived synthetically from starch.

DI: A prefix which means two.

DIABATIC COUPLING: Energy coupling between two potential-energy surfaces is sometimes known as diabatic coupling.

DIABATIC ELECTRON TRANSFER: Electron transfer process in which the reacting system has to cross over between different electronic surfaces in passing from reactants to products. For diabatic electron transfer, the electronic transmission factor is 1. The term **non-adiabatic electron transfer** has also been used and is in fact more widespread, but should be discouraged because it contains a double negation.

D

DIABETES INSIPIDUS (DI): An abnormal level of water in the body due to the deficiency of the kidney tubules to reabsorb enough. This may be due to inadequate secretion of anti-diuretic hormone (ADH) by the pituitary glands resulting in excessive production of dilute urine. It is characterised by excessive thirst and excretion of large amounts of highly diluted urine, with the fall in fluid intake having no effect on the latter. The most common type in humans is central diabetes insipidus, which is caused by a deficiency of arginine vasopressin (AVP), also known as antidiuretic hormone (ADH). The second common type is nephrogenic diabetes insipidus, which is caused by an insensitivity of the kidneys to ADH.

DIABETES MELLITUS: Relative or absolute lack of insulin leading to uncontrolled carbohydrate metabolism. In this case, the pancreas no longer produces enough insulin or the cells stop responding to the insulin that is produced, so that glucose in the blood cannot be absorbed into the cells of the body. The concentration of glucose in the blood, therefore, becomes dangerously high and the patient experiences thirst, weight loss and copious voiding along with degenerative changes in the capillary system. The symptoms include frequent urination, lethargy, excessive thirst and hunger. Diabetes can be controlled by a special diet helped in some cases by oral medications and regular injections of insulin.

DIADELPHOUS: A term used to describe stamen filaments that are fused into two groups.

DIAGENESIS: The physical, chemical or biological changes by which sediments become a sedimentary rock, after their initial deposition and during and after their lithification (converting sedimentary materials into solid rock by compacting or cementation), excluding weathering and metamorphism. These changes happen at relatively low temperatures and pressures and result in changes to the rock's original mineralogy and texture. After deposition, sediments are compacted as they are buried beneath successive layers of sediment and cemented by minerals that precipitate from solution. Grains of sediment, rock fragments and fossils can be replaced by other minerals during diagenesis. Porosity usually decreases, except in rare cases such as dissolution of minerals and dolomitisation (the process of converting into dolomite).

DIAGNOSE: To find out the nature of especially an illness by observing it or by making a careful examination, which may include a chemical test.

DIAGONAL: 1. In a polygon, the line segment joining a vertex with another non-adjacent vertex. 2. A line joining any two vertices of a polyhedron that are not in the same face. 3. Also known as **leading diagonal; main diagonal; primary diagonal;** or **principal diagonal**, it is a set of entries in a square matrix that runs from the upper left to lower right. 4. Also known as **off diagonal** or **secondary diagonal**, it is a set of entries in a square matrix that runs from the upper right to lower left elements. Alternatively, any other related sequence of elements of a matrix, such as the elements above the main diagonal.

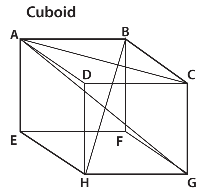


Fig. I: AG, AC and BH are diagonals

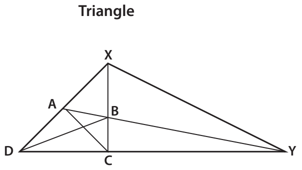


Fig. II: AC, BD and AY are diagonals of quadrilateral ABCDXY

DIAGONAL MATRIX: A matrix that has 0 entries along all nondiagonal entries, i.e., only the main diagonal may have non-zero values. An example is $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 5 \end{bmatrix}$.

DIAGONAL POINT: Any of the three points at which non-adjacent sides of a complete quadrangle intersect.

DIAGONAL PROCESS: The method of constructing a new member of a set from a list of given members by making its n^{th} term differ from the n^{th} term of the n^{th} member, so that the new member is distinct from every member on the given list and the set that includes the new member must have strictly larger cardinality than the original list. This method is used in the proof of Cantor's diagonal theorem and to show the uncountability of any proper interval of the real line.

DIAGONAL RELATIONSHIP (SIMILARITIES): A relationship within the Periodic Table by which certain elements in the second period have a close chemical similarity to their diagonal neighbours in the next group of the third period. For example, lithium (Li) and magnesium (Mg) show a diagonal relationship.

DIAGRAM: 1. A simplified drawing showing the appearance, structure or workings of something; a schematic representation. 2. A pictorial or graphical plan or figure drawn of some entities and their relations, used to explain an idea or statement. Examples are Argand or Venn diagrams. 3. The graphical representation of functions between structured sets, using arrows to specify the desired relationships.

DIAGRAM LEVEL: A level described by the removal of one electron from the configuration of the neutral ground state. These levels form a spectrum similar to that of a one-electron or hydrogen-like atom but, being single valency levels, have the energy scale reversed relative to that of single-electron levels. Diagram levels may be divided into valence levels and core levels according to the nature of the electron vacancy. Diagram levels with orbital angular momentum different from zero occur in pairs and form spin doublets.

DIAKINESIS: The final stage of the prophase in meiosis or the period at the end of the first prophase of meiosis when separation of homologous chromosomes is almost complete and crossing over has been achieved. It is characterised by shortening and thickening of the paired chromosomes, formation of the spindle fibres and disappearance of the nucleolus and degeneration of the nuclear membrane.

DIAL-UP LINE (SWITCHED LINE): A connection that provides PCs and other network devices access to a LAN or WAN via telephone lines. Types of dial up services include V.34 and V.90 modem as well as Integrated Services Digital Network (ISDN). Dial up systems utilise special-purpose network protocols like Point-to-Point Protocol (PPP).

DIALLEL CROSS: A breeding experiment in which each of a number of males is mated to each of a number of females.

DIALLER (DIALER): In computing, it is an Internet software package that makes the connection to the online service or Internet Service Provider. In Window systems, this is usually the WINSOCK.DLL file, with or without a front end (part of the program that interacts with the user) to make configuration easier. A dialler is also an electronic device that is connected to a telephone line to monitor the dialed numbers and alter them to seamlessly provide services that otherwise require lengthy National or International access codes to be dialed. it automatically inserts and modifies the numbers depending on the time of day, country or area code dialled, allowing the user to subscribe to the service providers who offer the best rates. A spyware dialler can be used maliciously to disconnect the legitimate telephone connection to the Internet and re-connects via a premium rate number (1-900 numbers), resulting in an extremely expensive phone bill for the account holder.

DIALOGUE BOX (DIALOG BOX): A small temporary window in a graphical user interface that appears in order to request information from the user. After the selections have been made, the user can typically click “OK” to enter the changes or “Cancel” to discard the selections.

DIALYSED IRON: A red colloidal solution of iron(III) hydroxide ($Fe(OH)_3$), used in medicine.

DIALYSIS: 1. A separation of particles passing through a semi permeable-membrane or a method by which large molecules such as starch or proteins and small molecules such as glucose in solution may be separated by selective diffusion through a semi permeable membrane. 2. A method used to provide an artificial replacement for lost kidney function in people with renal failure.

DIAMAGNETISM: Ability of certain materials to become magnetised in a magnetic field with a polarity opposite to the external magnetic field as a result of the applied external magnetic field inducing a reversed magnetic field in them, causing a repulsive force. In other words, diamagnetic materials are substances having a negative magnetic susceptibility which are weakly repelled by a magnet because they lack unpaired electrons and take a position at right angles to the lines of magnetic force. In a magnetic field, their induced magnetism is in a direction opposite to that of iron. They have a permeability of less than that of a vacuum (less than 1). All materials are naturally diamagnetic, although paramagnetism is stronger in materials with unpaired electrons. Examples of diamagnetic materials include bismuth (the most strongly diamagnetic material), copper, antimony, sodium chloride, gold and mercury.

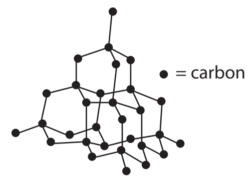
DIAMETER: 1. A straight line passing through the centre of a circle or a sphere and joining two points on its circumference. It divides the circle into two equal halves. Alternatively, the length of the segment of such a line of which the end-points are on the perimeter or surface of the figure. 2. Of a set in a metric space, the supremum of distances between pairs of points in the set.

DIAMETRAL: Located on or forming a diameter; for example, a diametral plane is one that includes a diameter of a given circle.

DIAMETRICAL: 1. Related to or located along a diameter. 2. In direct opposition; being at opposite extremes.

DIAMINOPIMELIC ACID: An amino acid found in bacteria and blue-green algae. It is involved in the biosynthesis of lysine in such organisms. It is also often found in the peptide side chains of the acetylmuramic acid molecules that partly make up the mucopeptides of bacterial cell walls.

DIAMOND: 1. An allotrope carbon that has crystallised in the cubic system under great pressure with a giant structure in which all the carbon atoms are tetrahedrally arranged around each other. They may be colourless and transparent or yellow, brown or black. It is less stable than graphite, but the conversion rate from diamond to graphite is negligible at ambient conditions. Diamond is a remarkable material with superlative physical qualities, most of which come from the strong covalent bonding between its atoms. It has the highest hardness and thermal conductivity of any bulk material. Diamond is also the least compressible and stiffest substance. It has an extremely low thermal expansion, is chemically inert with respect to most acids and alkalis, is transparent from the far infrared through the deep ultraviolet and is one of only a few materials with a negative work function (electron affinity). One consequence of the negative electron affinity is that diamonds repel water, but readily accept hydrocarbons such as wax or grease. Diamonds do not conduct electricity well, although some are semiconductors. It can burn if subjected to a high temperature in the presence of oxygen. It is extensively used in industry for cutting, grinding and polishing tools. However, industrial diamonds are increasingly being produced synthetically. It is highly prized as gemstones. 2. Also known as **rhombus**, it is a parallelogram with four equal sides and oblique angles or oblique-angled equilateral parallelogram with diagonals bisecting each other at right angles. Its area is half the product of the lengths of the two diagonals. A rhombus whose internal angles are 90° is called a square. A rhombus is thus a quadrilateral, because it has four sides and a parallelogram, because its opposite sides are parallel.



Structure of diamond (a gain covalent solid)

DIAMOND BY CHEMICAL VAPOUR DEPOSITION (DIAMOND BY CVD; CVD DIAMOND; LOW-PRESSURE DIAMOND): It is formed as crystals or as films from various gaseous hydrocarbons or other organic molecules in the presence of activated, atomic hydrogen. It consists of sp^3-hybridized carbon atoms with the three-dimensional crystalline structure of the diamond lattice. Diamond by CVD can be prepared in a variety of ways. Deposition parameters are: total (low) pressure, partial hydrogen pressure, precursor molecules in the gas phase, temperature for activation of the hydrogen and that of the surface of the underlying substrate. The energy supply for the hydrogen activation may be, for example, heat, radio frequency, microwave excitation (plasma deposition) or accelerated ions (e.g. Ar^+ ions). CVD diamond has also been obtained at atmospheric pressure from oxyacetylene torches and by other flame-based methods. Often CVD carbon films consist of a mixture of sp^2-hybridized and sp^3-hybridized carbon atoms and do not have the three-dimensional structure of the diamond lattice. In this case they should be called hard amorphous carbon or diamond-like carbon films.

DIAMOND RING EFFECT: A dazzling burst of light that happens a few seconds before and after totality during a solar eclipse. The effect is caused by the last bit of sunlight shining through valleys on the edge of the Moon.

DIAMOND-ANVIL CELL (DAC): A device for producing very high pressures. It is used to study the insulator-metal transition in such substances as iodine as the pressure is increased. It allows compressing a small (sub-millimetre sized) piece of material to extreme pressures, which can exceed 3,000,000 atmospheres (300 gigapascals). It has been used to recreate the pressure existing deep inside planets, creating materials and phases not observed under normal conditions.

DIAMOND-LIKE CARBON FILMS: Hard, amorphous films with a significant fraction of sp^3-hybridized carbon atoms, which can contain a significant amount of hydrogen. Depending on the deposition conditions, these films can be fully amorphous or contain diamond crystallites. These materials are not called diamond unless they have a full three-dimensional crystalline lattice. Diamond-like films without hydrogen can be prepared by carbon ion beam deposition, ion-assisted sputtering from graphite or by laser ablation of graphite. Diamond-like carbon films containing significant contents of hydrogen are prepared by chemical vapour deposition. The hydrogen content is usually over 25 atomic per cent. The deposition parameters are (low) total pressure, hydrogen partial pressure, precursor molecules and plasma ionisation. The plasma activation can be radio frequency, microwave or Ar^+ ions. High ionisation favours amorphous films while high atomic hydrogen contents favour diamond crystallite formation.

DIAMONDS DISEASE (SCOURING): A condition where an animal frequently passes liquid faeces.

DIANDROUS: A flower bearing two stamens.

DIANIONS: Molecular entities bearing two negative charges, which may be located on a single atom or on different atoms or may be delocalised.

DIAPAUSE: A period of delayed development that occurs in some species of insects characterised by greatly reduced metabolism and often timed to coincide with the insect’s chances of surviving adverse conditions such as temperature extremes, drought or reduced food availability. It is commonly observed in arthropods, especially insects and in the embryos of many of the oviparous species of fish in the order Cyprinodontiformes. It is considered a physiological state of dormancy.

DIAPHANOUS: Translucent or so thin as to transmit light.

DIAPHORECTICS (SUDORIFICS): Agents and substances containing properties which induce perspiration, helping the body to sweat and eliminate waste from the body. They enhance the entire circulatory and nervous system for proper blood circulation. There are two groups of diaphorectics: the **warming group** which induces sweating by improving circulation; and the **cooling group** eases the surface tension of the pores and nerves to stimulate sweating. The warming group is more powerful than the cooling group. Examples of warming diaphoretic substances include herbs such as chilli pepper, sage, thyme and ginger. Cooling diaphoretic herbs include peppermint, lemon balm, catnip, elderflower, and chamomile. They are said to be effective in treating ailments such as fever, joint pains, diarrhoea, dysentery, skin conditions, oedema, urinary tract infections and herpes.

DIAPHRAGM: 1. A sheet of muscle across the bottom of the rib-cage or a dome shaped muscular sheet separating the thorax from the abdomen in mammals that plays a part in the breathing system. The contractions of the diaphragm enlarge the thoracic cavity, allowing inspiration or inhalation to move air into the lungs. During expiration, the diaphragm returns to its original position. 2. A barrier contraceptive that is pushed into the vagina and fits over the cervix, preventing sperm from entering the uterus. They are made of latex (rubber) and formed like a shallow cup. Since vaginas vary in size, each patient will need to be fitted with a diaphragm that conforms to the shape and contour of the vagina as well as the strength of the muscles in the vaginal walls. 3. An opaque disk with a circular aperture at its centre, used to control the total light flux passing through an optical system or to reduce aberration by restricting the light. In this regard, it is placed in the light path of a lens or objective and the size of the aperture regulates the amount of light that passes through the lens. The centre of the diaphragm's aperture coincides with the optical axis of the lens system. The role of the diaphragm is to stop the passage of light, except for the light passing through the aperture. It is alternatively called an **aperture stop**, if it limits the brightness of light reaching the focal plane or a **field stop** or **flare stop** for other uses of diaphragms in lenses.

DIAPHRAGM CELL: An electrochemical cell for the production of chlorine and sodium hydroxide from brine.

DIAPHYSIS: The shaft of a limb bone, which, in immature animals is separated from the ends of the bone by cartilage. It is made up of cortical bone and usually contains bone marrow and adipose tissue or fat.

DIAPIRISM: A geological process in which a mass of material of high plasticity and low density such as rock salt, gypsum or magma, pushes upward into the heavier overlying strata. The resulting structure formed is called a **diapir** that may take the shape of anticlines, salt domes and other structures, in which oil and natural gas is often trapped.

DIARRHOEA: An illness in which the bowels are emptied too often and in too liquid a form. In this condition, one may have three or more loose or liquid bowel movements per day. The loss of fluids may result in dehydration and electrolyte imbalances. It is a major cause of death in developing countries and the second most common cause of infant mortality worldwide. Oral rehydration salts and zinc tablets are the treatment of choice.

DIASPORE (ALUMINIUM OXIDE HYDROXIDE): A mineral form of a mixed aluminium oxide and hydroxide, with the chemical formula AlO(OH), specific gravity is 3.4 and Mohs scale hardness is 6.5-7. It crystallises in the orthorhombic system and isomorphous with goethite. It is colourless or greyish-white, yellowish, sometimes violet in colour and varies from translucent to transparent and it may be easily distinguished from other colourless transparent minerals by a perfect cleavage and pearly lustre like mica, talc, brucite and gypsum and by its greater hardness. The mineral occurs typically as a final product of diagenesis in bauxite deposits formed by the tropical weathering of alumino-silicate rocks; from hydrothermal alteration of aluminous minerals; and as a hydrothermal mineral in some alkalic pegmatites. When heated before a blowpipe it decrepitates violently, breaking up into white pearly scales. It is a component of bauxite and an important ore of aluminium.

DIASTASE: A mixture of enzymes (α-, β- and/or γ-amylase) present in malt which converts starch by hydrolysis into maltose. This action forms the basis of the brewing process. Diastase was the first enzyme to be discovered by Anselme Payen in 1833. It has been used to aid the digestion of starch in some digestive orders.

DIASTEMA: 1. A gap between two adjacent teeth in the same dental arch as in the mouth of many herbivores, including sheep, which allows the tongue to manipulate food more easily. 2. A thin, yolk-free structure which is considered to play an essential role in the induction of the cleavage furrow.

DIASTEREOISOMER (DIASTEREOMER): Stereoisomerism other than enantiomerism (having two stereoisomers having molecules that are mirror images of each other). It is a stereoisomer not related as mirror images. Diastereoisomers are characterised by differences in physical properties and by some differences in chemical behaviour towards achiral as well as chiral reagents.

DIASTEREOISOMERIC UNITS: Two non-superposable configurational units that correspond to the same constitutional unit are considered to be diastereomeric if they are not mirror images.

DIASTEREOTOPIC: Constitutionally equivalent atoms or groups of a molecule, which are not symmetry related. Replacement of one of two diastereotopic atoms or groups results in the formation of one of a pair of diastereoisomers.

DIASTOLE: The phase of a cardiac cycle or heart beat, which occurs between two contractions of the heart during which the heart muscles relax and the two ventricles are dilated and filled with the blood flowing through them.

DIASTOLIC PRESSURE: The pressure that is exerted on the walls of the arteries around the body in between heart beats when the blood vessels are relaxed and the ventricles of the heart are filled with blood. Diastolic pressure represents the minimum pressure in the arteries. In blood pressure reading, the lower number is diastolic pressure and is at the bottom and the systolic pressure is the higher number at the upper side of tha reading. For example, in the reading of 120/80, 80 is the diastolic pressure and 120 is the systolic pressure. The normal diastolic pressure range for adults is 60 – 80 mmHg (millimetre of mercury).

DIATOMACEOUS EARTH (DIATOMITE; KIESELGUR): Fossilised deposits of diatoms or naturally occurring, soft, siliceous sedimentary that is easily crumbled into a fine white to off-white powder. The chemical composition of oven dried ones is 80 to 90% silica, with 2 to 4% alumina and 0.5 to 2% iron oxide and it has a particle size ranging from less than 1 micron to more than 1 millimetre, however, it often ranges from 10 to 200 microns. It is used for abrasives, polishes and as a filtering agent.

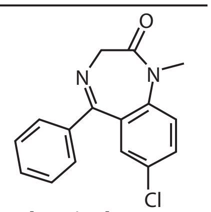
DIATOMIC MOLECULE: A molecule, which contains two atoms of either the same or different chemical elements. Examples are hydrogen, nitrogen, oxygen and carbon monoxide. Apart from the noble gases, most elements form diatomic molecules when heated, however, very high temperatures, sometimes reaching thousands of kelvins, are required.

DIATOMS: A large and diverse group of microscopic single-celled algae belonging to the class Bacillariophyceae having a silica cell wall called a **frustule**. They are found in great numbers in plankton in oceans and fresh waters and are important in the food chains. They are also widely distributed in soils. Because the silica cell walls do not decompose when they die, the silicified skeletons form diatomaceous earth. Although they can exist as colonies in the shape of filaments or ribbons (e.g., *Fragillaria*), fans (e.g., *Meridion*), zigzags (e.g., *Tabellaria*) or stellate colonies (e.g., *Asterionella*), most are unicellular. Diatom communities are a popular tool for monitoring environmental conditions, past and present and are commonly used in studies of water quality.

DIATROPISM: A tropism in which the plant part is aligned at right angles to the direction of the stimulus acting on it. A diatropic response to gravity (diageotropism) is seen in the horizontal growth of rhizomes and stolons. A diatropic response to light (diaphototropism) is often seen in the positioning of leaf blades at right angles to the incident light.

DIAUXIC GROWTH: Growth on a mixture of two carbon sources in which one carbon source is used up before the other one is mobilised. For example, in the presence of glucose and lactose, *Escherichia coli* (*E coli*) utilises the glucose before the lactose. During the first phase, cells preferentially metabolise the sugar on which it can grow faster, often glucose but not always. Only after the first sugar has been exhausted do the cells switch to the second. At the time of the "diauxic shift", there is often a lag period during which cells produce the enzymes needed to metabolise the second sugar.

DIAZEPAM (VALIUM): A benezodiazepine derivative drug used medically for treating anxiety, insomnia, seizures including status epilepticus, muscle spasms, restless legs syndrome, alcohol withdrawal, benzodiazepine withdrawal and Ménière's disease. Its action enhances the effect of the neurotransmitter gamma-aminobutyric acid (GABA) by binding to the benzodiazepine site on the GABA receptor leading to central nervous system depression. It may also be used before certain medical procedures such as endoscopy to minimise tension and anxiety and in some surgical procedures to induce amnesia. It also has anxiolytic, anticonvulsant, hypnotic, sedative, skeletal muscle relaxant and amnestic properties. It is also used as a recreational drug.



Chemical strucure of Diazepam

DIAZO COMPOUND: An organic compound that has two linked nitrogen atoms (azo) as a terminal functional group, it contains the functional group (-N=N-) and has the general formula $R_2C=N_2$. It is produced by the process of diazotisation and useful in the preparation of various dyes and drugs. Its electronic structure includes a positive charge on the central nitrogen and negative charge distributed between the terminal nitrogen and the carbon. A simplest example of a diazo compound is diazomethane, while a commercially relevant diazo compound is ethyl diazoacetate. Some of its most stable compounds are α-diazoketones and α-diazoesters (since the negative charge is delocalised into the carbonyl). A group of isomeric compounds with similar properties are the diazirines, where the carbon and two nitrogens are linked as a ring. Most alkyldiazo compounds are explosive.

DIAZONIUM COMPOUNDS: Organic compounds sharing a common functional group of the type $RN_2^+X^-$, where R can be any organic residue such as alkyl or aryl and X is an inorganic or organic anion such as a halogen. These compounds are highly unstable and decompose readily but are stable at low temperatures. They are used in the preparation of azo dyes, serving as important intermediates in the organic synthesis of dyes. Many of these compounds are explosive in the solid state.

DIAZONIUM SALT: An unstable salt containing positively charged nitrogen triply bonded to another nitrogen and an organic residue. They are formed by diazotisation and are useful ways of converting an amino group to other functional.

DIAZOOXIDES (DIAZOPHENOLS): Diazocyclohexadienones, which may also be considered as dipolar diazonio phenoxides, obtained by diazotizing aromatic primary amines having a hydroxy group in an *ortho* or *para* position.

DIAZOTISATION/DIAZOTIZATION (DIAZO PROCESS): The formation of diazo compounds by the interaction of sodium nitrite, an inorganic acid and a primary aromatic amine at low temperatures (below 5 °C, 278 K).

DIBASIC ACID: An acid which contains two replaceable hydrogen atoms per molecule or an acid that has two hydrogen ions to donate to a base in an acid-base reaction. An example is sulfuric acid (H_2SO_4). They undergo two successive steps of ionisation. Examples are $H_2SO_{4(aq)}$ $HSO_4^-{}_{(aq)} + H^+{}_{(aq)}$, and $HSO_4^-{}_{(aq)}$ $SO_4^{2-}{}_{(aq)}) + H^+{}_{(aq)}$.

DIBBLE (DIBBER): A hand-held pointed gardening tool that is used to make holes in soil, especially for planting bulbs or seedlings.

DIBIONTIC (DIPLOBIONTIC; DIPLOHAPLONTIC; HAPLODIPLONTIC): A life cycle in which there is alternation of generation or a life cycle that displays two distinct phases that differ in ploidy: haploid and diploid, a term primarily used to describe the life cycle of plants. All land plants have an alternation of multicellular haploid and diploid phases. In liverworts, mosses and hornworts, the dominant phase in the life cycle is the haploid gametophyte phase.

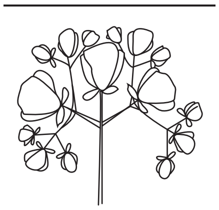
DICARBOXYLIC ACID (DIOIC ACID): An organic acid that contains two carboxyl groups or organic compounds that are substituted with two acid functional. They usually end with "-dioic", for example, hexanedioic acid. Their molecular formulae are generally given by HOOC-R-COOH, where R is usually an alkyl, alkenyl or akynyl group. They may be used in preparing copolymers such as polyamides and polyesters.

DICARPELLATE (BICARPELLATE): A flower having two carpels.

DICE: A small cube-shaped object with multiple resting attitudes, used for generating random numbers or other symbols. The sum of the numbers of opposite sides is seven.

DICER: An endoribonuclease in the RNase III family that splits double-stranded RNA (dsRNA) and pre-microRNA (miRNA) into short double-stranded RNA fragments known as small interfering RNA (siRNA), which is approximately 20-25 nucleotides long, often with a two-base overhang on the 3' end. It is a protein originally identified as the product of the Dicer gene in Drosphila, that trims double-stranded RNA molecules to form short interfering RNAs (siRNAs) or microRNAs (miRNAs). miRNAs are produced by eukaryotic cells as a means of suppressing gene activity. It contains two RNase III domains and one PAZ domain. The distance between these two regions of the molecule is determined by the length and angle of the connector helix and determines the length of the siRNAs it produces. Dicer catalyses the first step in the RNA interference pathway and initiates formation of the RNA-induced silencing complex (RISC), whose catalytic component argonaute is an endonuclease capable of degrading messenger RNA (mRNA), whose sequence is complementary to that of the siRNA guide strand. Dicer and other miRNA processing enzymes are useful in cancer prognosis. The human version is called Dicer 1.

DICHASIUM (DICHASIAL CYME): A cymose inflorescence in which the apices develop into flowers and two lateral branches develop below each apex from axillary buds at a common node. This gives a symmetrical appearance to the inflorescence, as seen in stitchwort (*Stellaria holostea*).

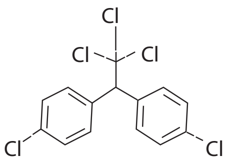


Dichasium cyme

DICHLORINE OXIDE (DICHLORINE MONOXIDE; CHLORINE(I) OXIDE; CHLORINE MONOXIDE): A strong oxidising orange or brownish-yellow gas at room temperature, with the molecular formula Cl_2O, density 0.893 g/cm^3, melting point -120.6 °C (153 K) and boiling point of 2.2 °C (275 K). It is very water soluble and it is the anhydride of hypochlorous acid. It is made by oxidation of chlorine using mercury(II) oxide. Preparation is by treating fresh yellow mercury(II) oxide with Cl_2 gas: $2\ Cl_2 + 2\ HgO \rightarrow HgCl_2{\cdot}HgO + Cl_2O$. It can alternatively be prepared by the reaction of Cl_2 gas with moist sodium carbonate, Na_2CO_3. It can explode in high concentrations when heated or sparked. Much of the dichlorine oxide manufactured industrially is used to make hypochlorites.

DICHLOROBENZENE: Any one of three isomeric liquid aromatic compounds: 2-Dichlorobenzene (orthodichlorobenzene), 1,3-dichlorobenzene (*meta*-dichlorobenzene, *m*-dichlorobenzene) and 1,4-dichlorobenzene (para-Dichlorobenzene or p-Dichlorobenzene) with the molecular formula $C_6H_4CI_2$. 1,2-Dichlorobenzene (orthodichlorobenzene) is a colourless liquid, which boils at 179 °C (452 K) and used as a solvent and chemical intermediate. 1,3-dichlorobenzene (*meta*-dichlorobenzene or *m*-dichlorobenzene) is a colourless liquid, which boils at 172 °C (445 K), soluble in alcohol and ether but insoluble in water. It is used as a solvent and chemical intermediate. 1,4-dichlorobenzene (para-Dichlorobenzene, p-Dichlorobenzene) are volatile white crystals, insoluble in water, soluble in organic solvents and used as a germicide, insecticide and chemical intermediate.

DICHLORODIPHENYLTRICHLOROETHANE; DDT; 1,1,1-TRICHLORO-2,2-DI(4-CHLOROPHENYL)ETHANE): A colourless organic, crystalline compound, with the molecular formula $C_{14}H_9Cl_5$, density 0.99 g/cm³ (solid), melting point 108.5 °C (381.65 K) that decomposes at a boiling point of 260 °C (533 K). It is made by the reaction of trichloromethanol with chlorobenzene. It is the best known of a number of chlorine-containing pesticides used extensively in agriculture. It has also been used to successfully control mosquitoes. It is, however, harmful to animal life as some animals can bioaccumulate it in their bodies. It is chemically stable and persists in the environment and in living tissues for several years. Its use has been banned in many countries.



Chemical structure of Dichlorodiphenyltrichloroethane (DDT)

DICHLOROMETHANE (DCM; METHYLENE CHLORIDE): A colourless, volatile, slightly toxic liquid with a moderately sweet aroma and the molecular formula CH_2Cl_2, density 1.33 g/cm^3, melting point -96.7 °C (176 K) and boiling point 39.6 °C (313 K). It has a characteristic odour similar to that of trichloromethane (chloroform), from which it is made by heating with zinc and hydrochloric acid. It is used as a refrigerant and solvent for paint stripping and degreasing, to decaffeinate coffee and tea, to prepare extracts of hops and other flavourings. Its volatility has led to its use as an aerosol spray propellant and as a blowing agent for polyurethane foams. Although it is not miscible with water, it is miscible with many organic solvents. It was first prepared in 1840 by Henri Victor Regnault, French chemist, who isolated it from a mixture of chloromethane and chlorine that was exposed to sunlight.

DICHOGAMY (SEQUENTIAL HERMAPHRODISM): The condition in which the male and female reproductive organs of a flower mature at different times, thereby ensuring that self fertilisation does not occur or the separation in time of gender expression in a hermaphroditic organism. This is a characteristic of most flowering plants and some fishes, gastropods and crustaceans. It is the opposite of simultaneous hermaphrodites or homogamy.

DICHOPATRIC SPECIATION: A form of allopatric specialisation in which portions of a pre-existing population become separated because of formation of geological barrier between them. Allopatric and allopatry are terms referring to organisms whose ranges are entirely separate, so that they do not occur in any one place together.

DICHOTOMOUS: 1. Any splitting of a whole into exactly two non-overlapping parts or a procedure in which a whole is divided into two parts or in half. 2. A term describing the type of branching in plants such as ferns and mosses that results, when the growing point (apical bud) divides into two equal growing points, which in turn divides after a period of growth, etc. 3. A partition of a whole (or a set) into two parts (subsets) that are: (i) **Jointly exhaustive**: everything must belong to one part or the other; and (ii) **Mutually exclusive**: nothing can belong simultaneously to both parts. The two parts thus formed are complements. 4. In logic, the partitions are opposites if there exists a proposition such that it holds over one and not the other.

DICHOTOMOUS KEY: A programme or chart for identifying an organism, consisting of contrasting statements and by a step-by-step elimination of the wrong statements leading to the clue of the identity of the specimen.

DICHOTOMOUS LINE SEARCH (BINARY LINE SEARCH; BISECTION METHOD): The iterative method of finding the maximum of a unimodal function that proceeds at each iteration by excluding one half of the remaining interval by testing two new function values, both near the midpoint of the present interval, one above and one below.

DICHOTOMOUS VARIABLE (DUMMY VARIABLE; INDICATOR VARIABLE): A variable that can take only two values. For example, gender can take only two variables, male and female; or in banking, bankrupt or solvent.

DICHOTOMY: 1. A division into two equal parts. 2. In botany, a mode of branching by constant forking, as, for example, in some stems and in the veins of leaves. 3. In astronomy, the phase of the moon or of an inferior planet when half of its disk is visible. 4. The classical paradox that motion can never be initiated because before a body can traverse a distance it must first complete the first half of the distance and before that the first quarter, etc. and so that a runner cannot start until he has made the last of this infinite sequence of steps.

DICHROIC MIRROR: A mirror used to reflect light selectively according to its wavelength.

DICHROISM: The property in some crystals, such as tourmaline, of selectively absorbing light vibration in one plane, while allowing light vibration at right angle to this plane to pass through. Light transmitted by thin plates of dark forms of tourmaline is almost completely polarised.

DICHROMATE: Salt containing the dichromate ion ($Cr_2O_7^{2-}$), used as an oxidising agent. They usually have an intense orange colour. An example is potassium dichromate ($K_2Cr_2O_7$).

DICLINY: The separation of male and female reproductive parts into different flowers. Diclinous plants may either have female and male unisexual flowers on the same individual (monoecy) or on different individuals (dioecy). The term diclinous may also refer to the flowers themselves, in which case the stamens are found in one flower and the pistil in another.

DICOTYLEDON (DICOT): One of the two subsets or classes of flowering plants distinguished by having two seed leaves, embryonic leaves or cotyledons within the embryo, the other being monocotyledons typically having one embryonic leaf (seed leaf). It is the major division of the angiosperm. Other characteristic features of the dicotyledons are possession of net-veins (net-venation), tap root, vascular cambium and large colourful flowers with parts in multiple of five. There are around 199,350 species within this group. Examples are beans, citrus, cowpea, okra and cotton.

DICOTYOSOME (GOLGI BODY): A cup-shaped array of flattened membranous vesicle found in plant cells and functionally equivalent to the golgi apparatus of animal cells. They modify proteins from the endoplasmic reticulum and may also polymerise sugar to polysaccharides.

DICTYOPTERA: An order of insects comprising three groups of polyneopterous insects: cockroaches (Blattaria), termites (Isoptera) and mantids (Mantodea). They occur mainly in tropical regions.

DICTYOSTELE: A type of dissected siphonostele in which the vascular tissue (as viewed in transverse section) is divided into a number of amphicribal vascular bundles called **meristeles**. These are separated by parenchymatous areas, which are usually associated with large closely placed leaf gaps. Dictyosteles are typical of many ferns such as Dryopteris and are essentially similar to solenosteles, except that the amphiphloic cylinder of the solenostele is reduced to a meshwork by the occurrence of numerous overlapping leaf gaps.

DICYCLIC: 1. In chemistry, it refers to a molecule that contains only two rings or atoms. 2. In botany, it refers to a flower which has two whorls. The petals form one whorl and the sepals form another whorl.

DIDYNAMOUS: Having two parts or pair or two long stamens and two short stamens, as do plants in the family Labiatae.

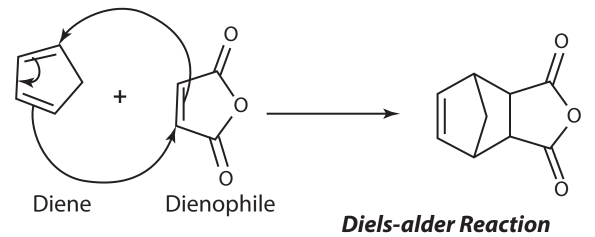
DIELECTRIC: Insulating materials such as glass, plastics and ceramics with no loosely bound electrons and so no current flows through them. When placed in an electric field, the positive and negative charges in the dielectric are displaced minutely in opposite directions, which reduce the electric field in the dielectric. Dielectrics are used in capacitors to reduce strong electric fields.

DIELECTRIC ABSORPTION: A capacitor which has been charged for a long time and then been completely discharged and has a small voltage on its terminal wires again, within seconds or minutes. This phenomenon has a particularly unfavourable effect in sample and hold applications in which charges are to be stored for comparison/ measuring purposes. The recharging comes from polarisation processes in the insulating material and is largely independent of the capacitance of the capacitor and the thickness of the dielectric.

DIELECTRIC BARRIER DISCHARGE (DBD): The electrical discharge between two electrodes separated by an insulating dielectric barrier.

DIELECTRIC CONSTANT (RELATIVE PERMITTIVITY): A measure of the resistance of a substance to an applied electric field, which is the ratio of the electric displacement in a medium to the intensity of the electric field producing or the ratio of the permeability of a substance to the permeability of a vacuum at the same magnetic field strength. Its symbol is μ_r. It is expressed as $\mu_r = \mu/\mu_o$, where μ is the permeability of the medium and μ_o is the permeability of free space or vacuum. It is an important parameter in characterising capacitors and materials with high dielectric constants are used in the manufacture of high-value capacitors. If all other factors such as temperature remain constant, the value of the dielectric constant increases as the electric flux density increases. This enables objects of a given size, such as sets of metal plates, to hold their electric charge for long periods of time and/or to hold large quantities of charge. Diamagnetic materials have constant relative permeabilities of slightly less than 1. Paramagnetic materials have constant relative permeabilities of slightly more than 1. The relative permeability of ferromagnetic materials increases as the magnetising field increases, reaches a maximum and then decreases. The reciprocal of magnetic permeability is magnetic reluctance.

DIELS-ALDER REACTION: The 1,4 addition of a conjugated diene to a compound, known as a dienophile, containing a double or triple bond to form a ring. The reaction can proceed even if some of the atoms in the newly-formed ring are not carbon. Some of these reactions are reversible, in which case the decomposition reaction of the cyclic system is called the Retro-Diels-Alder. When a Diels-Alder product is analysed via mass spectrometry these compounds are commonly observed.

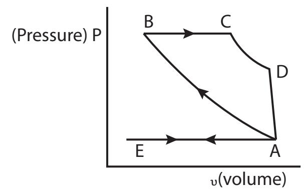


Diels-alder Reaction

DIENCEPHALON (BETWEENBRAIN; INTERBRAIN; THALAMENCEPHALON): The posterior part of the forebrain that connects the midbrain with the cerebral hemispheres, encloses the third ventricle and contains the thalamus and hypothalamus.

DIENE (DIOLEFIN): An organic compound or hydrocarbon which contains two carbon double bonds. They occur occasionally in nature but are often utilised in the polymer industry. Conjugated dienes are functional groups, with a general formula of C_4R_6.

DIESEL CYCLE: A thermodynamic cycle in which only air is admitted in the intake stroke. The air is then adiabatically compressed and fuel is injected into the hot air in the form of many small drops (not a vapour). Each drop burns over a small time, giving an approximation of an isobaric explosion. The explosion pushes the cylinder outwards. The power stroke, valve exhaust and exhaust stroke which follow are identical to those in the Otto cycle. In the diagram, the air is compressed adiabatically from A B to a very high temperature. The fuel burns from B C causing an expansion at constant pressure. CA is the rest of the working stroke, as an adiabatic expansion. Value opens at D and pressure falls to the atmospheric pressure. AE and EA represent the exhaust and charging strokes respectively.



Diesel cycle

DIESEL ENGINE: An internal combustion engine that operates on a thermodynamic cycle in which the ratio of compression of the air charge is sufficiently high to ignite the fuel subsequently injected into the combustion chamber. It burns a light weight fuel but relative to an Otto cycle operating engine, it uses a wider variety of fuels with a higher thermal efficiency, resulting in economic advantage under many service applications.

DIESEL FUEL: A type of liquid fuel used in a diesel engine or any liquid fuel used in diesel engines. It consists of alkanes obtained from petroleum. The most common one is a specific fractional distillate of petroleum fuel oil, however, alternatives such as biodiesel, biomass to liquid (BTL) or gas to liquid (GTL) diesel, that are not derived from petroleum, are increasingly being used. Petroleum-derived diesel is increasingly called petro diesel. Ultra-low sulfur diesel (ULSD) is the standard for defining diesel fuel with substantially lowered sulfur contents. In the UK, diesel fuel for on-road use is commonly abbreviated **DERV**, standing for **Diesel Engined Road Vehicle**, which carries a tax premium over equivalent fuel for non-road use.

DIET: The range or sum of foods consumed by an animal, human being or other organism. It sometimes implies the use of specific intake of nutrition for health or weight-management reasons. Dietary habits and choices play a significant role in health and mortality and can also define cultures and play a role in religion. A good diet requires proper ingestion, absorption of vitamins, minerals and food energy in the form of carbohydrates, proteins and fats.

DIETARY FIBRE (ROUGHAGE): A plant material that cannot be digested by human digestive enzymes but aids digestion, peristaltic movement and easy passage of free bowls. Even though it is indigestible in the small intestine, it is believed that it is partially digestible in the large intestine. Cellulose makes up most of the dietary fibres in the human diet. It is important in the diet because it relieves and prevents constipation. It also reduces the risk of colon cancer and plasma cholesterol levels and, therefore, the risk of heart disease. In addition, it slows gastric emptying and contributes to satiety. Sources of dietary fibre include whole grains, vegetables, nuts and fruits.

DIETETICS: The study of food, nutrition and health, especially when applied to food intake. It involves the integration and application of principles derived from several disciplines such as nutrition, biochemistry, physiology, food science and composition, management, food service and the behavioural and social sciences to achieve and optimise human health. **Dietitians** or **dieticians** translate the scientific evidence regarding human nutrition and use that information to help shape the food intake or choices of the public.

DIFFEOMORPHISM GROUP: The set of symmetry transformations that gives the relationship between events invariant in the general theory of relativity. It is the analogue in the general theory of Lorentz group in the special theory of relativity.

DIFFERENCE: 1. The result of subtracting one number or quantity from another, for example, the difference between 10 and 3 is 7. Also, the number or quantity that is added to one to yield the other. 2. Of two sets A and B, the set of elements of set A that are not elements of set B, written A\B or A - B. 3. **Symmetric difference** of two sets A and B is the set of elements which are in either of the sets and not in their intersection, usually written as A Δ B, A   B or A   B. It is therefore the union of the complement of A with respect to B and B with respect to A and corresponds to the XOR operation in Boolean logic. For example, the symmetric difference of set A = {1,2,3} and B = {3,4} is A   B = {1,2,4}, since 1, 2, 4 are each in one set but not both. The symmetric difference of the set of all students and the set of all females consists of all non-female students together with all female non-students.

DIFFERENCE EQUATION: A sequence relation between consecutive elements. The first difference is written as $D_2u(n)=u(n+1)-u(n)$, where $u(n)$ is the n^{th} element of sequence u. The second difference is written as follows:

$$D_2u(n)=D(Du(n))=(u(n+2)-u(n+1)-u(n))$$

$$=u(n+2)-2u(n+1)+u(n), \text{ etc.}$$

A recurrence relation, i.e., $u(n+2)+au(n+1)+bu(n)=0$ can be converted to a difference equation, in this case, a second order linear difference equation, $D_2u(n) + pD\, u(n) + q\, u(n) = 0$ and vice versa, where a, b, p, q are constants.

DIFFERENCE OF TWO SQUARES (THE DIFFERENCE OF PERFECT SQUARES): A way of clarifying expressions describing the difference between squared powers by means of factorisation. It is obtained by squaring a number or multiplying it by itself and then subtracting from it another squared number. The product of a binomial expression and its conjugate yields a difference of two squares. In mathematics, the difference of two squares, could be represented as: $a^2 - b^2 = (a - b)(a + b)$. For example, $x^2 - 9 = (x - 3)(x + 3)$.

DIFFERENCE PATTERN: The generation of a sequence by using differences between consecutive terms, for example, 2, 4, 6, 8…, has a regular difference pattern; therefore the next number is 10.

DIFFERENCE POLYNOMIAL: The polynomial, usually denoted d, over the ring, $[t_1, ..., t_n]$ defined as the product $\prod_{i<j}(t_1-t_2)$.

DIFFERENCE QUOTIENT: The slope of the secant line intersecting the graph of a function f at $f(x, f(x))$ and $(x + h, f(x + h))$. It is given by $\frac{f(x+h)-f(x)}{h}$. Forward difference quotient is positive and backward difference quotient is negative. The average of these produces the central differential quotient of the form $\frac{f(x+\Delta x)-f(x-\Delta x)}{2\Delta x}$.

DIFFERENCE SEQUENCE: A sequence of numbers constructed as the differences between the successive terms of a given sequence. Higher-order differences of a sequence $\{x_k\}$ are defined recursively. Thus the m^{th} forward difference sequence at x_k, is given by $^{m}X_k = \ ^{m-1}X_{k+1} - \ ^{m-1}X_k$, where $^{0}X_k = X_k$.

DIFFERENTIABLE: 1. Of a function or operator, possessing a well-defined derivative. 2. Of a function f of the real variables $x_1, ..., x_n$, is said to be differentiable at a point if its derivative exists at that point.

DIFFERENTIAL: 1. A term such as dx used in an expression such as $ydx - xdy$ to denote first-order small changes in the variable. Differentiation is the method by which a differential is found. 2. An increment in a given function, expressed as the product of the derivative of that function and the corresponding increment of the independent variable. If $F(x)$ is the given function, then $dF=\frac{dF}{dx}\times dx$. However, when dx_i is an increment in x_i, dF is not in general the increment in F. 3. An increment in a given function of two or more variables, expressed as the sum of the products of each partial derivative and the increment in the corresponding variable; if $F(x_1,...,x_n)$ is the given function, then $dF=\sum_{i=1}^{n}\frac{\partial F}{\partial x_i}\times dx_i$. However, when dx are the increments in x_1, dF is not in general the increment in F. 4. A mapping, df, derived from a given mapping, f, between two normed vector spaces, such that $\lim_{x\to 0}\frac{|f(x+h)-f(x)-df(x)h|}{|h|}=0$.

5. An arrangement of gears in the final drive of a vehicle's transmission that allows the wheels to spin at different speeds, especially when negotiating a corner.

DIFFERENTIAL CENTRIFUGATION: Separation of molecules, mixtures or organelles by sedimentation rate. In microbiology and cytology, for example, it is used to separate certain organelles from whole cells for further analysis of specific parts of cells. A tissue sample is first homogenised to break the cell membranes and mix up the cell contents. The homogenate is then subjected to repeated centrifugations, each time removing the pellet as the centrifugal force increases. Finally, purification may be achieved through equilibrium sedimentation and the desired layer is extracted for further analysis.

DIFFERENTIAL COEFFICIENT (DERIVATIVE): The limiting value of the ratio of the change in a function to the corresponding change in its independent variable. The derivative of a variable with respect to another variable, for example, $dy/dx = y'$ is the differential coefficient of y with respect to x.

DIFFERENTIAL DETECTOR: A device which measures the instantaneous difference in the composition of the column effluent.

DIFFERENTIAL EQUATION: A mathematical equation for an unknown function of one or several variables that relates the values of the function itself and its derivatives of various orders or an equation in which a derivative of 'y' with respect to 'x' appears as well as the dependent variable y. For example, given that $y = x^2 + 2x - 3$, then $\frac{dy}{dx}=2x+2$. The order of a differential equation is the order of its highest derivative. The degree of the equation is the highest power present of the highest-order derivative. Whenever a deterministic relation involving some continuously varying quantities (modelled by functions) and their rates of change in space and/or time (expressed as derivatives) is known or postulated, differential equation becomes indispensable and, therefore, plays an important role in engineering, physics, economics and other disciplines.

DIFFERENTIAL FORM: A part of the formalisation of the notion of surface integration that is central to a modern treatment of Stokes's theorem in which one talks of integrating k-forms over k-surfaces. A typical 1-form is xdy - ydx. More precisely, a differential form of degree r in n variables is a mapping from a domain in n-space into the set of r-covectors.

DIFFERENTIAL GEOMETRY: A branch of geometry concerned with the application of differential calculus to geometry. The concept may be applied to a small portion of a surface or curve around a point. An example is finding the tangent on a two-dimensional curve at a given point. It may also be used to calculate the curvature and length of a curve and to similar properties of surfaces in any number of dimensions.

DIFFERENTIAL MEDIUM: A medium with certain indicators such as colours or colony shapes, used to distinguish between different microorganisms during growth. An example of a differential medium is blood agar (nutrient culture medium that is enriched with real blood used for the growth of certain strains of bacteria) that is used to distinguish between bacterial species by their ability to break down the red blood cells included in the media.

DIFFERENTIAL OPERATOR (DEL; NABLA): 1. The differential vector operator, with the symbol Δ, used in vector analysis and defined as $i\frac{\partial}{\partial x}+j\frac{\partial}{\partial y}+k\frac{\partial}{\partial z}$, where i, j, k are unit vectors in the direction of the x-, y- and z-axes, respectively and $\partial/\partial x$, $\partial/\partial y$ and $\partial/\partial z$ are the respective partial derivatives of the given function of x, y and z. 2. Any operator involving derivatives.

DIFFERENTIAL ROTATION: 1. The variable rotation rate of a gassy body, such as a star or gas giant planet, with latitude. Gas near the equator spins around faster than gas near the poles. 2. The variable rotation in a disk-shaped structure, such as a galaxy, according to distance from its centre. Stars near the centre take less time to complete one rotation than those further away.

DIFFERENTIAL SCANNING CALORIMETRY (DSC): A thermoanalytical technique for measuring the temperature, direction and magnitude of transitions in a sample material by heating/cooling and comparing the amount of energy required to maintain its rate of temperature increase or decrease with an inert reference material under similar conditions. The reference sample should have a well defined heat capacity over the range of temperatures to be scanned. The technique was designed by E.S. Watson and M.J. O'Neill in 1960. The term DSC was used to describe this instrument which measures energy directly and allows precise measurements of heat capacity.

DIFFERENTIAL SPLICING (ALTERNATIVE SPLICING): The splicing of the RNA transcript of a gene in different ways by the cellular machinery during RNA processing to create distinct mature messenger RNAs (mRNA) encoding different proteins. This is responsible for the disparity between the relatively small numbers of human genes identified so far.

DIFFERENTIAL STRUCTURE (DIFFERENTIABLE STRUCTURE): A maximal continuously differentiable atlas; if it is r times continuously differentiable, the atlas is called a $C^{(r)}$ differential structure.

DIFFERENTIAL THERMAL ANALYSIS (DTA): A technique, similar to differential scanning calorimetry for observing the temperature, direction and magnitude of thermally induced transitions in a material by heating/cooling a sample and comparing its temperature with that of an inert reference material under similar conditions. The material under study and an inert reference are made to undergo identical thermal cycles and any temperature difference between sample and reference are recorded. This differential temperature is then plotted against time or against temperature (DTA curve or thermogram). Changes in the sample, either exothermic or endothermic, can be detected relative to the inert reference. DTA curve, therefore, provides data on the transformations, such as glass transitions, crystallisation, melting and sublimation. The area under a DTA peak is the enthalpy change and is not affected by the heat capacity of the sample.

DIFFERENTIAL THERMOMETER: A thermometer in which the differential thermal expansion of thin metals of different types, joined into a narrow strip and coiled into the shape of a helix or spiral, which serves as a pointer. It is used for accurate measurement of very small changes in temperature.

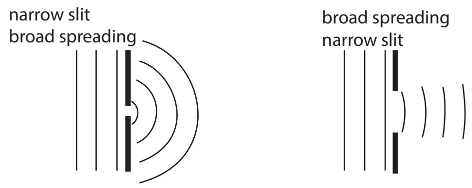
DIFFERENTIAL WEATHERING: A change in physical and chemical character of rocks in which rocks exposed to the same environment disintegrate at different rates. Variations in composition make softer, less weather resistant rocks wear away at a faster rate than harder, more weather resistant rocks, usually resulting in rugged or uneven surfaces as more resistant material protrudes above softer or less resistant parts.

DIFFERENTIALLY PERMEABLE MEMBRANE (SELECTIVELY PERMEABLE MEMBRANE; SEMI PERMEABLE MEMBRANE):: Cells that allow some materials such as water, molecules and nutrients to pass through, while keeping others out.

DIFFERENTIATION: 1. A mathematical process of a function or obtaining the gradient of the tangent to a curve of a function $f(x)$ at any point x. It may be done using algebraic manipulations that rely on three basic formulae and four rules of operation. The formulae used are:
(i) for algebraic functions, the derivative (D) of Dx^n = nx^{n-1}, in which n is any real number.
(ii) for trigonometric functions, the derivative (D) of D(sin$x$) is cos$x$ and
(iii) the derivative of the exponential function D(e^x) is itself (e^x). Where the functions are composed of combinations of algebraic, trigonometric and exponential functions, the basic rules for differentiating the sum, product, or quotient of any two functions $f(x)$ and $g(x)$ whose derivatives are known are: for sums, D($af + bg$) = aDf + bDg, known as the **sum rule**; for products, D(fg) = fDg + gDf, known as the **product rule**; and for quotients D(f/g) = (gDf − fDg)/g^2, known as the **quotient rule**. In all cases a and b are constants, and f and g are functions. Another basic rule is the **chain rule** that is used to differentiate a composite function which states that the derivative of a composite function is given by a product, i.e, $\mathrm{D}(f(g(x))) = \mathrm{D}f(g(x)).\mathrm{D}g(x)$. The derivation and exploration of these formulae and rules is the subject of differential calculus. To **differentiate** is to find the first derivative of a function. 2. In biology, it is the changes from simple to more complex forms undergone by developing tissues and organs so that they become specialised for particular functions or the process by which cells become increasingly different and specialised, giving rise to more complex structures that have particular functions. It occurs during embryonic development, regeneration in plants and meristematic activity.

DIFFERENTIATOR ANALOGUE COMPUTER: A device in the form of a computer whose variable output is proportional to the time differential of the variable input.

DIFFRACTION: The spreading or bending of waves as they pass through a small aperture or around a barrier in their path. In case of light waves, they spread into a pattern of bright and dark bands after diffraction. Diffraction is most noticeable if the wavelength is comparable to or greater than the size of the obstacle or aperture. If the wave length is much smaller than the obstacle or aperture, diffraction is difficult to detect. Diffraction of sound waves enables us to easily hear around corners and walls and through open windows and doors, even though you are not in line with the sound source. The diffracted waves subsequently interfere with each other, producing regions of reinforcement bright bands and weakening dark bands. Regions of reinforcement can be predicted by the use of Bragg's Law. Similar effects are observed when light waves travel through a medium with a varying refractive index or a sound wave through one with varying acoustic impedance. Diffraction occurs with all waves, including sound waves, water waves and electromagnetic waves such as visible light, x-rays and radio waves. It can be studied according to the principles of quantum mechanics.

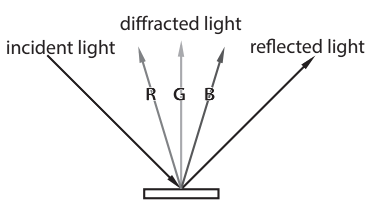


Diffraction through a narrow aperture with a broad wavefront *Diffraction through a broad aperture with a narrow wavefront*

DIFFRACTION ANALYSIS: The application of diffraction techniques (X-rays, electrons, neutrons) which are sometimes used to identify the presence of certain solid aerosols and dust particles through the characteristic diffraction patterns which result from each unique crystal structure.

DIFFRACTION FACTOR: For a specified frequency and a specified direction of sound incidence, the ratio of the sound pressure acting on the part of a transducer designed to receive sound, to the free-field sound pressure at that place in the absence of the transducer.

DIFFRACTION GRATING: An optical device consisting of periodic parallel grooves or lines either on a flat surface or on a concave surface, which by diffracting light, gives a large number of beams travelling in different directions which can interfere in such a way as to produce spectra. It can be thought of as a collection of narrow slits. Multiple slit interference can then be used to model the effects of a grating on incident light. It is used as the dispersive component in monochromators. Important characteristics are width of grating, groove density per unit length, distance between two successive grooves and blaze angle. By controlling the shape and size of the diffracting grooves, when producing a grating and by illuminating the grating at suitable angles, a beam of light can be thrown into a single spectrum, whose purity and brightness may exceed that produced by a prism. Gratings can now be made with much larger apertures than prisms and in such a form that they waste less light and give higher intrinsic dispersion and resolving power.



Diffraction Grating

DIFFRACTION PATTERN: In general, the interference pattern that results when diffracted waves are superposed. In astronomy, it is the image of a star formed by an optically perfect telescope. In a refractor, this consists of a small bright disk surrounded by concentric rings of light. In a reflector, the disk will have (typically) four spikes caused by diffraction from the secondary mirror supports.

DIFFUSE: To spread out freely through a tissue or substance in all directions; not concentrated or localised.

DIFFUSE COEVOLUTION: Multiple evolution of species that is influenced by the evolution of some other species, which itself may be evolving depending on some other factors. Coevolution is basically, the evolution of groups depending on each other, in order to survive.

DIFFUSE COMPETITION: The weak interactions between species that are ecologically and distantly related.

DIFFUSE GALACTIC EMISSION: Non-point-source gamma-ray emission from the plane of our Milky Way Galaxy. It is mostly due to interactions of cosmic rays with interstellar material.

DIFFUSE LAYER: The region in which non-specifically adsorbed ions are accumulated and distributed by the contrasting action of the electric field and thermal motion. Counter and co-ions in immediate contact with the surface are said to be located in the Stern layer. Ions farther away from the surface form the diffuse layer or Gouy layer.

DIFFUSED AIR AERATION: A technique in which air is compressed and discharged with force, below the surface of water. This is used in wastewater treatment establishments to increase oxygen transfer in the wastewater to maintain an aerobic condition, which improves water quality for recycling.

DIFFUSED JUNCTION SEMICONDUCTOR DETECTOR: A semiconductor detector in which the p–n or n–p junction is produced by diffusion of donor or acceptor impurities.

DIFFUSER: A porous plate or tube, commonly made of carborundum, alundum or silica sand, through which air is forced and divided into minute bubbles for diffusion in liquids.

DIFFUSION: 1. The random movement of molecules or particles from a region of high concentration to a region of lower concentration until they are evenly distributed. The concept is tied to the notion of mass transfer, driven by a concentration gradient, but diffusion can still occur when there is no concentration gradient, in which case there will be no net flux. The time dependence of the statistical distribution of the molecules or particles in space is given by the diffusion equation. 2. The scattering of a beam of light by reflection at a rough surface or by transmission through a translucent rather than transparent medium such as frosted glass. 3. The passage of elementary particles through matter when there is a high probability of scattering and a low probability of capture.

DIFFUSION BARRIER: Any material limiting penetration of atoms and molecules through the material. It is a thin layer, usually micrometres thick, of metal usually placed between two other metals. It is done to act as a barrier to protect either one of the metals from corrupting the other.

DIFFUSION BATTERY: An aerosol sizing instrument for particles with diameters below 0.2 μm. The fractionation is based on different diffusivities of the small particles and their deposition on the walls of the long parallel or circular channels, formed by equally spaced plates, bundles of small bore parallel tubes or sets of stainless wire screens.

DIFFUSION CLOUD CHAMBER: A type of Wilson cloud chamber developed by Cowan, Needels and Nielsen in 1950. The diffusion cloud chamber consists of a row of felt strips soaked in a suitable alcohol at the top of the chamber. The lower part of the chamber is cooled by solid carbon dioxide. The vapour continuously diffuses downwards and that in the centre is almost continuously sensitive to the presence of ions created by the radiation. A cloud chamber is used to detect particles of ionising radiation, developed by Charles T. R. Wilson (1869-1959), Scottish physicist and Nobel laureate in the late 1890s to study how water vapour and light interact.

DIFFUSION COEFFICIENT: A measure of the rate of movement of solutes or gases in response to a concentration gradient in the medium in which they are dissolved.

DIFFUSION CURRENT (DIFFUSION-CONTROLLED CURRENT): A faradaic current whose magnitude is controlled by the rate at which a reactant in an electrochemical process diffuses toward an electrode-solution interface and sometimes, by the rate at which a product diffuses away from that interface.

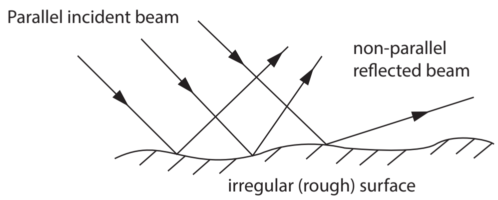
DIFFUSION EQUATION (HEAT EQUATION): The partial differential equation defined by $\nabla^2 u = c^2 \frac{\partial u}{\partial t}$, where ∇^2 is the

Laplacian in one, two or three dimensions, generally solved by using Fourier series. The solution of the heat equation gives the distribution of temperature in a region as a function of space and time when the temperature at the boundaries, the initial distribution of temperature and the physical properties of the medium are specified.

DIFFUSE-POROUS WOOD: Secondary xylem in which there is little or no seasonal variation in tracheary-element diameter as, for example, in the wood of yellow birch (*Betula lutea*). Growth rings are therefore difficult to discern.

DIFFUSION LAYER (CONCENTRATION BOUNDARY LAYER): The region in the vicinity of an electrode where the concentrations are different from their value in the bulk solution.

DIFFUSE REFLECTION (IRREGULAR/SCATTERED REFLECTION): Reflection of light by a rough surface resulting in an incident ray reflecting at many angles that can be described as casual, rather than at just one precise angle, which is the case of specular reflection. If a surface is completely non-specular, the reflected light will be evenly scattered over the hemisphere surrounding the surface. Examples of surfaces where diffused reflection can occur are leaves, walls and clothing.



Diffused Reflection

DIFFUSION LIMITED AGGREGATION (DLA): The process in which particles undergoing a random movement due to Brownian motion cluster together to form aggregates of such particles or the aggregation dominated by particles diffusing and having a nonzero probability of sticking together irreversibly when they touch. It can be observed in many systems such as electrodeposition, Hele-Shaw flow, mineral deposits and dielectric breakdown. This theory, proposed by Witten and Sander in 1981, is also applicable to aggregation in any system where diffusion is the primary means of transport in the system.

DIFFUSION OF GASES: A phenomenon in which there is a gradual mixing of molecules of one gas with the molecules of another by virtue of their kinetic properties.

DIFFUSION OF SOLUTION: The free movement of molecules or ions of a dissolved substance through a solvent, resulting in a complete evenly distribution of ions or molecules.

DIFFUSION PUMP (CONDENSATION PUMP; MOMENTUM TRANSFER PUMP): A vacuum pump in which oil or mercury vapour is diffused through a jet, which entrains the gas molecules from the container in which the pressure is to be reduced. The diffused vapour and the entrained gas molecules are condensed on the cooled walls of the pump. It uses a high speed jet of vapour to direct gas molecules in the pump throat down into the bottom of the pump and out of the exhaust. This pump is widely used in both industrial and research applications. Most modern diffusion pumps use silicone oil as the working fluid.

DIFFUSIONAL TRANSITION: A transition that requires the rearrangement of atoms, ions or molecules in a manner that cannot be accomplished by a cooperative atomic displacement; it may require the movement of atoms, ions or molecules over distances significantly larger than a unit cell. An example is the transition of graphite (hexagonal sheets of three-coordinated carbon atoms) to diamond (infinite three-dimensional framework of four-coordinated carbon atoms) at high temperature and pressure.

DIFFUSIONLESS TRANSITION: A transition that does not involve long-range diffusion of atomic species over distances significantly greater than a typical interatomic distance.

DIGAMMA FUNCTION (PSI FUNCTION): The logarithmic derivative of the gamma function, by the capital Greek letter Γ and defined as $\Psi(z) = \frac{\Gamma'(z)}{\Gamma(z)}$.

DIGESTER: A device or vessel in which chemical digestion takes place. An example is anaerobic digester, which is a series of biological processes in which microorganisms break down biodegradable material in the absence of oxygen to produce, for example, biogas, which is combusted to generate electricity and heat or can be processed into renewable natural gas and transportation fuels.

DIGESTIBILITY: The percentage of foodstuff taken into the digestive tract that is absorbed into the body. Digestibility is one of the main characteristics which determines the feeding value of herbage plants, i.e., the amount of the plant which is actually digested by the animal. Digestibility of starch is an important attribute for starch applications in food which use starch as the main ingredient.

DIGESTIBLE ENERGY (DE): It is the gross energy (GE) in feed less the energy lost in faeces.

DIGESTION: 1. Digestion is the process by which food eaten by an animal is broken down physically and chemically into simpler forms that can readily be absorbed and assimilated by the body. Food is chewed in the mouth, during which it is mixed with saliva, which begins to break down starches and kneaded by the tongue into a ball for swallowing. This is followed by peristalsis, which propels the food through the oesophagus and the rest of the alimentary canal. In the stomach, it mixes with acid and enzymes, which further break it down into a mixture, called **chyme**. It then enters the duodenum, the first part of the small intestine. Bile from the liver breaks up fat globules and enzymes from the pancreas and intestinal glands act on specific molecules, breaking carbohydrates down into simple sugars, proteins into amino acids and fats into glycerol and fatty acids, which are then absorbed by the bloodstream. Indigestible substances, such as fibre, pass into the large intestine, where water and ions are reabsorbed and faeces held for excretion. 2. The process through which complex compounds are broken down usually to liberate entrapped metals using concentrated mineral acids or a mixtures of it with the help of heat. It may also be achieved using a microwave.

DIGESTIVE SYSTEM: The system of organs and its associated glands, which are responsible for the digestion of food. It begins with the mouth and extends through the oesophagus, stomach, small intestine and large intestine, ending with the rectum and anus. The liver, pancreas, gallbladder, appendix and other organs are also included in this system.

DIGIMETIC: A cryptarithm in which digits represent other digits. A cryptarithm is a type of mathematical puzzle in which the digits are replaced by letters of the alphabet or other symbols.

DIGIT: 1. Any of the numbers from zero to nine in the decimal system or a symbol used in numerals to represent a single number in any system of numbering. Examples are the ten Arabic numerals, from 0 to 9, that are used to represent numbers in the decimal system. In a given number system, if the base is an integer, the number of digits required is always equal to the absolute value of the base. 2. In biology, a finger or toe of a human or a similar part on a terrestrial vertebrate.

DIGITAL: 1. Any system that is based on discrete or discontinuous data or values. In computing, a digital computer is one that processes data in discrete values of 1's and 0's as opposed to analogue systems that use a continuous range of values to represent information. Although digital representations are discrete, the information represented can be either discrete, such as numbers, letters or icons or continuous, such as sounds, images and other measurements of continuous systems. A digital system uses two-statements, either on/off or high/low voltage pulses to encode, receive and transmit information. 2. A measuring device that uses numbers. For example, a digital watch uses numbers to indicate the time, while a digital computer uses numbers (binary code) to solve problems.

DIGITAL AUDIO TAPE (DAT): A magnetic tape adapted for computer storage and backup and on which sound is recorded and played back in digital form. It was originally developed by Hewlett Packard and Sony as a solution for backing up information on a computer or network. DAT drives and tapes are available in two primary formats, DDS (Digital Data Storage) and DataDAT. DDS drives come in three different types, DDS-1, DDS-2 and DDS-3.

DIGITAL AUDIO WORKSTATION (DAW): A digital system such as audio hardware and audio software designed for recording and editing digital audio.

DIGITAL BROADCAST SATELLITES, DBS (DIRECT BROADCAST SYSTEM; DIRECT BROADCAST SATELLITES; DIRECT BROADCAST TELEVISION SATELLITE SYSTEMS): In computing, it is a combination of a small satellite dish and receiver which allows subscribers or end users to receive signals directly from geostationary satellites rather than through terrestrial broadcasting towers and repeaters. Signals are broadcast in digital format at microwave frequencies. A television that receives direct-broadcast satellites is known as **direct-broadcast satellite television** (**DBSTV**) or **direct-to-home television** (**DTHTV**).

DIGITAL CAMERA (DIGICAM): A camera that uses digital technology to record and store moving pictures or video with sounds and also still photographs in an automatically created folder called **Digital Camera Images** (**DCIM**) on the memory which contains the digital images. It records images by an electronic image sensor and stores it within the camera in a replaceable or fixed memory module, which is converted into a code of ones and zeroes that a computer can read and can also delete images to free storage space. Digital cameras can display images on a screen immediately after they are recorded. They can perform editorial jobs such as cropping and stitching pictures or image editing. They can be attached to astronomical probe devices such as the Hubble Space Telescope to record images in space.

DIGITAL COMPOSITING (COMPOSITING): In computing, it is computerized film editing in which multiple images are digitally assembled to make a final image, which can be used for print, motion pictures or screen display. For example, a sequence showing an actor hanging off the edge of a skyscraper may be put together out of footage of the actor in a safe location inserted into a shot looking down the side of the skyscraper, which might itself be a model.

DIGITAL COMPUTER: A computer that works on data supplied and stored in digital form. It processes data, including magnitudes, letters and symbols, that are expressed in binary form, using only the two digits 0 and 1. By counting, comparing and manipulating these digits or their combinations according to a set of instructions held in its memory, a digital computer can perform such tasks as to control industrial processes and regulate the operations of machines; analyse and organise vast amounts of business data; and simulate the behaviour of dynamic systems such as global weather patterns and chemical reactions in scientific research.

DIGITAL CITY: In computing, it is a locally focused online network, either text-based or graphical that uses the model of a city to make it easy for visitors and residents to find specific types of information such as community events, nightlife, localised yellow pages, entertainment, visitor's guide and e-commerce.

DIGITAL DATA TRANSMISSION: In computing, it is a means of transmitting data by converting all signals such as pictures, sounds or words into numeric codes, normally binary, a modem. At the receiving end, a similar system translates the incoming signal back to the original electronic data. This is done to eliminate any distortion or degradation of the signal during transmission, storage or processing.

DIGITAL DISPLAY: A method of indicating a reading of a measuring instrument, in which the appropriate numbers are generated on a fixed display unit by the varying parameter being measured rather than fixed numbers on a scale being indicated by a moving pointer or hand.

DIGITAL IMAGE PROCESSING: In computing, it is the process of using computer algorithms to perform pre-processing (extracting data, filtering, etc.), enhancement, display and information extraction of an image.

DIGITAL LIVING NETWORK ALLIANCE (DLNA): A standard protocol for wired and wireless multimedia devices to communicate with each other on a home network. It was created by Sony and other consumer electronic companies in 2003. It enables people, for example, to watch digital camera films on a big-screen TV, play an MP3 from a smartphone on a stereo, or send pictures from a photo album to a wireless printer via a tablet. Examples of DLNA devices are Windows PCs, tablets, Android phones, Blu-Ray disc players, wireless printers, camcorders, smartphones, and flat-screen TVs.

DIGITAL MONITOR: 1. In computing, it is a monitor accepts digital rather than analogue signals using standard cathode-ray tube (CRT) technology to convert a digital signal from the computer into an analogue one for displaying on the screen. All monitors except flat-panel displays use CRT technology, which is analogue. Examples of such monitors are the earlier MDA, CGA and EGA monitors used with PCs. 2. A video monitor controlled by a computer circuit that retains settings and resolutions in memory. Subsequent changes in resolutions do not require adjustments. Most computer monitors today are digitally controlled, but still accept analogue signals from the display adapter.

DIGITAL NERVOUS SYSTEM: In computing, the term was suggested by Bill Gates and is analogous to the biological autonomic nervous system of an organism and is Microsoft's way of describing how networks of computers in companies, particularly its own software products can help companies to operate more efficiently like the human body's nervous system, by passing information between them. This helps companies to have the technological capabilities necessary to sense, analyse, interpret and detect the most important signals from others. It is used to market Microsoft's ideas to people considered not technical enough to understand DNA, its Distributed interNetwork Applications architecture.

DIGITAL PUBLISHING (E-PUBLISHING; ELECTRONIC PUBLISHING): Distribution of information such as of e-books and digital magazines using computer-based media such as diskettes, CD-ROMs, DVDs, portable document files (PDF) or over the internet.

DIGITAL RECORDING: 1. A method of recording or transmitting sound in which the sound itself is not transmitted or recorded. Instead the pressure in the sound wave is sampled at least 30,000 times per second and the successive values represented by numbers which are then transmitted or recorded. 2. Magnetic recording in which the information is first coded in a digital form, generally with a binary code that uses two discrete values of residual flux.

DIGITAL ROOT: Take a number, *n*, add its digits, then add the digits of numbers derived from it and so on, until the remaining number has only one digit. This single digit result is called the **digital root** of *n*. For example, in the case of 53281: $5 + 3 + 2 + 8 + 1 = 19$; $1 + 9 = 10$; $1 + 0 = 1$; the digital root of 53281 is 1.

DIGITAL SIGNAL: An electronic signal that has discrete-line representation, i.e., it takes on discrete values over time as opposed to analogue which varies continuously over time.

DIGITAL SIGNAL PROCESSING, DSP (SIGNAL PROCESSING): In computer science, it is the process of manipulating analogue and digital signals using various mathematical and computational algorithms to filter, enhance image, compress data, etc., in order to improve the accuracy and reliability of communications. Examples include audio signal processing for electrical signals, such as speech or music; speech signal processing for processing and interpreting spoken words; image processing used in digital cameras, computers and various imaging systems; and video processing for interpreting moving pictures.

DIGITAL SIGNATURE: A scheme to establish or authenticate a digital message normally in the form of a digital file that is attached to an e-mail message or other electronic document. It is used to authenticate its origin and contents by means of associated encryption and decryption algorithms. Digital signatures can be used with PDF, e-mail messages and word processing documents.

DIGITAL SUBSCRIBER LINE (DSL): A technology used for transferring data over regular phone lines that significantly increases the digital capacity; it can be used to connect to the Internet. Like a cable modem, a DSL circuit is much faster than a regular phone connection, even though the wires it uses are copper like a typical phone line. The two main types of DSL are: (i) **Asymmetric DSL** (**ADSL**), which is used for Internet access with download speeds of up to about 1.5 megabits (not megabytes) per second and upload speeds of 128 kilobits per second; and (ii) **Symmetric DSL** (SDSL) designed for connections that require high speeds of up to 9 megabits per second and upload speeds of up to 640 kilobits per second in both directions.

DIGITAL SUBSCRIBER LINE ACCESS MULTIPLEXER (DSLAM): A network device used by Internet Service Providers (ISPs) to route incoming DSL connections to the Internet. It receives signals from multiple customer Digital Subscriber Line (DSL) connections and connects to high-speed digital communications channels using multiplexing techniques.

DIGITAL VERSATILE DISK (DVD): A type of optical disk or disk format introduced in 1996, which is similar in technology to a compact disk (CD) but contains much more data and is more precise. They are made in a Read-Only Memory (ROM) format as well as in erasable (DVD-E) and recordable (DVD-R) formats. DVD players can read CDs, however, CD players cannot read DVDs. Like a CD drive, a DVD drive uses a low-power laser to read digitised (binary) data that have been encoded onto the disc in the form of tiny pits. A double-sided, dual-layer version can store about 30 times as much information as a standard CD. It is projected that DVDs will eventually replace CDs, especially for multimedia workstations.

DIGITAL WATERMARK (FORENSIC WATERMARK; WATERMARK): Data embedded into intellectual property (IP) that is used to identify the source of illegal users. Digital watermarks provide copyright protection to digital intellectual property (IP), such as audio, video or image data. Digital watermarks are undetectable to the naked eye, or in the case of audio clips, inaudible but serve as signals when copyrighted materials are downloaded or reproduced.

DIGITAL-TO-ANALOGUE CONVERTER (DAC): A device or circuit that converts a digital (usually binary) code to an analogue signal (current, voltage or electric charge). An example is the processing by a modem, of computer data into audio-frequency (AF) tones that can be transmitted over a twisted pair telephone line.

DIGITALIS: 1. A genus of about 20 species of herbaceous perennials, shrubs and biennials, belonging to the family Plantaginaceae. The best-known species is the common foxglove, *Digitalis purpurea*. The flowers are tubular, produced on a tall spike and vary in colour with species, from purple to pink, white and yellow. It is often grown as an ornamental plant due to its vivid flowers. 2. A preparation of dry leaves or seeds of the foxglove (*Digitalis lanata*) used historically as a heart stimulant.

DIGITATE: Veined, lobbed or divided from a common point like the fingers of a hand or spreading from the centre like fingers from a hand: palmate but with narrower segments.. Examples can be found in flowers and leaves. **Digitation** means finger-like or digit-like division.

A digitate leaf

DIGITIGRADE: Animals that stand or walk on their digits or toes or the gait of most fast-running animals, including dogs and cats in which only the toes are on the ground and the rest of the foot is raised off ground. Digitigrades are generally quicker and move more quietly than other mammals and it is responsible for the distinctive hooked shape of their legs. **Plantigrade** locomotion refers to mammals such as humans who usually walk with the soles of their feet on the ground.

DIGITIZE: In computing, to convert analogue signals such as pictures, sound, video or film into a binary data, which devices such as a computers and digital cameras can process, by manipulating, storing, transmission, etc. For example, a scanner captures an image such as texts and converts them to an image file, such as a bitmap. Another example of digitisation is optical character recognition (OCR) program, which analyses a text image for light and dark areas in order to identify each alphabetic letter or numeric digit and converts each character into an ASCII code.

DIGITRON (NIXIE TUBE): An electronic gas-discharge glass tube that provides a digital display of numerals or other characters, used in devices such as calculators and counters. It consists of a wire-mesh anode and several cathodes, shaped like numerals or other symbols. When power is applied to one cathode it displays an orange glow discharge depicting a character.

DIGOXIN: A drug derived from digitalis (*Digitalis lanata*) that increases the strength and efficiency of heart contractions by inhibiting the activity of an enzyme (ATPase) that controls movement of calcium, sodium and potassium into heart muscle. It is used to treat heart failure and to control the abnormal rhythm of the heart. Side effects include loss of appetite, nausea, vomiting, diarrhoea and dizziness.

DIGRAPH (DIRECTED GRAPH): In graph theory, a graph or a set of vertices or nodes that are connected by edges. These play an important role in network optimisation problems.

DIGYNOUS: A flower that has two pistils.

DIHEDRAL: 1. Having or formed by the intersection of two planes. 2. **Dihedral angle**, also known as **dihedron** or **torsion angle** is an angle formed by the intersection of two planes or the angle between two intersecting planes on a third plane normal to the intersection of the two planes. 3. In chemistry, in a nonlinear chain of atoms A-B-C-D, it is the angle between the plane containing atoms ABC and the plane containing BCD. The torsion angle can have any value from 0° to 180°. If the chain is viewed along the line BC, the torsion angle is positive if the bond AB would have to be rotated in a clockwise sense (less than 180°) to eclipse or align with the bond CD. If the rotation of AB has to be in an anticlockwise sense, the torsion angle would be negative. In proteins, the two torsion angles phi and psi describe the rotation of the polypeptide chain around the two bonds on both sides of the Ca atom. **Dihedral group**, denoted by D_n or dih (n), where n is the number of sides of the polygon, is the group of symmetries of a regular polygon.

DIHYBRID: A hybrid individual that is heterozygous for two genes or two characters.

DIHYBRID CROSS: A genetic cross between individuals, controlled by genes at different loci or a cross between F_1 offspring (first generation offspring) of two individuals that differ in two traits of particular interest. For example, if a true-breeding plant with green pod colour (GG) and yellow seed colour (YY) is cross-pollinated with a true-breeding plant with yellow pod colour (gg) and green seeds (yy), the resulting offspring will all be heterozygous for green pod colour and yellow seeds (GgYy). It is very useful in testing for dominant and recessive genes in two separate characteristics. It also has different applications in Mendelian genetics.

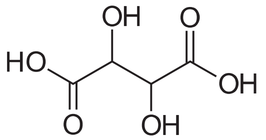
DIHYDRATE: A hydrate (chemical) whose molecules have two associated molecules of water of crystallisation. An example is sodium dichromate, $Na_2Cr_2O_7 \cdot 2H_2O$. The chemical state of the water varies widely between hydrates.

DIHYDRIC: Describing alcohols containing two hydroxyl groups (OH). An example is ethylene glycol, the simplest and most important dihydric alcohol. Dihydric alcohols are dihydroxy derivatives of alkanes with the general formula $C_nH_{2n+2}O_2$. They are classified as α, β, γ... glycols, according to the relative position of two hydroxyl groups; α is 1,2 glycol, β is 1,3 glycol. When alcohol contains only one hydroxyl group it is called as **monohydric alcohol**. An example is ethanol (CH_3CH_2OH). Alcohol with three hydroxyl groups attached is called **trihydric**. An example is glycerol (CH_2OH-CHOH-CH_2OH).

DIHYDROGEN: The normal form of molecular hydrogen, H_2 used to distinguish it from hydrogen atoms.

DIHYDROXYPHENYLALANINE (DOPA): A derivative of amino acid, with the chemical formula $C_9H_{11}NO_4$ formed from tyrosine in the liver during melanin and epinephrine biosynthesis. It occurs widely in animals and plants. It is a precursor in the synthesis of dopamine noradrenaline. Its levorotatory isomer (L-dopa) is used as a drug in treating Parkinson's disease.

DIHYDROXYSUCCINIC ACID (TARTARIC ACID): A colourless crystalline diprotic organic acid, with the chemical formula $C_4H_6O_6$, which is a dihydroxy derivative of succinic acid. It occurs naturally in many plants, particularly grapes, bananas and tamarinds and is one of the main acids found in wine. When added to other foods it gives a sour taste and is used as an antioxidant. Salts of tartaric acid are referred to as tartrates.



Chemical structure of Dihydroxysuccinic Acid

DIISOTACTIC POLYMER: An isotactic polymer that contains two chiral or prochiral atoms with defined stereochemistry in the main chain of the configurational base unit.

DIKARYON: A cell of a fungal hypha or mycelium composed of two haploid nuclei of different strains which associate in pairs but do not fuse or the state in certain fungi in which each compartment of a hypha contains two nuclei, each derived from a different parent. The cell formed, therefore is not completely diploid.

DIKARYOSIS: The commonest form of heterokaryosis. It is an abnormal maturation seen in exfoliated cells that have normal cytoplasm but hyperchromatic nuclei or irregular chromatin distribution. It may be followed by the development of a malignant neoplasm.

DIKARYOTIC: Having two different and distinct nuclei per cell. Examples of dikaryotic organisms are *Dikaryomycota*, a phylum of fungi which has an extended dikaryon period in their life cycle; and *Basidiomycetes*, which includes both poisonous and edible mushrooms. Dikaryotic cells are formed when compatible nuclei from two different cells break down the wall between the cells to "cohabit" a compartment without sexual reproduction, although they are sexually compatible.

DIKETONES: Organic compounds that have two carbonyl (ketone) groups. They are three kinds: 1,2-Diketones, also called α-diketones, R.CO.CO.R'; 1,3-Diketones or β-diketones, R.CO.CH_2; and 1,4-Diketones or γ-diketones, R.CO.CH_2.CH_2.CO.R'. The simplest diketone is diacetyl, also called 2,3-butanedione. Examples of 1,2-, 1,3- and 1,4-diketones are diacetyl, acetylacetone and hexane-2,5-dione respectively, while dimedone is an example of a cyclic diketone.

DILATANCY: The tendency of a substance or material to have its viscosity increased when affected by an outside force or agitation, occurring when closely packed particles are combined with enough liquid to fill the gaps between them. A liquid, which becomes more viscous the faster it moves is called **dilatant**. Such liquids are called non-Newtonian fluids. At low velocities, the liquid acts as a lubricant and the dilatant flows easily. At higher velocities, the liquid is unable to fill the gaps created between particles and friction greatly increases, causing an increase in viscosity. An example of a dilatant material is sand soaked completely with water.

DILATATION (DILATION): 1. An increase in the volume of a biological structure or organ such as a membrane and pupils of the eye, as a result of relaxation of the blood vessels within them. 2. In geometry, a transformation in which a figure is enlarged or reduced by a given factor, called a scale factor around a given central point; preserving the same shape as the original, but a different size. The distance the points move depends on the scale factor. If the scale factor is greater than 1, the image is an enlargement; but if the scale factor is between 0 and 1, the image is a reduction. A dilatation that is not just a translation is called a central dilatation because all lines joining corresponding points of a figure and its image are concurrent. 3. A **dilatation at P with ratio *k*** is a mapping of a space into itself that is defined by $f(x) = kx + (1 - k)p$, where k is a non-zero ral number and p is the set of the point P, the centre of the dilatation. Positive or negative dilatation, depends on the position of k.

DILATIONAL TRANSITION (DILATATIONAL TRANSITION): A transition in which the crystal structure is dilated or compressed along one or more crystallographic direction(s), while the symmetry about that direction is retained. Examples are: (i) the transition of a C_sCl (Caesium chloride) type structure to a rock salt structure in which dilation occurs along the three-fold axis; (ii) the transition T_D in quenched NiS in which volume expansion occurs without change of symmetry on going from a metallic state ($T > T_D$) to a semiconducting state ($T < T_D$).

DILATOMETER: A device used to measure the thermal expansion and dilation in solids and liquids (cubic expansivities of liquids).

DILEAD(II) LEAD(IV) OXIDE (TRIPLUMBIC TETROXIDE; RED LEAD; RED LEAD OXIDE; MINIUM; LEAD TETROXIDE): A red amorphous powder, with the chemical formula Pb_3O_4, relative density 9.1, which decomposes at 500 °C (773 K) to lead(II) oxide. It is prepared by heating lead(II) oxide to 400 °C (673 K) and has the unusual property of being black when hot and red-orange when cold. It is used in the manufacture of storage batteries, lead glass and red pigments. A paint made with red lead is often used to protect iron and steel from rusting. Its use in the paint industry has largely been discontinued because of the toxicity of lead.

DILEMMA: In logic, a form of argument in which one of the premises is the conjunction of two conditional statements and the other affirms the disjunction of their antecedents and the conclusion is the disjunction ("either . . . or") of their consequents. In logic, the symbols , or signify "if . . . then"; signifies "either . . . or". Symbolically, therefore, a dilemma is an argument of the form A C, B C, A B, C. This can be translated as "one or both of A or B is known to be true, but they both imply C, so regardless of the truth values of A and B, the conclusion is C.

DILOGARITHM: In polylogarithm, the function $\mathrm{Li}(z)=\sum_{k=1}^{\infty}\frac{z^k}{k^n}$, defined for the integral $n \geq 2$ and z in the unit disk; for which $\mathrm{Li}_2(z)$ is the dilogarithm and $\mathrm{Li}_3(z)$ the trilogarithm.

DILUENT (FILLER; DILUTANT; THINNER): A solvent used to reduce the strength of a solution. Certain fluids are too viscous to be pumped easily or too dense to flow from one particular point to the other, hence the need to dilute it.

DILUENT GAS: A gas of known quality introduced for analytical purposes so that it quantitatively lowers the concentration of the components of a gaseous sample; this may also be the complementary gas.

DILUTE: To reduce the concentration of a solution by the addition of more solvent to it. This leads to fewer numbers of particles per unit volume, but the total number of moles of solute remains constant.

DILUTE SOLUTION: A solution which contains a relatively low concentration of solute or solution in which the sum of amount fractions of all the solutes is small compared to 1.

DILUTE SPIN SPECIES: A type of nucleus in which it is statistically unlikely that there will be more than one such nucleus in a molecule, for example, ^{13}C since it has a natural abundance of 1.1%.

DILUTION: The process of reducing the concentration of a solution by the addition of solvent.

DILUTION PLATE COUNT METHOD: A method of estimating the number of viable microorganisms in a sample.

DILUTION RATE: The ratio of the ingoing volume flow rate and the culture volume. In continuous fermentation, a measure of a rate at which the existing medium is replaced with fresh medium, is the reciprocal of the hydraulic retention time.

DIMENSION: 1. Any directly measurable physical quantity, such as mass (m), length (l) or time (t). 2. Each of a set of independent and mutually perpendicular or orthogonal, directions in which a Euclidean space may be measured. 3. Also called **Hamel dimension**, the minimal number of mutually independent vectors that generate the given space, that is, in terms of linear combinations of which every element of the space can be canonically expressed. 4. A coordinate used with others to locate a point in space and time. 5. The product or quotient of the basic physical quantities raised to the appropriate powers in a derived physical quantity.

DIMENSION OF A MATRIX: The number of rows and the number of columns of a matrix. For example, the dimension of the matrix

$A = \begin{bmatrix} 2 & 4 \\ -5 & 7 \\ 1 & 3 \end{bmatrix}$ is 3 by 2.

DIMENSIONAL ANALYSIS: A method of checking correctly the validity of an equation or a solution to a problem by analysing the dimensions in which it is expressed. In mathematics and science, it is a tool which helps to understand the properties of physical quantities independent of the units used to measure them. Every physical quantity is some combination of mass, length, time, electric charge and temperature, (denoted by M, L, T, Q and Θ, respectively). For example, speed, which may be measured in metres per second (m/s) or some other units, has dimension expressed as L/T or alternatively LT^{-1}. Dimensional analysis is routinely used to check the validity of derived equations and computations. It is also useful for establishing the form but not the numerical coefficients of an empirical relationship.

DIMENSIONAL STABILITY: Ability of paper and other substrates to retain their exact size despite the influence of temperature, moisture or stretching. Paper that maintains its original dimensions has a high degree of dimensional stability.

DIMENSIONAL TOLERANCE: The total amount that a specific dimension is permitted to vary. The tolerance defines the maximum and minimum limits of the dimension.

DIMENSIONALITY: The combination of basic dimensions associated with a physically significant quantity. These dimensions are those of mass M, length L and time T and any combinations of their powers. For instance, momentum has dimensions of mass × velocity (MLT^{-1}) and energy has the dimensions of force × distance (ML^2T^{-2}). Equations describing physical quantities can be checked by ensuring they have the correct dimensions. Dimensionless quantities are significant as they are entirely independent of the conventions used to define mass, length and time.

DIMENSIONLESS QUANTITIES: Numbers which may be integers or exceptionally half-integers, when obtained by counting, for example, number of molecules, quantum numbers, rational numbers such as ratios, factors and fractions when obtained as ratios of two quantities of the same kind or real numbers when obtained by taking logarithms, for example, absorbance, power levels, etc.

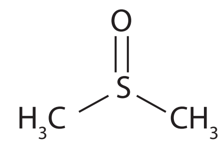
DIMER: A chemical or biological entity consisting of two structurally similar subunits (monomers) which are joined by bonds. The bonds could be strong or weak.

DIMERIC ION: An ion formed either when a chemical species exists in the vapour as a dimer and can be detected as such or when a molecular ion can attach to a neutral molecule within the ion source to form an ion such as $[M_2]^+$, where M represents the molecule.

DIMERISATION/DIMERIZATION: 1. The transformation of a molecular entity A to give a molecular entity A_2. 2. A chemical union of two identical molecules.

DIMEROUS (2-MEROUS): A term used to describe flowers in which the parts of each whorl are inserted in twos, as in the enchanter's nightshades (Circaea).

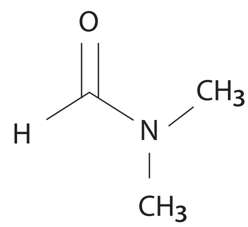
DIMETHYL SULFOXIDE (DMSO): A colourless liquid, with the molecular formula $(CH_2)_3SO$ used as an industrial solvent and antifreeze. It is a by-product in the processing of wood to paper. It is an important polar aprotic solvent that dissolves both polar and nonpolar compounds and is miscible in a wide range of organic solvents as well as water. It can penetrate the skin very readily and rubbed into the skin to reduce swelling and inflammation.



Chemical strucure of Dimethyl Sulphoxide

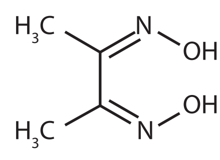
DIMETHYLBENZENES (XYLENES; XYLOL): Three benzene derivatives having two methyl group substituents. They may be described as ortho-xylene (1,2-dimethyl benzene), meta-xylene (1,3-dimethyl benzene) and para-xylene (1,4-dimethyl benzene), with the chemical formula C_8H_{10}, $C_6H_4(CH_3)_2$ or $C_6H_4C_2H_6$ respectively. Xylene is a clear, colourless, sweet-smelling liquid that is very flammable. It is usually refined from crude oil in a process called alkylation. It is also produced as a by-product from coal carbonisation derived from coke ovens, extracted from crude benzole from gas or by dehydrocyclodimerization and methylating of toluene and benzene. They are useful in pharmaceuticals, dye preparation and insecticide manufacture and as solvents in the printing, rubber and leather industries. They are also used as an inhalant drug because of its intoxicating properties.

DIMETHYLFORMAMIDE (DMF; *N,N*-DIMETHYLMETHANAMIDE): A colourless organic liquid, with the chemical formula $(CH_3)_2NC(O)H$, relative density 0.944, melting point -61°C (212 K) and boiling point 153 °C (426 K). It is widely used as a solvent for organic compounds. It is miscible with water and majority of organic liquids. DMF is a common solvent for chemical reactions. When pure it is odourless whereas in technical grade or degraded form it has a fishy smell due to impurity of dimethylamine.



Chemical structure of dimethylformamide

DIMETHYLGLYOXIME (DMG): A colourless solid with the chemical formula $(CH_3)_2C_2(NOH)_2$ which sublimes at 215 °C (488 K) and slowly polymerises if left to stand. It is used in chemical tests for palladium and nickel, with which they form a dark-red complex.



Chemical strucure Dimethylglyoxime

DIMICTIC LAKE: A lake stratified by a thermocline that is not permanent but circulates twice a year.

DIMIDIATE: Splitting into two halves with only one half developed.

DIMORPHISM: The existence of two distinctly different types of individuals within a species. Examples include: (i) **sexual dimorphism**, which is the differences in the physiology of a species based only on sex; (ii) **non-sexual dimorphism**, which is two clearly distinct physiologies present in a species not based on sex; (iii) **nuclear dimorphism**, when a cell's nuclear apparatus is composed of two structurally and functionally differentiated types of nuclei switching between two cell-types. For example, the fungus *Candida albicans* infects host tissue by switching from its usual unicellular yeast-like form into an invasive, multicellular filamentous form; and (iv) **frond dimorphism**, which is differing forms of fern fronds between the sterile and fertile fronds.

DINI DERIVATIVES: Of a real-valued function on real n-space, the four directional quantities defined, for a point, x and a direction, h, as the limits superior and the limits inferior, as t tends to 0 both from above and from below, of the quotient $f(x + th) - f(x)$.

D

DINI'S THEOREM: The result that a monotone decreasing sequence of continuous functions defined on a compact set that converges pointwise to a continuous limit actually exhibits uniform convergence.

DINITRO COMPOUNDS: A group of chemicals including dinocap, which is effective in controlling powdery mildews and dinitro-orthocresol (DNOC), used as a fungicide, herbicide and insecticide.

DINITROGEN: The normal form of molecular nitrogen, N_2 used to distinguish it from nitrogen atoms.

DINITROGEN OXIDE (NITROUS OXIDE; "LAUGHING GAS"): A colourless gas, with the chemical formula N_2O, density 1.97 g dm^{-3}, melting point -90.8 °C (182 K) and boiling point -88.5 °C (184 K). It is soluble in water, ethanol and sulfuric acid. It is prepared by the controlled heating of ammonium nitrate which is chloride free to 250 °C (523 K) and passing the gas produced through solutions of iron(II) sulfate to remove impurities of nitrogen monoxide. It is relatively unreactive, being inert to halogens, alkali metals and ozone at normal temperatures. It decomposes on heating above 520 °C (793 K) to nitrogen and oxygen and will support the combustion of many compounds. Dinitrogen oxide is used as an anaesthetic gas and as an aerosol propellant.

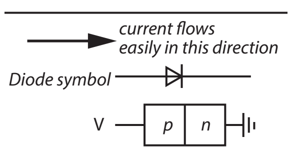
DINITROGEN TETRAOXIDE: The International Union of Pure and Applied Chemistry's (IUPAC) name for **nitrogen tetraoxide** or **nitrogen peroxide**, with the chemical formula N_2O_4. It is a colourless solid which melts at 9 °C (282 K) to give a pale yellow liquid. When it forms an equilibrium mixture with nitrogen dioxide it is referred to as dinitrogen tetroxide or nitrogen dioxide. Dinitrogen tetroxide is a powerful oxidiser, highly toxic and corrosive. It is hypergolic with various forms of hydrazine, they burn on contact without a separate ignition source, making them popular bipropellant rocket fuels. It is a useful reagent in chemical synthesis.

DINOFLAGELLATES (DINOMASTIGOTA): Single-celled colonial protists characterised by two flagella, with one girdling the cell and the other trailing the cell or a large group of flagellate protists. Most are marine plankton, but they can also be found in fresh water habitats. The distribution of their populations depends on temperature, salinity or depth. About 50% of all dinoflagellates are photosynthetic and these make up the largest group of eukaryoticalgae aside from the diatoms. Since they are primary producers they form an important part of the aquatic food chain. Some species, such as the zooxanthellae, are endosymbionts of marine animals and protozoa and play an important part in the biology of coral reefs. Others are colourless predators on other protozoa and a few are parasitic.

DINOSAUR: An extinct terrestrial reptile belonging to a group that constituted the dominant land animals of the Mesozoic era or the Jurassic and Cretaceous period, 190-65 million years ago. Some were the largest known land animals.

DINUCLEOTIDE: A compound consisting of two nucleotides. An example is nicotinamide adenine dinucleotide (NAD+), a coenzyme found in all living cells.

DIODE: An electronic device with two electrodes that allows an electric current to flow in one direction only. It contains a piece of p-n semiconductor material joined to two connecting wires. A small positive potential difference enables electrons from the cathode (n junction) to the anode (junction). Most modern diodes are solid state devices built up from semiconducting materials such as silicon and germanium. Diodes are used as rectifiers in electronic equipments such as radios, television sets, computers, etc. The Diode Valve is so called because it has just two electrodes – the cathode and the anode. The Cathode is the electrode that is heated and emit electrons and the anode is the electrode that collects the electrons. Since electrons are negatively charged particles, they will only be attracted to the anode, if this is given a positive potential with respect to the cathode. This explains why the valve will only conduct in one direction, from cathode to anode and not vice versa. It also explains the choice of the word valve to describe the device since a valve, is by definition, a one-way device. The Americans call it a vacuum tube, due to this.



Diode

DIODE LASER: A solid state laser in which the active medium is the p-n junction between p-type and n-type semiconductor host materials. This term is preferred over the sometimes-used term semiconductor laser.

DIOECIOUS: A plant or animal having separate reproductive organs in different individuals of the same species. An example is the pawpaw, *Carica papaya.*

DIOL (DIHYDRIC ALCOHOL; GLYCOL): An organic compound containing two hydroxyl groups (-OH groups). Diols usually have a sweet taste. An example is ethane-1,2-diol (ethylene glycol with the formula $C_2H_6O_2$), which is widely used as a solvent and antifreeze agent. Diols in which the hydroxyl groups are borne on the same carbon atom are referred to as gem diols and are very unstable. Examples include methanediol $H_2C(OH)_2$ and 1,1,1,3,3,3-hexafluoropropane-2,2-diol $(F_3C)_2C(OH)_2$, the hydrated form of hexafluoroacetone. Another form of diol is the vicinal diol with two hydroxyl groups in vicinal positions that is attached to adjacent atoms. Some examples are 1,2-ethanediol or ethylene glycol HO-$(CH_2)_2$-OH, a common ingredient of antifreeze products; and propane-1,2-diol or propylene glycol, HO-CH_2-CH(OH)-CH_3.

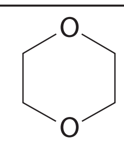
DIOPHANTINE EQUATION: An indeterminate polynomial equation that is to be solved in integers only. They define an algebraic curve, algebraic surface or more general object and ask about the lattice points on it. Linear diophantine equations take the form $a_1x_1 + a_2x_2 + ... + a_nx_n = c$, where c, a_1, a_2, ..., a_n are integers and where integer solutions are sought for the unknowns x_1, x_2, x_3,..., x_n. These can be found by applying the extended Euclidean algorithm. If a diophantine equation has an additional variable or variables occurring as exponents, it is an exponential diophantine equation. One example is the Ramanujan–Nagell equation, $2^n - 7 = x^2$.

DIOPTRE (DIOPTER; RECIPROCAL OF METER): An optical unit in which the power of a lens or curved mirror is expressed. It is equal to the reciprocal of the focal length measured in metres, 1/L. For example, a 2-dioptre lens brings parallel rays of light to focus at $^1/_2$ metres. The same unit is also sometimes used for other reciprocals of distance, especially radii of curvature and the vergence of optical beams.

DIORITE: A grey to dark grey granular intermediate intrusive igneous rock made up basically of plagioclase feldspar typically andesine, biotite, hornblende and/or pyroxene. It may contain small amounts of quartz, microcline and olivine. It is used for surfacing roads.

DIOSPHENOLS: Cyclic α-diketones, which exist predominantly in an enolic form.

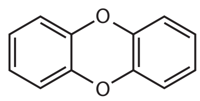
DIOXANE (1,4-DIOXANE; 1,4-DIOXACYCLOHEXANE): A colourless toxic, heterocyclic organic compound, that has a six-membered ring containing four CH_2 groups and two oxygen atoms at opposite corners. It is classified as ether. Its chemical formula is $C_4H_8O_2$, with a density of 1.033 g/cm^3, melting point 11.8 °C (285.95 K) and boiling point 101.1 °C (374.25 K). It has a faint sweet odour similar to that of diethyl ether. It is a liquid at



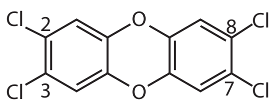
Structure of 1,4-Dioxane

room temperature and pressure. It can be made from ethane-1,2-diol. It is used as a solvent for a variety of practical applications as well as in the laboratory and used mainly as a stabiliser for the solvent trichloroethane. It is also used as a solvent for fats, greases and resins and in various products including paints, lacquers, glues, cosmetics and fumigants.

DIOXIN (POLYCHLORINATED DIB-ENZO-PARA-DIOXIN): A member of a family of highly toxic chlorinated aromatic hydrocarbons which is found in a number of chemical products such as herbicide 2,4,5-T as lipophilic contaminants that are highly persistent environmental pollutants. Their basic chemical structure consists of two benzene rings connected by a pair of oxygen atoms. When substituents on the rings are chlorine atoms, the molecules are particularly toxic. The most widely used dioxin, is 2,3,7,8-tetrachlorodibenzo-*p*-dioxin (2,3,7,8-TCDD). It is extremely stable chemically; it does not dissolve in water but dissolves in oils and accumulates in body fat. They are considered human carcinogens and can also cause severe reproductive and developmental problems. They are formed as by-products of industrial processes such as incineration.



Chemical strucure of dibenzo-1, 4-dioxin



Chemical strucure of 2,3,7,8 - Tetrachlorodibenzodioxin

DIOXONITRATE(III) ACID (NITROUS ACID): A weak acid known only in solution and in the gas phase, with the chemical formula HNO_2. It is prepared by the action of acids upon nitrites, preferably using a combination that removes the salt as an insoluble precipitate, for example, barium nitrite, $Ba(NO_2)_2$ and sulfuric acid, H_2SO_4. The solutions are unstable and decompose on heating to give nitric acid and nitrogen monoxide. It is widely used for the preparation of diazonium compounds in organic chemistry.

DIOXONITRATE(III) ION: The International Union of Pure and Applied Chemistry's (IUPAC) name for nitrite ion, with the chemical formula HNO_2^-. It is any salt or ester of nitrous acid (HNO_2). The salts are inorganic compounds with ionic bonds, containing the nitrite ion (NO_2^-) and any cation. The esters are organic compounds with covalent bonds, having the structure R-O-N-O, in which R represents a carbon-containing combining group and the bonding is from carbon to oxygen.

DIOXYGEN: The normal form of molecular oxygen, O_2, used to distinguish it from oxygen atoms or from ozone (O_3).

DIOXYGENYL COMPOUNDS: Compounds containing the positive oxygen ion (O_2^+), for example, dioxygenyl hexafluoroplatinate $O_2PtF_6^-$, an orange substance that sublimes in vacuum at 100 °C (373 K).

DIP: 1. A chemical which is dissolved in water and used for dipping animals especially sheep, to remove lice and ticks. To plunge an animal into a dip, for about thirty seconds. A **dipper** is a deep trench into which sheep are guided to be dipped. **Dipping** is the process of treating animals for external parasites by walking them or directing them to swim through a medicated bath. 2. The steepest angle of a tilted plane of rock (such as a bed, joint, cleavage or fault). The dip is measured from the horizontal, 90° being vertical, using a **clinometer**.

DIP ANGLE (ANGLE OF DIP; MAGNETIC DIP; MAGNETIC INCLINATION): The angle made with the horizontal by the Earth's magnetic field lines at any specific location or by a compass needle when the compass is held in a vertical orientation. It varies at different points on the Earth's surface, for example, it is 0° at the magnetic equator and 90° at each of the magnetic poles. If the value is positive at the point of measurement, the magnetic field of the Earth is pointing downward, into the Earth; and if negative, it is pointing upward.

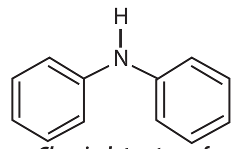
DIP NEEDLE (CLINOMETERS; INCLINOMETER): An instrument for measuring magnetic dip, consisting of a magnetic needle mounted free to pivot in the vertical plane with a graduated circle. The instrument is carefully levelled and oriented into the magnetic meridian.

DIPEPTIDE: A compound consisting of two amino acid units joined at the amino end of one and the carboxyl end of the other. By convention, the amino acid component retaining a free amine group is drawn at the left end (the N-terminus) of the peptide chain and the amino acid retaining a free carboxylic acid is drawn on the right (the C-terminus). Examples are aspartame (*N*-L-α-aspartyl-L-phenylalanine 1-methyl ester), carnosine (*beta*-alanyl-L-histidine), which is highly concentrated in muscle and brain tissues, anserine (*beta*-alanyl-*N*-methyl histidine) found in the skeletal muscle and brain of mammals.

D

DIPETALOUS (BIPETALOUS): Possessing two petals.

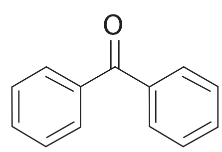
DIPHENYLAMINE (N-PHENYLBENZENAMINE; N-PHENYL ANILINE; ANILINOBENZENE; (PHENYLAMINO)BENZENE): A colourless crystalline aromatic compound, with the chemical formula $(C_6H_5)_2NH$, density 1.2 g/cm³, melting point 53 °C (326 K) and boiling point 302 °C (575 K). It is made by heating phenylamine (aniline) with phenylamine hydrochloride and used as a stabiliser for plastics and in the manufacture of dyes, explosives, pesticides and pharmaceuticals.



Chemical structure of Diphenylamine

DIPHENYLAMINE TEST: A presumptive test for nitrates. The reagent is a solution of diphenylamine ($[C_6H_5]_2NH$) in sulfuric acid.

DIPHENYLMETHANONE (BENZOPHENONE; PHENYL KETONE; DIPHENYL KETONE): A colourless solid organic compound, with the chemical formula $C_6H_5COC_6H_5$, density 1.11 g/cm³ (solid), melting point 49 °C (322 K) and boiling point 305.4 °C (578.55). It is made from benzene and benzoyl chloride using the Friedel-Crafts reaction with aluminium chloride as catalyst. It is often used as a building block in organic chemistry, being the parent diarylketone. In the printing industry, it can also be used as a photo initiator in UV-curing applications such as inks, imaging and clear coatings. It prevents ultraviolet light from damaging scents and colours in products such as perfumes and soaps and can also be added to the plastic packaging as a UV blocker. Its use allows manufacturers to package their products in clear glass or plastic. Without it, opaque or dark packaging would be required.



Chemical structure of Diphenylmethanone

DIPHOSPHANE (DIPHOSPHINE): A yellow liquid, with the chemical formula P_2H_4, melting point -99°C (174 K) and boiling point 52 °C (325 K), which is spontaneously flammable in air. It is obtained by hydrolysis of calcium phosphide. Many of the references to the spontaneous flammability of phosphine (PH_3) are in fact due to traces of P_2H_4 as impurities.

DIPHTHERIA: An acute and highly infectious bacterial disease affecting the mucous membranes of the nose and throat and also forming a thick membranous substance in the throat, obstructing breathing and swallowing. It also produces a powerful bacterial toxin in the blood that damages the heart and nerves. It is caused by the infection of the bacterium, *Corynebacterium diphtheriae*. Signs and symptoms include formation of a thick membranous substance in the throat, sore throat, fever, swollen glands and weakness. If not treated early it can damage the heart, kidneys and nervous system. It can also affect the skin; this type is called skin (cutaneous) diphtheria, causing pain, redness and swelling. It can be treated and also there are vaccines for immunisation to prevent it.

DIPHYODONT: Describing a type of dentition that is characterised by two successive sets of teeth: the deciduous (milk) teeth, which are followed by the permanent teeth. Mammals such as humans have this kind of dentition.

DIPLANETIC: A term used to describe organisms that successively produce two different types of zoospore (planospore), as do certain species of Saprolegnia.

DIPLECOLOBAL: Having incumbent cotyledons folded two or more times.

DIPLOBIONTIC: A term used to describe a life cycle in which two types of vegetative plant are formed, one haploid and the other diploid (i.e. a life cycle in which there is an alternation of generations). The life cycles of ferns are diplobiontic.

DIPLOBLASTIC: Having a body wall composed of two layers. The outer layer is the ectoderm and an inner layer known as endoderm. Examples are shown by coelenterates. The endoderm allows them to develop true tissue, while the ectoderm gives rise to the epidermis, the nervous tissue and if present, nephridia.

DIPLOCOCCUS: A genus of spherical or ovoid, nonmotile, gram-positive bacteria, which occurs in pairs. Pneumonia, meningitis and gonorrhoea are caused by types of diplococcus bacteria.

DIPLOID: Possessing two matched sets of chromosomes in the cell nucleus, one set from each parent; or a nucleus, cell or organism in which the chromosomes occur in homologous pairs. All animal or plant cells, except gametes (sex cells) and cells of the endosperm are diploid since gametes contain half the diploid number (haploid) of chromosomes as a result of meiosis and endosperms are triploid because they contain three sets of chromosomes. Diploid number of chromosomes is represented as 2x.The diploid number for humans is 46 and the haploid number is 23.

DIPLOID NUMBER: The number of chromosomes in each cell of the vegetative body of most organisms. The chromosomes are arranged in pairs, with one chromosome inherited from the female parent and the other from the male parent. The diploid number for humans is 46 and the haploid number is 23. However, many organisms have more than two sets of homologous chromosomes and are called polyploid. Ploidy refers to the number of sets of chromosomes in a biological cell.

DIPLONTIC: A term used to describe a life cycle in which the diploid phase predominates and where the haploid stage is limited to the gametes. Such life cycles are seen in the diatoms and members of the Fucales.

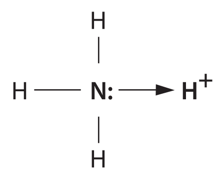
DIPLOSTEMONOUS: A term used to describe stamens that are inserted in two whorls with the outer part aligned with the sepals and the inner side aligned to the petals.

DIPLOTENE: The period in the first prophase of meiosis, when paired homologous chromosomes begin to move apart. They remain attached at a number of points, being kept from falling apart by the chiasmata.

DIPLURA: An order of small to medium-sized wingless insects, sometimes known as two-pronged bristletails with elongated bodies, prominent paired tail-like cerci and slender paired antennae. Around 800 species are known to exist.

DIPNOI: A subclass of fresh water bony fishes that contains the lungfishes, which have lungs and breathe air. Lungfishes retain characteristics primitive within the Osteichthyes, including the ability to breathe air and structures primitive within Sarcopterygii, including the presence of lobed fins with a well-developed internal skeleton.

DIPOLAR BOND (COORDINATION BOND; COORDINATE COVALENT BOND; DATIVE COVALENT BOND): A type of covalent bonding between two atoms in which both electrons shared in the bond come from the same atom.



Dipolar Bond

DIPOLAR COMPOUNDS: Electrically neutral molecules carrying a positive and a negative charge in one of their major canonical descriptions. In most dipolar compounds the charges are delocalised; however the term is also applied to species where this is not the case. 1,2-dipolar compounds have the opposite charges on adjacent atoms. The term 1,3-dipolar compounds is used for those in which a significant canonical resonance form can be represented by a separation of charge over three atoms (in connection with 1,3-dipolar cycloadditions).

DIPOLE: 1. Two equal and opposite charges that are separated by a distance. 2. An aerial rod usually one-half wavelength or a whole wave length long. 3. An uneven distribution of magnetic or electrical characteristics within a molecule or substance such that it behaves as though it possesses two equal but opposite poles or classes, a finite distance apart.

DIPOLE ANTENNA: A type of antenna commonly used in wireless networking devices. It has a signal range of 360° horizontally and 75° vertically.

DIPOLE LENGTH: Electric dipole moment divided by the elementary charge.

DIPOLE MAGNET: Any magnet with a north and south pole. In an accelerator, the dipole magnets are used to steer a particle beam to the left or right by placing one pole above and the other below the beam pipe.

DIPOLE MOMENT: The vector sum of the bond moments in a molecule; a measure of molecular polarity. It is given by the magnitude of the partial charges on a molecule in coulombs times the distance between them in metres. A measured dipole moment refers to the dipole moment of an entire molecule.

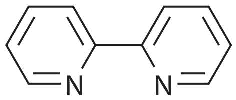
DIPOLE RADIATION: The interaction between a quantum-mechanical system, such as an atom and an electromagnetic field, in which only the electric dipole moment is considered.

DIPOLE-DIPOLE FORCE: The intermolecular attraction between oppositely charged poles of nearby polar molecules. The more polar the molecules of a solid or liquid are, the stronger the attractive forces between them and the harder it is to separate them. Intermolecular forces become significant at molecular separations of about 1 nanometre or less, but are much weaker than the forces associated with chemical bonding. Highly polar compounds tend to have higher boiling and melting points than less polar compounds. Water, methanol and ammonia are polar molecules that have dipole-dipole interactions. Examples of dipole-dipole forces are hydrogen bonding and dispersion forces. They are important, however, because they are responsible for many of the physical properties of solids, liquids and gases. These forces are also largely responsible for the three-dimensional arrangements of biological molecules and polymers.

DIPOLE-DIPOLE INTERACTION: Attractive interaction of two polar systems such as atoms and molecules by their dipole moments. The energy of the dipole-dipole interaction depends on the relative orientation and the strength of the dipoles. A water molecule has a permanent dipole moment; hence dipole-dipole interaction results when two water molecules approach each other. Although isolated atoms do not have permanent dipole moments, a dipole moment can be induced by the presence of another atom near it, thus leading to induced dipole-dipole interactions. Dipole-dipole interactions are responsible for Van der Waal forces and surface tension in liquids.

DIPTERA: An order of insects comprising the two winged flies. It is one of the major insect orders both in terms of ecological and human importance. They possess a pair of wings on the mesothorax and a pair of halteres, derived from the hind wings, on the metathorax. Examples are gnats, mosquitoes and midges. The possession of a single pair of wings differentiates the true flies from other insects with the word "fly" in their name such as mayflies, dragonflies, damselflies, stoneflies, whiteflies, fireflies, alderflies, dobsonflies, snakeflies, sawflies, caddisflies, butterflies and scorpionflies. Some true flies are secondarily wingless, for example, the Hippoboscoidea and some social insects. The diptera, especially mosquitoes (Culicidae), are disease transmitters, acting as vectors for malaria, dengue, yellow fever, encephalitis and other infectious diseases. An insect with two wings such as a fly is described as **dipterous**.

DIPYRIDYL (2,2′-BIPYRIDINE; BIPYRIDYL): A compound formed by linking two pyrine rings by a single C-C bond. Its chemical formula is $(C_5H_4N)_2$, with a melting point of 70-73 °C (343-346 K) and a boiling point of 273 °C (546 K). Various isomers are possible depending on the relative positions of the nitrogen atoms.



Chemical strucure of Dipyridyl

DIPYRRINS: Compounds containing two pyrrole rings linked through a methine, –CH=, group.

DIRAC CONSTANT (RATIONALIZED PLANCK CONSTANT; REDUCED PLANCK CONSTANT): A constant with the symbol ħ, used in quantum mechanics. It is equal to Planck's constant divided by 2π or $\bar{h} = \frac{h}{2\pi} =$ 1.054 589 x 10^{-34} J s. It is used when frequency is expressed in terms of radians per second or angular frequency instead of cycles per second. The energy of a photon with angular frequency ω, where ω **=** 2πν, is given by $\bar{h} = \frac{h}{2\pi}$.

DIRAC DELTA FUNCTION (δ FUNCTION; DELTA FUNCTION): The function, δ(x), introduced by theoretical physicist, Paul Dirac that has the value zero for all non-zero real x, except at the origin, where it is infinite, with integral of this function having the value 1. It satisfies the identity $\int_{-\infty}^{\infty} \delta(x)\,dx = 1.$

DIRAC EQUATION: A version of the nonrelativistic Schrödinger equation taking special relativity theory into account, which was formulated by the British physicist, Paul Dirac in 1928. The Dirac equation is needed to discuss the quantum mechanics of electrons in heavy atoms and, more generally, to discuss fine structure features of atomic spectra. It provides a description of elementary spin-½ particles, such as electrons, consistent with both the principles of quantum mechanics and the theory of special relativity. The equation demands the existence of antiparticles. It can be solved exactly in the case of the hydrogen atom but can only be solved using approximation techniques for more complicated atoms. The Dirac equation is given by $i\bar{h}\frac{\partial\varphi}{\partial t} = \frac{\bar{h}c}{i}\left(\alpha_1\frac{\partial\varphi}{\partial x^1} + \alpha_2\frac{\partial\varphi}{\partial x^2} + \alpha_3\frac{\partial\varphi}{\partial x^3}\right)\alpha_4 mc^4\varphi,$ where $\bar{h}$ is h-bar, c is the speed of light, φ is the wavefunction, m is the mass of the particle and α_i are the Dirac matrices.

DIRAC, PAUL ADRIEN MAURIC (1902-84): British theoretical physicist, who was educated at the universities of Bristol and Cambridge and professor of mathematics at Cambridge and also professor of physics at Florida State University. He is remembered for his prediction of the existence of the positron or antielectron, which was confirmed in 1932, when the American physicist Carl Anderson discovered the positron. He shared the 1933 Nobel Prize in Physics with Erwin Schrödinger for developing Schrödinger's non-relativistic wave equations to take account of relativity. This modified equation predicted the existence and properties of the positron. Dirac also invented, independently of Enrico Fermi, the form of quantum statistics known as Fermi-Dirac statistics.

DIRADICALS: Molecular species with two electrons occupying two degenerate molecular orbitals (MO). They are known by their higher reactivities and shorter lifetimes. In other words, diradicals are even-electron molecules that have one bond less than the number permitted by the standard rules of valence. The electrons can pair up with opposite spin in one MO leaving the other empty. This is called a singlet state. Alternatively, each electron can occupy one MO with spins parallel to each other. This is called a **triplet state**.

DIRECT: Of a relationship, relating two variables in such a way that an increase in the value of one is associated with an increase in the value of the other.

DIRECT (RADIOCHEMICAL) ISOTOPE DILUTION ANALYSIS: Isotope dilution analysis used for the determination of a non-radioactive element with the aid of one of its radionuclides.

DIRECT ACCESS STORAGE DEVICE (DASD): A mass storage medium on which a computer stores data and is directly connected to the computer.. An example, is internal disk drive or an external disk drive that connects through an interface such as USB or Firewire.

DIRECT AMPLIFICATION: A type of amplification reaction, where the constituent to be determined is amplified directly and finally measured.

DIRECT BROADCAST SATELLITE (DBS): A high-powered communications satellite that transmits or retransmits signals, such as television, radio or telephone signals, intended for direct reception by the public. The signals are transmitted to a small earth station or dish mounted on homes or other buildings.

DIRECT BROADCAST SYSTEM, DBS (DIGITAL BROADCAST SATELLITES; DIRECT BROADCAST SATELLITES; DIRECT BROADCAST TELEVISION SATELLITE SYSTEMS): In computing, it is a combination of a small satellite dish and receiver which allows subscribers or end users to receive signals directly from geostationary satellites rather than through terrestrial broadcasting towers and repeaters. Signals are broadcast in digital format at microwave frequencies. A television that receives direct-broadcast satellites is known as **direct-broadcast satellite television** (**DBSTV**) or **direct-to-home television** (**DTHTV**).

DIRECT COMPARISON: The process of comparing objects without using a measuring tool, for example, by lining up, matching, visually estimating, etc.

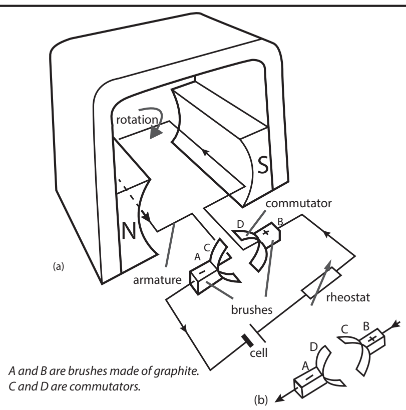
DIRECT COMBINATION: A method of making a simple salt by heating its two constituent elements together. For example, iron(II) sulfide can be made by heating iron and sulfur.

DIRECT CONNECTION: In computing, it is the connection between two computers via a cable to transfer files without the intermediary of a network or online service. Each computer must be running communications software using the same protocols for file transfers. If the computers are in the same room, they can be connected using a special type of serial cable known as a **null modem cable**; if they are connected via telephone lines each must have a modem so that one can dial the other.

DIRECT COUNT: Determining the number of microorganisms present in a given mass of soil, using direct microscopic examination.

DIRECT CURRENT (DC): An electric current in which the net flow of charge is in one direction only and does not reverse its flows like the alternating current. Alternating current (AC) and direct current (DC) are the two primary methods of electrical transmission. Direct current is used in any device such as personal computers (PCs) and other elctronic devices that has a circuit board because the chips within these devices require a steady, unidirectional flow of electrons to operate and store data. Such devices often have devices known as **inverters** for changing AC current into DC. DC current is also required to run the majority of electric motors such as those used in optical disk drive and the spinning of its hard disk in a computer and the movements of a robotic arm at a manufacturing plant. Basic electrical generators generate DC, which is then changed via transformer into AC for transmission. The DC motor has a split-ring commutator which is a device which changes the direction of the current in the coil every half cycle. Each time the coil turns through the position at the right angles to the magnetic field, the coil connections become reversed, so that the current changes direction in the coil and the coil is pulled round for another half-turn. The d.c. motor usually has several coils on the same spindle, spaced at equal angles. The commutator has a corresponding number of segments, each pair connected to a coil.

In this way, the brushes are at any time in contact with one of the coils, so that a more even turning effect is produced. In the diagram above, brush A touches split-ring C and brush B touches split-ring D and the flow of current rotates B, the armature coil. Half a cycle later, brush A (graphite plate) will be in contact with split-ring D and brush B (graphite plate) touches split-ring C. The torque, due to the current in the armature coil is always in the same direction and maintains the rotation. The d.c. motor has many applications in appliances such as tape recorders, washing machines and in operation of electric trains and electric fans.



A direct current electric motor

DIRECT DRILLING: A form of minimal cultivation, where the seed is sown directly into the field without previous cultivation.

DIRECT FISSION YIELD: The fraction of fissions giving rise to a particular nuclide before any nuclear decay has occurred.

DIRECTED GRAPH (DIGRAPH): In graph theory, a graph or a set of vertices or nodes that are connected by edges. These play an important role in network optimisation problems.

DIRECT IMPACT: An impact that occurs between two bodies having a velocity that is along a common normal.

DIRECT MEMORY ACCESS (DMA): A method of accessing data from the computer's RAM by some hardware subsystems of the computer without processing it through the CPU. While most data that is input or output from a computer is processed by the CPU, some data does not require processing or can be processed by another device. In these situations, DMA can save processing time and is a more efficient way to move data from the computer's memory to other devices.

DIRECT MOTION: The apparent motion of a planet from west to east, as seen from the Earth against the background of the stars or the anticlockwise rotation of a planet as seen from its North Pole.

DIRECT PRODUCT: An operation taking place in two groups with the construction of a new group. It is analogous to the direct sum of groups, but with multiplication as the operation. For example, the order of a direct product G × H is the product of the orders of the sets G and H:G x H = G x|H|. This can be followed from the formula available for the cardinality of the Cartesian product of the sets. Direct product is defined for a number of classes of algebraic objects, including sets, groups, rings and modules and the product of topological spaces. In each case, the direct product of an algebraic object is given by the Cartesian product of its elements, considered as sets and its algebraic operations are defined componentwise. For example, the direct product of two vector spaces of dimensions m and n is a vector space of dimension $m + n$.

DIRECT PROOF: A way of proving a given statement of the form "if p then q" or "p implies q", usually written as p q by assuming that the original statement (in this case p) is true and proceeding by deriving premises from previous statements, to conclude that another statement (in this case q) is true. This is in contrast to **indirect proof** that eliminates the possibility of the falsehood of the hypothesis.

DIRECT PROPORTION (DIRECT VARIATION): In mathematics, a variation in which an increase in one variable is associated with an increase in its corresponding variable. Two quantities are said to be proportional if they vary in such a way that one of the quantities is a constant multiple of the other or equivalently if they have a constant ratio. The symbol ‵ ′ is used to indicate that two values are proportional. For example, x y (i.e. x is directly proportional to y).

DIRECT REACTION: A chemical process in which the reactive complex has a lifetime that is shorter than its period of rotation. In a molecular-beam experiment the reaction products of a direct reaction are scattered, with reference to the centre of mass of the system, in preferred directions rather than at random. Some direct reactions are impulsive which means that there is an energy exchange that is very fast compared to the vibrational time scale.

DIRECT RESEEDING: The process of sowing grass seed without a cover crop.

DIRECT SCALING (MAGNITUDE ESTIMATION): A method of evaluating stimuli above threshold that involves directly assigning a number to represent the intensity of the sensation according to the apparent magnitudes of the stimuli. For example, in an experiment measuring perceived brightness of light, the experimenter presents a series of lights. The subject then assigns a number to each stimulus, depending on how bright he thinks each light is. If the first light was moderately bright, the subject could assign a value of 5. If the next one was perceived to be twice as bright, it could be given a value of 10. If the next was just a flicker, it could be a 1, and so on.

DIRECT STANDARDISATION: The process of adjusting a crude mortality or morbidity rate estimate for one or more variables, by using a known reference population, when the study population is large enough that age-specific rates within the population are stable.

DIRECT SUM: The decomposition of the vector space or group as a sum of subspaces or subgroups such that each element has a unique representation as a finite sum of elements, one from each subspace or subgroup. Let U, W be subspaces of V, then V is said to be the direct sum of U and W, written as $V = U \quad W$, if $V = U + W$ and $p = \frac{dy}{dx} = f(x, y)$, A finite direct sum may be identified with the Cartesian product of sets that are disjoint apart from identity. Direct sum can be defined for different mathematical objects including subspaces, matrices, modules and groups. For modules, it is a construction that combines several modules into a new, larger module.

DIRECT TRANSFER GAS FLOW SYSTEM: A system in which the purge gas transports the analyte directly to the sampling or excitation sources.

DIRECT TRANSITION: A transition between two electronic energy levels in a molecule that proceeds directly between two levels. An indirect transition between two electronic energy levels in a molecule involves either changes in a vibrational and/or energy level crossing in potential curves.

DIRECT VARIATION: A variation in which an increase in one variable is associated with an increase in its corresponding variable. For example, x varies directly as y can be written as x y, which is $x = ky$, where k is a constant.

DIRECT-INJECTION BURNER: A burner in which oxidant and fuel emerge from separate ports and are mixed above the burner orifices through their turbulent motion.

DIRECT-INJECTION ENTHALPIMETRY: An analytical method in which a reactant is injected into a calorimetric vessel containing another reagent. The enthalpy change of the ensuing reaction is measured and directly related to the amount of the limiting reagent (usually the analyte). If the experiment determines information other than amounts of analytes, an acceptable synonym is batch injection calorimetry.

DIRECTED: Of a number, line, angle, etc., having an orientation or direction distinguished from an opposite or direction, by the use of plus and minus signs. Thus two points A and B determine not one but two directed lines $\overline{AB}$ and $\overline{BA}$, lying in the same position, which is its direction, but with opposite orientation, its sense.

DIRECTED MOLECULAR EVOLUTION: Processes used to produce molecules exhibiting properties that conform to a user-defined goal. A large starting population of molecules (typically nucleic acids) that varies randomly in base sequence and shape is subjected to replication with variation, followed by selection. After several cycles of replication and selection, the population of molecules will evolve toward one containing a high proportion of molecules well adapted to the selection criterion applied.

DIRECTED MUTAGENESIS (DIRECTED MUTATION): A hypothesis, first proposed in 1988 by John Cairns, which states that organisms can respond to environmental stresses through directing mutations to certain genes or areas of the genome. It is an intentional alteration in a DNA sequence that can be genetically inherited. While studying *Escherichia coli* (*E. Coli*) that lacked the ability to metabolise lactose, John Cairns grew these bacteria in media in which lactose was the only source of energy and discovered that the rate at which the bacteria evolved the ability to metabolise lactose was many orders of magnitude higher than would be expected if the mutations were truly random. This led him to conclude and propose this hypothesis.

DIRECTED NUMBER (DIRECT NUMBERS): An integer or a number with a positive (+) or negative (-) sign attached, for example, +4 or -4.

DIRECTED RATIO: A ratio of directed quantities, in contrast to their absolute magnitudes.

DIRECTED SET: A partially ordered set with a transitive and reflexive relation, ≥, such that, given any two points a and b in the set, there exists another point c in the set with $a \geq c$, $c \geq b$. In other words, a partially ordered set with the property that for every pair of elements a,b in the set, there is a third element which is larger than both a and b. The relation then directs the set. For example, the finite subsets of an infinite set are directed by inclusion.

DIRECTION: 1. The line along which anything, for example, lies, faces, moves, etc., with reference to the point or region toward which it is directed. 2. Position on a line extending from a specific point toward a point of the compass or toward the nadir or the zenith. 3. The orientation of a line in space, distinguished from its sense in the case of a directed line.

DIRECTION ANGLE: Any of the triple angles that a line in space or a vector makes with the positive directions of the three coordinate axes (x-, y- and z- axes) respectively, which determines the orientation of the line or vector.

DIRECTION COSINE: Any of the triple cosines of the direction angles made by a given line or vector that determines its orientation.

DIRECTION FIELD: A geometrical interpretation of the set of lineal elements consisting of a system of independent differential equations. If the equation is of the form $p = \frac{dy}{dx} = f(x, y)$, the lineal elements are (x, y, p).

DIRECTION FINDER: A device used to locate the direction of an incoming radio signal. Usually, a loop antenna is rotated until maximum reception strength is achieved, giving the line of the transmission. If this is repeated from a different position, the transmitting station may be located. An example of direction finder is a radio compass used in air and sea navigation. Position is determined by finding the directions of two or more transmitters.

DIRECTION NUMBER (DIRECTION RATIO): Any sequence of numbers proportional to the direction cosines of a line or vector that determines its orientation relative to the three coordinate axes (x-, y- and z- axes).

DIRECTIONAL ANTENNA: An antenna that radiates or receives radio waves in a narrow beam.

DIRECTIONAL DERIVATIVE: The limit of the divided differences of a given function in a given direction, $\mathbf{h}: f'(x,h) = \lim_{t \to 0} \frac{f(x+t\mathbf{h}) - f(x)}{t}$. for x in n-space. One-sided directional derivatives are also defined for t tending to zero from above or below.

DIRECTIONAL SELECTION: Natural selection that favours the establishment of one particular advantageous mutation within a population, resulting in a change in phenotype in that direction. Increase in darker forms of the peppered moth is an example. In population genetics, directional selection occurs when natural selection favours a single phenotype and therefore allele frequency continuously shifts in one direction. Under directional selection, the advantageous allele will increase in frequency independently of its dominance relative to other alleles, even if the advantageous allele is recessive, it will eventually become fixed.

D

DIRECTIVITY: The ratio of the maximum radiative power to the average radiation intensity of an antenna. It varies directly as the effective aperture of the antenna and inversely as the square of the wavelength.

DIRECTORY: In computing, it is a list of file names, together with relevant information. In other words, it is an index of files stored on a computer disk, which may have many separate directories, called **directory tree** containing different types of files which can be retrieved from storage.

DIRECTORY SERVICE: A network service that identifies all resources on the network and make the resource accessible to user and application.

DIRECTORY TREE: In computing, it is a collective name for a directory called the **parent directory** or **top level directory** and all its subdirectories. Modern computer operating systems use directory trees for organising files.

DIRECTRIX: A fixed line on the convex side of a conic section that, together with a point called the focus and the eccentricity, serves to define a conic section as the locus of points whose distance from the focus is proportional to the horizontal distance from the directrix. If the ratio $r = 1$, the conic is a parabola, if $r < 1$, it is an ellipse and if $r > 1$, it is a hyperbola.

DIRECTX: A set of Application Program Interface (APIs) developed by Microsoft for programming graphics and multimedia effects such as games or active Web pages in Windows-based software. It includes tools that let a developer create or integrate graphic images, overlays, and other game elements, including sound. It enables users to utilise multimedia and graphics accelerator features more efficiently; and can enhance graphics, sounds and performance to improve playability.

DIRICHLET CONDITIONS: Conditions for a real-valued, periodic function $f(x)$ to be equal to the sum of its Fourier series at each point where f is continuous.

DIRICHLET SERIES: Any one of a class of series of the form: $\sum_{n=1}^{\infty} a_n n^{-s}$, where s is complex and a is a complex sequence. The series are of great importance in number theory. The most important series are the Dirichlet L-series of which the simplest are the zeta function and the primitive L-series modulo three: $1^{-s} - 2^{-s} + 4^{-s} - 5^{-s} + \ldots$

DIRICHLET, JOHANN PETER GUSTAV LEJEUNE (1805-1859): French-born German mathematician who made great contributions to number theory, complex analysis and to the theory of Fourier series.

DIRICHLET'S KERNEL: The collection of functions, important in Fourier analysis that is defined as the sum

$$\frac{1}{2} + \sum_{k=1}^{n} \cos kt = \frac{\sin \frac{(2n+1)t}{2}}{2 \sin \frac{t}{2}},$$

for t not a multiple of 2π.

DIRICHLET'S PRINCIPLE (DRAWER PRINCIPLE; LETTER-BOX PRINCIPLE; PIGEON-HOLE PRINCIPLE): It states that if n items are put into m containers, with $n > m$, then at least one container must contain more than one item. It was introduced in 1834 by German mathematician, Peter Gustav Lejeune Dirichlet (1805-1859).

DIRICHLET'S PROBLEM: The partial differential equation problem that looks for solutions to Laplace's equation in a region subject to boundary conditions. These are typically either requirements of Dirichlet type that the solutions agree with a given continuous function on the boundary of the region or of Neumann type that require the normal derivative to satisfy a boundary condition.

D

DIRICHLET'S TEST: 1. A test for the convergence of an infinite series: if $\{a_n\}$ and $\{b_n\}$ are sequences such that Σa_n has bounded partial sums and $\{b_n\}$ is strictly decreasing and converges to zero, then $\Sigma a_n b_n$ is convergent. This test is often useful in testing the convergence of of a power series on the boundary of its circle of convergence. The alternating series test is a special case of this result. 2. A test for the uniform convergence of an infinite series: if $\{a_n(z)\}$ and $\{b_n(z)\}$ are sequences of complex functions on a compact set K, such that $\Sigma a_n(z)$ has partial sums that are uniformly bounded in K, $\Sigma\{b_n(z) - b_{n+1}(z)\}$ is uniformly absolutely convergent in K and $\{b_n(z)\}$ converges uniformly to zero in K, then $\Sigma a_n(z) b_n(z)$ is uniformly convergent in K.

DIRICHLET'S THEOREM (DIRICHLET PRIME NUMBER THEOREM): In number theory, it states that for any two positive co-prime integers a and d, there are infinitely many primes of the form $a + nd$, where $n \geq 0$. In other words, there are infinitely many primes which are congruent to a modulo d. The numbers of the form $a + nd$ form an arithmetic progression. It is named after Dirichlet Johann Peter Gustav Lejeune (1805-1859), French-born German mathematician.

DIRIGIBLE (AIRSHIP): A powered and navigable aircraft that is lighter than air.

DIURNAL TEMPERATURE RANGE (DTR): The difference between the maximum and minimum temperature that occurs during the same day. These changes can be caused by factors such as cloud cover, urban heat, land use change, aerosols, water vapour and greenhouse gases. Different regions are affected by different factors.

DISACCHARIDES: Compounds in which two monosaccharides are joined by a glycosidic bond. Examples are maltose, sucrose and lactose.

DISARTICULATING: Separating at a joint.

DISASTER RECOVERY PLAN: A set of procedures and processes laid out in a document on how to protect and recover IT infrastructure and assets of a business.

DISCHARGE LAMP: A type of lamp that produces high intensity light by passing an electric current through a heated gas or artificial light sources that generate light by sending an electrical discharge through an ionised gas, such as plasma. Examples are the orange sodium street light and the lamp. The character of the gas discharge critically depends on the frequency or modulation of the current. Such lamps often use a noble gas (argon, neon, krypton and xenon) or a mixture of them. The lamps are frequently filled with additional materials such as mercury, sodium and/or metalhalides. When current is passed through, the gas is ionised and free electrons, accelerated by the electrical field in the tube, collide with gas and metal atoms. Some electrons circling around the gas and metal atoms are excited by these collisions, which bring them to a higher energy level. When the electron falls back to its original level, it gives a photon, resulting in visible light or ultraviolet radiation. For some lamp types, ultraviolet radiation is converted to visible light by a fluorescent coating on the inside of the lamp's glass surface. Gas-discharge lamps offer long life and high light efficiency.

DISCHARGE TUBE: A glass envelope containing gas at low pressure, which gives light due to the collision of electrons with gas molecules. Discharge lamps, filled with one or more of various gases, are used to give coloured light: mercury vapour gives purple, sodium vapour gives yellow and neon gives red. Early discharge tubes, called Geissler tubes, were used as toys. A fluorescent lamp is a discharge tube coated internally with a fluorescent substance.

DISCHARGE VELOCITY: The rate of discharge of water through a porous medium per unit of total area perpendicular to the direction of flow.

DISCICRISTATES: An assemblage of eukaryotic protists whose membranes have dicer shaped cristae in their mitochondria and are shown to be related by molecular systematics. They include the euglenids, kinetoplastids and amoebomastigotes plus the acrasid cellular slime moulds.

DISCOIDAL CLEAVAGE: A meroblastic cleavage in which the cytoplasmic divisions are restricted to the animal pole, resulting in the formation of a disc of cells confined at the animal pole or on top of the yolk. It is found in birds, fishes and reptiles, whose eggs contain a large amount of yolk that does not participate in cleavage.

DISCOMFORT THRESHOLD: The lowest value at which a sensation of discomfort is perceived. It is a measure which varies from person to person.

DISCOMITOCHONDRIA: In some classifications, a phylum of protoctists consisting of motile single-celled organisms characterised by disc-shaped cristae in their mitochondria and the absence of sexual reproduction in their life cycle.

DISCOMYCETES: A class of the Ascomycotina containing those ascomycetes that produce a pothecium (the exception being the Tuberales, where the hymenium is enclosed). It contains over 3000 species in some 425 genera, which are distributed among the orders Phacidiales, Helotiales, Ostropales, Pezizales and Tuberales. The majority of fungi that form lichens are Discomycetes.

DISCONECTED: Not topologically or graphically connected. A topological space X is disconnected if there are two non-overlapping non-empty open sets that cover X. For example, any punctured neighbourhood in the reals is disconnected, as is the cut interval (-1,1) \ {0}.

DISCONTINUITY: A point or value at which a function is not continuous. It is at that point or value that the independent variable of a function is not equal to its limit as the value of the independent variable approaches that point or where it is not defined. A **discontinuous function** is a function that has a discontinuity that for certain values or between certain values of the variable does not vary continuously as the variable increases. The discontinuity may, for example, consist of an abrupt change in the value of the function or an abrupt change in its law of variation or the function may become imaginary.

DISCONTINUOUS FUNCTION: A function, which for certain values or between certain values of the variable does not vary continuously as the variable increases; or a function for which, intuitively, small changes in the input result in small changes in the output. The discontinuity may, for example, consist of an abrupt change in the value of the function or an abrupt change in its law of variation or the function may become imaginary.

DISCONTINUOUS INDICATION: Indication related to the concentration during intervals of time which are not continuous.

DISCONTINUOUS MEASURING CELL: A measuring cell which operates intermittently and not necessarily at fixed time intervals.

DISCONTINUOUS PRECIPITATION: A diffusional reaction in a multi-component system in which structural and compositional changes occur in regions immediately adjacent to the advancing interface. The parent phase remains unchanged until swept over by the interface; the transition is complete in regions over which the interface has passed.

DISCONTINUOUS REPLICATION: The synthesis of a new strand of a replicating DNA molecule as a series of short fragments that are subsequently joined together.

DISCONTINUOUS SIMULTANEOUS TECHNIQUES: The application of coupled techniques to the same sample when sampling for the second technique is discontinuous. For example, discontinuous thermal analysis and gas chromatography, when discrete portions of evolved volatile(s) are collected from the sample situation in the instrument used for the first technique.

DISCONTINUOUS VARIATION (QUALITATIVE VARIATION): Clearly defined differences in a characteristic (traits) that can be observed in a population with no intermediates between them. Examples are blood groups in humans, length of style, sex in animals and plants.

DISCRETE: 1. Completely separate and unconnected or a value that is not broken up into smaller units. 2. Mathematical elements or variables that are distinct, unrelated and have a finite number of values. For example, a discrete variable is a variable that can only have an integer value (i.e. 0, 1, 2, 3, ...). **Discrete mathematics** is the study of mathematical structures that are fundamentally discrete rather than continuous. In contrast to real numbers that have the property of varying "smoothly", the objects studied in discrete mathematics such as integers, graphs and statements in logic do not vary smoothly in this way, but have distinct, separated values. It deals with countable sets or sets that have the same cardinality as subsets of the integers, including rational numbers but not real numbers.

DISCRETE DATA: Data that can take only whole numbers or fractional values. Examples include population data. The opposite is **continuous data** which can take all in between values. Example of continuous data is the measurement of time.

DISCRETE DISTRIBUTION FREQUENCY: A distribution made up of integer values. Such a distribution is normally represented by a frequency line histogram. In this case the samples of a discrete sinusoid occur. Just as in its continuous-time counterpart, the discrete time signal has a time axis, conventionally denoted by n. The variable n can take on only discrete integral values and thus is not a continuous variable.

DISCRETE FOURIER TRANSFORM, DFT (FINITE FOURIER TRANSFORM): In mathematics, the problem of determining the coefficients the unique polynomial p of degree n that interpolates given values a_i; at w^i for $i = 0, 1, ..., n$ where w is a primitive $(n + 1)^{th}$ root of unity either in the complex field or in a finite field. This finds application in fields such as digital signal processing, where it is widely used in signal processing and analysis. It converts a finite list of equally spaced samples of a function into the list of coefficients of a finite combination of complex sinusoids ordered by their frequencies, that has those same sample values. It can be said to convert the sampled function from its original domain (often time or position along a line) to the frequency domain. They are extremely useful because they reveal periodicities in input data as well as the relative strengths of any periodic components.

DISCRETE MATHEMATICS: The study of mathematical structures that are fundamentally discrete rather than continuous. In contrast to real numbers that have the property of varying "smoothly", the objects studied in discrete mathematics such as integers, graphs and statements in logic do not vary smoothly in this way, but have distinct, separated values. It deals with countable sets or sets that have the same cardinality as subsets of the integers, including rational numbers but not real numbers.

DISCRETE MODEL OF AREA: Imagining area as filling a figure with unit squares and counting them. **Discrete model of volume** is imagining volume as filling a figure with unit cubes and counting them.

DISCRETE RADIO SOURCE: An object such as a planet or the jet of an active galaxy that can be observationally separated at radio wavelengths from its local background.

DISCRETE RANDOM VARIABLE: A random variable which can only take on values from a discrete list. The probability function (or density) lists the probability that the variable will take on each of the possible values. The sum of the probabilities for all possible values must be 1.

DISCRETE TOPOLOGY: The topology on a given space that consists of its entire power set.

DISCRETE VARIABLES: Variables made up of only integers. Examples include the following: (i) number of students in a class; (ii) number of books in a library; (iii) and number of players in a team.

DISCRETE-TIME MARKOV CHAIN: A probabilistic model of a randomly evolving system whose future is independent of the past if the present state is known.

DISCRETISATION/DISCRETIZATION: 1. The discrete approximation to a continuous or other non-discrete object, often for the purposes of computation, for example, in replacing differentials by difference quotients or in quadrature. 2. The process of transferring continuous functions, models and equations into discrete counterparts; usually carried out as a first step toward making them suitable for numerical evaluation and implementation on digital computers.

DISCRIMINANT: A term used in determining the nature of a quadratic equation. The discriminant of a quadratic equation $ax^2 + bx + c = 0$ is $b^2 - 4ac$. The quadratic discriminant has the following properties (i) if $b^2 - 4ac > 0$, then the quadratic equation has real and unequal roots; (ii) if $b^2 - 4ac = 0$, then the roots are real and equal or repeated; (iii) if $b^2 - 4ac < 0$, then the roots are imaginary numbers; (iv) if $b^2 = 4ac \geq 0$, then the roots are real; (iv) if $b^2 - 4ac \geq 0$ then the roots are real.

DISCRIMINANT ANALYSIS: A statistical technique to find a set of descriptors which can be used to detect and rationalise separation between activity classes.

DISCRIMINATOR: A basic function unit comprising an electronic circuit, which gives an output pulse for each input pulse whose amplitude lies above a given threshold value.

DISCUTIENTS: Agents and substances containing properties that can dissolve and remove tumours, cysts, glandular muscles and abdominal growths. They include herbs such as garlic, red clover, and black walnut bark.

DISEASE: 1. Any condition that results in lowered health in a living thing or that impairs the normal state of an organism or one or more of its organs or systems. Some diseases such as pneumonia can be acute, producing severe symptoms that terminate after a short time; some, such as diabetes mellitus are chronic disorders, lasting over a long time; others such as malaria return periodically and are termed recurrent. Diseases are classified according to cause. External factors that produce diseases are infectious agents, including both microscopic organisms such as bacteria, viruses and protozoans and macroscopic ones such as fungi and various parasitic worms. Some other causes are chemical and physical agents such as drugs, poisons and radiation, which can be encountered in specific work situations, deficiency of nutrients in the environment and physical injury. Diseases that arise from internal or endogenous causes include hereditary abnormalities or disorders inherited from one or both parents. Congenital diseases are disturbances in the development of a normal embryo. Allergies are hypersensitive reactions to substances in the environment. Some others are endocrine disorders resulting from a defective endocrine gland; circulatory disorders or diseases of the heart and blood vessels, etc. Degenerative diseases occur as a result of the natural aging of the body tissues. Emotional disturbances such as psychoses and neuroses result in disturbed behaviour. 2. In agriculture science, it is injury from biotic stress resulting from infection by fungi, oomycetes, nematodes, bacteria, or viruses.

DISEASE TRIANGLE: A term that is applied to the relationship of the host, pathogen and the environment in disease development.

DISH: An object shaped like a dish or bowl, especially the large concave reflector of a radio telescope

D

DISHABITUATION: The recovery of a habituated or accustomed-to-habit response in an animal following the application of a different stimulus. For example, if the siphon of the sea slug is repeatedly tapped, the reflex withdrawal of the siphon is gradually reduced. But if the animal's tail is then given a strong electric shock, further tapping of the siphon will cause a withdrawal response that is nearer normal.

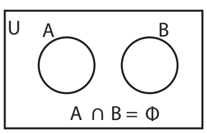
DISINFECTANT: An agent that kills or prevents the growth of disease-causing bacteria and other microorganisms and their spores or a chemical used to kill microbes outside the body, for example, on surfaces such as floors. Chemical disinfectants include iodine and chlorine. Also, mercuric chloride, carbolic acid (phenol), chlorinated lime, etc., are used to decontaminate objects. Formaldehyde is used as a deodorant and as airborne disinfectant or fumigant. Mercury toluene sulfonanilide is used for disinfecting seeds to protect them against soilborne plant diseases. Disinfectants such as chlorine and chlorine compounds, are used as preservatives for leather and other materials. Boiling also disinfects some contaminated objects. To **disinfect** means using a disinfectant to treat, kill or prevent the growth of disease-causing bacteria and other microorganisms.

DISINTEGRATION: 1. Any process in which an atomic nucleus breaks up spontaneously into two or more fragments in a radioactive decay or in collision with a high energy particle or nuclear arrangement. 2. The wearing away or breakdown of rock and mineral particles into smaller particles produced by physical forces such as atmospheric action, frost, and ice.

DISINTEGRATION CONSTANT (DECAY CONSTANT; DECAY COEFFICIENT; DISINTEGRATION CONSTANT; RADIOACTIVE DECAY CONSTANT; TRANSFORMATION CONSTANT): The probability of a radioactive decay of an atomic nucleus per unit time. It is given by the constant c in the equation $I =$ $I = I_0 e^{-ct}$, for the time dependence of rate of decay of a radioactive species, where, I is the number of disintegrations per unit time.

DISINTEGRATION ENERGY (REACTION ENERGY): The energy released or the negative of the energy absorbed, during a nuclear or particle reaction.

DISJOINT SET: A set whose members do not overlap or have anything in common. For example, the set of intervals {[1,5], [4,7], [8,9]} is not disjoint since [1,5] overlaps [4,7]; or two sets A and B are said to be disjoint if A and B have nothing in common. That is A B = ϕ, where ϕ is a null set. Two sets are disjoint if and only if the properties of membership are mutually exclusive and in a Venn diagram.



Disjoint Sets: Sets A and B are disjoint sets

DISJOINT UNION: A binary operator that combines the set of all elements of a pair of given sets, when all those elements are regarded as distinct. This can be done by first indexing the elements of the sets to ensure they are disjoint and then taking the union of the indexed sets. The disjoint union of A and B is written as $A \cup^* B = (A \times \{0\}) \cup (B \times \{1\}) \equiv A^* \cup B^*$.

DISJUNCT (ALTERNANT): In logic, either of a pair of propositions or formulae related or operated upon by disjunction.

DISJUNCTION: 1. In genetics, it is the separation of homologous chromosomes during meiosis. 2. Also known as **alternation** or **logical sum**, it is the proposition that presents two or more alternative terms, with the assertion that only one or at least one is true. It is an OR statement of the form A OR B. It is true if either A OR B is true. Its symbol is " ". The truth set table is given in the diagram. 3. The sentence formed from the proposition that presents two or more alternative terms, for example, the inclusive disjunction of P and Q is written P Q.

DISJUNCTION NORMAL FORM (DNF): In logic, the form to which every statement of sentential calculus can be reduced, consisting of a disjunction (sequence of ORs) of conjunctions (AND) in which each of the conjuncts is either an atomic formula or its negation. In other words, it is OR of ANDs. For example, the DNF of (A or B) and C is (A and C) or (B and C).

DISJUNCTIVE SYLLOGISM: In logic, a form of argument in which one of the premises is the disjunction of two statements and the other the negation of one of these statements. It is of the form "P or Q, not P, therefore Q", which can also be written as P Q, ¬P Q. P and Q may represent any proposition or any other formula. An example of disjunctive syllogism argument form is: Either the Sun orbits the Earth or the Earth orbits the Sun. The Sun does not orbit the Earth. Therefore, the Earth orbits the Sun.

DISK: 1. A common medium for storing large volumes of data on computer programs. It is a thin circular magnetic material. Information can be recorded on or read from a disk as it spins. 2. In botany, it is an enlargement of the receptacle around the ovary as in the flower head of Compositae plants. An example of such plants is the daisy. 3. **Galactic disk** is the flattened component of a spiral galaxy in which, together with many ordinary stars such as the Sun, lie the biggest and brightest of stars – the O stars and B stars. The disk is also home to large tracts of interstellar material from which new stars are continually being made. 4. **Disk star** is a star that lies within the galactic disk of a spiral galaxy such as our own.

DISK DRIVE: Commonly called a **drive**, it is a device that reads and/or writes data to a disk. The most common type of disk drive is a hard drive or hard disk drive. Examples of different drives in a computer or that may be accessible by a computer are Bernoulli drive, CD-ROM drive, CD-R drive, CD-RW drive, DVD drive, Hard drive, Local drive, LS120 drive or SuperDisk, Network drive, RAM disk and Solid-State Drive or Solid-State Disk (SSD).

DISK FORMATTING: Preparing a blank magnetic disk in a way that data can be stored on it.

DISK OPTIMIZER (DISK COMPRESSION): A software designed to organise or compress data to maximize free disk space using a variety of techniques such as defragmentation of the disk. Disk optimisation brings all the files together in one place, leaving the rest of the hard disk empty and will sort the files, for example, alphabetically. An example of a disk compression software program is a Microsoft Windows utility known as double space.

DISK PARTITION (PARTITION): In computers, it is a segment of the hard drive that is separated from other portions of the hard drive. Partitions help enable users to divide a computer hard drive into different drives or into different portions for multiple operating systems to run on the same drive.

DISK PLATTER: A magnetic metal, plastic, ceramic or glass disk located inside the computer hard drive and coated on both sides with a high precision magnetic material that is used to store data.

DISKETTE (FLOPPY): A flat, circular flexible plastic disk with a magnetic coating encased in a stiff envelope. It is used to store information in a small computer system.

DISLOCATION: Separation or displacement of bones at a joint so that it can no longer be moved. These conditions affecting the joint often result from trauma which causes adjoining bones to no longer align with each other. An incomplete or partial dislocation is referred to as a **subluxation**.

DISODIUM HYDROGENPHOSPHATE(V) (DIOSODIUM ORTHOPHOSPHATE): A colourless crystalline solid with the chemical formula Na_2HPO_4, which is soluble in water and insoluble in ethanol. It is used in treating boiler feed water and in the textile industry.

DISODIUM OXIDE (SODIUM OXIDE): A chemical compound with the formula Na_2O, density 2.27 g/cm^3, melting point 1132 °C (1405 K) and decomposes at 1950 °C (2223 K). Treatment with water results in the formation of sodium hydroxide: $Na_2O + H_2O \rightarrow 2\ NaOH$. It is used in ceramics and glasses.

DISODIUM TETRABORATE (BORAX; SODIUM BORATE): A soft colourless crystal, with the chemical formula $Na_2B_4O_7.10H_2O$, density 1.73 g/cm^3, melting point 743 °C, 1016 K (anhydrous) and boiling point 1575 °C (1848 K) that dissolves easily in water. It is an important boron compound, a mineral and a salt of boric acid. It is used in the manufacture of many detergents, cosmetics, enamel glazes, fire retardant, anti-fungal compound for fibreglass, insecticides, flux in metallurgy, texturing agent in cooking, etc. It is also used in standardising acids, preparing buffer solutions and as a precursor for other boron compounds.

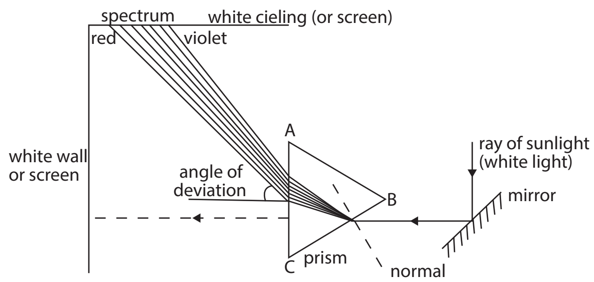
DISORDERED SOLID: A material that neither has the structure of a perfect crystal lattice nor of a crystal lattice with isolated defects. In a random alloy, one type of disordered solid is formed by introducing a high concentration of defects, with the defects distributed randomly throughout the solid, while in an amorphous solid, such as glass, there is a random network of atoms with no lattice.

DISPENSER: A device used to deliver a measured amount of material.

DISPERSAL: A process in which seeds, gametes, eggs or offspring move away from the parent into other areas. Seeds, for example, are dispersed by wind, water and animals.

DISPERSED PHASE: The solute-like species in a colloid or a substance microscopically dispersed evenly throughout another one.

DISPERSION: 1. In chemistry, a mixture in which one substance of fine particles called **dispersed phase** is scattered throughout another substance called **dispersion medium**. Dispersion occurs in suspensions, colloids or solutions. In solutions, the particles are molecular or ionic, those in colloids are larger but too small to be observed with an ordinary microscope and those particles in suspensions can be observed with microscopes or with the naked eye. 2. Subdivision of aggregates. Dispersion increases the specific surface of the particle; hence, it results in an increase in viscosity and gel strength. 3. In statistics, it is also called **variability**, **scatter** or **spread** and is the amount by which a set of observations deviate from the mean. When the values of a set of observations are close to the mean, the dispersion is less than when they are spread out widely from the mean. Common examples of measures of statistical dispersion are variance, standard deviation, inter-quartile range and the range. 4. Dispersion of light is the separation of a beam of light into its component wavelengths or colours. The dispersion is possible because the speed of light in a medium such as glass varies slightly with wavelength. The dispersion of light by glass is clearly demonstrated by the formation of spectrum using glass prisms. For a refracting, transparent substance, such as a prism of glass, the dispersion is characterised by the variation of refractive index with change in wavelength of the radiation. Newton used a small hole in a window shade and a glass prism to disperse sunlight into a visible spectrum, from violet through red. Using a second prism, he showed that no further decomposition of any of the spectral colours could be achieved. In other words, it is the splitting up of a ray of light of mixed wavelengths by refraction into its components or the splitting of white light into a spectrum when, for example, it passes through a prism, a raindrop or a diffraction grating.



Producing the colours of the rainbow from sunlight using dispersion

DISPERSION COLLOID SOLUTION (MOLECULAR DISPERSED SOLUTION): A colloidal solution in which the dispersed phase can be concentrated by centrifugation.

DISPERSION FORCE (LONDON FORCE; VAN DER WAALS FORCE OR INTERACTION): The force of attraction that exists between molecules or between parts of the same molecule other than those due to covalent bonds or to the electrostatic interaction of ions with one another or with neutral molecules that have no permanent dipole. It was named after the Dutch scientist, Johannes Diderik van der Waals. The dispersion force includes the force between two permanent dipoles (Van der Waals-Keesom force), the force between a permanent dipole and a corresponding induced dipole (Van der Waals-Debye force) and the force between two instantaneously induced dipoles (London dispersion force or Van der Waals-London force).

DISPERSION MEDIUM (CONTINUOUS PHASE): The solvent-like liquid in a colloid in which solids are suspended or droplets of another liquid are dispersed or the liquid in a dispersed system in which solids are suspended or droplets of another liquid are dispersed. In metallurgy, it refers to the matrix or background phase of a multiphasic alloy (external phase).

DISPERSIVE POWER: A measure of the power of a medium to separate different colours of light. It is equal to $\frac{(n_2 - n_1)}{(n-1)}$, where n_1 and n_2 are the indices of refraction at two specified widely differing wavelengths and n is the index of refraction for the average of these wavelengths or for the D line of sodium.

DISPERSIVE REPLICATION: A form of DNA replication in a cell, virus, program, etc., which the original DNA chain breaks and recombines in a random fashion before the double helix structure unwinds and separates to act as a template for messenger RNA synthesis.

DISPLACEMENT: 1. Also known as **displacement vector**, it is the position of one point with reference to another point. It is the change in an object's position. It is a vector quantity. Its magnitude is a straight line distance between the two points; and its direction is the angle between the straight line joining the two points and the reference direction. Its SI unit is the metre and the symbol is m. 2. The weight of fluid such as water pushed aside by floating objects. 3. The ratio of the charge displaced in a medium subjected to an electric field to the area of an imagined plane perpendicular to the potential gradient. 4. A chemical reaction in which one atom or a group of chemicals takes the place of another in a compound. 5. The distance that a point in a geologic fault has moved in relation to a corresponding point on the other side. 6. The total volume displaced by the pistons in an internal combustion engine.

DISPLACEMENT ACTIVITY: In animal behaviour, an action that is performed out of its normal context, while the animal is in a state of stress, frustration or uncertainty. Birds, for example, often peck at grass when uncertain whether to attack or flee from an opponent; similarly, humans scratch their heads when nervous.

DISPLACEMENT CHROMATOGRAPHY: A procedure in which the mobile phase contains a compound (the displacer) more strongly retained than the components of the sample under examination. The sample is fed into the system as a finite slug.

DISPLACEMENT CURRENT: In electromagnetism, it is a quantity appearing in Maxwell's equations that is defined in terms of the rate of change of electric displacement field. It has the units of electric current density and an associated magnetic field just as actual currents do. However, it is not an electric current of moving charges, but a time-varying electric field. In materials, there is also a contribution from the slight motion of charges bound in atoms and dielectric polarisation.

D

DISPLACEMENT GRADIENT: The gradient with respect to the reference configuration, of the displacement of a given body; thus the displacement gradient is equal to the deformation gradient minus the identity.

DISPLACEMENT METHOD: A method for estimating the volume of an object by submerging it in water and then measuring the volume of water it displaces, which is especially useful for finding the volume and density of irregular objects. This is based on a similar method used by Archimedes of Syracuse (287–212 B.C.) to solve a problem relating to the mass of a king's crown by noticing how his body displaced water in a bathtub and applying the same method to the crown. The solution is based on the fact that a submerged object displaces a volume of liquid equal to the volume of the object and the density of an object is equal to its mass divided by its volume.

DISPLACEMENT REACTION: 1. A substitution reaction in which an atom or ion in one substance displaces an atom or ion in another. 2. A reaction in which a less reactive element is replaced in a compound by a more reactive one. For example, iron reducing copper ions or the addition of iron to copper(II) tetraoxosulfate(VI) displacing copper metal. It may be classified as a single-displacement or double displacement reaction. A single displacement reaction can be represented by $X + YZ \rightarrow XZ + Y$. Double displacement reactions include precipitation and acid-base reaction and can be represented by $WX + YZ \rightarrow WZ + Y$.

DISPLACIVE TRANSITION: A transition in which a displacement of one or more kinds of atoms or ions in a crystal structure changes the lengths and/or directions of bonds, without severing the primary bonds. Examples include the transitions of the low-temperature polymorphs of SiO_2 (quartz, tridymite and cristobalite) to their respective high-temperature polymorphs, which involve distortions or rotations of the SiO_4 tetrahedra.

DISPLAY BEHAVIOUR: A stereotyped movement that serves to influence the behaviour of another animal or ritualised activity by which an animal conveys specific information. For example, the male peacock spreads its tail in courtship. Aggressive or agonistic displays make it unnecessary to chase intruders from a territory or for animals to injure each other in competition for mates. One type of defensive display deceives predators or lures them away from vulnerable young ones.

DISPLAY MONITOR: A term used for an electronic device for the visual presentation of data or images. Examples include Liquid-Crystal Display (LCD), Monitors, Flat-Panel displays and projectors.

DISPOSAL: 1. The act or process of getting rid of something. 2. A general term applied to radioactive wastes which require disposal, for which there is no intention of recovery.

DISPROPORTIONATION (DISMUTATION): The change in a substance in which it is simultaneously oxidised and reduced into two or more very different substances. It describes two particular types of chemical reactions. The first type is: 2A → A′ + A″ where A, A′ and A″ are different chemical species. Not all are redox reactions. For example, $2H_2O \rightarrow H_3O^+ + OH^-$ is a disproportionation but is not a redox reaction. The second type is a reversible or irreversible reaction in which a species is simultaneously reduced and oxidised so as to form two different products.

DISPROPORTIONATION REACTIONS: Oxidation-reduction reactions in which the oxidising and the reducing agents are of the same species. In other words, it is a redox reaction in which a chemical species of a particular oxidation state is simultaneously oxidised and reduced.

DISRUPTIVE SELECTION (DIVERSIFYING SELECTION): Natural selection that favours the extremes of a phenotype in a population. It describes changes in population genetics in which extreme values for a trait are favoured over intermediate values. It occurs if the very large and very small individuals in a population are favoured and the average-sized individuals are selected against. Examples are high temperature in summer and low temperature in winter with no intermediate forms. In this case the variance of the trait increases and the population is divided into two distinct groups.

DISSECT: 1. Cut open a dead body, a plant, etc., to reveal internal parts in order to study its structure. 2. Examine a theory, an event, etc., in great detail. 3. To divide (an interval) into a number of subintervals the union of which is the given interval and the only common points of which are the endpoints of the subintervals. For example,

$\left[0,\frac{1}{3}\right],\left[\frac{1}{3},1\right]$ is a dissection of [0,1].

DISSEPIMENT (SEPTUM): A partition formed within organs, such as ovaries and fruits, dividing them into chambers.

DISSIMILATORY NITRATE REDUCTION: One of the biological pathways in nitrate reduction to ammonia. It is the direct reduction from nitrate to ammonium that uses the substrate as a place to dump electrons and generate energy. The other pathway is assimilatory pathway.

DISSIPATIVE SYSTEM: A thermodynamically open system that involves irreversible processes. All real systems are dissipative. This is in contrast to such idealised systems as the frictionless pendulum, which is invariant under time reversal. In a dissipative system the system is moving towards a state of equilibrium, which can be regarded as moving towards a point attractor in phase space. This is equivalent to moving towards the minimum of the free energy, *F*.

DISSOCIATION: 1. In chemistry, it is the breaking up of a compound into smaller simpler molecules or ions or the process whereby a single compound splits into two or more smaller products which may be capable of recombining under other conditions to form the reactant. Dissociation is usually a reversible reaction, i.e., A   B + C. 2. The partial or complete disruption of the normal integration of a person's conscious or psychological functioning. It can be a response to trauma or drugs and perhaps allows the mind to distance itself from experiences that are too much for the psyche to process at that time. Since they are normally unanticipated, they are typically experienced as startling, autonomous intrusions into the person's usual ways of responding or functioning.

DISSOCIATION CONSTANT: A specific type of equilibrium constant that measures the propensity of a larger object to separate or dissociate reversibly into smaller components, such as when a complex falls apart into its component molecules or when a salt splits up into its component ions. The dissociation constant is usually denoted by K_d and is the inverse of the association constant. In the special case of salts, the dissociation constant can also be called an **ionisation constant**. For a general reaction: $A_xB_y \quad {}_xA + {}_yB$ in which a complex A_xB_y breaks down into ${}_xA$ subunits and ${}_yB$ subunits, then $K_d = [A]^x \times [B]^y / [A_xB_y]$, where [A], [B] and $[A_xB_y]$ are the concentrations of A, B and the complex A_xB_y, respectively.

DISSOCIATION ENERGY: Energy required to dissociate a molecule into two parts. Subscripts 0 and e, i.e. D_o and D_e are used to denote the dissociation from the ground state and potential energy minimum, respectively.

DISSOCIATION PRESSURE: The pressure of gas or gases in equilibrium with the solid at a given temperature, when a solid compound dissociates to give one or more gaseous products. For example, when calcium carbonate is maintained at a constant high temperature in a closed container, the dissociation pressure at that temperature is the pressure of carbon dioxide from the equilibrium $CaCO_{3(s)}\quad CaO_{(s)} + CO_{2(g)}$.

DISSOCIATIVE ADSORPTION (DISSOCIATIVE CHEMISORPTION): Adsorption with dissociation into two or more fragments, both or all of which are bound to the surface of the adsorbent. The process may be homolytic, as in the chemisorption of hydrogen.

DISSOCIATIVE IONISATION: An ionisation process in which a gaseous molecule decomposes to form products, one of which is an ion.

DISSOLUTION: The process in which a solute in gaseous, liquid, or solid phase dissolves in a solvent to form a solution. For the dissolution of solids, it is the breakdown of the crystal lattice into individual ions, atoms or molecules in a solvent such as water. For example, the dissolution of sodium chloride may be represented by the following equation: $NaCl\ (s)\quad Na^+(aq) + Cl^-(aq)$.

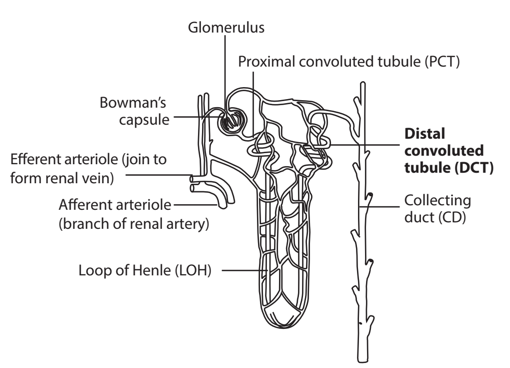
DISSOLVE: Mix a solid or gas with a liquid solvent to form a solution. When a substance dissolves, it spreads evenly throughout the solvent.

DISSOLVED OXYGEN: A relative measure of the amount of oxygen that is dissolved in a given medium, usually water. Dissolved oxygen is one of the vital indicators of the health status of a river, lake, etc. For example, a fish requires at least 5 parts per million (ppm) of dissolved oxygen to sustain life. It can be measured with a dissolved oxygen probe such as an oxygen sensor or an optode in liquid media, usually water.

DISSYMMETRY: 1. A lack of symmetry. 2. The relation between two objects when one is the mirror image of the other, that is, when one is a reflection of the other in some axis of symmetry.

DISTAL: A term describing the part of an organ that is farthest from the organ's point of attachment to the rest of the body. For example, the feet are at the distal ends of the body.

DISTAL CONVOLUTED TUBULE (DCT; SECOND CONVOLUTED TUBULE): The part of a nephron that leads from the thick ascending limb of the loop of Henle and drains into a collecting duct. It is responsible for the resorption of sodium, water and secretion of hydrogen and potassium ions.



Distal convoluted tubule of the nephron and other parts

DISTANCE: 1. The space between two points or places. In Euclidean space, the distance between two points is the length of the shortest line segment joining the given points, measured as the square root of the sums of the squares of the differences between the coordinates of the two points. if A and B are respectively the points (a_1, a_2) and (b_1, b_2) in the Cartestan plane, then the length $|AB| = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2}$, where AB is the hypotenuse of a right-angled triangle whose other sides are the differences of the coordinates of A and B. 2. Any length measured along a curve, especially straight a line or great circle. 3. The distance between sets in metric space is the infimum of the distance between points in one set and points in the other. 4. Amount of space between two points or places separated by time. 5. In physics, it is the total amount travelled regardless of direction. It is a scalar quantity.

DISTANCE BETWEEN TWO POINTS: The distance between two points $A(x_1,y_1)$ and $B(x_2,y_2)$ is defined as $|AB| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$.

For example, if $A(3,4)$ and $B(6, 8)$, then the distance between A and B is given by $|AB| = \sqrt{[(3-6)^2 + (4-8)^2]}$. This implies that $|AB| = \sqrt{25} = 5$.

DISTANCE FROM A POINT TO A LINE (PERPENDICULAR DISTANCE FROM A POINT TO A LINE): The perpendicular distance from a point, P to a given line, L is the shortest distance from P to L. Given the point $p(x_1,y_1)$ and the equation of the line $ax + by + c = 0$, the perpendicular distance is expressed as $\left|\frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}}\right|$ where a, b and c are constants. This is distance of $p(x_1,y_1)$ from the line $ax + by + c = 0$.

DISTANCE RATIO: The distance moved by the input force or effort into a machine divided by the distance moved by the output force or load. This ratio indicates the movement magnification achieved and is equivalent to the machine velocity ratio.

DISTANCE-TIME GRAPH: A graph that shows the distance that a body travels on the y-axis against the time taken on the x-axis. The gradient/slope on the graph indicates that the body is moving at a steady speed and indicates its speed at any point on the graph. The average speed can be determined from the slope/gradient of the graph. The steeper the gradient, the higher the average speed. A horizontal line on the graph indicates that the object is stationary.

DISTELY: A special case of polystely in which the vascular system is divided into two separate steles. This condition is exhibited in certain species of *Selaginella*, for example, *Selaginella kraussiana*.

DISTENDED: To swell out or stretch.

DISTICHOUS: A term used to describe a form of alternate leaf arrangement in which successive leaves arise on opposite sides of the stem so that two vertical rows of leaves result. It is seen, for example, in grasses. The term also describes a form of opposite leaf arrangement in which the pairs of leaves all arise in the same plane, again giving two rows of leaves.

DISTILLATION: The process of vapourising a solvent followed by condensing and collecting the vapour. It is a process by which a solvent can be recovered from a solution or a mixture. The technique is used to purify liquids or to separate mixtures of liquids possessing different boiling points.

DISTILLED WATER: Water, which has undergone distillation to increase its purity. For example, tap water and rain water contain dissolved salts and gases and are often distilled to increase their purity.

DISTINCT: 1. In botany, it refers to like parts that are not joined together. 2. In mathematics, of a pair of entities, not numerically identical.

DISTINCTIVENESS RATIO: In statistics, the ratio of the relative frequency of some event in a given sample to that in the general population or another relevant sample.

DISTONIC RADICAL CATION: A radical cation in which charge and radical sites are separated.

DISTORTION: 1. The bending, twisting, stretching or forcing of something out of its usual or natural shape. 2. The altering of something such as a television or radio signal to the extent that it becomes unrecognisable or unclear. In a distortionless communications system, the output must be proportional to a delayed version of the input, requiring a constant-amplitude response and a phase characteristic that is a linear function of frequency. 3. An alteration in an image in which the original proportions are changed, resulting from a defect in an optical system or lens.

DISTORTION ABERRATION: The faulty formation of an image arising because the magnification of the peripheral part of an object is different from that of the central part when viewed through a lens.

DISTRIBUEND: The substance that is distributed between two immiscible liquids or liquid phases.

DISTRIBUTE: To apply or, of an operator, to obey a distributive law, for example, multiplication distributes over addition.

DISTRIBUTED COMPONENT OBJECT MODEL (DCOM): In computing, it is the network version of Microsoft's Component Object Model, which provides a set of interfaces allowing clients and servers to communicate within the same computer.

DISTRIBUTED COMPUTING ENVIRONMENT (DCE): In network computing, it is a suite of technology services software developed by The Open Group for setting up and managing computing and data exchange in a system of distributed computers. DCE services include: Remote Procedure Calls (RPC), Security Service, Directory Service, Time Service, Threads Service and Distributed File Service. DCE is a popular choice for very large systems that require robust security and fault tolerance. DCE is typically used in a larger network of computing systems that include different size servers scattered geographically. DCE uses the client/server model. Using DCE, application users can use applications and data at remote servers. Application programmers need not be aware of where their programs will run or where the data will be located.

DISTRIBUTED DATABASE: A database in which the storage devices are not all attached to the same processing unit, but is transparent to the user and appears as as one logical unit.

DISTRIBUTED PRINT SYSTEM: A computer system that interchanges print data across different computing environments, thereby allowing data to be printed on a system other than the one at which the print request was generated. For example, in host-to-LAN distributed printing, data that resides on the host is printed on printers attached to a local area network (LAN).

DISTRIBUTED PROCESSING: A variety of computer systems whose processing power is in different locations but appears logical as one unit to the user. They coordinate their advanced distribution management system (ADMS) through messages.

DISTRIBUTION: 1. In statistics, the set of values of a set of data, grouped into classes, together with their frequencies or relative frequencies. In the case of random variables the distribution is the set of possible values together with their probabilities in the discrete case and the probability density function in the case of a continuous variable. 2. Also known as **generalised function**, a generalisation of the concept of a function, defined as continuous linear functionals over spaces of infinitely differentiable functions, introduced so that all continuous functions possess partial distributional derivatives that are again distributions. 3. In ecology, area(s) taken together where a species lives and reproduces or where something is located.

DISTRIBUTION CONSTANT: The concentration of a component in or on the stationary phase divided by the concentration of the component in the mobile phase. Since in chromatography a component may be present in more than one form, for example, associated and dissociated forms, the analytical condition used here refers to the total amount present without regard to the existence of various forms. These terms are also called the **distribution coefficients**. However, the present term conforms more closely to the general usage in science.

DISTRIBUTION FUNCTION: A normalised function giving the relative amount of a portion of a polymeric substance with a specific value or a range of values, of a random variable or variables.

DISTRIBUTION RATIO: The ratio of the total analytical concentration of a solute in the extract (regardless of its chemical form) to its total analytical concentration in the other phase.

DISTRIBUTIVE LAW: An axiom or theorem of a particular formal system which states that for a given pair of operators one distributes over the other. In simpler words, it states that multiplying a number by a group of numbers added together is the same as doing each multiplication separately. For example, $a(b + c) = ab + ac$ is the distributive law for arithmetic multiplication over addition.

DISTRIBUTIVE PROPERTY: A trait of two mathematical operations that allows one operation to be distributed over the other. In **distributive property of multiplication over addition**, it relates multiplication to a sum of numbers by distributing a factor over the terms in the sum. For example,
$2 \times (5 + 3) = (2 \times 5) + (2 \times 3) = 10 + 6 = 16$. In symbols, for any numbers a, b, and c: it is stated as: $a(b + c) = ab + ac$. In this case, multiplication is distributed over addition. The monomial factor a, is distributed or separately applied to each term of the polynomial factor $b + c$, resulting in the product $ab + ac$. In other words, the result of first adding several numbers and then multiplying the sum by some number is the same as first multiplying each separately by the number and then adding the products. In **distributive property of multiplication over subtraction**, it relates multiplication to a difference of numbers by distributing a factor over the terms in the difference. For example, $2\times(5-3)=(2\times5)-(2\times3)=10-6=4$. In symbols, for any numbers a, b, and c: $a \times (b - c) = (a \times b) - (a \times c)$ or $a(b - c) = ab - ac$.

DISTRIBUTOR: A device in the ignition system of a piston engine that distributes pulses of high voltage electricity to the spark plugs in the cylinders.

DISTURBANCE REGIME: In ecology, it is natural pattern of periodic disturbances, such as frequency, intensity, and types of disturbances, such as fires, pest outbreaks, floods, and droughts, followed by a period of recovery.

DISULFIDE: Sulfide containing two atoms of sulfur, as carbon disulfide, CS_2.

DISULFIDE BRIDGE: A covalent linkage formed by oxidation between two SH (thiol) groups either in the same polypeptide chain or in a different polypeptide chain. The linkage is also called an SS-bond or disulfide bridge. For example, the mild oxidation of cysteine leads to formation of the compound cystine which contains the disulfide bridge. The overall connectivity is, therefore, R-S-S-R.

DISULFUR DICHLORIDE (SULFUR MONOCHLORIDE): An orange-red liquid, with the chemical formula S_2Cl_2, density 1.688 g/cm^3, melting point -80°C (193 K) and boiling point 137°C (410 K). It is readily hydrolysed by water and is insoluble in benzene and ether. It is prepared by passing chlorine over molten sulfur. In the presence of iodine or metal chlorides sulfur dichloride, SCl_2, is also formed. In the vapour phase S_2Cl_2 molecules have Cl–S–S–Cl chains. The compound is used as a solvent for sulfur and can form higher chlorosulfanes of the type Cl–(S)n–Cl ($n < 100$), which are of great value in vulcanisation processes. A different isomer of S_2Cl_2 is $S{=}SCl_2$. This isomer forms transiently when S_2Cl_2 is exposed to UV-radiation.

DISULFURIC(VI) ACID (PYROSULFURIC ACID): A colourless hygroscopic crystalline solid, which is an oxoacid of sulfur, with the chemical formula $H_2S_2O_7$. It is commonly encountered mixed with sulfuric acid as it is formed by dissolving sulfur trioxide in concentrated sulfuric acid. The resulting fuming liquid, called oleum or Nordhausen sulfuric acid, is produced during the contact process and is also widely used in the sulfonation of organic compounds. It is prepared by reacting excess sulfur(VI) oxide (SO_3) with sulfuric acid: $H_2SO_4 + SO_3 \rightarrow H_2S_2O_7$. It is a strong acid and protonates sulfuric acid in the (anhydrous) sulfuric acid solvent system. Related acids have the general formula $H_2O.(SO_3)_x$ though none has so far been isolated.

DISYNDIOTACTIC POLYMER: A syndiotactic polymer that contains two chiral or prochiral atoms with defined stereochemistry in the main chain of the configurational base unit.

DITACTIC POLYMER: A tactic polymer that contains two sites of defined stereoisomerism in the main chain of the configurational base unit.

DITHERING: A process that uses digital noise to smooth out colours in digital graphics and sounds in digital audio.

DITHIOCARBAMATE FUNGICIDE: Any fungicide based on the organic sulfur compound dithiocarbamic acid. Such compounds include the commonly used fungicides ziram, thiram, zineb and maneb.

DITHIONITE (HYPOSULFITE): Salt derived from dithionic acid ($H_2S_2O_6$). Dithionates often form stable compounds that are not readily oxidised or reduced. Strong oxidants oxidise them to sulfates and strong reducing agents reduce them to sulfites and dithionites. Aqueous solutions of dithionates are quite stable and can be boiled without decomposition.

DITHIONITE ANION: A sulfur oxoanion derived from dithionic acid ($H_2S_2O_6$). Its chemical formula, $S_2O_6^{2-}$, is sometimes written in a semi structural format as $[O_3SSO_3]^{2-}$. Due to the presence of the S-S bond, the sulfur atoms of the dithionate ion are in the +5 oxidation state.

DIURESIS: Increased excretion of urine due to excessive intake of fluids or caused by a disease or a drug.

DIURETIC: A drug or other agent that increases the rate of urine formation and hence the rate at which water and certain salts are lost from the body. It promotes removal of excess water, salts, poisons and metabolic wastes to help relieve oedema, kidney failure or glaucoma. It acts mainly by decreasing the amount of fluid that is reabsorbed by the kidney's nephrons and passed back into the blood. Those that allow the body to retain potassium are used for patients with hypertension or congestive heart failure.

DIURNAL: A term denoting an event that happens once every 24 hours. A **Diurnal Cycle** is the apparent rising/culmination/setting of a heavenly body, for example, solar day, lunar day, etc, as observed from a particular site on the Earth's surface. This arises due to the Earth's daily spin on its axis. The diurnal cycle is one of the most basic forms of climate patterns. The most familiar of such pattern is the diurnal temperature variation. Such a cycle may be approximately sinusoidal or include components of a truncated sinusoid (due to the sun's rising and setting).

DIURNAL RHYTHM (CIRCADIAN RHYTHMS): An endogenous rhythm in which physiological responses occur at 24-hourly intervals. For example, some plants exhibit a characteristic change in leaf position on a daily cycle. The opening and closing of stomata and flowers and the frequency of cell divisions also exhibit a diurnal rhythm. Such rhythms will persist for several days if the plant is kept in continual darkness. It also appears that, in some species, the photoperiodic response is governed by diurnal rhythms.

DIURNAL VARIATION: Variations which follow a distinctive pattern which recurs with a daily cycle.

DIVALENT (BIVALENT): An atom or molecule which has a valence of two and thus can form two covalent bonds with other ions or molecules. Divalent anions are atoms or radicals with two additional electrons when compared to their elemental state (that is, with 2 more electrons than protons). Divalent cations are atoms or radicals with two electrons less than the neutral atom. An example of a divalent anion is the sulfide anion (S^{2-}), while iron(II), Fe^{2+}, is the divalent cationic form of iron (Fe). The ions present in hard water are Ca^{2+} and Mg^{2+} and they are responsible for the hardness in water.

DIVARICATE: To diverge at a wide angle. This is a term used in identifying plants, especially in describing the pattern of branching.

DIVERGE TO ZERO: Of an infinite product of non-zero complex numbers, having partial products tending to a zero limit as n tends to infinity. If a sequence is finitely zero, one determines its convergence by considering convergence of the non-zero tail, although the value of the product will be zero in both the convergent and the divergent case. These conventions allow one to convert products securely into series by taking logarithms.

DIVERGENCE: 1. The branching of the terminal axon of a neuron which results in the connection with numerous target cells. For example, each horizontal cell in the retina of the eye has a greatly branched axon ending that can transmit signals to many other cells in the retina. 2. The simultaneous outward movement of both eyes away from each other, usually in an effort to maintain single binocular vision when viewing an object. 3. The divergence of a vector field f (written as $\cdot f$) is defined as the scalar $\cdot f = \frac{\partial f_x}{\partial x} + \frac{\partial f_y}{\partial y} + \frac{\partial f_z}{\partial z}$. It can be thought of as the dot product of the operator (del) with the field f.

DIVERGENCE THEOREM (GAUSS' THEOREM; OSTROGRADSKY'S THEOREM): It states that the outward flux of a vector field through a closed surface is equal to the volume integral of the divergence on the region inside the surface. It shows the relationship between the total flux of a vector out of a surface which surrounds the volume and the vector inside the volume. Gauss' law of electric fields is a particular case of the divergence theorem. The divergence theorem is used in the mathematics of engineering, especially in electrostatics and fluid dynamics. Mathematically, suppose V is a subset of R^n (in the case of $n = 3$, V represents a volume in 3 dimensional space) which is compact and has a piecewise smooth boundary S, F is a continuously differentiable vector field defined on the neighbourhood of V, then divergence theorem is stated as $\iiint_V (\ .\mathrm{F})\, dV = \oint\!\!\oint (\mathrm{F}.n)\, dS$. The left side is a volume integral over the volume V, the right side is the surface integral over the boundary of the volume V.

DIVERGENT EVOLUTION (ADAPTIVE RADIATION; BRANCHING EVOLUTION; CLADOGENESIS): The formation of several species with adaptations to different ways of life or evolution of different species in different directions from one from a parent species. This is likely to occur whenever members of a species migrate to exploit new habitats and food sources and new populations emerge, each adapted to its particular environment. Eventually, these become sufficiently distinct to be recognised as new species. Examples are the 13 species of Darwin's finches on the Galapagos Islands, each filling a different niche on different islands but evolved from one ancestral species. Also, early placental mammals, for example, gave rise to modern burrowing, climbing, flying, running and swimming organisms. The vertebrate limb is another example of divergent evolution. The limb in many different species has a common origin, but has diverged somewhat in overall structure and function.

DIVERGENT PLATE MARGIN (CONSTRUCTIVE PLATE BOUNDARY): In plate tectonics, it is the boundary between two plates that move away from each other. Magma rises and solidifies to create a new crust that forms mid-ocean ridges such as the Mid-Atlantic Ridge located along the floor of the Atlantic Ocean or within continents to form rift valleys such as the Great Rift Valley. Divergent boundaries also form volcanic islands that occur when the plates move apart to produce spaces filled by rising magma.

DIVERGENT SEQUENCE: A sequence in which the difference between the nth term and the one after it, is constant or increases as n increases. For example, {4, 6,8,10 . . .} is a divergent sequence. A divergent sequence has no limit.

DIVERGENT SERIES: An infinite series that is not convergent or the infinite sequence of the partial sums of the series does not have a limit.

DIVERGING LENS (CONCAVE LENS): A lens in which a parallel beam of light that is close to the principal axis diverges after passing through it. It has a negative focal length. Examples are biconcave, diverging meniscus and plano concave.

DIVERS' DISEASE (BENDS; CAISSON DISEASE; DECOMPRESSION SICKNESS): A condition that is caused by a sudden and substantial change in atmospheric pressure. It is mostly experienced by deep-sea divers who surface too quickly, causing nitrogen bubbles to be dissolved into the bloodstream and tissues because of a sudden drop in the surrounding pressure. It is characterised by breathing difficulties, joint and muscle pain, cramps, numbness, nausea and paralysis.

D

DIVERSIFYING SELECTION (DISRUPTIVE SELECTION): Natural selection that favours the extremes of a phenotype in a population. It describes changes in population genetics in which extreme values for a trait are favoured over intermediate values. It occurs if the very large and very small individuals in a population are favoured and the average-sized individuals are selected against. Examples are high temperature in summer and low temperature in winter with no intermediate forms. In this case the variance of the trait increases and the population is divided into two distinct groups.

DIVERSION DAM: A structure or barrier that diverts water from a stream or river to a different course for purposes such as irrigation.

DIVERSITY: 1. In biology, the degree of variation of living things present in a particular ecosystem. Broadly, it defines the variation of life forms present in different ecosystems, for example, genetic diversity, crop diversity and ecosystem diversity. 2. In logic, the relation that holds between two entities when and only when they are not identical; the property of being numerically distinct.

DIVERTICULUM: A sac-like or tubular outgrowth from a tubular or hollow internal organ or a small pouch or sac formed in the wall of a major organ such as the oesophagus, small intestine or large intestine. It may occur as a normal structure or abnormally from a weakened area of the organ.

DIVIDE: 1. To calculate the multiplier of a given number required to yield a product equal to another number. In other words, to perform the operation of division to separate into equal parts. The divisor divides the dividend to yield a quotient. $\frac{x}{y}$ The symbols usually used, for example, x divided by y are x ÷ y, or For example, in 10 divided by 2 equals 5, written as 10 is the dividend, 2 the divisor and 5 the quotient. 2. An elevated boundary between areas that are drained by different river systems.

DIVIDED: In botany, being lobed into distinct parts, extending to the base or the midrib, as seen in the diagram.

DIVIDED DIFFERENCE (FIRST DIVIDED DIFFERENCE SEQUENCE): A sequence of difference quotient of the form $\frac{f(x_k+1)-f(x_k)}{x_k+1-x_k}$. The second and higher divided differences are defined recursively: the $(n+1)^{th}$ divided difference sequence is the first divided difference sequence of the n^{th} divided difference sequence with respect to the same points.

DIVIDERS: An instrument, similar to a compass with two arms hinged together at the top, each ending with a point. It is used to divide lines or to measure distances and transfer the measurements from one place to another, especially on maps. For example, by placing the points of the dividers on two points of a map and then transferring them to a scale, one can read off the distance between them.

DIVING BELL: A hollow structure providing a dry environment for underwater workers.

DIVING REFLEX: A developed reflexive response to diving that includes selectively shutting down parts of the body in order to conserve energy for survival. It is found in most aquatic mammals and birds, for example, seals, otters, dolphins, and diving birds, such as ducks and penguins. It also occurs in humans, for example, when human baby is exposed to cold water, the heart slows down and blood is shifted away from the peripheral muscles to conserve oxygen for the brain and heart, and they typically hold their breath until breathing resumes to delay potential brain damage.

DIVISIBLE BY: If the larger of two counting numbers can be divided by the smaller with no remainder, then the larger is divisible by the smaller. For example, 28 is divisible by 7, because 28/7 = 4 with no remainder. If a number n is divisible by a number d, then d is a factor of n. Every counting number is divisible by itself.

DIVISIBILITY RULE: A shortcut for determining whether a given integer is divisible by a fixed divisor or if it (given integer) can be divided by 2, 3, 4, 5, 9, and 10, without actually doing the division. For example, a number is divisible by 5 if the digit in the ones place value is 0 or 5. A number is divisible by 3 if the sum of its digits is divisible by 3.

DIVISIBLE: Of a number or quantity, able to be exactly divided by another with the quotient an exact whole number.

DIVISION: 1. Splitting into parts, sharing of something or the act of separating or splitting something into parts. 2. An operation used to calculate the number of times one number is contained in another. The symbols usually used, for example, x divided by y are $x \div y = z$ (called the obelus method), x/z (called the slash method) and by placing the dividend over the divisor with a horizontal line $\frac{x}{y}=z$.

Another method, called the **long division** is usually used for dividing simple or complex numbers by breaking them into simple steps. A **dividend** is a number or quantity to be divided by another number or quantity. A number that divides another number is called a **divisor**. **Quotient** is the result of the division. For example, in 10 ÷ 2 = 5, 10 is the dividend, 2 the divisor and 5 is the quotient. **Divisible** denotes how many times one number contains or is contained in another number. For example, 30 is divisible by 3 to give 10. 3. The process of separating the root mass of a perennial plant into smaller pieces that are used to grow new plants. 4. The process by which a cell divides to form two new cells either to produce identical cells, mitosis or to produce cells with half the number of chromosomes, meiosis. 5. A classification category, which groups plants and animals with similar characteristics together or a major category in the taxonomic classification of plants comprising a group of classes. The corresponding category in animal classification is called phylum. 6. A fallacy in which it is argued that what is true of a whole collectively is true of any of its parts.

DIVISION ALGEBRA: An algebra over a field in which all non-zero elements have multiplicative inverses. The only commutative and associative division algebras over the real field are the reals (of dimension 1) and the complex numbers (of dimension 2). The quaternions are a non-commutative associative four-dimensional division algebra and the Cayley algebra is a noncommutative and non-associative algebra of dimension 8.

DIVISION ALGORITHM: 1. The theorem in number theory that for any two integers a and b, there are two unique others, q and r, such that $a = qb + r$ and r < b. This can be written as the sum of the product of two integers, one a given positive integer, added to a positive integer smaller than the given positive integer. This also holds in a Euclidean domain by virtue of the existence of a gauge. 2. Any systematic process for calculating the quotient of two numbers.

DIVISION OF A SEGMENT: In geometry, the construction of a point that divides a line segment in some desired proportion.

DIVISION OF FRACTIONS PROPERTY (INVERT AND MULTIPLY RULE): A rule for dividing that says division by a fraction is the same a multiplication by the reciprocal of the fraction. For example,

$$5 \div 8 = \frac{1}{8} = \frac{5}{8}$$

$$15 \div \frac{3}{5} = 15 \times \frac{5}{3} = \frac{75}{3} = 25$$

$$\frac{1}{2} \div \frac{3}{5} = \frac{1}{2} \times \frac{5}{3} = \frac{5}{6}.$$

In symbols, for any a and nonzero b, c, and d:

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}.$$

If b = 1, then $\frac{a}{b}$ = a and the property is applied as in the first two examples above.

DIVISION RING: A ring in which every non-zero element, a, has an inverse a^{-1} such that $aa^{-1} = e = a^{-1}a$, where e is the (multiplicative) identity element.

DIVISOR FUNCTION: The function, $d(n)$, that counts the number of divisors of an integer n, including 1 and n. When p is prime, $d(p^a)$ is a + 1 and since d is multiplicative, the value for any other argument can be easily computed from its prime factorisation. The divisor function $d(n)$ is odd if and only if n is a square number.

DivX (DIGITAL VIDEO EXPRESS): 1. Abbreviation for Digital Video Format, a type of DVD-ROM format for a movie or other data loaded onto a DVD-ROM, which is playable only during a specific time frame, typically two days. The Divx player is connected to a telephone outlet and communicates with a central server to exchange billing information. **DivX HD** is the high definition (HD) version of the DivX format and supports a 6-channel, 5.1 Dolby Digital surround sound and delivers 720p HD resolution video at 4Mbps. DivX HD content can be viewed on HDTV with PC or with a DivX HD DVD player. 2. *DivX*, usually in italics is a family of video codecs, formats and software from DivXNetworks that makes a movie 8 to 12 times smaller with little quality loss. The DivX software provides tools for encoding in DivX and MPEG-4, as well as a player for DivX, MPEG-4 and other video formats. It is widely being used for compressing computer video files and even DVD files to fit onto a standard CD and DivX HD can reduce an HD movie to fit on a DVD.

DIZYGOTIC TWINS (FRATERNAL TWINS): Twins produced by two separate ova fertilised simultaneously by separate sperms. Genetically, such twins are no more alike than siblings and may be of the same or opposite sexes.

DM-INTERFERENCE: Interference in which the analytical signal is influenced by a different mechanism than the analyte, for example, the formation of non-dissociating species in emission spectroscopy.

DNA (DEOXYRIBONUCLEIC ACID): A nucleic acid molecule in the form of a twisted double strand helix that is the major component of chromosomes and carries genetic information. DNA is found in all living organisms except some viruses, reproduces itself and is the means by which hereditary characteristics pass from one generation to the next. Because it comes from the nucleus and is acidic in nature, scientists first called it nucleic acid. DNA occurs in chloroplasts and mitochondria. Part of the molecule of DNA is a sugar called ribose, which lacks an atom of oxygen molecule, giving it the name DNA. The DNA together with some proteins make up the chromosome in every nucleus and transmits genetic information from parent to offspring. It carries its messages with four kinds of symbols called nucleotide bases: Adenine (A), Cytosine (C), Guanine (G) and Thymine (T). The bases form pairs. Because of their shapes and chemical forces they can only fit together in one way. (A) combines with (T) and (C) combines with (G). The pairs form the steps or rungs of a very long ladder-like molecule. The sides of the ladder are made of the sugar units (deoxyribose) and phosphate units (the acid part). The ladder is twisted into a helix, resembling a spiral staircase. The sequence of the base-pairs that make up the steps of the DNA ladder form the genetic code, which directs the types of amino acid used in the synthesis of proteins in the ribosomes.

DNA CLONING: A genetic engineering technique used in producing exact copies (clones) of particular DNA material (genes). This may be done by inserting specific individual segments of DNA into a vector and propagating it in a host organism thereby generating a large number of copies of that sequence.

DNA FILTER ASSAY: A method used to determine the presence of recombinant DNA in cloned cells. It involves immobilising the DNA on a filter and putting it in a solution that contains radioactively-labelled probe DNA or RNA molecules. This, as a culture, could contain many cells that may not have been transformed.

DNA FINGER PRINTING (GENETIC FINGER PRINTING; DNA PROFILING; DNA TESTING; DNA TYPING): A technique employed by forensic scientists to assist in the identification of individuals on the basis of their respective DNA profiles. The individual's DNA is analysed, usually using small samples of body tissues such as hair, blood, semen or vaginal secretions to reveal the pattern of repetition of particular short nucleotide sequences called variable number tandem repeats, throughout a particular genome. This pattern is usually very unique to the individual concerned, hence has found widespread application for identification purposes in forensic science and paternity disputes, as well as in veterinary science. Although 99.9% of human DNA sequences are the same in every person, the small percentage, which is different can distinguish one individual from another.

DNA HYBRIDISATION: A method of determining the similarity of DNA from different sources. Single strands of DNA from two sources, for example, different bacteria species are put together and the extent to which double hybrid strand are formed is estimated.

DNA LIBRARY (GENE BANK; GENOMIC LIBRARY): 1. A collection of cloned DNA fragments representing the entire genetic material of an organism. This facilitates screening and isolation of any particular gene. 2. A mixture of clones, each containing a cloning vector and a segment of DNA from a source of interest.

DNA LIGASE: An enzyme that is able to join together two portions of DNA and therefore plays an important role in DNA repair. Purified DNA ligase is used in gene cloning to join DNA molecules together.

DNA METHYLATION: The chemical reactions that place a methyl group (a combination of one carbon atom and three hydrogen atoms) at a particular spot on DNA during organism's development or the addition of methyl group to constituent bases of DNA. The effect of this process is probably to "turn off" various genes during the process of cellular differentiation, causing the cell to develop into a specific type. In both prokaryotes and eukaryotes certain bases of the DNA generally occur in a methylated form.

DNA MICROARRAY (DNA CHIP): A microarray containing numerous small DNA molecules used to analyse gene transcript or to detect mutation of specific genes. This involves measuring the amount of complementary DNA, synthesised from the mRNAs that binds to complementary nucleotides on the microarray. It consists of an arrayed series of thousands of microscopic spots of DNA oligonucleotides, called features, each containing picomoles (10^{-12} moles) of a specific DNA sequence, known as probes or reporters. Since an array can contain tens of thousands of probes, a microarray experiment can accomplish many genetic tests in parallel, thus arrays have dramatically accelerated many types of investigation.

DNA PHOTOLYASE: An enzyme found in bacteria and other organism that repairs damage to DNA induced by exposure to ultraviolet light.

DNA POLYMERASE: An enzyme that catalyses the formation of 3' - 5' phosphodiester bonds from deoxyribonucleotide triphosphates or an enzyme that catalyses the polymerisation of deoxyribonucleotides into a DNA strand. It is best-known for its role in DNA replication, in which the polymerase "reads" an intact DNA strand as a template and uses it to synthesise the new strand. This process copies a piece of DNA. The newly-polymerised molecule is complementary to the template strand and identical to the template's original partner strand.

DNA PROBE: A short sequence of DNA labelled isotopically or chemically that is used for the detection of a complementary nucleotide sequence.

DNA REPAIRS: The way in which a cell corrects potentially damaging or mutagenic errors in its DNA. It is a variety of mechanisms that help to ensure that the genetic sequence as expressed in the DNA is maintained and that errors occurring during DNA replication by mutation are not allowed to accumulate.

DNA REPLICATION: The duplication of the parent DNA molecules into two genetically identical daughter cells during mitosis, each of which has exactly the same sequence of bases as the parent molecule. As the basis for biological inheritance, it is a fundamental process occurring in all living organisms to copy their DNA. Cellular proofreading and error toe-checking mechanisms ensure near perfect fidelity for DNA replication. DNA replication can also be performed in vitro or outside a cell.

DNA SEQUENCING (GENE SEQUENCING): A process by which the sequence of nucleotides along a strand of DNA is determined. Now, polymerase chain reaction (PCR)-based methods have obviated the need to clone the DNA and automated DNA sequencing using column chromatographic separation has become commonplace. To carry out the sequencing of the human genome, scientists cut the DNA up into short fragments, sequence these fragments simultaneously and then assemble the entire genome by using sophisticated computer techniques to match the fragments to each other.

DNA SUPERCOILING: The coiling of double helix DNA strands upon itself.

DNA-BINDING DOMAIN: Domain in a protein that is responsible for binding to DNA, usually with sequence specificity or an independently folded protein domain that contains at least one motif that recognises double- or single-stranded DNA. Thus many transcription factors would be expected to have such domains. Examples are in zinc-finger proteins, the helix turn–helix proteins and the leucine zipper proteins. It can recognise a specific DNA sequence (a recognition sequence) or have a general affinity to DNA. Some DNA-binding domains may also include nucleic acids in their folded structure.

DNA-BINDING PROTEINS: Proteins that are composed of DNA-binding domains and thus have a specific or general affinity for either single or double stranded DNA. Proteins, such as histones, interact with DNA in both eukaryotes and prokaryotes usually to pack or modify the DNA hence acting either as activators or repressors of gene expression by controlling the binding of RNA polymerase to DNA during the process of transcription. Among those that are DNA sequence specific, there are several specific DNA-binding domains (characteristic structural motifs) believed to be essential for specificity.

DNASE (DNAASE; DEOXYRIBONUCLEASE): A type of nuclease or an enzyme that catalyses the hydrolytic phosphodiester linkages in the DNA backbone. DNase I is a digestive enzyme, secreted by the pancreas that degrades DNA into shorter nucleotide fragment. The known types differ in their chemical mechanisms, substrate specificities and biological functions.

DÖBEREINER'S TRIADS: A set of triads of chemically similar elements noted by Johann Döbereiner (1780–1849) in 1817, which he put forward as the law of triads. Each of Dobereiner's triads was a group of three elements. It was observed that when each triad was arranged in order of increasing atomic mass, then the mass of the central member was approximately the average of the values for the other two. The chemical and physical properties were similarly related. His discovery gave other scientists a clue that relative atomic masses were important when arranging the elements. The triads are now recognised as consecutive members of the groups of the periodic table. Examples are lithium, sodium, potassium, calcium, strontium, barium, chlorine, bromine and iodine.

DOBSON UNIT: A unit of measurement for ozone in the atmosphere. One Dobson unit represents the amount of atmospheric ozone that would form a uniform layer 0.01 millimetre (10 micrometres) thick at standard temperature (0 °C) and pressure.

DOCK: 1. Remove or shorten the tails of certain animals. It also refers to the solid part of an animal's tail as distinguished from the hair; also, the part of an animal's tail left after it has been shortened. 2. Any of temperate large weedy plants of the genus Rumex having long taproots and greenish or reddish flowers sometimes used as potherbs.

DOCKING PROTEIN (SIGNAL RECOGNITION PARTICLE RECEPTOR): A receptor on the membrane of rough endoplasmic reticulum that specifically binds to the signal recognition particle.

DOCKING STATION: In computing, a docking station or port replicator provides a simplified way of connecting an electronic device such as a laptop computer to common peripheral devices.

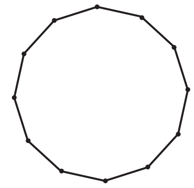
DOCKING STUDIES: Computational techniques for the exploration of the possible binding modes of a substrate to a given receptor, enzyme or other binding site.

DOCUMENT: Data associated with a particular application in computing.

DOCUMENT IMAGE PROCESSING, DIP (DOCUMENT IMAGING): Scanning documents for storage as digital images that can be displayed, indexed, retrieved, processed, transmitted and printed.

DOCUMENT OBJECT MODEL (DOM): Specification of a standard framework for creating object-oriented documents in HTML, XHTML and XML that defines the logical structure of documents and the way documents are accessed and manipulated. DOM allows a Web page document to be treated as a container for holding a number of objects. Each object is seen as a node in a tree, with each document having a parent or root node and any number of child nodes.

DODDER: A rootless, leafless parasitic plant of the morning glory family (convulvulaceae) of the genus *Cuscuta*. It has a reddish twining stem and lacks chlorophyll and have slender, twining, yellow or reddish stems and small whitish flowers.



Dodecagon

DODECAGON: A plane figure with 12 sides.

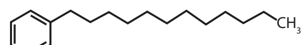
DODECAHEDRO-: An affix used in names to denote eight atoms bound into a dodecahedron with triangular faces.

DODECAHEDRON: A solid or polyhedron with 12 plane faces and 12 vertices. For a regular polyhedron, all the faces are pentagonal and is one of the five platonic solids.

DODECANOIC ACID (LAURIC ACID): A white crystalline fatty acid, with the chemical formula $CH_3(CH_2)_{10}COOH$, density 0.88 g/cm^3, melting point 43.2 °C (316.35 K) and boiling point 298.9 °C (572). It has a faint odour of bay oil. Glycerides of the acid are present in natural fat and oils such as coconut and palm-kernel oil. It is also found in human milk (6.2% of total fat), cow's milk (2.9%) and goat's milk (3.1%). It is solid at room temperature but melts easily in boiling water, so in its liquid form, it can be treated with various solutes and used to determine their molecular masses. It also has antimicrobial properties. It is inexpensive, has a long shelf-life and is non-toxic and safe to handle, thus, it is often used in laboratory investigations of freezing-point depression.

DODECENE (1-DODECENE; DODEC-1-ENE; ADACENE 12; ALPHA.-DODECENE; N-DODEC-1-ENE; DODECYLENE; ALPHA-DODECENE): A straight chain alkene with the molecular formula $CH_3(CH_2)_9CH=CH_2$, molar mass 168.319 g mol^{-1}, density 0.7584 g/cm^3, melting point -35.2 °C and boiling point 213.8°C. It is a colourless liquid with a mild, pleasant odour, which is insoluble in watere, but soluble in ethanol, ethyl ether and acetone. It is obtained from petroleum and used in making dodecylbenzene.

DODECYLBENZENE (1-PHENYLDODECANE; DETERGENT ALKYLATE; LAURYLBENZENE; n-DODECYLBENZENE; PHENYLDODECANE): A hydrocarbon which is a blend of isomeric (mostly monoalkyl) benzenes with saturated side chains averaging 12 carbon atoms, with the chemical formula $CH_3(CH_2)_{11}C_6H_5$. It is manufactured by a Friedel–Crafts reaction between dodecene and benzene. It can be sulfonated and the sodium salt of the sulfonic acid is the basis of common detergents such as alkyl amyl sulfonate type of detergents.



Chemical Structure of Dodecylbenzene

DOLABRIFORM: In botany, it refers to having the shape of an axe. In the case of such a leaf, it does not have divisions.

DOLBY DIGITAL (AC-3): A perceptual digital audio coding technique that reduces the amount of data needed to produce high-quality sound. Perceptual digital audio coding takes advantage of the fact that the human ear screens out a certain amount of sound that is perceived as noise. Reducing, eliminating or masking this noise significantly reduces the amount of data that needs to be provided.

DOLDRUMS (EQUATORIAL CALMS): An area of low atmospheric pressure along the equator. It is in the inter-tropical convergence zone, where the North East and South East Trade Winds converge. They are characterised by calm or very light winds during which there may be sudden squalls and stormy weather. The low pressure is caused by the heat at the equator, which makes the air rise and travel north and south, high in the atmosphere, until it subsides again in the horse latitudes, some of which returns to the doldrums through the trade winds. This process can lead to light or variable winds and more severe weather, in the form of heavy squalls, thunderstorms and hurricanes.

DOLIPORE SEPTUM: Specialised cross-wall with a swelling around the central pore that separates cells (septa) within the hypha of fungi belonging to the genus, Basidiomycota.

DOLOMITE: A common sedimentary carbonate or mineral containing calcium carbonate ($CaCO_3$) and magnesium carbonate ($MgCO_3$) or calcium magnesium carbonate ($CaMg(CO_3)_2$).

DOLOMITISATION (DOLOMITIZATION): The process of converting limestone into dolomite.

DOMAIN: 1. Also known as **maximal domain**, it is the set of all real numbers for which a function is defined. In other words, it is a set from which all the first elements of the ordered pair that constitute the function or relation are drawn. It is composed of the set of x-values that give rise to real y-values. 2. Also known as **essential domain**, the domain of a function is the set of all values of the independent variable for which the function is defined. 3. A taxonomic category above the kingdom level; the three domains of biological organisms are Bacteria, Eukarya, and Archaea. 4. A segment of a folded protein structure showing conformational integrity. A domain can contain the entire protein or a fraction of it. Some proteins, such as antibodies, contain many structural domains. 5. In computer system, a domain contains a group of computers that can be accessed and administered with a common set of rules. 6. A small area in a magnetic material that behaves like a tiny magnet.

DOMAIN NAME: A series of alphanumeric strings separated by periods, such as www.jfkl.com, which is an address of a computer network connection that identifies the owner of the address. **Domain name system (DNS)** is a system for converting hostnames and domain names into IP addresses on the Internet or on local networks that use the TCP/IP protocol. For example, when a Web site address is given to the DNS either by typing a URL in a browser or behind the scenes from one application to another, DNS servers return the IP address of the server associated with that name.

DOMAIN OF DEFINITION (RANGE OF SIGNIFICANCE): The set of subjects for which a given predicate is intelligible.

DOMAIN OF DISCOURSE (UNIVERSE OF DISCOURSE; UNIVERSE OF INTERPRETATION): In logic, the complete set of individuals able to be referred to or quantified over in an interpreted theory.

DOMAIN SUFFIX: The last part of a domain name and is often referred to as a "top-level domain" or TLD. Popular domain suffixes include ".com," ".net," and ".org," and several others approved by ICANN.

DOME: A geological feature that is the reverse of a basin. It consists of anticlinally folded rocks that dip in all directions from a central high point like an interval but usually has an irregular shape. The highest part is at the centre, from which the rocks dip in all directions. An example is the Harlech Dome of North Wales.

D

DOMINANCE HIERARCHY: A linear pecking order of animals, where position dictates characteristic social behaviours, so that one organism dominates all below it, the next all below it and so on down to the organism dominated by all. An example is found in the pecking order of apes, seals, barnyard hens and other species.

DOMINANT: 1. Dominant trait is the allele that is expressed in the phenotype or functional when two different alleles of a gene are present in the cells of an organism or one of a pair of alleles which is always expressed in a phenotype with the other being described as recessive or the greater influence by one of a pair of genes (alleles) that affect the same inherited trait. For example, if an individual pea plant that has one allele for tallness and one for shortness is the same height as an individual that has two alleles for tallness, the tallness allele is said to be **completely dominant**. If such an individual is shorter than an individual that has two tallness alleles but still taller than one that has two shortness alleles, the tallness allele is said to be partially or **incompletely dominant** and the shortness allele is said to be **recessive**. 2. The most conspicuously abundant and characteristic species in a community. 3. An animal that is allowed priority in access to food, mates, etc., by others of its species because of its success in previous aggressive encounters. Less dominant animals frequently show appeasement behaviour towards more dominant individuals so that over aggressive behaviour is minimised.

DOMINATED: 1. Of a subset in a partial order, possessing an upper bound that is then said to dominate the subset. 2. Of a sequence of positive terms, such that each element is less than the corresponding member of a given second sequence; that is, $\{a_i\}$ is dominated by $\{c_i\}$ if for every i, $a_i \le c_i$. 3. Of a sequence of positive terms, such that the sequence of absolute values of the terms of a given sequence of real numbers or moduli of the terms of a given sequence of complex numbers, is dominated in the preceding sense by a given second sequence.

DOMINATED CONVERGENCE THEOREM: The theorem of Lebesgue integration that allows one to evaluate the limit of the integrals of a sequence of functions as the integral of the pointwise limit of the functions as soon as the sequence of functions is dominated in absolute value by an integrable function. For example, if $\{f_n\}$ is a sequence of integrable functions, convergent almost everywhere to f and if there is an integrable g such that $f_n < g$ for all n, then f is integrable and

$$\int f\,d\mu = \lim_{x\to\infty}\int f_n d\mu.$$

DOMINO: Two congruent squares joined along an edge.

DOMINO RULE: A term used to describe events happening one after the other or in sequence.

DONATIONWARE: A software that is free to use, but encourages users to make a donation to the developer. Some donationware programs request a specific amount, while others allow users to determine what the program is worth and send in an appropriate donation.

DONGLE: A small hardware device that can be plugged into the serial or USB port of a computer to add an additional functionaity to the computer. It provides a way of adding RJ-45 and RJ-11 jacks to cards that do not have the jacks built in; and used as an adapter for Ethernet, Bluetooth or some other wireless technology that enables wireless access from a computer to an external Wi-Fi device such as a mobile phone, to the Internet via high-speed broadband or to enable wireless connectivity in a printer or other peripheral. A **Bluetooth dongle** or **Bluetooth adapter** is a USB device that plugs into the USB port and transmits and receives Bluetooth wireless signals such as Bluetooth mice and keyboards. A **PC Card dongle** connects a network or telephone cable to a PC Card (PCMCIA). A **cellular dongle** plugs into the USB port of a laptop or desktop computer to add 3G/4G cellular service. An **HDMI dongle** plugs into the HDMI port of a TV set and provides Wi-Fi streaming from the home network.

DONNAN EMF (DONNAN POTENTIAL): The potential difference at zero electric current between two identical salt bridges, usually saturated KCl bridges (conveniently measured by linking them to two identical electrodes) inserted into two solutions in Donnan equilibrium. Its symbol is E_D.

DONNAN EQUILIBRIUM (GIBBS-DONNAN EQUILIBRIUM): The equilibrium set up when two coexisting solutions are separated by a membrane permeable to some, but not all of the ions in the solutions or when two coexisting phases are subject to the restriction that one or more of the ionic components cannot pass from one phase into the other, i. e. this restriction is caused by a membrane which is permeable to the solvent and small ions but impermeable to colloidal ions or charged particles of colloidal size. An electrical potential develops between the two sides of the solutions with varying osmotic pressure.

DONNAN EXCLUSION: Reduction in concentration of *mobile* ions within an ion exchange membrane due to the presence of *fixed* ions of the same sign as the mobile ions.

DONOR: An individual whose tissues or organs are medically transferred to another (the recipient). Donors may provide blood for transfusion or a kidney or heart for transplanting.

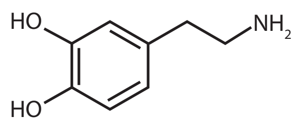
DONOR ATOM: The atom in a ligand that is bonded directly to the metal atom or an atom, molecule or ion that provides a part to combine with an acceptor, especially an atom that provides two electrons to form a bond with another atom.

DONOR IMPURITIES: 1. Impurities that provide conduction electrons to semiconductors. They are pentavalent atoms (atoms with five valence electrons). Examples include phosphorus (P), arsenic (As), bismuth (Bi) and antimony Sb). 2. An element introduced into a semiconductor with a negative valence greater than that of the pure semiconductor.

DONOR NUMBER (DN): A quantitative measure of Lewis basicity.

DOPA (DIHYDROXYPHENYLALANINE): A derivative of amino acid, with the chemical formula $C_9H_{11}NO_4$ formed from tyrosine in the liver during melanin and epinephrine biosynthesis. It occurs widely in animals and plants. It is a precursor in the synthesis of dopamine noradrenaline. Its levorotatory isomer (L-dopa) is used to treat Parkinson's disease.

DOPAMINE: A catecholamine, produced in several areas of the brain, including the substantia nigra and the ventral tegmental area, that is a precursor in the synthesis of noradrenaline and adrenaline. It functions as a neurotransmitter in the brain by activating the five types of dopamine receptors (D_1, D_2, D_3, D_4 and D_5) and their variants. It occurs in a wide variety of animals, including both vertebrates and invertebrates. It is also a neurohormone released by the hypothalamus. Its main function as a hormone is to inhibit the release of prolactin from the anterior lobe of the pituitary.



Chemical Structure of Dopamine

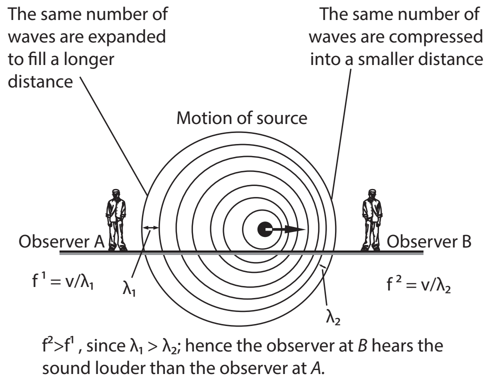
DOPANT (DOPING AGENT): A substance or chemical element (impurity) such as arsenic or antimony that is added in small quantities to a semiconductor material in order to modify its electrical properties. For example, dopants are added during the manufacture of transistors and semiconducting diodes. Doping with an element of valence 5 donates electrons (negative ions) to the germanium or silicon to form n-type semiconductor. Doping with an element of valence 3 donates holes (positive ions) and gives rise to a p-type semiconductor. It is used in crystal diodes and transistors.

DOPING: 1. The use of a drug or blood product to improve athletic performance. 2. The process of adding impurity atoms to intrinsic or extremely pure silicon or germanium to improve the conductivity of the semiconductor material. Lightly and moderately doped semiconductors are referred to as **extrinsic**. A semiconductor doped to such high levels that it acts more like a conductor than a semiconductor is referred to as **degenerate**.

DOPPLER BOOSTING: An effect, predicted by the special theory of relativity, that enhances the radiation from material that is moving toward the observer at nearly the speed of light and hides material moving in the opposite direction at such speeds. For example, if there are many quasars with radio jets moving at relativistic speeds, the strongest radio sources will be those that are most nearly pointed in our direction. This is important in understanding blazars and superluminal radio sources.

DOPPLER BROADENING: The broadening of a spectral line caused by the Doppler Effect. The Doppler Effect is the apparent variation in frequency of any emitted wave, such as a wave of light or sound, as the source of the wave approaches or moves away, relative to an observer. When the radiating atoms, molecules or nuclei do not all have the same velocity they may each give rise to a different Doppler shift.

DOPPLER EFFECT: Apparent change in observed frequency or wavelengths of waves due to relative motion between the wave source and the observer; or the apparent change in the observed frequency of a wave as a result of relative motion between the source and the observer, i. e. the total Doppler effect may, therefore, result from motion of the source, motion of the observer or motion of the medium. Thus, each of these effects could be analysed separately. For example, the sound made by a car as it approaches an observer appears to fall in pitch as it passes and moves away from the observer. In fact, the frequency of the engine remains constant but as the car is approaching, more sound waves per second impinge on the ears of the observer and as it recedes, fewer sound waves per second impinge on the ear of the observer. The received frequency is higher compared to the emitted frequency during the approach, it is identical at the instant of passing by and it is lower during the recession. For waves that propagate in a medium such as sound waves, the velocity of the observer and of the source are relative to the medium in which the waves are transmitted. For waves which do not require a medium such as light or gravity in general relativity only the relative difference in velocity between the observer and the source needs to be considered. With sound, the immediate observation is a change of pitch. The relevant equation for sound in still air is $\frac{f_o}{f_s} = \frac{V \pm V_s}{V \mp V_o}$, where f_o is the observed or apparent frequency, f_s the source frequency, V the speed of sound in still air, V_o speed of observer relative to the air and V_s the speed of source relative to the air. The upper signs in the righthand side of the equation are used, when the source and the observer move towards each other. However, when the source and the observer recede, the lower signs on the righthand side of the equation are used. With light or any other electromagnetic wave, it is usual to measure the wavelength rather than its frequency. The Doppler effect is then expressed as the wavelength change, which is expressed as $\frac{\Delta\lambda}{\lambda} \pm \frac{V}{C}$, where λ is the wavelength of light, V the speed of light and C the speed of the observer. When the source is moving towards the observer the negative sign is used. This shortens the wavelength and the colour shift is towards the blue. However, when the source is moving away from the observer, the positive sign in the equation is used and the wavelength increases and the colour shift is towards the red. The Doppler Effect was named after the Austrian physicist Christian Doppler, who proposed it in 1842.



Doppler Effect

DOPPLER RADAR: Radar that measures the velocity of a moving object by measuring the shift in carrier frequency of the return signal. The shift is proportional to the velocity with which the object approaches or recedes from the radar station.

DOPPLER SHIFT: The change in position of spectral lines due to the Doppler Effect. If an object is approaching, its light is compressed and any lines in its spectrum appear at shorter wavelengths than if the object were at rest. If the object is receding, the opposite effect occurs and the spectral lines are shifted toward the longer wavelength or red end of the spectrum.

DOPPLER TRACKING: The most common method of tracking the position of vehicles in space. It involves measuring the Doppler shift of a radio signal sent from a spacecraft to a tracking station on Earth, this signal either coming from an onboard oscillator or being one that the spacecraft has coherently transponded in response to a signal received from the ground station.

DOPPLER, CHRISTIAN ANDREAS (1803–1853): Austrian physicist and mathematician famous for his discovery of the Doppler Effect. He was educated at Vienna Polytechnic Institute, where he excelled in mathematical and other studies and graduated in 1825. After this, he attended philosophy lectures at the Salzburg Lyceum, then went to the University of Vienna, where he studied higher mathematics, mechanics and astronomy, where he became professor of physics in 1850. In 1842, at a meeting of the Royal Bohemian Society in Prague, he presented the work that was to make him world famous. It was entitled "On the coloured light of the double stars and certain other stars of the heavens"; the paper described (applied to light) the shift of frequency, which was named after him. The theory was later proved experimentally and extended for any electromagnetic and acoustic waves. It has found applications in astronomy, physics, aviation, meteorology and health science.

DORADO: A southern constellation that is distinguished only by containing most of the Large Magellanic Cloud and having, in Beta Dor, one of the brightest Cepheid variables.

DORMANCY: Phase of reduced physiological activity exhibited by certain buds, seeds and spores or an inactive period in the life of an animal or plant during which growth slows or completely ceases, though their morphological and physiological activities continue. For example, annual plants survive the winter as dormant seeds while many perennial plants survive as dormant tubers, rhizomes or bulbs.

DORMANT VOLCANO: An active volcano that has not erupted recently but capable of erupting again. Volcanoes go dormant because magma from the Earth's mantle can no longer reach the volcano.

DORSAL: 1. A term describing the features of, on or near the surface of an animal closest to the backbone, farthest from the ground or normally directed upwards, although in upright mammals such as humans and kangaroo, it is backward directed (posterior surface). 2. Pertaining to the surface most distant from the axis such as the back of an outer face of an organ or lower side of a leaf; abaxial. **Dorsal Side** describes back or abaxial side or the lower side of a perianth part.

DORSAL AORTA: The artery in vertebrate embryos that transports blood from the aortic arches to the trunk and limbs. The dorsal aortæ give branches to the yolk-sac and are continued backward through the body-stalk as the umbilical arteries to the villi of the chorion. In adult fish it is a major artery that carries oxygenated blood from the efferent branchial arteries to branches that supply the body organ; in tetrapods it rises from the systemic arch. The two dorsal aortae combine to become the descending aorta in later development.

DORSAL ROOT (POSTERIOR ROOT): The part of a spinal nerve that enters the spinal cord on the dorsal side and contains only sensory fibres. The lateral division of the dorsal root contains lightly myelinated and unmyelinated axons of small diameter, which transmit pain and temperature sensation from the body. The medial division of the dorsal root contains myelinated axons of larger diameter. These transmit information of discriminative touch, pressure, vibration and conscious proprioception originating from spinal levels C2 through S5. If the dorsal root of a spinal nerve were severed it would lead to numbness in certain areas of the body.

DORSIFIXED: Describing an anther that is joined to the filament for some distance along its dorsal edge.

DORSIVENTRAL: To have upper and lower surfaces that are structurally different.

DOS (DISK OPERATING SYSTEM): A computer operating system specifically designed for use with disk storage. It was the first operating system used by IBM-compatible computers. DOS uses a command line or text-based interface, that allows the user to type commands. By typing simple instructions such as "pwd" for "print working directory" and "cd" for "change directory", the user can browse the files on the hard drive, open files and run programs. While the commands are simple to type, the user must know the basic commands in order to use DOS effectively (similar to Unix). This made the operating system difficult for novices to use, which is why Microsoft later bundled the graphic-based Windows operating system with DOS.

DOSE: 1. A quantity of something such as chemical, physical or biological agent that may impact an organism biologically; the greater the quantity, the larger the dose. 2. A measure of the extent to which matter has been exposed to ionising radiation. The magnitude of radiation exposures is specified in terms of the radiation dose. There are two categories of radiation dose: (i) The **absorbed dose,** also known as the **physical dose**, defined by the amount of energy deposited in a unit mass in human tissue or other media. The SI unit is the gray (Gy) [1 J/kg], where 1 gray = 100 rad = 0.01 J/kg. The material absorbing the radiation can be human tissue or silicon microchips or any other medium such as air, water and lead shielding. The **roentgen**, is used to quantify the number of rad deposited into a target when it is exposed to radiation. The **maximum permissible dose** is the recommended upper limit of absorbed dose that a person or organ should receive in a specified period according to the International Commission on Radiological Protection. (ii) The **biological dose** expressed in rems; with the SI unit of sievert (Sv is a measure of the biological effectiveness of the radiation exposure. It reflects the fact that the biological damage caused by a particle depends not only on the total energy deposited but also on the rate of energy loss per unit distance traversed by the particle. The biological impact is specified by the **dose equivalent** or **equivalent dose** which is a quantity that expresses the probability that exposure to ionising radiation will cause biological effects. It is the product of the absorbed dose (in rad or gray) and a quality factor specific to the type and energy of the radiation delivering that dose. Its unit is the rem if the absorbed dose is in rads and the sievert (Sv) if the absorbed dose is in grays. 1 Sv = 100 rem. 3. Specified quantity of a therapeutic agent, such as a drug or medicine, prescribed to be taken at one time or at stated intervals. 4. The amount of a specific nutrient in a person's diet or in a particular food, meal or dietary supplement. 5. The amount of a harmful agent such as a poison, carcinogen, mutagen or teratogen, to which an organism is exposed. For humans, most doses of micronutrients and medications are measured in milligrams (mg.), but some are measured in micrograms because of their potency. Non-medicinal poisons span the measurement scale. Even water is toxic when consumed in large enough quantities.

D

DOSE EQUIVALENT: Product of quality factor of the radiation and absorbed dose that is used to assess the radiation damage that actually occurs in tissues. Its unit of measurement is H. It is measured in sieverts (SI unit) or rems.

DOSE RATE: A term applied for the rate at which ionising radiation is being absorbed by a medium such as tissue.

DOSIMETER (RADIATION DOSIMETER): An instrument used to measure the dose of ionising radiation exposure for human radiation protection and for measurement of dose in both medical and industrial processes. **Dosimetry** is the study, measurement, calculation and assessment of the ionising radiation dose absorbed by the human body. It is measured in units of gray (Gy) for mass, and dose equivalent in units of sieverts (Sv) for biological tissue, where 1 Gy or 1 Sv is equal to 1 joule per kilogram.

DOT: 1. A tiny round mark made by or as if by a pointed instrument; a spot. 2. A period · as used in URLs and e-mail addresses, to separate strings of words, for example, john@yahoo.com. In the MS-DOS and OS/2 operating systems, it is the character that separates a filename from an extension as in, for example, TEXT.DOC. 3. The basic unit of composition for an image produced by a device that prints text or graphics on paper, for example, a resolution of 300 dots per inch. 4. In mathematics, it is the symbol (. or ·) for conjunction, decimal point and multiplication. For example, 2 × 4 can be written as 2 · 4 or 2 . 4.

DOT MATRIX: A 2-dimensional array of dots used to represent characters, symbols and images. It may be selectively lit or coloured to display or print letters, numbers and other symbols. Typically, the dot matrix is used in older computer printers and many digital display devices.

DOT PITCH (PIXEL PITCH): A measurement that defines the sharpness of a monitor's display. It measures the distance between the dots that display the image on the screen. This distance is very small and is typically measured in fractions of millimetres. The smaller the dot pitch, the sharper the picture.

DOT PLOT: A graphical display of data above a number line diagram, where each data point (i.e., dot) is plotted above the corresponding value on the number line. It is used to compare frequency counts within categories or groups.

DOT PRODUCT (INNER PRODUCT; SCALAR PRODUCT): The dot product of U with unit vector V, denoted $U \cdot V$, is defined as the projection of U in the direction of V or the amount that U is pointing in the same direction as unit vector V. This operation that can be defined algebraically as $U \cdot V = U_1V_1 + U_2V_2 + U_3V_3$ or geometrically as $U \cdot \mathrm{V} = |U||V|\cos\theta$, where $|U|$ and $|V|$ are the magnitudes of the vectors **U** and **V** respectively and θ is the angle between them.

DOT-BLOT (SLOT BLOT): A method for detecting biomolecules or a specific sequence of neucleotides in a DNA or RNA molecule. It represents a simplification of the northern blot, southern blot or western blot methods. In a dot blot the biomolecules to be detected are not first separated by chromatography, rather, a mixture containing the molecule to be detected is applied directly on a membrane as a dot. This is then followed by detection by either nucleotide probes for a northern blot and southern blot or antibodies for a western blot.

DOTS PER INCH (DPI): A term used to refer to the measure of the resolution of an image both on screen and in print. DPI measures how many dots fit into a linear inch. The higher the DPI, the more detail can be shown in an image.

DOUBLE: In botany, it refers to having more than the usual number of petals, mostly in a crowded arrangement.

DOUBLE ANGLE FORMULAE: Trigonometrical or hyperbolic formulae expressing the value of the function at twice one value in terms of that value. For trigonometric functions of $2x$ in terms of functions of an angle x, it is expressed as follows:

$$\sin(2x) = 2\sin x \cos x,$$

$$\cos(2x) = \cos^2 x - \sin^2 x = 2\cos^2 x - 1 = 1 - 2\sin 2x,$$

$$\tan(2x) = \frac{2(\tan x)}{1-\tan^2 x}.$$

For hyperbolic functions, it is as follows:

$$\sinh(2x) = 2\sinh x \cosh x$$

$$\cosh(2x) = 2\cosh^2 x - 1$$

$$\tanh(2x) = \frac{2\tanh x}{1-\tanh^2 x}.$$

DOUBLE BETA DECAY: The simultaneous decay of two neutrons to form two protons and emit two beta particles and two neutrinos. Elements undergoing double beta decay include germanium-76, selenium-82 and uranium-238. Beta decay involves a neutron transforming into a proton by emitting an electron and an antineutrino.

DOUBLE BLIND EXPERIMENT: An experimental procedure in which neither the human subjects nor the conductors of the experiment know its critical aspects until after the study is completed and analysed. This is to eliminate biased results carried by an experiment's subjects and conductors. This procedure can be used in testing a new drug where participants in the study will take, for example, a pill, but only some of them will receive the real drug under investigation. The rest of the subjects will receive an inactive pill, called a placebo. Both the placebo and the active or actual medicine look, and taste, the same. With a double-blind study, the participants and the experimenters have no idea who is receiving the real drug and who is receiving the placebo. By keeping both the experimenters and the participants in the dark until after the study is completed, the true effects of the pill are known. Also, psychologically, if some sick people believe that they are receiving a medicine, even if it is just a placebo, without any medical composition, they show some signs of improvement in health.

DOUBLE BOND: A chemical bond which contains two shared pairs of electrons or two covalent bonds between adjacent atoms, as in the alkenes (-C=C-) and ketones (-C=O). Double bonds are stronger than single bonds and they are also shorter. The bond order is two and they are also electron-rich and that makes them reactive. Many types of double bonds between two different elements exist, for example, in a carbonyl group with a carbon atom and an oxygen atom (-C=O). The most common double bond between two carbon atoms is found in alkenes (-C=C-). Other common double bonds are found in azo compounds (N=N), imines (C=N) and sulfoxides (S=O). In skeletal formula, the double bond is drawn as two parallel lines (=) between the two connected atoms.

DOUBLE CIRCULATION: The type of circulatory system that occurs in terrestrial vertebrates (amphibians, birds and mammals) in which the blood passes through the heart twice before completing a full circuit of the body. Blood is pumped through the heart to the lungs and returns to the heart before being distributed to the other organs and tissues of the body. In the first circuit, the blood is pumped to the lungs, where it acquires oxygen. It then returns to the heart and enters the second circuit, going to the rest of the body, eventually, after having been deoxygenated, returning once again to the heart. Fishes have a single circulation system because they lack lungs. Most animals living above the water require a double circulatory system to allow the added benefit of direct oxygenation from a developed pulmonary circuit. Embryology of the human circulatory system is an advanced form of the double circulatory system as the distinction between right heart and left heart is founded.

DOUBLE CLICKING: Clicking the left mouse button quickly two times over a selected item, which is recognised by a computer as a specific command. It is used to perform a variety of actions, such as opening a program, opening a folder or selecting a word of text.

DOUBLE CROPPING: A type of multi-cropping, in which two crops are grown alternately within a year on a piece of land by seeding or transplanting one after the harvest of the other.

DOUBLE DATA RATE (DDR): A clocking technique that increases the transfer speeds of synchronous memories by using both the leading and trailing edges of the clock signal to transfer data, effectively doubling the transfer rate or bandwidth.

DOUBLE DECOMPOSITION (METATHESIS REACTIONS; METATHESIS): A type of reaction between two compounds, which can either be ionic or covalent in which the first and second parts of one reactant are united, respectively, with the second and first parts of the other reactant. It can also be defined as the exchange of bonds between the two reacting chemical species, which results in the creation of products with similar or identical bonding affiliations. An example is $KCl + AgNO_3 \rightarrow KNO_3 + AgCl$. Metathesis of alkenes is an important type of reaction in synthetic organic chemistry. It involves exchanges of groups. Reactions of this type are catalysed by metal alkylides containing an $M = CR_2$ grouping and the intermediate is a four-membered ring containing the metal ion. The catalysts most often used are the Schrock catalysts based on molybdenum and the Grubbs catalysts based on ruthenium. The American chemists Richard Schrock and Robert Grubbs shared the Noble price for Chemistry in 2005 for work in this field.

DOUBLE FERTILISATION: A process unique to flowering plants in which two male gametes of the pollen tube separately fuse with different female nuclei in the embryo sac. One male gamete nucleus fuses with the female gamete to form a diploid zygote while the other fuses with a diploid nucleus to form a triploid nucleus or endosperm nucleus.

DOUBLE FLOWER: A flower having more than the usual number of petals usually due to the transformation by a mutation of stamens into petals. Extremely double flowers, in which the carpels have also become petals, are termed **flore pleno**.

DOUBLE HELIX: The structural arrangement of DNA, which looks like an immensely long ladder twisted into a helix or coil. An example is actin, a fibrous protein found in most eukaryotic cells takes the form of a double helix. Double helices are, however, relatively rare configurations. Much more common is the single helix (α-helix) characteristic of globular protein.

DOUBLE INTEGRAL: An integral used to treat two functions that are of different variables. More than two such functions are treated together by use of a multiple integral. This is denoted by $dxdy$.

DOUBLE LAYER (ELECTRICAL DOUBLE LAYER): In chemistry, the electric charges that appear on the surface of an object such as an electrode when it is immersed into an electrolytic solution. The double layers are two parallel layers of opposite charge surrounding the object. The first layer is the surface charge which is either positive or negative, which are ions adsorbed onto the object due to chemical interactions. The second layer is composed of ions attracted to the surface charge via the Coulomb force, electrically screening the first layer.

DOUBLE NEGATION: In logic, the occurrence of two forms of negation in the same sentence according to which a statement is equivalent to or can be derived from, the negation of its negation. For example, "It is not the case that John is not here" and "John is here" are related in this way. Intuitionist logic denies that this relationship holds in both directions: it permits the derivation of its double negation from a given sentence, but not vice versa.

DOUBLE ORDINATE: A line segment between two points on a curve and parallel to a coordinate axis.

DOUBLE PENDULUM: A two-dimensional dynamical system. It has two degrees of freedom. In dynamical systems, it is a pendulum with another pendulum attached to its end and is a simple physical system that exhibits rich dynamic behaviour with a strong sensitivity to initial conditions. The motion of a double pendulum is governed by a set of coupled ordinary differential equations. For certain energies its motion is chaotic.

DOUBLE POINT: A point at which a curve intersects itself, such as a crunode.

DOUBLE PRECISION: In computing, using twice the usual number of bits to represent a number, giving greater arithmetic accuracy.

DOUBLE RADIO SOURCE: A radio galaxy in which the bulk of radio emission comes from two sources on opposite sides of the visual galaxy. The radiation is due to the violent expulsion from the nucleus of the parent galaxy of high speed energetic particles in two opposite directions. About one-third of all known radio galaxies are double sources.

DOUBLE RECESSIVE: A diploid individual homozygous for or containing two copies of the same recessive allele of a gene, as indicated by the expression of the recessive allele in the phenotype.

DOUBLE REFRACTION (BIREFRINGENCE): The property of forming two refracted rays from a single incident ray, possessed by certain crystals such as calcite crystals or boron nitride. When the incident ray passes through these materials, depending on the polarisation of the light, they decompose into ordinary and extraordinary ray. The birefringence magnitude can be calculated by the expression: $\Delta n = n_e - n_o$, where n_e and n_o are the refractive indices for polarisations parallel (extraordinary) and perpendicular (ordinary) to the axis of anisotropy respectively. The reason for birefringence is the fact that in anisotropic media the electric field vector $\bar{E}$ and the dielectric displacement $\bar{D}$ can be nonparallel (namely for the extraordinary polarisation), although being linearly related. Birefringence can also arise in magnetic, not dielectric, materials, but substantial variations in magnetic permeability of materials are rare at optical frequencies. Liquid crystal materials as used in liquid crystal displays (LCDs) are also birefringent.

DOUBLE ROOT: One of a pair of equal roots of the same polynomial or equation that occurs when the polynomial has $(x - a)^2$ as a factor, where a is the double root.

DOUBLE RULED SURFACE: A surface in which for every one of its points there are two distinct lines that lie on the surface. In other words, the surface has two distinct ruled patches on it, so that a ruling of one patch does not belong to the other patch. Examples are the hyperbolic paraboloid and the hyperboloid of one sheet, because they admit two one-parameter families of lines.

DOUBLE SALT: Salts containing more than one cation. They are formed when more than one salt is dissolved in a liquid and when together they crystallise in a regular pattern. An example is alum ($KAl(SO_4)_2.12H_2O$), which contains two cations (potassium and aluminium) and a sulfate anion. Potassium sodium tartrate and bromlite are other examples. A double salt differs from a complex salt in that when double salt is dissolved in water it dissociates into simple ions completely while complex salts do not dissociate into simple ions and maintain the property of complexes.

DOUBLE SEQUENCE: A sequence of certain elements indexed by two indices, for example, $a_{n,m} = (-1)^{n+m}(n+m)$.

DOUBLE SLIT INTERFERENCE (YOUNG'S DOUBLE SLIT EXPERIMENT): Interference of light from two parallel slits, which are illuminated by light from a single slit, which in turn is illuminated by a source; the interference can be seen by letting the light fall on a screen, which then shows a series of parallel fringes.

DOUBLE STAR (BINARY STAR): A pair of stars which revolve around their common centre of mass under mutual gravitational attraction. The brighter star is called the primary and the other is its companion star.

DOUBLE SUCKLING: A method of raising beef calves, where a second calf is placed with the cow's own calf and allowed to suckle.

DOUBLE TANGENT: 1. A line that is tangent to a curve at two distinct points. 2. A pair of distinct but coincident tangents at the same point of a curve, for example, at a cusp.

DOUBLE-BEAM SPECTROMETER: A luminescence spectrometer in which both the excitation and emission monochromators scan the excitation and emission spectra simultaneously, usually with a fixed wavelength difference between excitation and emission. Double (spectral) beam spectrometers are used, where two samples are to be excited by two different wavelengths.

DOUBLE-EYE CUTTING: A term used when a plant has leaves that are opposite.

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DOUBLE-FOCUSING MASS SPECTROMETER: An instrument which uses both direction and velocity focusing and therefore an ion beam of a given mass/charge is brought to a focus when the ion beam is initially diverging and contains ions of the same mass and charge with different translational energies. The ion beam is measured electrically.

DOUBLE-GLAZING (INSULATED GLASS): A window made from two or more glass panes separated by a narrow air gap. Double glazing reduces the heat lost through the window of a building. These units use the thermal and acoustic insulating properties of a gas or vacuum contained in the space formed by the unit. They give good insulation without sacrificing transparency. Single glazed tinted and reflective glasses can provide similar thermal insulation, but for the same insulation performance are harder to see through and provide little protection against unwanted sound.

DOUBLE-LAYER CURRENT: The non-faradaic current associated with the charging of the electrical double layer at an electrode solution interface.

DOUBLE-SERRATE (BISERRATE): A rough or coarsely serrate margin with smaller teeth on the margin of larger teeth.

DOUBLE-STEM PLOT: A plot used to compare two distributions side-by-side in which each stem is split into two parts. Each data value is split into a "leaf" (usually the last digit) and a "stem" (the other digits). In double stem and leaf plots, numbers on the original stem ending in 0 through 4 are plotted on one half of the split, and numbers ending in 5 through 9 are plotted on the other half. Double-stem plots are useful if the original stem-and-leaf plot has many leaves falling on few stems.

DOUBLE-STRAND CHAIN: A chain that comprises constitutional units connected in such a way that adjacent constitutional units are joined to each other through three or four atoms, two on one side and either one or two on the other side of each constitutional unit.

DOUBLE-STRAND COPOLYMER: A copolymer the macromolecules of which are double-strand macromolecules.

DOUBLE-STRAND MACROMOLECULE: A macromolecule that comprises constitutional units connected in such a way that adjacent constitutional units are joined to each other through three or four atoms, two on one side and either one or two on the other side of each constitutional unit.

DOUBLE-WAVELENGTH SPECTROSCOPY: The effect of spectral background due to impurities, solvent or radiation scattering may be reduced if the difference in the absorbances of a sample measured at two selected wavelengths is obtained. This is often achieved by repetitively switching from one wavelength to the other. Double-wavelength spectroscopy does this automatically by allowing two beams of radiation of different wavelengths to pass through the cell. One beam is fixed at a longer wavelength and the other measures absorbance while being scanned over a limited wavelength range at shorter wavelengths.

DOUBLING TIME: 1. The time needed for a certain population to double in number. 2. The time taken for a cell in culture to double.

DOUBLES FACT: The sum (or product) of a 1-digit number added to (or multiplied by) itself. Examples are 4 + 4 = 8 or 3 × 3 = 9. A doubles fact does not have a turn-around fact partner.

DOUBLET: 1. A pair of optical lenses of different shapes made of different materials used together so that the chromatic aberration produced by one is largely cancelled by the reverse aberration of the other. 2. A pair of associated lines in certain spectra.

DOUBLET STATE: In quantum mechanics, it is a quantum state of a system with a spin of $\frac{1}{2}$, such that there are two allowed values of the spin component, $-\frac{1}{2}$ and $+\frac{1}{2}$.

DOUBLING STRATEGY: To count, add, or subtract by doubling a given number, for example, adding 8 and 9 by doubling 8 and adding 1 to it.

DOUBLING THE CUBE (DELIAN ALTAR PROBLEM; DELIAN PROBLEM): The traditional geometric problem of constructing a cube having a volume which is double that of a given cube, using a straight-edge and compasses only. It was named after the oracle at Delos. The problem was solved using conics by Apollonius in third century BC, but it was not shown to be impossible using Euclidean constructions until the 18th century, since $\sqrt[3]{2}$ is not a constructible number.

DOUBLING TIME: The time in minutes and hours required for a cell population to double either the number of cells or the active cell mass. Its symbol is t_D.

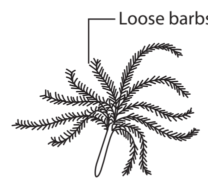
DOUBLY PERIODIC FUNCTION: A function $f(z)$ with two distinct periods ($\omega_1$and ω_2) whose ratio is not a real number and are periodic in two directions. A function $f(z)$ of complex variable is doubly periodic if there exist two constants ω_1 and ω_2 such that $f(z) = f(z + \omega_1) = f(z + \omega_2)$ for all z. Doubly periodic functions are periodic in the horizontal (real-axis) and vertical (imaginary-axis) directions.

DOUBLY STOCHASTIC MATRIX: A square matrix, $A = (a = (a_{ij}))$ such that $a_{i,j} \geq= 0$, whose columns and rows are non-negative and sum to unity.

DOUCHING: Applying water or liquid to the body for medical or hygienic purposes. An example is vaginal douching which is washing or flushing the vagina with water or other fluids, usually with water mixed with vinegar, baking soda, or iodine.

DOW'S PROCESS: 1. A process of extracting magnesium from sea water by adding calcium hydroxide to precipitate magnesium hydroxide. 2. The electrolytic method of bromine extraction from brine. 3. Benzene is also easily converted to chlorobenzene by the Dow process. Chlorobenzene is hydrolysed by a strong base at high temperatures to give a phenoxide salt, which is acidified to phenol.

DOWN FEATHERS (PLUMULES): Small soft feathers that cover and insulate the whole body of a bird, as shown in the diagram.



Down feather

DOWN PROCESS: A process used in the extraction of sodium by the electrolysis of molten sodium chloride. Potassium chloride and fluoride are added to the sodium chloride to reduce the melting point; the fused mixture is electrolysed, with sodium forming at the cathode and chlorine at the anode.

DOWN-CALVER: A cow or heifer about to calve.

DOWN-TIME: The loss of time that should be available for analysis. This might be due to breakdown, maintenance or other factors.

DOWN'S SYNDROME: A congenital form of mental retardation due to chromosome defect in which there are three copies of chromosome number 21 or trisomy instead of the usual two. It affects mental ability and appearance. For example, the affected individual may have a short broad face and slanted eyes, short fingers, weak muscles and sometimes epicanthal folds, moderate intellectual disability, heart or kidney malformations and abnormal fingerprint patterns. Victims age prematurely and have a short life expectancy of about 55 years. The danger of giving birth to children with the disorder increases with the mother's age. It can be detected in the foetus by amniocentesis.

DOWNCONVERSION (PARAMETRIC DOWN CONVERSION): A process by which a photon with frequency interacts with a non-linear medium and splits into two simultaneously emitted photons with frequencies so that the energy is conserved.

DOWNDRAUGHT (DOWNDRAFT): Downward-moving of air currents that occur during a thunderstorm. It also occurs as part of the turbulence that is created when air passes over topographic barriers such as mountains.

DOWNHOLE MEASUREMENT: Any one of a number of experiments performed by instruments lowered down a bore hole.

DOWNLINK: In telecommunications, it is the radio signal received from a spacecraft, as distinct from the uplink which is the signal sent to the spacecraft from the ground. Generally, the downlink of a network connection is used to receive signals from a remote server into a mobile device.

DOWNLOAD: The process in which data is sent over the internet to a computer or information received from the internet to a computer. The time it takes to download data depends on file size and network speed. Uploading is the opposite of downloading and sends information to a bigger computer.

DOWNRANGE: The horizontal distance covered by a launch vehicle or missile, with respect to the Earth's surface, after launch and before entering orbit.

DOWNWASH: As applied to the action of chimney gases, it is the downward motion of the chimney gases brought on by eddies which form in the lee of a chimney when wind is blowing. It may result in bringing the flue gases to the ground prematurely.

DOWNY: A term used to describe small, soft feathers or something that resembles such feathers; or covered or filled with small, soft feathers or something like them.

DOWNY MILDEW: A plant disease caused by fungi of the family Peronosporaceae in the order Peronosporales. The pathogen penetrates the tissues more deeply than powdery mildews and produces a white or grey mealy growth on the leaves and stems of the host in humid conditions.

DOWNWELLING: The process of accumulation and sinking of higher density material beneath lower density material, such as cold or saline water beneath warmer or fresher water or cold air beneath warm air.

DOWSING: The detection, usually with a divining rod, of hidden, usually underground, resources of water, metals, etc. Laboratory tests have shed little light on dowsing, though some dowsers have remarkable records of success. According to one theory, the perception of magnetic fields may be involved.

DOXASTIC LOGIC: The branch of modal logic concerned with reasoning about beliefs.

***DRACUNCULUS medinensis* (GUINEA WORM): T**he large female nematode, Dracunculus medinensis that causes the disease guinea worm. A native to warm tropical regions, it is a long thin worm that lives as a parasite under the skin of people and animals. The adult female is larger than the adult male and can grow to several centimetres in length. It attacks several parts of the body including the legs and feet, the head, torso, upper extremities, buttocks and genitalia, forming painful ulcerating blisters.

DRAFT: 1. The distance between the water line of a ship and the lowest part of its hull, which is the minimum depth of water it requires in order to float. 2. Animals used for pulling loads.

DRAFT EWE: An ewe sold from a breeding flock of sheep while still young enough to produce lambs.

DRAFT SEQUENCE: In bioinformatics, preliminary assembly of data subsets to form a semblance of a complete base sequence of an entire chromosome or genome. A draft sequence is usually corrected and refined to form the finished sequence.

DRAG: The force acting against the motion of an object moving through a fluid or the resistance to motion that a body experiences when passing through a fluid such as a gas or liquid. It is caused by friction between the outside of the moving object and the fluid surrounding it, for example, between a bullet and the air through which it is moving or between a person and the water in which he/she may be swimming. If the object is spherical and its speed equals the terminal speed, provided that the flow is streamlined, the magnitude of the drag is given by the Stokes' Law ($F = 6\pi r\eta v$). The drag is dramatically increased by the onset of turbulence. Drag increases with speed and it is lower for streamlined objects such as an airplane and also greater in a liquid than in a gas. The drag is dramatically increased by the onset of turbulence. As a result of this, a considerable attention is paid to this matter in vehicle designs, hence their streamlined bodies.

DRAG AND DROP (DRAG-AND-DROP): In computing, it refers to the action within a graphical user interface by moving the cursor over an object, selecting it and moving it to a new location.

DRAG COEFFICIENT: A dimensionless number used in aerodynamics to describe the drag of a shape. It is the ratio of the drag on a body moving through air to the product of the velocity and the surface area of the body. It is independent of the size of the object and is usually determined in a wind tunnel. Its symbol is C_d.

DRAG FORCE: A force on an electrically conducting fluid arising from inelastic collisions of electrons and ions and proportional to the fluid velocity.

DRAG FORM FACTOR: The drag coefficient, C_d, of an object multiplied by its cross-sectional area, A. It is used to scale the drag value for a particular object from the dimensionless C_d.

DRAGENDORFF TEST: A test for alkaloids which involves two solutions: bismuth nitrate in acetic acid, followed by sodium nitrate solution. A reddish brown deposit indicates the presence of alkaloids.

DRAGNET: 1. A net with weights on it used when searching for something at the bottom of a river or pond. It is often used when trawling for fish at sea. 2. A net that is drawn across the ground and used to trap small game.

DRAGON CURVE: Any member of a family of self-similar fractal curves, which can be approximated by recursive methods such as Lindenmayer systems. The curve can be constructed by representing a left turn by 1 and a right turn by 0.

DRAGONFLY (ANISOPTERA): A large brightly coloured carnivorous insect with a long thin body, large thin wings and toothed mandibles belonging to the order Odonata. There are two suborders: Anisoptera and Zygoptera (damselflies). Currently there are eight recognised families and 124 genera of dragonflies. Eggs are laid in or near water and immature dragonflies called nymphs, are fully aquatic having adaptations such as gills that allow them to spend the first few months to several years as aquatic nymphs, preying on other small aquatic organisms. As adults, dragonflies are agile fliers that prey on other insects.

DRAINAGE: The natural or artificial removal of surface and sub-surface water from an area. Most agricultural soils need drainage to improve production or to manage water supplies.

DRAINAGE BASIN (CATCHMENT AREA): An area of land that gathers precipitation water and directs it to a particular stream, river, lake, or other body of water. Its edge is known as the **watershed**.

DRAWBAR: A metal bar at the back of a tractor, used to pull trailed implements. In some tractors, the drawbar is attached to the hydraulic linkage.

DRAWER PRINCIPLE (DIRICHLET'S PRINCIPLE; LETTER-BOX PRINCIPLE; PIGEON-HOLE PRINCIPLE): It states that if n items are put into m containers, with $n > m$, then at least one container must contain more than one item. It was introduced in 1834 by German mathematician, Peter Gustav Lejeune Dirichlet (1805-1859).

DRENCHING: The administration of medicines in the form of solutions or suspensions in water through the mouth with a drench bottle, gun or funnel.

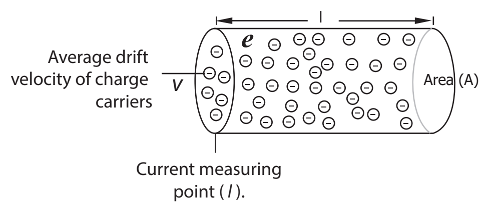
DRESSING: In agriculture, a process of coating seeds with chemicals before sowing, to control disease.

DRIFT: Small particles carried away from larger rocks by glacial meltwater. 2. Continental drift is the gradual movement of continents from each other and formation of new ones across the Earth's surface through geological time.

DRIFT NET: A long straight net suspended from the water and kept upright by weights at the bottom and floats at the top so as to drift with the current or tide. Although it is used in commercial fishing to catch fish such as herrings, sardines and tuna, it is indiscriminate in what it catches and bigger fish such as dolphins, sharks, turtles and other marine animals can become trapped in it and drown. For this and other reasons, fishing by drift net has become controversial.

DRIFT SCAN: A method of conducting sky surveys, including some Search for Extraterrestrial Intelligence (SETI) programmes, which involves a radio telescope remaining stationary and allowing the rotation of the Earth to sweep its beam in a narrow circular path across the sky each day.

DRIFT SPEED: The mean speed of charge carriers through a conductor in an applied electric field (when current flows through a conductor). For a metal of length l and a cross-sectional area A, if a current I flows through the conductor that has a charge density η with each charge carrier having a charge e and moving with a drift speed v, then the current through the conductor is expressed as: $I = \eta Ave$. In semiconductors, charge carriers have higher drift speed because the carrier density η is smaller than in conductors.



t = l/v
I =(Q/t) =(ηeAl)/(l/v))
I = ηeAv

η = charge density
Q = total charge carriers in the conductor of length l
Q = ηeAl
t = time taken for the charge carriers to move through the conductor length l with mean speed V

Drift Speed

DRIFT-TUBE ACCELERATOR (LINEAR ACCELERATOR; LINAC): An electron, a proton or a heavy-ion accelerator in which the paths of the particles accelerated are linear and imparts a series of relatively small increases in energy to subatomic particles as they pass through a sequence of alternating electric fields set up in a linear structure. Linacs are used in the generation of X-rays for medicinal purposes, as an injector for higher-energy accelerators and the investigation of the properties of subatomic particles. An example of linac is cathode ray tube.

DRILL: A tool or machine with a rotating cutting tip or reciprocating hammer or chisel, used for boring or making holes in a material or structure such as wood or a wall.

DRILL PLANTER: A machine which plants seeds in narrow rows at a high population.

DRIP IRRIGATION (TRICKLE/MICRO IRRIGATION): An irrigation method which saves water and fertiliser by allowing water to drip slowly to the roots of plants, either onto the soil surface or directly onto the root zone, through a network of valves, pipes, tubing and emitters. It is the backbone of Israel's agricultural economy; especially in hydroponically grown tomatoes that are exported to Europe..

DRIPLINE: 1. One of the limits on a chart in which the number of protons in a nucleus is plotted against the number of neutrons that indicate where nuclei can exist. There are two types of dripline: proton dripline and neutron dripline. 2. The dripline is the area directly located under the outer circumference of the tree branches, where the tiny rootlets are located that take up water for the tree and where they should be watered but not by the base of the trunk or the tree may develop root rot.

DRiPS (DEFECTIVE RIBOSOMAL PRODUCTS): Peptides produced in cells by enzymic digestion of the natural waste matter of protein synthesis, such as misfolded proteins and prematurely terminated polypeptides arising from errors in the synthetic machinery. They play a role in the immune system as an important source of self antigen for presentation at the cell surface in combination with MHC class I proteins.

DRIVE: 1. In computer science, it is also known as **disk drive**, which is a medium that is capable of storing and reading information. Examples of different drives in a computer or that may be accessible by a computer are Bernoulli drive, CD-ROM drive, CD-R drive, CD-RW drive, DVD drive, Hard drive, Local drive, LS120 drive or SuperDisk, Network drive, RAM disk and Solid-State Drive or Solid-State Disk (SSD). 2. In psychology, it is an innate, biologically determined urge to attain a goal or satisfy a need. 3. To operate the mechanism and controls and direct the course of, for example, a motor vehicle.

DRIVER (DEVICE DRIVER): In computing, it refers to a program that controls a device that is attached to the computer or internal function to the operating system and providing for activation of all device functions. Drivers may be required for internal components, such as video cards and optical media drives, as well as external peripherals, such as printers and monitors.

DRIZZLE: Liquid precipitation that is less than 0.02 cm in diameter. It is is normally produced by low stratiform clouds and stratocumulus clouds.

DRONE: 1. A reproductive male stingless bee in a colony of social bees (especially honeybees), whose sole function is to mate with the queen. 2. Also known as **remotely piloted vehicle**, it is a pilotless craft such as an aircraft, a ship and land vehicle guided by remote control. Underwater vessels, both piloted and pilotless are usually called **submersibles**. 3. In music, a drone is a harmonic or monophonic effect or accompaniment, where a note or chord is continuously sounded throughout most or all of a piece. 4. An abbreviation referring to one of two psychoactive drugs: Mephedrone and 4-Methoxymethcathinone, commonly known as methedrone.

DROOPING : In botany, any part, which is sagging or hanging down

DROPBOX (ONLINE BACKUP SERVICE): An online free cloud storage service on a computer that stores and synchronises files such as photos, documents and videos making it available on multiple computers.

DROP-DOWN MENU (DROP-DOWN LIST; POP-DOWN MENU): The common type of menu used with a graphical user interface (GUI), which is a horizontal list of options that each contain a vertical menu. Clicking a menu title causes the menu items to appear to drop down from that position and be displayed. Options are selected either by clicking the menu item or by continuing to hold the mouse button down and letting go when the item is highlighted. The most common type of drop down menu is a menu bar. On Windows systems, the menu bar is typically located at the top of each open window.

DROPLET: A small liquid particle, whose size encountered in the atmosphere extends over a widerange; for example, liquid aerosol solutions which make up the fine particle fraction of continental tropospheric aerosol are usually 2 μm in diameter. Cloud water droplets usually have diameters in the range of 5 - 70 μm, while rain droplets commonly have diameters ranging from 0.1 - 3 mm.

DROPPER: A small glass or plastic tube with a rubber bulb at one end that is used to suck up liquid and release it one drop at a time.

DROPPING: Excreta of animals and birds.

DROSERA: 1. An insect-eating plant belonging to the family, Droseraceae but commonly called the sundew that produces a rosette of hairy sticky leaves that are used to trap, capture and digest insects. The insects are used to supplement the poor mineral nutrition obtained from the soil in which they grow. The sundew comprises one of the largest genera of carnivorous plants, with at least 188 species, which are native to Australia and New Zealand. Various species, which vary greatly in size and form, can be found growing natively on every continent except Antarctica. 2. A homeopathic remedy, prepared from the sundew plant that is used to treat coughs and sore throats.

DROSOPHILA: A genus of small two-winged fruit flies, also called vinegar flies, especially *D. melanogaster* often used in genetic and developmental biology research because the larvae possess giant chromosome. They have conspicuous salivary gland, easy to raise and have a short life cycle.

DROUGHT: A long period of extremely dry weather when there is not enough rain or the replenishment of water supplies for the successful growing of crops. It involves water shortages, crop damage, stream flow reduction and depletion of groundwater and soil moisture. They occur when evaporation and transpiration exceed precipitation for a considerable period. Drought is the most serious hazard to agriculture in nearly every part of the world. Efforts have been made to control it by seeding clouds to induce rainfall. Four kinds of drought are recognised: permanent, typical of desert and semi-arid regions; seasonal, in climates with well-defined dry and rainy seasons; unpredictable, an abnormal failure of expected rainfall; and invisible, when even frequent showers do not restore sufficient moisture.

DRUG: Any of the range of chemicals voluntarily or involuntarily introduced into the bodies of humans and animals in order to enhance or suppress a biological function. Most drugs are medicines used to prevent or treat diseases or to relieve their symptoms. Examples are antibiotics, sedatives and pain relievers. Drugs can be beneficial or harmful, legal or illegal. Harmful drugs include alcohol in drinks, nicotine in tobacco, cocaine, heroin and caffeine in tea and coffee. Some, including antibiotics, stimulants, tranquilisers, antidepressants, analgesics, narcotics and hormones, have generalised effects. Others, including laxatives, heart stimulants, anticoagulants, diuretics and antihistamines, act on specific systems. Vaccines are sometimes considered drugs. Drugs may be given by mouth, by injection, by inhalation, rectally or by insertion through the anus or through the skin.

DRUG ABUSE (SUBSTANCE ABUSE): The use of drugs improperly or irresponsibly or the habitual consumption of substances which affect the nervous system. Some of the drugs most often associated with drug abuse are alcohol, amphetamines, barbiturates, benzodiazepines, cocaine, methaqualone and opioids. The use of these drugs may lead to criminal penalty (depending on local jurisdiction) in addition to possible physical, social and psychological harm.

DRUM: 1. A cylindrical container or receptacle. 2. A percussion instrument sounded by being struck with sticks or the hands, typically cylindrical, barrel-shaped or bowl-shaped, with a taut membrane such as leather stretched over one or both ends. 3. Also called **tambour**, it is a circular or polygonal wall supporting a dome or cupola. 4. Any of the cylindrical stone blocks that are stacked to form the shaft of a column. 5. A large bony, carnivorous, generally bottom-dwelling freshwater or saltwater fish, belonging to the Sciaenidae family, that emits a repeated rhythmic sound by moving strong muscles attached to the air bladder, which acts as a resonating chamber, amplifying the sounds. There are about 160 species and they have two dorsal fins and are usually silvery. The weakfishes, sea trouts and squeteagues (genus *Cynoscion*) have a large mouth, jutting jaws and canine teeth, but most drums have an under slung lower jaw and small teeth. The largest species, the totuava, weighs up to 100 kg, but other species are much smaller. Many drums are food or game fishes. 6. To produce a short burst of sound like a drum roll by rapidly beating with the beak or wings as observed in some birds.

DRUMLIN: An egg-shaped elongated hill composed largely of glacial deposits believed to have been formed by the streamlined movement of glacial ice sheets across rock debris, or till.

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DRUPE (STONE FRUIT): A fleshy indehiscent fruit containing one or more seeds which are surrounded by a hard, protective layer. Examples are coconut, peach, plum and cherry.

DRUPECETUM: An aggregation of drupelets, as in Rubus.

DRUSE (SPHAERORAPHIDE): A globular mass of needle-like crystals found either attached to the cell wall or free in the cytoplasm.

DRY ADIABATIC LAPSE RATE: The rate at which dry air cools as it rises through the atmosphere. The rate is 9.8°C per kilometre.

DRY-BULB TEMPERATURE (DBT): Temperature of the air as measured by a thermometer that is freely exposed to the atmosphere, but shielded from radiation and moisture.

DRY CELL: A primary or secondary cell, in which the chemicals or electrolytes are in the form of a paste, which does not spill. It consists of two dissimilar substances, a positive electrode and a negative electrode, that conduct electricity and a third substance, an electrolyte, that acts chemically on the electrodes. The two electrodes are connected by an external circuit such as a piece of copper wire; the electrolyte functions as an ionic conductor for the transfer of the electrons between the electrodes. The voltage or electromotive force, depends on the chemical properties of the substances used, but is not affected by the size of the electrodes or the amount of electrolyte. They are used in devices such as radio, torchlight, remote control and calculator.

DRY COW: A cow that is not producing milk, the period before the next calving and lactation.

DRY FARMING: A system of extensive agriculture, producing crops in areas of limited rainfall and without irrigation. It is achieved by planting drought-resistant crops and maintaining a fine surface tilth or mulch that protects the natural moisture of the soil from evaporation.

DRY ICE (CARDICE; CARD ICE): A solid carbon(IV) oxide made by cooling carbon dioxide gas. Dry ice sublimes at -78°C (195 K) standard temperature and pressure rather than melting and is used as a simple and convenient refrigerant to keep things cold. It is used as a cooling agent, because it is 78.5°C (351.65 K) colder than frozen water under normal conditions. The convenience of its sublimation in contrast to the melt-water left by warming ice outweighs other costs.

DRY MASS: The mass of a biological sample after the water content has been removed, usually by placing the sample in an oven.

DRY MATTER (DRY WEIGHT): 1. A measurement of the mass of something when completely dried. 2. Material left after all water is removed from a feed material. The dry matter of food, for example, would include carbohydrates, fats, proteins, vitamins, minerals and antioxidants such as thiocyanate, anthocyanin and quercetin. Dry matter content is an important parameter in the sugar industry for controlling the crystallisation process. It is often measured by microwave density metres.

DRY MATTER INTAKE (DMI): The amount of food which does not contain water or moisture consumed daily by an animal. It is influenced by factors such as food type, size, body condition, environment, stage and level of production.

DRY PERIOD: The period between the end of one lactation and the beginning of the next of a cow, which is six to eight weeks or the period when a cow is rested from giving milk.

DRY ROT: A plant disease in which there is disintegration of the tissues and the affected cells crumble into a powdery mass. Dry rot of stored potatoes is caused by the fungus *Fusarium solani* var. caeruleum and that of timber by *Serpula lacrimans*.

DRY-WEIGHT PERCENTAGE: The ratio of the weight of any constituent of a soil to the oven-dry weight of the soil. Oven-dry is a method of determining soil moisture content by taking a soil sample and drying it in an oven at a temperature of between 105 °C and 110 °C for 24 hours.

DRYING: 1. Removal of liquid water from a substance without altering its chemical composition. 2. A process of oxidation, in which a liquid such as linseed oil changes into a solid film. 3. An operation in which a liquid, usually water, is removed from a wet solid in an equipment called a dryer.

DRYING AGENT (DESSICANT): Soluble or insoluble chemical substance that has such a high affinity for water that it removes water from many fluid materials. Examples of soluble chemicals which serve as drying agents are calcium chloride, concentrated sulfuric acid and glycerol; insoluble chemicals used as drying agents are bauxite and silica gel.

DRYING CONTROL CHEMICAL ADDITIVE (DCCA): Co-solvent included to facilitate the rapid drying of gels without cracking.

DRYING OIL: Natural oil, such as linseed oil, that hardens on exposure to the air. It can be prepared from various non-drying fish oils such as sardine and herring oils and from whale oil. They are used in paints, varnishes, lacquers and printer's ink. Cottonseed oil, corn oil, soybean oil and tung oil are referred to as semi-drying oils.

DRYLAND PRODUCTION SYSTEM: Growing a crop in a region with less than 500 mm of annual precipitation and where annual potential water evaporation exceeds annual precipitation without supplemental water or irrigation.

DRYOPITHECUS: A genus of the extinct apes, fossils of which have been found in Eastern Africa, Eurasia and India and date as far back as 30–40 million years. It includes the common ancestor of the lesser apes, gibbons and siamangs and the great apes. The Y-5 arrangement of its molar teeth, which is a five-cusp and juvenile fissure pattern, is typical of the dryopithecids and of hominoids in general.

DSL (DIGITAL SUBSCRIBER LINE): A technology used for transferring data over regular phone lines that significantly increases the digital capacity; it can be used to connect to the Internet. Like a cable modem, a DSL circuit is much faster than a regular phone connection, even though the wires it uses are copper like a typical phone line. The two main types of DSL are: (i) **Asymmetric DSL** (**ADSL**), which is used for Internet access with download speeds of up to about 1.5 megabits (not megabytes) per second and upload speeds of 128 kilobits per second; and (ii) **Symmetric DSL** (SDSL) designed for connections that require high speeds of up to 9 megabits per second and upload speeds of up to 640 kilobits per second in both directions.

DUAL CORE: In computer science, it is a term used for a single chip that contains two separate processors that work simultaneously.

DUAL FLUORESCENCE: Systems or molecular species showing two fluorescence bands. Several non-trivial molecular properties may be at the basis of a dual fluorescence, such as conformational equilibrium in the groundstate and a mixture of excited states.

DUAL ISOMORPHISM: An isomorphism between a space and its dual, especially important in Galois theory.

DUAL NORM: The norm placed on the dual of a given normed space by $\|f\|=\sup\{|f(x)|:\|x\|=1\}$, for each continuous linear functional. This is the operator norm on the dual space. This space is necessarily complete and so is a Banach space.

DUAL PROCESSOR (DP): In computer science, it is a term used for a computer that contains two CPUs. DP systems have two independent CPU chips and differ from a dual core processor, which has two processors built into the same CPU integrated circuit or chip. In dual-core processor, each processor has its own cache and controller, which enables it to function as efficiently as a single processor. However, because the two processors are linked together, they can perform operations up to twice as fast as a single processor can.

DUAL-VIEW DISPLAYS: LCD screens that use a parallax barrier technology to show two different images from the left and right angles of the same screen. Applications allow drivers to view information that differs from what the passenger can see. For example, a driver may be able to see a scene captured by a rearview camera, while the passenger views a DVD.

DUAL-MODE PHOTOCHROMISM: Photochromism occurring in complex systems and triggered alternatively by two different external stimuli, such as light and an electric current. In such a case photochromism and electrochromism are mutually regulated.

DUALITY: 1. The state or quality of being two or in two parts; dichotomy. For example, in quantum mechanics, the particle and wave nature of both light and electrons is a duality. 2. In the theory of phase transitions in certain model systems there is a duality in which quantities in the low-temperature phase are related to quantities in the high-temperature phase to give information about the transition point. 3. The interchangability of two types of entity in a given theory. In projective geometry, there is duality between straight lines and points: there is interchangeability of the roles of the point and the plane in statements and theorems. If the dual of X is Y, then the dual of Y is X. Such involutions sometimes have fixed points, so that the dual of X is X itself.

DUALITY THEORY OF LINEAR PROGRAMMING: The assertion that a dual pair of linear programs is in strong duality, if both are feasible.

DUBBIN: A thick oily substance made of natural wax, oil or tallow, which is used for softening condition and to waterproof leather.

DUBNIUM (UNNILPENTIUM): A radioactive transactinide element, with the symbol Db and atomic number 105. It is in Group 5 (VB) of the Periodic Table and it can be made by bombarding californium-249 nuclei with nitrogen-15 nuclei. Each dubinium atom has a very large nucleus or central mass containing positively charged particles called protons and neutral particles called neutrons. The large number of particles in the nucleus makes the atom unstable and causes the atoms to split apart into smaller components soon after creation. The most stable isotope is ^{268}Db with a half life of 28 hours. It is the longest lived transactinide isotope, being a reflection of the stability of the Z = 108 and N = 162 closed shells and the effect of odd particles in nuclear decay.

DUCK: 1. Any of various wild or domesticated swimming birds of the family Anatidae with short legs, webbed digits and wide beak that are bred or hunted for meat, eggs and its soft feathers. The term also applies to a female duck. The male is a drake. A young duck is called a **duckling**. 2. A bivalve mollusc with a white thin shell found off the Atlantic coasts of America.

DUCT: 1. An often, enclosed channel, tube or pipe, through which a substance can be conveyed. An example is a tube or pipe for enclosing electrical cables or wires. 2. A narrow tubular exit passageway in a bladder or gland through which fluid passes. Examples are salivary or tear ducts in mammals.

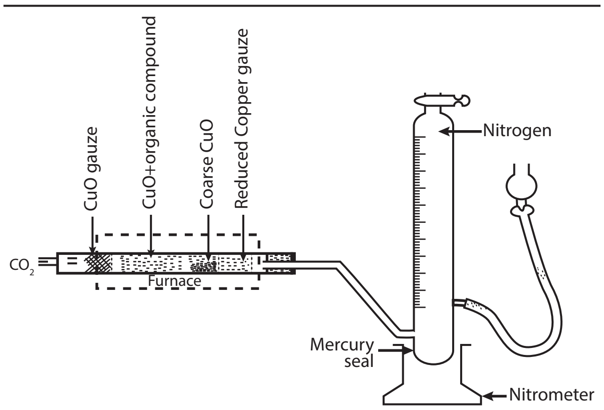
DUCTILITY: The ability of certain metals to retain their strength, when their shape is changed permanently under tensile stress without cracking or breaking. Ductile materials can be stretched and drawn into thin wires, bent or spread in response to stress without fracture. Examples of such materials include gold, silver, copper and platinum. The most ductile metal is platinum. Malleability is a material's ability to deform under compressive stress. For instance, lead is only malleable while gold is both ductile and malleable.

DUCTLESS GLAND: An endocrine gland such as the thyroid gland, testis and ovaries which secretes necessary hormones directly into the blood for normal growth and development, reproduction and homeostasis. Secretion is regulated either by regulators in a gland that detects high or low levels of a chemical and inhibits or stimulates secretion or by a complex mechanism involving the hypothalamus and the pituitary.

DUCTUS ARTERIOUS: A channel that connects the pulmonary artery with the aorta in the mammalian foetus and, therefore, allows blood to bypass the inactive lungs of the foetus to distribute oxygen received through the placenta from the mother's blood. It normally closes after birth. Pre-natal closure leads to circulatory problems. If the ductus stays open after birth, oxygenated and deoxygenated blood mix. Patent ductus arteriosus (PDA) is a heart defect that occurs in infants when the ductus arteriosus does not close at birth.

DULONG AND PETIT'S LAW: Formulated in 1819 by the French scientists Pierre Dulong (1785–1838) and Alexis Petit (1791–1820), it states that for a solid element, the product of the relative atomic mass and the specific heat capacity is a constant equal to about $25\,J\,mol^{-1}\,K^{-1}$. They found out, experimentally, that the heat capacity for a number of substances was close to constant, when divided by the presumed weights of the substance atoms, as defined by Dalton. In modern terms, it states that the molar heat capacity of a solid element is approximately equal to $3R$, where R is the gas constant. The law is only approximate but applies with fair accuracy at normal temperatures to elements with a simple crystal structure.

DUMA'S METHOD: A method for finding the nitrogen content of an organic compound by converting the nitrogen in whatever form it occurs to nitrogen gas and measuring the volume of the gas produced from which the mass of nitrogen can be calculated. The nitrogen in the organic compound is first converted to an oxide by heating it with copper(II) oxide and reducing the oxide over hot copper to produce the nitrogen gas. The method was first described by Jean-Baptiste Dumas (1800-1884), a French chemist.



Apparatus For Estimating Nitrogen By Duma's Method

DUMB TERMINAL: In computing, it is a terminal that has no processing capabilities and relies entirely on the main computer.

DUMMY VARIABLE: A variable in an expression or equation in which the letter being used as a variable can be replaced with another variable without changing the meaning of the equation or expression.

For example, if $\int_2^4 x^3\,dx$ and $\int_2^4 u^3\,du$ represent the same integral, then x and u are dummy variables.

DUMMY SUFFIX CONVENTION (SUMMATION CONVENTION): A shorthand notation used in manupulating components of vectors and tensors, in accordance with which the summation symbol Σ is omitted and the sum indicated by repetition of an index. For example, the scalar product $a \cdot b = a_1b_1 + a_2b_2 + a_3b_3$ may be more compactly written as $a_i b_i$.

DUNES: Wind-built ridges and hills of sand formed, especially on the sea coast or in a desert. They are of different sizes. Those found at the seaside are formed by the action of wind and water, while the dunes in a desert are formed purely by the action of wind.

DUNG: Solid excreta from animals, especially cattle, often used as fertiliser.

DUODECIMAL SYSTEM: A system of arithmetic notation using 12 as a base. Under base twelve, the place value changes from 10 to 12. Quantities are explained in terms of twelves, such as dozens, grosses and great-grosses, rather than tens, hundreds and thousands. Examples include 12 months in a year, 12 hours on a clock face and 12 items to a dozen.

DUODENAL: Of or pertaining to the duodenum.

DUODENAL GLANDS (BRUNNER'S GLANDS): Any of the small, branched, coiled tubular glands situated deeply in the submucosa of the duodenum that secrete mucus and an alkaline fluid that neutralises the acidic chyme leaving the stomach.

DUODENAL PAPILLA: Any of two small elevations on the mucosa of the duodenum. It consists of the major papilla located at the entrance of the conjoined pancreatic and common bile ducts and the minor papilla at the entrance of the accessory pancreatic duct.

DUODENUM: A short length of the alimentary canal, which is the first section of the small intestine, where carbohydrates, fats and protein are partially digested. In mammals, it is the principal site for iron absorption. It is formed between the stomach and the small intestine in higher vertebrates (mammals, reptiles and birds). It precedes the jejunum and ileum and it is the shortest part of the small intestine, where most chemical digestion takes place. In humans, the duodenum is a hollow jointed tube about 25-30 cm long linking the stomach to the jejunum, starting with the duodenal bulb and ending at the ligament of Treitz. In fishes, the divisions of the small intestine are not distinct and the terms anterior intestine or proximal intestine are used instead of duodenum.

DUPLET: A pair of electrons shared between two atoms in a covalent bond.

DUPLEX: 1. In telecommunications, a **full-duplex** is transmission of data along a communications channel in both directions at the same time, such as over a telephone line. A **half-duplex** is data transmission that can be used in either direction, but not at the same time. 2. Consisting of pairs of units or components that perform the same machine function but operate independently. 3. A biological molecule comprising two cross-linked polymeric chains oriented lengthwise, beside one another. The term is applied particularly to the double-stranded structure of DNA.

DUPLICATION (GENE DUPLICATION; CHROMOSOMAL DUPLICATION; GENE AMPLIFICATION): The doubling or repetition of part of chromosomes in meiosis. It may occur as an error in homologous recombination, a retrotransposition event or duplication of an entire chromosome. The other copy of the gene may be free from selective pressure, mutations of it have no negative effects on its host organism. Occasionally, this type of chromosome mutation may have beneficial effects on a population. It accumulates mutations faster than a functional single-copy gene over generations of organisms.

DEPURATIVE: Agents and substances containing properties which purify and cleanse the blood, wounds and sores. They include herbs such as dandelion and strawberry.

DUQUENOIS-LEVINE TEST: A presumptive test used for tetrahydrocannibanol and other cannabinoids. The test has a solution of 2% vanillin and 1% ethanol. Concentrated hydrochloric acid and chloroform are added. A purple colour in the chloroform layer indicates a positive result.

DURA MATER: The toughest, fibrous membrane which is the outmost of the three membranes that surround the central nervous system in invertebrates. It protects the delicate inner parts. It lies adjacent to the skull, covering the brain and the spinal cord and lining the inner surface of the skull.

DURALUMIN (DURALUMINIUM; DURAL): The trade name of one of the earliest types of aluminium alloys, which contains copper and traces of magnesium, manganese and silicon. The alloy is much harder and stronger than the pure metal. It is used for the production of aircraft frames and cooking utensils. A modern equivalent which is often used is AA2024, which contains 4.4% copper, 1.5% magnesium, 0.6% manganese and 93.5% aluminium by weight. Its yield strength is 450 MPa, with variations depending on the composition and temper.

DURAMEN (HEARTWOOD): The central part of the secondary xylem in some woody plants containing nonfunctional tracheary elements, often blocked by tyloses and infiltrated with organic compounds such as resins, tannins, gums and many aromatic substances and pigments.

DURBIN-WATSON TEST: A test that the residuals from linear regression or multiple regressions are independent. The test is given by D = $\frac{\sum_{i=2}^{n}(r_i - r_{i-1})^2}{\sum_{i=1}^{n} r_i^n}$, where r is the residuals.

DUST: Small, dry, solid particles projected into the air by natural forces, such as wind, volcanic eruption and by mechanical or manmade processes such as crushing, grinding, milling, drilling, demolition, shovelling, conveying, screening, bagging and sweeping. Dust particles are usually in the size range from about 1 to 100 µm in diameter and they settle slowly under the influence of gravity. Dust from smoke is a serious urban problem because it creates air pollution. Radioactive dust creates an even more serious hazard to all forms of life.

DUST COLLECTOR: A device for monitoring dust emissions. Also, it is the equipment used to remove and collect dust from process exhaust gases; this may employ simply sedimentation such as dustfall jars, coated slides and papers; inertial separation such as cyclones, impactors and impingers; precipitation (thermal and electrostatic) or filtration.

DUST CORE: A core made by mixing finely powdered magnetic material consisting of high purity electrolytic iron or iron alloy with an insulating binder and moulding under pressure to form a rod-shaped core that can be moved into or out of a coil or transformer to vary the inductance or degree of coupling for tuning purposes. It is used in power sources of TVs, audio equipments, copiers, printers, etc. Its function is to absorb noise emitted from electronics or electrical devices and also acts as a choke body.

DUST FALL: Solid particles in the air which fall to the ground under the influence of gravity.

DUST MULCH: A fine loose, or powdery layer on the surface of the soil, usually produced by shallowly cultivating the top soil, and also by deposition. This may prevent quick evaporation of soil moisture.

DUTCH METAL: An alloy of copper (84%) and zinc (16%), which is very malleable and ductile and can be beaten into very thin sheets and used as imitation gold leaf. It spontaneously inflames in chlorine.

DVD (DIGITAL VERSATILE DISK): A type of optical disk or disk format introduced in 1996, which is similar in technology to a compact disk (CD) but contains much more data and is more precise. They are made in a Read-Only Memory (ROM) format as well as in erasable (DVD-E) and recordable (DVD-R) formats. DVD players can read CDs, however, CD players cannot read DVDs. Like a CD drive, a DVD drive uses a low-power laser to read digitised (binary) data that have been encoded onto the disc in the form of tiny pits. A double-sided, dual-layer version can store about 30 times as much information as a standard CD. It is projected that DVDs will eventually replace CDs, especially for multimedia workstations.

DVD-RAM (DVD-RANDOM ACCESS MEMORY): A rewritable DVD optical disc storage technology on which media can be rewritten 100,000 times. It provides exceptional data integrity, data retention and damage protection and can be used for basic data storage, archiving data and data backup. Double-sided discs can store up to 9.4GB. DVD-RAM drives typically read DVD-Video, DVD-ROM and all types of CD media.

DVD-ROM (DVD-READ-ONLY MEMORY): A read-only DVD disc used to permanently store data files. DVD-ROM discs are widely used to distribute large software applications that exceed the capacity of a CD-ROM disc. DVD-ROM discs are read in DVD-ROM and DVD-RAM drives in computers, but typically not in DVD drives connected to TVs and home theatre systems. However, most DVD-ROM drives will play DVD movie discs.

DVD-RW (DIGITAL VERSATILE DISK REWRITABLE): A DVD disc for both movies and data, which can be erased and written to again. DVD-RWs can hold 4.7GB on one side.

DVD+R (DVD+RECORDABLE; DIGITAL VERSATILE DISK+ RECORDABLE): A recordable DVD format, which is write-once and read only. In other words, it can only record data once and then the data becomes permanent on the disc and can be read only. DVD+Rs hold up to 4.7GB of data per side and can be read by DVD-Video players and computer DVD-ROM drives. **DVD-RW** (**Digital Versatile Disk Rewritable**) is a rewritable DVD disc for both movies and data. Like CD-RWs, DVD-RWs must be erased in order for new data to be added. It can store 4.7GB per side and can be rewritten 1,000 times.

DVORAK KEYBOARD: A keyboard layout having the most common letters in home row. Since nearly all words have vowels, the home row on the Dvorak keyboard begins with the letters A, O, E, U, and I followed by D, H, T, N, and S. This keyboard has all the consonants on the left hand side and vowels and punctuation marks on the right hand side. This arrangement increases speed and reduce error in typing but has failed to replace the standard QWERTY layout. It was designed in the 1930s by August Dvorak and his brother-in-law, William Dealey.

DWARF: A plant or animal that is much shorter than others of its species, usually as a result of selective breeding. **Dwarf shoots** or spurs are shoots that develop from preformed buds which have very short internodal lengths or intervals.

DWARF NETTLE (ANNUAL NETTLE; SMALL NETTLE; *Urtica urens*): An annual stinging temperate weed, *Urtica urens* of the genus Urtica. It has medicinal and therapeutic uses and important in the diet of many farmland birds.

DWARF STAR: A star, such as the Sun, that typically has a surface temperature of 5456.85 °C (5730 K), a radius of 690,000 kilometres, a mass of 2 × 1033 grams and a luminosity of 4 × 1033 ergs per second. It lies on the main sequence Hertzsprung–Russell Diagram.

DWARFISM: A genetic condition in which an individual person or animal is undersized. For humans, it is when an individual is less than 127 cm in height. Some dwarfs have been less than 64 cm tall when fully grown.

DYAD: 1. A pair of sister chromatids. 2. A couple or a group of two, for example, in botany, it refers to pollen grains occurring in clusters of two. 3. In chemistry, it refers to an atom with a valency of two. 4. A term, now superseded by **tensors** for a pair of vectors written without an operator being indicated and which may form part of either a scalar product or a vector product. If **uv** is the dyad, **u**(**v.w**) is defined to be **u**(**v.w**),and **uv** x **w** is **u**(**v** x **w**).

DYAD SYMMETRY: Property of a structure that can be rotated through 180° to produce the same structure.

DYADIC RATIONAL (DYADIC FRACTION): A rational number whose denominator is a power of two, a number of the form $\frac{a}{2^b}$, where a is an integer and b is a natural number; for example, $\frac{1}{2}$ or $\frac{3}{8}$, but not $\frac{1}{3}$. These are precisely the numbers whose binary expansion is finite.

DYE: A chemical which can be mixed or react with materials to make a coloured fabric, plastics, etc. or a substance, which, when applied in solutions to fabrics, leather, paper and other materials imparts a colour resistance to washing. Most dyes are made from coal tar and petrochemicals. Their chemical structure is comparatively easy to modify, so many new colours and types of dyes have been synthesised. Fibre-reactive dyes form a covalent bond with the fibre. Other dyes require prior application of a mordant, an inorganic material that causes the dye to precipitate as an insoluble salt. Another method is vat dyeing, in which a soluble colourless compound is absorbed by the fibres, then oxidised to the insoluble coloured compound, which makes it to become remarkably resistant to the fading effects of washing, light and chemicals.

DYE LASER: A type of laser which uses an organic dye as the lasing medium. In this case, the active material is the organic dye dissolved in a suitable solvent such as Rhodanine G in methanol. A dye can usually be used for a much wider range of wavelengths, compared to gases and most solid state lasing media. Due to the wide bandwidth, they are very good for pulsed and tunable lasers. Although it usually requires replacing other optical components in the laser as well, the dye can be replaced by another type in order to generate different wavelengths with the same laser.

DYNAMIC BEHAVIOUR: The behaviour of a system or component under actual operating conditions, including acceleration and vibration.

DYNAMIC DATA EXCHANGE (DDE): In computing, it is a form of interprocess communication used in Microsoft Windows that uses shared memory as a common exchange area to provide exchange of commands and data between two applications. For example, when a form is changed in a database program or a data item in a spreadsheet program, they can be set up to also change these forms or items anywhere they occur in other programs that may used. A few of the thousands of applications using DDE include Microsoft's Excel, Word, Lotus 1-2-3, AmiPro, Quattro Pro and Visual Basic.

DYNAMIC EQUILIBRIUM: An equilibrium in which processes occur continuously at the same rate with no net change. This happens when a reversible reaction stops changing its ratio of reactants/products, but substances move between the chemicals at an equal rate, implying that there is no net change. It is a typical example of a system in a steady state.

DYNAMIC FIELD(S) MASS SPECTROMETER: A mass spectrometer in which the separation of a selected ion beam depends essentially on the use of fields or a field, varying with time. These fields are generally electric.

DYNAMIC HOST CONFIGURATION PROTOCOL (DHCP): A protocol that enables a server to automatically assign a temporary IP addresses to connected devices from a defined range configured for a network. The same temporary address may remain with a machine for a long time unless a conflict arises with other devices on the network.

DYNAMIC INTERNET PROTOCOL ADDRESS (DYNAMIC IP ADDRESS): A temporary automatically configured IP numeric identification that is assigned to a computing device or node by a Dynamic Host Configuration Protocol (DHCP) server, when it is connected to a network for the first time.

DYNAMIC LINK LIBRARY (DLL): A executable file used with MicrosoftWindows, Windows software and Windows drivers that allows programs to share code and other resources necessary to perform particular tasks. A DLL usually has a file extension ending in .dll. Other file extensions are .drv and .ocx. They are not loaded into RAM until they are needed and thus help reduce memory overhead. Applications that require DLL data receive it as required, which also helps manage memory.

DYNAMIC LOAD: A load associated with the elastic deformations of a structure subjected to time-dependent external forces.

DYNAMIC MEMBRANE FORMATION: A process in which an active layer is formed on the membrane surface by the deposition of substances contained in the fluid being treated.

DYNAMIC METEOROLOGY: The branch of meteorology concerned with motions in the atmosphere.

DYNAMIC PRESSURE: The pressure of a fluid, such as air that results from its motion or the pressure exerted on a body, because of its motion through a fluid. Dynamic pressure is equal to one-half the fluid density, ρ, multiplied by the square of the fluid's velocity: ($\frac{1}{2}\rho V^2$). In incompressible flow, the dynamic pressure is the difference between the total pressure and the static pressure. It is designated by q.

DYNAMIC PROGRAMMING: A mathematical technique in which very complex problems are broken down into smaller ones and solved in incremental steps so that, at any given stage, optimal solutions are known to sub-problems. When the technique is applicable, this condition can be extended incrementally without having to alter previously computed optimal solutions to subproblems. Eventually the condition applies to all of the data and, if the formulation is correct, this together with the fact that nothing remains untreated gives the desired answer to the complete problem.

DYNAMIC RANDOM ACCESS MEMORY (DRAM): A type of computer memory that stores each bit of data on only one transistor and one storage capacitor, thereby requiring less physical space to store the same amount of data than if it was stored statically. Therefore, a DRAM chip can hold more data than an SRAM (static RAM) chip of the same size. Its disadvantage is that the capacitors in DRAM must be refreshed often to keep their charge and so DRAM requires more power than SRAM. It is the most common type of computer memory.

DYNAMIC RANGE: The ratio between the maximum usable indication and the minimum usable indication (detection limit). A distinction may be made between the linear dynamic range, where the response is directly proportional to concentration and the dynamic range, where the response may be non-linear, especially at higher concentrations.

DYNAMIC REACTION PATH (DRP): A classical trajectory method based on molecular orbital calculations, which determines atomic accelerations, velocities and positions using the energy gradient and does not require prior knowledge of the potential energy surface.

DYNAMIC RESPONSE: The time-varying motion of a given structure to a given force input. For example, for a space vehicle that is exposed to side-acting winds during flight, the dynamic response would consist of a rigid body rotation of the vehicle plus a bending deformation, both changing with time.

DYNAMIC STABILITY: The property of a body, such as a rocket or plane, which, when disturbed from an original state of steady flight or motion, dampens the oscillations set up by the restoring movements and thus gradually returns the body to its original state.

DYNAMIC THERMOMECHANOMETRY: A technique in which the dynamic modulus and/or damping of a substance under oscillatory load is measured as a function of temperature, while the substance is subjected to a controlled temperature programme.

DYNAMIC VISCOSITY: For a laminar flow of a fluid, the ratio of the shear stress to the velocity gradient perpendicular to the plane of shear.

DYNAMICAL FRICTION (GRAVITATIONAL DRAG): The drag force between electrons and ions drifting with respect to each other. It is a term in astrophysics related to loss of momentum and kinetic energy of moving bodies through a gravitational interaction with surrounding matter in space.

DYNAMICAL METHOD: A method of measuring the mass of an object, such as a star, a galaxy or a cluster of galaxies, that makes use of the gravitational interactions of two or more bodies.

DYNAMICAL SYSTEM: A concept in mathematics, where a fixed rule describes the time dependence of a point in a geometrical space. Examples include the mathematical models that describe the swinging of a clock pendulum and the flow of water in a pipe.

DYNAMICAL TIME: The timescale used in calculations of orbital motions within the Solar System. The most widely used form of it, known as Terrestrial Time (TT), uses a fundamental 86,400 S.I. seconds (one day) as its fundamental unit and is related to International Atomic Time (TAI), by TT = TAI + 32.184 seconds.

DYNAMICAL TIMESCALE (FREEFALL TIMESCALE): The characteristic time it takes a cloud to collapse and release its gravitational energy.

DYNAMICS (KINETICS): The branch of mechanics concerned with the motion of bodies or the mathematical and physical study of the behaviour of bodies under the action of forces that produce changes of motion in them.

D

DYNAMITE: Any of a class of powerful explosives composed of nitroglycerin or ammonium nitrate dispersed in an absorbent medium with a combustible dope, such as wood pulp and an antacid, such as calcium carbonate, used in blasting and mining. It was devised by Alfred Nobel in 1867.

DYNAMO: A simple generator or machine for transforming mechanical energy into electrical energy. It consists of a powerful magnet field in which a coil of wire called the armature is rotated. The magnetic lines of force are cut by the rotating coil of wire, inducing a direct current to flow through the wire.

DYNAMO ACTION: The generation of electrical current and magnetic field by the motion of an electrically conducting fluid. It is generally believed that the magnetic fields of the Earth and the Sun are produced by dynamo action in the molten iron-nickel core of the Earth and in the plasma of the solar interior.

DYNAMIC PARTITION: A partition configured at the time of program execution according to the storage requirements of the application program or program to which the partition is allocated.

DYNAMO THEORY: A theory which proposes a mechanism by which a celestial body such as the Earth or a star generates a magnetic field. It describes the regular daily variations in the Earth's magnetic field in terms of electrical currents in the lower ionosphere, generated by tidal motions of the ionised air across the Earth's magnetic field.

DYNAMOMETER: 1. An instrument, often a spring balance, used to measure force, moment of force (torque) or the output power of an engine or motor. The power produced by an engine, motor or other rotating prime mover can be calculated by simultaneously measuring torque and rotational speed (rpm). 2. A variety of current balances for measuring electric current. 3. An instrument for measuring draft of tillage implements, and for measuring resistance of soil to penetration by tillage implements.

DYNASET: Abbreviation for **dynamic set**, a database sub-table that selects and sorts records as specified by a question and automatically reflects changes in underlying tables and makes changes to those tables.

DYNE: A centimetre–gram–second system (cgs) unit of force which is defined as the force that will accelerate a mass of one gram by one centimetre per second. Its symbol is dyn.

DYNEIN: A protein found in many eukaryotic cells, that possesses adenosine triphosphate ATPase activity and associated with the microtubules of undulipodia (slender cell extension used mainly for feeding and locomotion). They convert the chemical energy contained in adenosine triphosphate (ATP) into the mechanical energy of movement. They transport various cellular substances along cytoskeletal microtubules towards the minus-end of the microtubule, which is usually oriented towards the cell centre.

DYNORPHIN: A class of opioid peptides that arises from the precursor protein prodynorphin. They are produced as a neurotransmitter in various parts of the brain and spinal cord that binds to kappa opioid receptors. They are therefore in the class of endorphins and enkephalins known as endogenous opioids, causing pain relief among other effects. Although they are found widely distributed in the CNS, they have the highest concentrations in the hypothalamus, medulla, pons, midbrain and spinal cord.They are stored in large (80-120 nm diameter) dense-core vesicles that are considerably larger than vesicles storing neurotransmitters.

DYSENTERY: An infection of the large intestine characterised by abdominal cramps, pain, bloody mucous diarrhoea and often a fever. It may be caused by infection of bacteria, viruses, parasitic worms or protozoa. Amoebic dysentery is caused by an infection by the amoeba *Entamoeba histolytica*, which is found mainly in tropical areas and may lead to liver damage. Dysentey caused by shigellosis, an infection by bacteria of the genus Shigella is called bacillary dysentery. Dysentery causes severe dehydration and can be fatal, if not treated.

DYSLEXIA (WORD BLINDNESS): A malfunction in the brain's synthesis and interpretation of written information, characterised by difficulties in acquiring and processing language that is typically manifested by a lack of proficiency in reading, spelling and writing, although the person may excel in other areas, for example, in mathematics. A similar disability with figures is called **dyscalculia**. Acquired dyslexia may occur as a result of brain injury or disease.

DYSPEPSIA (INDIGESTION): A discomfort or pain in the upper abdomen, often with symptoms that include nausea and vomiting, bloating, heartburn, loss of appetite, heartburn, regurgitation of food or acid and belching. Some of the major causes include the following: ulcer, diabetes, thyroid disease and drugs such as ibuprofen, antibiotics and oestrogens.

DYSPHAGIA: Difficulty in swallowing, characterised by pain while swallowing, choking when eating, coughing or gagging when swallowing, drooling, recurrent heartburn, hoarseness, regurgitation, etc. It may be caused by circulatory or respiratory diseases, neurologic conditions such as stroke, muscle damage and a tumour or swelling that partially obstructs the passage of food.

DYSPNOEA: Difficulty in breathing or shortness of breath, often characterised by breathing rapidly and shallowly. It is mostly caused by circulatory or respiratory diseases.

DYSPROSIUM: A rare earth element with a metallic silver lustre, which is among the most magnetic of all known substances. Its symbol is Dy and atomic number 66. It has a great capacity to absorb neutrons. It is never found in nature as a free element, though it is found in various minerals, such as xenotime. The naturally occurring one is composed of 7 isotopes, the most abundant of which is 164Dy. Its insoluble salts are considered non-toxic while its soluble salts are mildly toxic. Due to its high thermal neutron absorption cross-section it is used in making control rods in nuclear reactors. Also, as a result of its high magnetic susceptibility to magnetisation it is used in data storage devices and as a component of Terfenol-D.

DYSTETIC MIXTURE: A mixture of substances that has a constant maximum melting point or a mixture of two or more substances that has the highest possible melting point compared to the melting point of the individual substances.

DYSTROPHIC: A body of water, such as a lake, that contains large amounts of undecomposed organic matter derived from terrestrial plants. They are poor in dissolved nutrients; contain very acidic brown or tea-coloured water, with the colour being the result of high concentrations of humic substances and organic acids suspended in the water. These bodies of water vary greatly in terms of both pH and productivity; however, they generally lack oxygenand are unable to support plant or animal life due to the excessive humus content and acidity.

DYSTROPHIN: A structural protein found in small amounts (0.002 %) in normal skeletal muscles, which is absent or present in excess in individuals with muscular dystrophy. It is important for the proper functioning of the muscle.

E

e: 1. In mathematics, it is used to denote the mathematical exponent function e^x. 2. Often called **Euler's number** or **Napier's constant**, it is the mathematical constant, with the approximate value of 2.718281828, which is a real and irrational number. It is one of the most important numbers in mathematics and has many definitions; a few are as follows:

(i) the e constant is the limit $e = \lim_{x\to\infty}\left(1+\frac{1}{n}\right)^n$;

(ii) in terms of infinite series as $e = \sum_{n=0}^{\infty}\frac{1}{n!} = \frac{1}{0!}+\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\ldots$;

(iii) in terms of the derivative of the exponential function as the exponential function $(e^x)' = e^x$;

(iv) in terms of the derivative of the natural logarithm function as the reciprocal function $(\log_e x)' = (\text{In}\, x)' = 1/x$;

(v) and in terms of the exponential function as $f(x) = \exp(x) = e^x$.

3. In IT, it is used to indicate electronic or digital format or version, for example, e-book and e-commerce.

4. The notation for eccentricity in conics.

E: 1. The number 14 in hexagonal system. 2. In duodecimal notation, it is the number 11. 3. The complete elliptic integral of the second kind, a special function with the following formula: $\int_0^{\pi/2}\left(1-k^2\sin^2\theta\right)^{\frac{1}{2}}\,d\theta$, which can be used to derive the arc length of an ellipse. 4. In statistics, it is the operator that yields the expected value of a random variable.

E-BOOK (ELECTRONIC-BOOK): Any book, which is published in a digital form and readable on a desktop computer, laptop, smartphone or e-book reader. **E-book reader** is a small, portable device onto which the contents of a book in electronic format can be downloaded and read. Electronic bookmarks make referencing easier and e-book readers may allow the user to annotate pages.

E-BUSINESS (ELECTRONIC BUSINESS; E-COMMERCE; ELECTRONIC COMMERCE): Businesses or commercial transactions conducted over the Internet. Examples are web sites such as Amazon.com, Buy.com and eBay.

E-CELL: A computer stimulation of a living cell based on information obtained from genomics, proteomics, metabolomics and other data drawn from real cells.

E-CLASS ASTEROID: A rare category of asteroid, slightly red in colour and spectrally similar to M-class asteroids and P-class asteroids, but of higher albedo (in the range 0.25-0.60). E-class asteroids dominate the Hungaria group of the main belt, though some, including the two largest, (44) Nysa Angelina, are found scattered further out. Their flat reflectance spectra in the 0.3 to 1.0 micron region and high albedos suggest a connection with the aubrite (otherwise known as enstatite achondrite, hence the "E") category of meteorite. However, this link is problematic because aubrites are clearly of igneous origin, whereas some E-class asteroids (including Nysa) show an absorption feature at 3 microns that points to water- and/or hydroxyl-bearing minerals and a non-igneous history.

E. coli*:** Abbreviation for ***Escherichia coli, a species of gram-negative aerobic flagellated bacteria in vertebrate intestine also occurring in freshwater bodies and soil; and existing as numerous strains, some of which are responsible for diarrhoea. It is the bacteria most widely used in microbiological and genetic research. Some types can cause food poisoning, but proper cooking of meat and washing of produce can prevent infection from contaminated food sources.

E-COMMERCE (ELECTRONIC-COMMERCE; E-BUSINESS; ELECTRONIC-COMMERCE): Businesses or commercial transactions conducted over the Internet. Examples are web sites such as Amazon.com, Buy.com and eBay.

E-CYCLING (ELECTRONICS RECYCLING): The practice of donating old computers and electronic devices such as cell and TVs phones that can be re-used, saving parts that can be reused or recycling components rather than discarding them at the end of their life cycle. Many computer, TV and cell phone manufacturers, as well as electronics retailers offer some kind of take back program or sponsor recycling events.

E-MAIL (ELECTRONIC MAIL): Correspondence sent via a computer network. It includes messages, documents, texts, graphics, sounds, animated images, etc., from one user of a computer system to another. E-mail is part of the standard TCP/IP set of protocols. Sending messages is typically done by SMTP (Simple Mail Transfer Protocol) and receiving messages is handled by POP3 (Post Office Protocol 3) or IMAP (Internet Message Access Protocol). IMAP is the newer protocol, allowing users to view and sort messages on the mail server, without downloading them to the hard drive.

E-MAIL BOMB (MAIL BOMB): A form of internet abuse, where a user or group of users send a massive amount of e-mails to a single e-mail address or computer network, usually filling up the recipient's disk space or causing a server or system to crash.

E-MAIL SERVER (MAIL SERVER): In computing, it is a computer or software in a network that stores incoming e-mail and distributes it to users and also sends outgoing mail.

E-PUBLISHING (DIGITAL PUBLISHING; ELECTRONIC PUBLISHING): Distribution of information such as of e-books and digital magazines using computer-based media such as diskettes, CD-ROMs, DVDs, portable document files (PDF) or over the internet.

E-READER (E-BOOK READER): A special handheld hardware device for reading digital publications such as e-books, electronic magazines and digital versions of newspapers. Since textual data does not require a lot of storage space, most e-readers can store thousands of books and other publications. Just like an iPod can store an entire music library, a single e-reader can store a large collection of books.

E-WASTE: Abbreviation for **electronic waste**, which refers to electronic equipment such as monitors, computers and cell phones that are thrown away or otherwise disposed off without any consideration of e-cycling. This may cause environmental problems, as electronic equipment frequently contains hazardous substances.

E-Z CONVENTION (E-Z NOTATION): The IUPAC's method of describing the stereochemistry of double bonds having three or four substituents or of describing the geometric isomeric alkenes to the cis-trans convention. It is applied when the groups attached to the double bonds are different. The groups attached to each double bond are ranked according to the mass of the atoms directly bonded to them and following a set of defined rules (Cahn-Ingold-Prelog priority rules), each substituent on a double bond is assigned a priority. If the bulky groups attached to the two carbons are on the same size of the double bond, then that double bond is assigned *Z*. If the bulky groups are on opposite sides of the double bond, the bond is assigned *E*. The letters *E* and *Z* are conventionally printed in italic type, within parentheses and separated from the rest of the name with a hyphen. They are always printed as full capitals (not lower case or small capitals). The letters are used in the names of compounds, for example, (*E*)-butenedioic acid (fumaric acid) and (*Z*)-butenedioic acid (maleic acid). In compounds containing two or more double